THE THIRD LARGEST NORMALIZED VOLUME OF KLT SINGULARITIES

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Abstract. We prove that a klt singularity of dimension at least four which is neither smooth nor an ordinary double point has normalized volume at most that of the product of the affine line and the ordinary double point of dimension one less. Since smooth points and ordinary double points have the largest and the second largest normalized volumes, the normalized volume of the product is the third largest normalized volume of klt singularities in each dimension at least four.

Contents

1. Introduction 1
2. Preliminaries 7
3. Hypersurfaces, complete intersections and products 13
4. Reduction to Gorenstein cones 18
5. Klt singularities of dimension four 22
6. Large stabilizers and jet semigroups 40
7. The Taylor degeneration and order-two slices 48
8. The fixed locus in the Taylor degeneration 61
9. Graded algebras with a small fixed locus 69
10. Proof of the main theorem 77
Appendix A. An exponential bound in four variables 78
References 82

1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

The normalized volume [Li18] of a klt germ $(X \ni x)$ of dimension d is the infimum of $a(v,X)^d \operatorname{vol}(v)$ over the real valuations v of $\mathbb{C}(X)$ centered at x with $a(v,X) < +\infty$, where $a(v,X)$ is the log discrepancy of v and $\operatorname{vol}(v)$ is the volume of v (see Definition 2.2). We denote it by $\widehat{\operatorname{vol}}(X \ni x)$. The largest value is attained only at smooth points: $\widehat{\operatorname{vol}}(X \ni x) \leq d^d$, and equality holds if and only if x is a smooth point of X [LX19, Theorem 1.6].

The second largest value is the subject of a conjecture of Spotti–Sun. The ordinary double point of dimension d is the germ of $\{z_0^2 + \cdots + z_d^2 = 0\} \subset \mathbb{A}^{d+1}$ at the origin (see Definition 2.3). Spotti–Sun proposed the following conjecture while studying Gromov–Hausdorff limits of Kähler–Einstein Fano manifolds.

Conjecture 1.1 ([SS17, Conjecture 5.5])**.** *Let $(X \ni x)$ be a klt germ of dimension d which is not smooth. Then $\widehat{\operatorname{vol}}(X \ni x) \leq 2(d-1)^d$, and equality holds only if $(X \ni x)$ is analytically an ordinary double point.*

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Spotti–Sun also formulated a metric version of Conjecture 1.1 for Ricci-flat Kähler cones [SS17, Conjecture 1.2]. Conjecture 1.1 appears as an open problem in [XZ26, Section 3.2, Problem 33]. It was known in the following cases.

(1) $d = 2$ [LL19, Proposition 4.10].
(2) $d = 3$ [LX19, Theorem 1.3(1)].
(3) $(X \ni x)$ is a local complete intersection germ, in any dimension [Liu22, Theorem 2.12]. Moreover, a local complete intersection klt germ $(X \ni x)$ of dimension d satisfies $\widehat{\mathrm{vol}}(X \ni x) = 2(d-1)^d$ if and only if $(X \ni x)$ is analytically an ordinary double point [Liu22, Theorem 2.12].
(4) $(X \ni x)$ is a $\mathbb{Q}$-Gorenstein toric germ. If $d \geq 3$ and $(X \ni x)$ is not smooth, then $\widehat{\mathrm{vol}}(X \ni x) \leq 16d^d/27$, and equality holds if and only if $(X \ni x)$ is isomorphic to the germ of $\{z_0^2 + \cdots + z_3^2 = 0\} \times \mathbb{A}^{d-3}$ at the origin [MS24, Theorem 2(2)]. For $d \geq 4$ the bound $16d^d/27$ is less than $2(d-1)^d$ (see Lemma 2.7(3)).
(5) $(X \ni x)$ is the germ at the vertex of the affine cone

$$C(Y, L) := \mathrm{Spec} \bigoplus_{m \geq 0} H^0(Y, mL),$$

where Y is a K-semistable Fano manifold and L is an ample line bundle on Y with $-K_Y \sim rL$ for an integer $r > 0$ [LM25, Theorem 4.17].

Conjecture 1.1 was proved in dimension four [OAI26a, Theorem 1.1] and in all dimensions [OAI26b, Theorem 1.1], including the equality case. The proofs make no assumption of isolatedness, of the Gorenstein property, or of being a hypersurface or toric germ. Thus, for any $d \geq 2$, the largest normalized volume of a klt germ of dimension d is d^d, attained only at smooth points, and the second largest is $2(d-1)^d$, attained only by germs which are analytically an ordinary double point.

By [XZ24, Theorem 1.2], for a fixed d, the normalized volumes of klt germs of dimension d accumulate only at 0. Hence it makes sense to ask for the largest normalized volume of a klt germ of dimension d which is neither smooth nor analytically an ordinary double point.

We now explain what was known below the ordinary double point. In dimension three, the A_k singularity, that is, the germ of $\{z_0^2 + z_1^2 + z_2^2 + z_3^{k+1} = 0\} \subset \mathbb{A}^4$ at the origin, has normalized volume 16, 125/9, and 27/2 for $k = 1$, $k = 2$, and $k \geq 3$, respectively [Li18, Section 5], [LL19, Example 4.7], [LX20, Example 7.1(2)] (see [Liu25, Example 5.2(1)]). By [Liu25, Theorem 1.2, Proposition 4.3, and Example 5.2] and the formula $\widehat{\mathrm{vol}}(\mathbb{A}^3/G \ni 0) = 27/|G|$ for a finite group $G \subset \mathrm{GL}_3(\mathbb{C})$ acting freely in codimension one [Liu25, Example 2.7], the third largest normalized volume of a klt germ of dimension three is 125/9, attained by the A_2 singularity. Indeed, a klt germ $(X \ni x)$ of dimension three with $\widehat{\mathrm{vol}}(X \ni x) \geq 9$ is either a hypersurface germ of type cA_m with $m \leq 2$ (a smooth germ counts as type cA_0 [Liu25, Section 2.2]) or a cyclic quotient singularity of type $\frac{1}{2}(1,1,1)$, $\frac{1}{3}(1,1,0)$, $\frac{1}{3}(1,1,1)$, or $\frac{1}{3}(1,1,2)$ [Liu25, Theorem 1.2]. The cyclic quotient singularities among them have normalized volume 27/2 or 9. The A_k singularities above are of type cA_1 [Liu25, Section 2.2]. A germ of type cA_2 has normalized volume at most 32/3 [Liu25, Proposition 4.3], and a germ of type cA_1 which is not an A_k singularity has normalized volume 27/2 [Liu25, Example 5.2(2)]. Moreover, the normalized volumes of klt germs of dimension three which are at least 9 are exactly the ten values listed in [Liu25, Question 5.3] [Liu26, Theorem 1.3], and a klt germ of dimension three which is not a finite quasi-étale quotient of a local complete intersection germ has normalized volume at most $(46 + 13\sqrt{13})/12$, with equality only for the vertex of the anti-canonical cone over the first Hirzebruch surface [Liu26, Theorem 1.4].

In dimension $d \geq 4$, the known bounds below $2(d-1)^d$ depend on the structure of the germ. For an integer $d \geq 4$ let

$$Q_d := 2d^d \Big(\frac{d-2}{d-1}\Big)^{d-1}.$$

For hypersurface and local complete intersection klt germs $(X \ni x)$ of dimension $d \geq 4$ which are neither smooth nor analytically an ordinary double point, the bound $\widehat{\mathrm{vol}}(X \ni x) \leq Q_d$ is implicit in [Liu22] (see Remark 3.3).

Let $(X \ni x)$ be a klt germ of dimension $d \geq 4$ such that $\dim_x \operatorname{Sing} X \geq 1$. By [Liu22, Theorem 2.16], there exists a klt germ $(Y \ni y)$ of dimension $d-1$ which is not smooth and satisfies $\widehat{\mathrm{vol}}(X \ni x)/d^d \leq \widehat{\mathrm{vol}}(Y \ni y)/(d-1)^{d-1}$ (see also Lemma 2.5). Since $\widehat{\mathrm{vol}}(Y \ni y) \leq 2(d-2)^{d-1}$ by [OAI26b, Theorem 1.1] in dimension $d-1$, we obtain

$$\widehat{\mathrm{vol}}(X \ni x) \leq d^d \cdot \frac{2(d-2)^{d-1}}{(d-1)^{d-1}} = Q_d,$$

and for $d = 4$ the bound $\widehat{\mathrm{vol}}(X \ni x) \leq Q_4$ is [Liu22, Corollary 2.18]. If $\dim_x \operatorname{Sing} X \geq d-3$, then $\widehat{\mathrm{vol}}(X \ni x) \leq 16d^d/27$ [Liu22, Corollary 2.18]. The sequence $(Q_d/d^d)_{d\geq 4}$ is strictly increasing and $Q_4/4^4 = 16/27$ (see Lemma 2.7(1)(3)). Thus $16d^d/27 \leq Q_d$, with equality only for $d = 4$. In particular, for $d = 4$ the toric bound $16 \cdot 4^4/27$ of [MS24, Theorem 2(2)] is equal to Q_4.

The bound Q_d is attained. For a klt germ $(Y \ni y)$ of dimension $d-1$,

$$\widehat{\mathrm{vol}}(Y \times \mathbb{A}^1 \ni (y,0)) = \frac{d^d}{(d-1)^{d-1}} \widehat{\mathrm{vol}}(Y \ni y)$$

[LZ18, Lemma 4.4], [Liu22, Proposition 2.17]. If $(Y \ni y)$ is the ordinary double point of dimension $d-1$, then $\widehat{\mathrm{vol}}(Y \ni y) = 2(d-2)^{d-1}$, and hence $\widehat{\mathrm{vol}}(Y \times \mathbb{A}^1 \ni (y,0)) = Q_d$. The germ $(Y \times \mathbb{A}^1 \ni (y,0))$ is a hypersurface germ, and $\dim_{(y,0)} \operatorname{Sing}(Y \times \mathbb{A}^1) = 1$. Thus the bound for hypersurface germs implicit in [Liu22] and the bound for germs whose singular locus has positive dimension are both attained. A related sharp bound holds in any dimension: for a klt germ $(X \ni x)$ of dimension $d \geq 2$, $\widehat{\mathrm{vol}}(X \ni x)$ is at most d^{d-1} times the minimal log discrepancy of X at x [Han26, Theorem 1.1].

The klt germs of dimension $d \geq 4$ which remain are those with an isolated singularity at x which are not local complete intersections, and for them no equations are available. For klt germs without equations, the method of [OAI26a] and [OAI26b] degenerates $(X \ni x)$ to an isolated Gorenstein cone with a torus action and studies integral gradings of the cone close to the grading that minimizes the normalized volume. The arguments of [OAI26a] and [OAI26b] are written for the threshold of the ordinary double point: [OAI26a, Proposition 2.2] constructs the cone under the assumption $\widehat{\mathrm{vol}}(X \ni x) \geq 162$, and [OAI26b, Sections 4–7] assume that the normalized volume of the cone at the vertex is at least $2(d-1)^d$. The threshold in this paper is lower. Indeed, $Q_4 = 4096/27 < 162$, and in general $Q_d < 2(d-1)^d$, where the ratio

$$\frac{Q_d}{2(d-1)^d} = \left(1 + \frac{1}{d-1}\right)\left(1 - \frac{1}{(d-1)^2}\right)^{d-1}$$

is equal to $2048/2187$ for $d = 4$, increases with d, and tends to 1 as $d \to +\infty$ (see Lemma 2.7(1)(3)(5)). The formula for the ratio takes the value $2048/2187$ at $d = 4$. By Bernoulli's inequality, $(1-(d-1)^{-2})^{d-1} \geq 1-(d-1)^{-1}$, so the formula for the ratio implies $1 - \frac{1}{(d-1)^2} \leq \frac{Q_d}{2(d-1)^d}$, and the ratio is less than 1 by Lemma 2.7(5). Hence $(1-(d-1)^{-2})^{d-1} \to 1$, and the ratio tends to 1.

The goal of this paper is to determine the largest normalized volume of a klt germ of dimension $d \geq 4$ which is neither smooth nor analytically an ordinary double point (see Theorem 1.2).

1.1. **Main results.** In the following, a klt germ $(X \ni x)$ consists of a normal variety X and a closed point $x \in X$ such that X is klt near x with zero boundary, and $\widehat{\mathrm{vol}}(X \ni x)$ denotes the normalized volume (see Definition 2.2). Ordinary double points of dimension d, and germs that are analytically ordinary double points, are defined in Definition 2.3.

Theorem 1.2. *Let $d \geq 4$ be an integer and let*

$$Q_d := 2d^d \left(\frac{d-2}{d-1}\right)^{d-1}.$$

(1) Let $(X \ni x)$ be a klt germ of dimension d. Assume that $(X \ni x)$ is neither smooth nor analytically an ordinary double point. Then $\widehat{\mathrm{vol}}(X \ni x) \leq Q_d$.
(2) Let $T := \{z_1^2 + \cdots + z_d^2 = 0\} \subset \mathbb{A}^d$ and let $o \in T$ be the origin. Then $(T \times \mathbb{A}^1 \ni (o,0))$ is a klt germ of dimension d which is neither smooth nor analytically an ordinary double point, and $\widehat{\mathrm{vol}}(T \times \mathbb{A}^1 \ni (o,0)) = Q_d$.

The germ $(T \ni o)$ is the ordinary double point of dimension $d-1$, and $\widehat{\mathrm{vol}}(T \ni o) = 2(d-2)^{d-1}$ by [Liu22, Theorem 2.12]. Thus

$$\frac{Q_d}{d^d} = \frac{2(d-2)^{d-1}}{(d-1)^{d-1}} = \frac{\widehat{\mathrm{vol}}(T \ni o)}{(d-1)^{d-1}},$$

and Theorem 1.2(1) is the inequality

$$\frac{\widehat{\mathrm{vol}}(X \ni x)}{d^d} \leq \frac{\widehat{\mathrm{vol}}(T \ni o)}{(d-1)^{d-1}}$$

for any klt germ $(X \ni x)$ of dimension d that is neither smooth nor analytically an ordinary double point. By [Liu22, Proposition 2.17], the product germ of Theorem 1.2(2) attains the bound (Proposition 3.5).

Combining Theorem 1.2 with Theorem 2.4, we obtain the third largest value.

Corollary 1.3. *Let $d \geq 4$ be an integer. Then $Q_d < 2(d-1)^d < d^d$, and the normalized volumes of klt germs of dimension d that are at least Q_d are exactly Q_d, $2(d-1)^d$, and d^d.*

We prove Theorem 1.2 and Corollary 1.3 in Section 10.

Remark 1.4. Theorem 1.2 does not hold for $d = 3$. Let $V := \{z_1 z_2 + z_3^2 + z_4^3 = 0\} \subset \mathbb{A}^4$. The germ $(V \ni 0)$ is isomorphic to the A_2 singularity of dimension three. It is neither smooth nor analytically an ordinary double point, and $\widehat{\mathrm{vol}}(V \ni 0) = 125/9$ by [Liu25, Example 5.2(1)]. The value at $d = 3$ of the formula defining Q_d is $27/2 < 125/9$.

Indeed, $z_1 z_2 + z_3^2 + z_4^3 \in \mathfrak{m}^2$, where $\mathfrak{m} = (z_1, z_2, z_3, z_4)$, so 0 is a singular point of V. The tangent cone of V at 0 is $\{z_1 z_2 + z_3^2 = 0\} \subset \mathbb{A}^4$, which is singular along the z_4-axis. On the other hand, the tangent cone of the ordinary double point of dimension 3 is $\{z_0^2 + \cdots + z_3^2 = 0\} \subset \mathbb{A}^4$, which is singular only at the origin. Since the tangent cone is determined by the completed local ring, which is also the completion of the analytic local ring [Ser56, Proposition 3], it depends only on the analytic germ, and hence $(V \ni 0)$ is not analytically an ordinary double point.

The substitution $z_1 = y_0 + iy_1$, $z_2 = y_0 - iy_1$ transforms $z_1 z_2 + z_3^2 + z_4^3$ into $y_0^2 + y_1^2 + z_3^2 + z_4^3$, so $(V \ni 0)$ is isomorphic to the A_2 singularity of dimension three, and the germ at 0 of the Cartier divisor $V \cap \{z_4 = 0\}$ of V is isomorphic to the ordinary double point of dimension two. Hence $(V \ni 0)$ is a klt germ by inversion of adjunction [KM98, Theorem 5.50] (we omit the proofs that V is normal with K_V Cartier, being a hypersurface in $\mathbb{A}^4$ singular only at 0 [Har77, II, Proposition 8.23], that the ordinary double point of dimension two is canonical, hence klt [KM98, Theorem 4.20], and that V is klt near 0 once $(V, V \cap \{z_4 = 0\})$ is plt near 0 [KM98, Corollary 2.35(1)]).

The germ $(V \ni 0)$ is a hypersurface singularity whose quadratic part $z_1 z_2 + z_3^2$ has rank 3. Proposition 3.1 proves the inequality of Theorem 1.2(1) for klt germs of dimension $d \geq 4$ with embedding dimension at most $d+1$, including those whose quadratic part is degenerate, and the proof of Proposition 3.1 uses the hypothesis $d \geq 4$.

Sketch of the proof. Theorem 1.2(2) is Proposition 3.5. Suppose that $(X \ni x)$ violates Theorem 1.2(1). Construction-Proposition 4.1 degenerates $(X \ni x)$ to a Gorenstein cone $(C \ni o)$ with a torus action, an isolated singularity, $\widehat{\mathrm{vol}}(C \ni o) = \widehat{\mathrm{vol}}(X \ni x)$, and $\mathrm{edim}(C \ni o) \geq \mathrm{edim}(X \ni x)$, where edim denotes the embedding dimension (Definition 2.3). The cone $(C \ni o)$ is not a local complete intersection (Construction-Proposition 4.1(4)).

(1) If $d = 4$, Propositions 5.5 and 5.7 imply $\mathrm{lct}(C \ni o; \mathfrak{m}_o) > 3/2$ (Proposition 5.9), so $\mathrm{edim}(X \ni x) \leq \mathrm{edim}(C \ni o) \leq 5$ (Proposition 5.10), which contradicts Proposition 3.1.
(2) If $d \geq 5$, Propositions 6.4 and 6.10 find a point of $C \backslash \{o\}$ with a large stabilizer, at which the leading terms of Taylor expansions are exactly the points of the lattice they generate in the cone they span (Proposition 6.10(3)).
(3) Taylor expansions at that point degenerate C to $\operatorname{Spec} D \times \mathbb{A}^1$ with D reduced, $\mathrm{Hilb}_D(t) = (1-t)^{1-d}$, and non-trivial stabilizers μ_2 on $\operatorname{Spec} D$ (Construction-Propositions 7.1 and 7.4, Lemma 7.3, Proposition 7.10).
(4) The fixed subscheme Z of $-1 \in \mathbb{G}_m$ on $\operatorname{Spec} D$ is a curve (Theorem 8.6), so D is a complete intersection at the vertex (Theorem 9.1), and so is C (Lemma 7.6), a contradiction.

The key new step is Lemma 8.1: by two hyperplane sections and the concavity

$$p_{d-2} + p_d < 2p_{d-1} = 2Q_d/d^d,$$

where $p_j := 2((j-1)/j)^j$ for $j \geq 2$ (Lemma 2.7(1) and (4)), the inequality $\widehat{\mathrm{vol}}(C \ni o) > Q_d$ makes the singular loci of the slices of Construction-Proposition 7.7 at most one-dimensional, whence $\dim Z \leq 2$ (Construction-Proposition 8.3(2) and (3)).

The argument is not an induction on d: lower-dimensional volume bounds enter only through Theorem 2.4(2). Of the volume arguments of [OAI26b], only [OAI26b, Theorem 4.8] for $d = 5$ and [OAI26b, Theorem 7.7] (replaced by Theorems 8.6 and 9.1) fail at Q_d. On the route followed here, the other places where [OAI26b] uses its volume assumption, namely [OAI26b, Proposition 2.5], the first inequality of [OAI26b, Lemma 4.3], and [OAI26b, Lemmas 6.3 and 7.2], are replaced under $\widehat{\mathrm{vol}}(X \ni x) > Q_d$ by Construction-Proposition 4.1, Lemma 6.8, Lemma 7.3, and Proposition 7.9 (see Remarks 6.3 and 6.6). For $d = 4$ the new arguments are Propositions 5.5 and 5.7.

The hypothesis that $(X \ni x)$ is not analytically an ordinary double point enters only through Proposition 3.1 for $d = 4$, and only through Construction-Proposition 4.1(4) for $d \geq 5$, namely in the final contradiction and in Proposition 6.4. In the proof of Proposition 6.4, if all stabilizers were small, then [OAI26b, Theorem 3.1] would imply that $(C \ni o)$ is smooth or an ordinary double point. In [OAI26b, Proof of Theorem 7.7], the ordinary double points are excluded by their normalized volume $2(d-1)^d$, which does not contradict $\widehat{\mathrm{vol}}(C \ni o) > Q_d$, so we use Construction-Proposition 4.1(4) instead.

(a) The canonical vector for $d = 5$. To show that the canonical vector $c = (r, 1, \ldots, 1)$ of Construction-Proposition 6.7, where r is the degree of a homogeneous generator of the canonical module of C, is in the interior of the cone spanned by the Taylor leading terms, the proof of [OAI26b, Theorem 4.8] compares the constant $C_* = \frac{3}{4}e^{4/3} - \frac{e}{3}$ of [OAI26b, Theorem 4.4] with a lower bound 2 provided by the volume assumption. At the threshold Q_d, the lower bound is $2\rho_d$ (Lemma 6.8) with $\rho_d = Q_d/(2(d-1)^d)$ (Lemma 2.7(1)), and $2\rho_5 = 253125/131072 < C_*$. The constant C_* is a supremum over the number of variables, approached as the number of variables tends to infinity [OAI26b, Theorem 4.4], whereas $d = 5$ involves four variables. In four variables, Lemma 6.9 improves C_* to $137/72 < 2\rho_5$, and it is proved in Appendix A by reduction to a function of one variable. For $d \geq 6$, $C_* < 2\rho_6 \leq 2\rho_d$ (Lemma 2.7(3)), and [OAI26b, Theorem 4.4] suffices.

(b) Slices at points with stabilizer μ_2. For the slice $(Y \ni y)$ of Construction-Proposition 7.7 at a point of $\operatorname{Spec} D$ with stabilizer μ_2, a singular klt germ of dimension d (Proposition 7.9),

Proposition 7.10 implies

$$\frac{\widehat{\mathrm{vol}}(C \ni o)}{d^d} \leq \frac{p_d}{2} + \frac{\widehat{\mathrm{vol}}(Y \ni y)}{2d^d}.$$

Using [OAI26b, Lemma 7.6], the proof of [OAI26b, Theorem 7.7] bounds $\widehat{\mathrm{vol}}(Y \ni y)/d^d$ by the supremum M of $\widehat{\mathrm{vol}}/d^d$ over singular klt germs of dimension d and obtains $M \leq p_d/2 + M/2$, that is, $M \leq p_d$. At the threshold Q_d, the inequality $M \leq p_d/2 + M/2$ yields nothing, because the fixed point p_d of $t \to p_d/2 + t/2$ is larger than $p_{d-1} = Q_d/d^d$. Instead, if $\dim_y \mathrm{Sing}(Y) \geq 2$, Lemma 8.1 cuts $(Y \ni y)$ twice through its singular locus (Lemma 2.5), and Theorem 2.4(2) in dimension $d-2$ implies $\widehat{\mathrm{vol}}(Y \ni y)/d^d \leq p_{d-2}$, so $\widehat{\mathrm{vol}}(C \ni o) < Q_d$ by Lemma 2.7(4).

To exclude two-dimensional components of Z, we use parity. If $\mathrm{Spec}\, D$ is a hypersurface germ at a point $z \in Z$ other than the vertex, then $\dim_z Z$ is odd (Lemma 8.5). The points of $\mathrm{Spec}\, D$ of embedding codimension at least 2 form a locus of dimension at most 1 near Z (Lemma 8.2, Construction-Proposition 8.3(4)), so $\dim Z = 1$ (Theorem 8.6).

(c) Recognition of D. If the stabilizers on $\mathrm{Spec}\, D$ outside the vertex are trivial, then $\mathcal{O}(1)$ is invertible on $\mathrm{Proj}\, D$, and D is a polynomial ring by [OAI26b, Lemma 6.5]. Here $\mathrm{Spec}\, D$ always has points with stabilizer μ_2 (Lemma 7.5), and Theorem 9.1 is the order-two version of [OAI26b, Lemma 6.5]. As $\dim Z = 1$, we take an odd k such that some element of D_{2k} vanishes at no point of Z other than the vertex, and use it to form a double cover that removes the stabilizers μ_2 (Construction-Proposition 9.3). Then [Fuj82, Theorem (2.1)] shows that D_1 has a finite base locus on $\mathrm{Proj}\, D$ (Lemma 9.4), which would be empty if all stabilizers outside the vertex were trivial and is one point for the quadric hypersurfaces of Theorem 9.1. After cutting down to dimension one by general elements of D_1, Lemma 9.5 shows that D is generated by D_1 and one element of degree 2, and comparing Hilbert series leaves at most one quadratic relation.

(d) Dimension four. The argument for $d \geq 5$ depends on [OAI26b, Theorem 3.1], which assumes $d \geq 5$ (Remark 6.3), and [OAI26b] takes the case $d = 4$ from [OAI26a]. The stabilizer bound in [OAI26a, Proof of Proposition 3.4] uses a margin provided by $\widehat{\mathrm{vol}}(X \ni x) \geq 162$, independent of the cone. Here we take the largest constant $\alpha = Q_4/64 = 64/27$ for which $\widehat{\mathrm{vol}}(X \ni x) > Q_4$ implies $\widehat{\mathrm{vol}}(C \ni o)/\alpha > 64$ (Set-up 5.2), in place of the constant $5/2$ of that proof, so that the interval $(0, 2\alpha/3)$ of weights in which Proposition 5.7 finds a non-zero homogeneous element is as wide as possible. The margin $\widehat{\mathrm{vol}}(C \ni o)/\alpha - 64$ can then be arbitrarily small, and Proposition 5.5 works with the fixed cone. The interval $(0, 2\alpha/3)$ shrinks from $(0, 5/3)$ in [OAI26a, Proposition 4.3] to $(0, 128/81)$ in Proposition 5.7. The test functions of [OAI26a, Proof of Proposition 4.3] still work, after recentering the one in dimension three, and Lemma 5.6 proves the inequalities needed for the one in dimension four. The last step uses Proposition 3.1 for $(X \ni x)$, where [OAI26a] uses [Liu22, Theorem 2.12].

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce Li's normalized volume, recall the upper bounds d^d for all klt germs [LX19, Theorem 1.6] and $2(d-1)^d$ for singular ones [OAI26b, Theorem 1.1], prove a refined form of Liu's inequality for hyperplane sections through the singular locus [Liu22, Theorem 2.16] (Lemma 2.5), collect the numerical constants built from the normalized volumes of ordinary double points (Lemma 2.7), and recall the Reeb vector and the weight valuations of a cone with a torus action.

In Section 3 we bound the normalized volumes of hypersurface singularities and of local complete intersection singularities, and compute the normalized volume of the product of the ordinary double point of dimension $d-1$ with $\mathbb{A}^1$.

Section 4 degenerates a klt germ $(X \ni x)$ violating Theorem 1.2(1) to a Gorenstein cone with a torus action whose normalized volume at the vertex equals $\widehat{\mathrm{vol}}(X \ni x)$, and whose vertex is an isolated singularity and is not a local complete intersection.

Section 5 proves Theorem 1.2(1) for $d = 4$.

For $d \geq 5$, Section 6 finds points of the cone whose stabilizers under the $\mathbb{G}_m$-action of a one-parameter subgroup of the torus are large, and shows that the leading terms of the Taylor expansions of homogeneous functions at such a point form a saturated semigroup.

Section 7 degenerates the cone to a graded ring, cuts the graded ring by one function to obtain a graded quotient ring of dimension $d-1$ with the Hilbert series of a polynomial ring in $d-1$ variables, and studies the points of the quotient ring with non-trivial stabilizers under the $\mathbb{G}_m$-action given by the grading.

Section 8 shows that the fixed locus of the involution of the quotient ring given by the grading is a curve.

Section 9 shows that a graded algebra satisfying the conditions obtained in Section 7, whose points other than the vertex have stabilizers of order at most 2 and whose fixed locus of the involution given by the grading is at most a curve, is either a polynomial ring or a hypersurface defined by a quadratic form.

In Section 10 we combine the results of Sections 3–9 to prove the main theorem.

In Appendix A we prove an inequality on exponentially distributed random variables used in Section 6.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Pairs and normalized volumes.

Definition 2.1. A *pair* $(X/Z \ni z, B)$ consists of a contraction $\pi : X \to Z$ between normal quasi-projective varieties, a point $z \in Z$, and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier over a neighborhood of z. If π is the identity morphism, then we may drop Z and say that $(X \ni z, B)$ is a pair. If $(X/Z \ni z, B)$ is a pair for any $z \in Z$, then we say that $(X/Z, B)$ is a pair and also say that (X, B) is a pair. If we do not require that $B \geq 0$, then we say that (X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \operatorname{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B). For any divisorial valuation $v = c \operatorname{ord}_E$ for some real number $c > 0$, we denote by $a(v, X, B) := ca(E, X, B)$.

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X.

The normalized volume of a klt germ was introduced in [Li18].

Definition 2.2. Let X be a normal quasi-projective variety of dimension d and $x \in X$ a closed point. We say that $(X \ni x)$ is a *klt germ* if $(X \ni x, 0)$ is a pair and $(X, 0)$ is klt over a neighborhood of x. Since K_X is a Weil divisor, K_X is then $\mathbb{Q}$-Cartier near x [Kol13, Proposition 2.21].

Let $(X \ni x)$ be a klt germ of dimension d. For a real valuation v of $\mathbb{C}(X)$ centered at x, we denote by $a(v, X)$ the log discrepancy of v [Li18, Section 2.1]. If v is divisorial, then

$a(v,X)=a(v,X,0)$. For a real number $t>0$ let $\mathfrak{a}_t(v):=\{f\in\mathcal{O}_{X,x}\mid v(f)\geq t\}$, and let

$$\mathrm{vol}(v):=\lim_{t\to+\infty}\frac{\dim_{\mathbb{C}}\mathcal{O}_{X,x}/\mathfrak{a}_t(v)}{t^d/d!}$$

be the *volume* of v (the limit exists, see [Li18, Section 2.1] and the references there). The *normalized volume* of $(X\ni x)$ is

$$\widehat{\mathrm{vol}}(X\ni x):=\inf_v a(v,X)^d\,\mathrm{vol}(v),$$

where v runs through the real valuations of $\mathbb{C}(X)$ centered at x with $a(v,X)<+\infty$.

Definition 2.3. Let $d\geq 2$ be an integer. The *ordinary double point of dimension d* is the germ $(\{z_0^2+\cdots+z_d^2=0\}\ni 0)$, where $\{z_0^2+\cdots+z_d^2=0\}\subset\mathbb{A}^{d+1}$. A germ $(X\ni x)$ of dimension d is *analytically an ordinary double point* if the complex analytic germ of X at x is isomorphic to the complex analytic germ of $\{z_0^2+\cdots+z_d^2=0\}$ at 0.

For a scheme X of finite type over $\mathbb{C}$ and a closed point $x\in X$ with maximal ideal $\mathfrak{m}_x\subset\mathcal{O}_{X,x}$, we denote by $\mathrm{edim}(X\ni x):=\dim_{\mathbb{C}}\mathfrak{m}_x/\mathfrak{m}_x^2$ the embedding dimension of X at x, and we call $\mathrm{edim}(X\ni x)-\dim X$ the *embedding codimension* of X at x. We say that $(X\ni x)$ is a *hypersurface germ* if the completion of $\mathcal{O}_{X,x}$ is isomorphic to the quotient of a complete regular local ring by a non-zero element.

2.2. **Known bounds and hyperplane sections.** Part (1) of the following theorem is [LX19, Theorem 1.6], and part (2) is [OAI26b, Theorem 1.1].

Theorem 2.4 ([LX19, Theorem 1.6] and [OAI26b, Theorem 1.1])**.** *Let $d\geq 2$ be an integer and let $(X\ni x)$ be a klt germ of dimension d. Then the following hold.*

(1) $\widehat{\mathrm{vol}}(X\ni x)\leq d^d$, and equality holds if and only if x is a smooth point of X.
(2) If $x\in\mathrm{Sing}(X)$, then $\widehat{\mathrm{vol}}(X\ni x)\leq 2(d-1)^d$, and equality holds if and only if $(X\ni x)$ is analytically an ordinary double point.

Together with Theorem 2.4(2) in dimension $d-1$, the following lemma bounds the normalized volume of a klt germ $(X\ni x)$ whose singular locus has positive dimension at x. Construction-Proposition 4.1 and Lemma 8.1 use it in this way, and Lemma 8.1 also uses the conclusion on $\mathrm{Sing}(H)$.

Lemma 2.5. *Let $d\geq 3$ be an integer, let $(X\ni x)$ be a klt germ of dimension d, and let V be an irreducible component of $\mathrm{Sing}(X)$ such that $x\in V$ and $\dim V\geq 1$. Then there exist a closed point $x'\in V$ and a hyperplane section H of an affine neighborhood of x in X such that $x'\in H$, $(H\ni x')$ is a klt germ of dimension $d-1$, $V\cap H\subset\mathrm{Sing}(H)$, and*

$$\frac{\widehat{\mathrm{vol}}(X\ni x)}{d^d}\leq\frac{\widehat{\mathrm{vol}}(H\ni x')}{(d-1)^{d-1}}.$$

In particular $x'\in\mathrm{Sing}(H)$ and $\dim_{x'}\mathrm{Sing}(H)\geq\dim V-1$.

Proof. For any open neighborhood U of x in X, $V\cap U$ is an irreducible component of $\mathrm{Sing}(U)$ through x. Since $(X,0)$ is klt over a neighborhood of x, possibly replacing X with an affine open neighborhood of x, we may assume that X is a closed subvariety of $\mathbb{A}^N$ for some integer N and that $(X,0)$ is klt. Since $\dim V\geq 1$, there exists an integral curve $C\subset V$ such that $x\in C$ (take an irreducible component through x of the intersection of V with $\dim V-1$ general hyperplanes through x). Let $\ell:X\to\mathbb{A}^1$ be the restriction of a general linear function on $\mathbb{A}^N$. Since $\mathrm{Sing}(X)$ has finitely many irreducible components, $\ell|_C$ is not constant, and ℓ is not constant on any irreducible component of $\mathrm{Sing}(X)$ of positive dimension. For any $c\in\mathbb{A}^1$ let $H_c:=\ell^{-1}(c)$, which is a hyperplane section of X.

We show that there exists a non-empty open subset $W\subset\mathbb{A}^1$ such that for any $c\in W$, H_c is normal, K_{H_c} is $\mathbb{R}$-Cartier, and $(H_c,0)$ is klt. Since X is klt, X has rational singularities, hence X is Cohen–Macaulay [KM98, Theorem 5.22 and Theorem 5.10]. Since H_c is a Cartier divisor

on X, H_c is Cohen–Macaulay [Mat89, Theorem 17.3(ii)], and in particular H_c satisfies S_2. For general c, $H_c \setminus \mathrm{Sing}(X)$ is smooth by [Har77, III, Corollary 10.9 and Remark 10.9.2]. For general c, H_c does not contain any isolated point of $\mathrm{Sing}(X)$. Since moreover ℓ is not constant on any irreducible component of $\mathrm{Sing}(X)$ of positive dimension, and $\dim \mathrm{Sing}(X) \leq d-2$ as X is normal, we have

$$\dim(H_c \cap \mathrm{Sing}(X)) \leq d-3$$

for general c. Thus $\mathrm{Sing}(H_c) \subset H_c \cap \mathrm{Sing}(X)$ is of codimension at least 2 in H_c, so H_c satisfies R_1, and H_c is normal by Serre's criterion [Har77, II, Theorem 8.22A]. By adjunction, $K_{H_c} = (K_X + H_c)|_{H_c}$ holds on $H_c \setminus \mathrm{Sing}(X)$. Since H_c is normal and $H_c \cap \mathrm{Sing}(X)$ is of codimension at least 2 in H_c, $K_{H_c} = (K_X + H_c)|_{H_c}$ holds on H_c, and in particular K_{H_c} is $\mathbb{R}$-Cartier. Finally, the hyperplane sections H_c, $c \in \mathbb{A}^1$, are the members of a base point free linear system on X, so $(H_c, 0)$ is klt for general c by [KM98, Lemma 5.17]. Since ℓ is general and $\dim X \geq 3$, by [Laz04, Theorem 3.3.1] applied to $X \subset \mathbb{A}^N \subset \mathbb{P}^N$ (for general ℓ, the hyperplanes $\{\ell = c\}$ with c in a non-empty open subset of $\mathbb{A}^1$ are general), we may also assume that H_c is irreducible for $c \in W$.

Let $\nu : C' \to C$ be the normalization of C. Since $\ell|_C$ is not constant, $\ell \circ \nu : C' \to \mathbb{A}^1$ is dominant. Thus there exists a non-empty open subset $T \subset C'$ such that $\ell \circ \nu|_T : T \to \mathbb{A}^1$ is étale and $\ell(\nu(T)) \subset W$. Let $\mathcal{X} := X \times_{\mathbb{A}^1} T$, where $X \to \mathbb{A}^1$ is ℓ and $T \to \mathbb{A}^1$ is $\ell \circ \nu|_T$, let $g : \mathcal{X} \to X$ and $f : \mathcal{X} \to T$ be the projections, and let $\sigma : T \to \mathcal{X}$ be the section of f defined by $\sigma(t) := (\nu(t), t)$. Since X is integral and ℓ is dominant, ℓ is flat [Har77, III, Proposition 9.7], hence f is flat. The morphism g is étale, as a base change of $\ell \circ \nu|_T$, so $\mathcal{X}$ is normal [Mat89, Corollary to Theorem 23.9]. For any closed point $t \in T$, g induces an isomorphism from $f^{-1}(t)$ to $H_{\ell(\nu(t))}$ which sends $\sigma(t)$ to $\nu(t)$. Since K_X is an $\mathbb{R}$-Cartier Weil divisor, K_X is $\mathbb{Q}$-Cartier by [Kol13, Proposition 2.21] applied to $(X, 0)$, and hence $K_{\mathcal{X}} = g^* K_X$ is $\mathbb{Q}$-Cartier. Since $\ell(\nu(T)) \subset W$, the closed fibers of f are normal varieties and klt. Moreover, $\mathcal{X}$ is klt, since for a log resolution $h : X' \to X$ of X the base change $h_{\mathcal{X}} : \mathcal{X}' \to \mathcal{X}$ of h along g is a log resolution of $\mathcal{X}$, and $K_{\mathcal{X}'} - h_{\mathcal{X}}^* K_{\mathcal{X}}$ is the pullback of $K_{X'} - h^* K_X$ under the étale morphism $\mathcal{X}' \to X'$. Since f is flat of finite type, it is open [Har77, III, Exercise 9.1], and since T is irreducible, two connected components of $\mathcal{X}$ would meet a common closed fiber, which is irreducible. Thus $\mathcal{X}$ is connected, hence a normal variety (the local rings of $\mathcal{X}$ are domains), all fibers of f are connected, and the generic fiber of f, a localization of $\mathcal{X}$, is normal and klt. Therefore f together with σ is a $\mathbb{Q}$-Gorenstein flat family of klt singularities over the smooth curve T [BL21, Definition 7]. By [Liu22, Proposition 2.17], there exists a non-empty open subset $T_0 \subset T$ such that for any closed point $t \in T_0$,

$$\frac{\widehat{\mathrm{vol}}(\mathcal{X} \ni \sigma(t))}{d^d} = \frac{\widehat{\mathrm{vol}}(H_{\ell(\nu(t))} \ni \nu(t))}{(d-1)^{d-1}}.$$

Since g is étale and $g(\sigma(t)) = \nu(t)$, the completions of $\mathcal{O}_{\mathcal{X},\sigma(t)}$ and $\mathcal{O}_{X,\nu(t)}$ are isomorphic, and X and $\mathcal{X}$ are klt, so $\widehat{\mathrm{vol}}(\mathcal{X} \ni \sigma(t)) = \widehat{\mathrm{vol}}(X \ni \nu(t))$ by [Liu22, Lemma 2.15].

By [BL21, Theorem 33(2)] for the klt pair $(X, 0)$ and the irreducible subvariety C, for a very general closed point $z \in C$ we have

$$\widehat{\mathrm{vol}}(X \ni z) = \sup_{y \in C} \widehat{\mathrm{vol}}(X \ni y) \geq \widehat{\mathrm{vol}}(X \ni x).$$

Since $\nu(T_0)$ contains a non-empty open subset of C and $\mathbb{C}$ is uncountable, there exists a closed point $t \in T_0$ such that $\nu(t)$ is a very general closed point of C. Let $x' := \nu(t)$ and $H := H_{\ell(x')}$. Then $x' \in C \subset V$, $x' \in H$, and

$$\frac{\widehat{\mathrm{vol}}(X \ni x)}{d^d} \leq \frac{\widehat{\mathrm{vol}}(X \ni x')}{d^d} = \frac{\widehat{\mathrm{vol}}(\mathcal{X} \ni \sigma(t))}{d^d} = \frac{\widehat{\mathrm{vol}}(H \ni x')}{(d-1)^{d-1}}.$$

Since $\ell(x') \in W$, H is normal, K_H is $\mathbb{R}$-Cartier, and $(H, 0)$ is klt. Since H is normal and irreducible, H is a normal variety, and by Krull's principal ideal theorem $\dim H = d-1$. Thus $(H \ni x')$ is a klt germ of dimension $d-1$.

It remains to show that $V \cap H \subset \mathrm{Sing}(H)$. Let $y \in V \cap H$ be a closed point, let $\mathfrak{m}_y$ be the maximal ideal of $\mathcal{O}_{X,y}$, and suppose that H is regular at y. Since $\mathcal{O}_{H,y} = \mathcal{O}_{X,y}/(\ell - \ell(x'))$ is a regular local ring of dimension $d-1$, $\mathfrak{m}_y$ is generated by $\ell - \ell(x')$ and lifts of $d-1$ generators of the maximal ideal of $\mathcal{O}_{H,y}$. Thus $\mathfrak{m}_y$ is generated by $d = \dim \mathcal{O}_{X,y}$ elements, so $\mathcal{O}_{X,y}$ is regular [Mat89, Section 14], which contradicts $y \in V \subset \mathrm{Sing}(X)$. Since $\mathrm{Sing}(H)$ is closed and the closed points of $V \cap H$ are dense in $V \cap H$, we have $V \cap H \subset \mathrm{Sing}(H)$. In particular $x' \in \mathrm{Sing}(H)$. Since H is defined by one equation in X and $x' \in V \cap H$, by Krull's principal ideal theorem any irreducible component of $V \cap H$ through x' is of dimension at least $\dim V - 1$, hence $\dim_{x'} \mathrm{Sing}(H) \geq \dim V - 1$. □

Remark 2.6. Lemma 2.5 refines [Liu22, Theorem 2.16]. The additional conclusions $x' \in V$ and $V \cap H \subset \mathrm{Sing}(H)$ also appear in the proof of [Liu22, Corollary 2.18].

The constants in the following lemma are the equality values of Theorem 2.4(2) in dimensions $d-1$ and d, divided by the corresponding powers of the dimension. Part (4) is used only in Lemma 8.1.

Lemma 2.7. *For an integer $j \geq 2$ let $p_j := 2((j-1)/j)^j$, and for an integer $d \geq 4$ let $q_d := p_{d-1}$, $\rho_d := q_d/p_d$, and $Q_d := 2d^d((d-2)/(d-1))^{d-1}$. Then the following hold.*

(1) $Q_d = d^d q_d$ and $\rho_d = Q_d/(2(d-1)^d)$ for every $d \geq 4$, and $Q_4 = 4096/27$.
(2) $((d-1)/d)^{d-1} \leq 1/2$ for every $d \geq 2$.
(3) The sequences $(q_d)_{d\geq 4}$ and $(\rho_d)_{d\geq 4}$ are strictly increasing, and

$$q_4 = \frac{16}{27}, \qquad \rho_5 = \frac{253125}{262144} > \frac{5}{6}, \qquad 2\rho_6 = \frac{95551488}{48828125}.$$

(4) $p_{d-2} + p_d < 2p_{d-1}$ for every $d \geq 5$.
(5) $Q_d < 2(d-1)^d$ for every $d \geq 4$.

Proof. (1) By definition,

$$Q_d = 2d^d\Big(\frac{d-2}{d-1}\Big)^{d-1} = d^d p_{d-1} = d^d q_d.$$

Since $d^d p_d = 2(d-1)^d$, we have

$$\rho_d = \frac{q_d}{p_d} = \frac{d^d q_d}{d^d p_d} = \frac{Q_d}{2(d-1)^d}.$$

Moreover, $Q_4 = 2 \cdot 4^4 \cdot (2/3)^3 = 4096/27$.

(2) By the binomial theorem,

$$\left(\frac{d}{d-1}\right)^{d-1} = \left(1 + \frac{1}{d-1}\right)^{d-1} \geq 1 + (d-1) \cdot \frac{1}{d-1} = 2.$$

(3) For a real number $t > 1$ let $g(t) := t\log(1 - 1/t)$. Then $p_j = 2e^{g(j)}$ for any integer $j \geq 2$, so $q_d = 2e^{g(d-1)}$ and $\rho_d = e^{g(d-1)-g(d)}$. A direct calculation shows

$$g'(t) = \log\left(1 - \frac{1}{t}\right) + \frac{1}{t-1}, \qquad g''(t) = -\frac{1}{t(t-1)^2} < 0.$$

Let $z := 1/(t-1)$. Then $1 - 1/t = 1/(1+z)$, so $g'(t) = z - \log(1+z) > 0$. Thus g is strictly increasing on $(1, +\infty)$, and $(q_d)_{d\geq 4}$ is strictly increasing. Since g is strictly concave,

$$\frac{\rho_{d+1}}{\rho_d} = e^{2g(d)-g(d-1)-g(d+1)} > 1,$$

so $(\rho_d)_{d\geq 4}$ is strictly increasing. For the values, $q_4 = p_3 = 2(2/3)^3 = 16/27$,

$$\rho_5 = \frac{p_4}{p_5} = \frac{2(3/4)^4}{2(4/5)^5} = \frac{3^4 \cdot 5^5}{2^{18}} = \frac{253125}{262144},$$

and $6 \cdot 253125 = 1518750 > 1310720 = 5 \cdot 262144$, so $\rho_5 > 5/6$. Finally,

$$2\rho_6 = \frac{2p_5}{p_6} = \frac{2(4/5)^5}{(5/6)^6} = \frac{2^{11} \cdot 6^6}{5^{11}} = \frac{95551488}{48828125}.$$

(4) For a real number $t \geq 3$ let $P(t) := 2e^{g(t)} = 2(1-1/t)^t$, and let $z := 1/(t-1)$, so that $0 < z \leq 1/2$. Then $P''(t)/P(t) = g'(t)^2 + g''(t)$. Since

$$g'(t) = z - \log(1+z) = \int_0^z \frac{u}{1+u} du,$$

we have $0 < g'(t) < z^2/2$. Moreover, $-g''(t) = z^3/(1+z)$. Hence

$$\frac{P''(t)}{P(t)} < \frac{z^4}{4} - \frac{z^3}{1+z} = \frac{z^3\big(z(1+z)-4\big)}{4(1+z)} < 0,$$

where the last inequality holds because $z(1+z) \leq 3/4$. Thus P is strictly concave on $[3, +\infty)$. If $d \geq 5$, then $d-2 \geq 3$, and the strict concavity of P implies

$$p_{d-2} + p_d = P(d-2) + P(d) < 2P(d-1) = 2p_{d-1}.$$

(5) Let $m := d-1 \geq 3$. By (1),

$$\frac{Q_d}{2(d-1)^d} = \rho_d = \frac{(m+1)^{m+1}(m-1)^m}{m^{2m+1}} = \left(1 - \frac{1}{m^2}\right)^m \left(1 + \frac{1}{m}\right).$$

By Bernoulli's inequality,

$$\left(1 - \frac{1}{m^2}\right)^{-m} = \left(1 + \frac{1}{m^2-1}\right)^m \geq 1 + \frac{m}{m^2-1} > 1 + \frac{1}{m}.$$

Hence $Q_d/(2(d-1)^d) < 1$. □

2.3. **Cones, gradings, and Reeb vectors.**

Definition 2.8. Let $C = \operatorname{Spec} S$ be a normal affine variety with a faithful action of an algebraic torus $\mathbb{T}$. Let $\mathfrak{t}_{\mathbb{Z}} := \operatorname{Hom}(\mathbb{G}_m, \mathbb{T})$ and $\mathfrak{t} := \mathfrak{t}_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{R}$, and let $S = \bigoplus_\chi S_\chi$ be the decomposition of S into $\mathbb{T}$-eigenspaces, where χ runs through the characters of $\mathbb{T}$ and

$$S_\chi := \{f \in S \mid f(g \cdot p) = \chi(g) f(p) \text{ for any } g \in \mathbb{T} \text{ and any closed point } p \in C\}.$$

Assume that $S_0 = \mathbb{C}$ and that

$$\mathfrak{t}^+ := \{\xi \in \mathfrak{t} \mid \langle \chi, \xi \rangle > 0 \text{ for any } \chi \neq 0 \text{ with } S_\chi \neq 0\}$$

is non-empty. The *vertex* of C is the closed point $o \in C$ defined by the ideal $\bigoplus_{\chi \neq 0} S_\chi$. The elements of $\mathfrak{t}^+$ are the *Reeb vectors* of C, and $\mathfrak{t}^+$ is the *Reeb cone*. For $\xi \in \mathfrak{t}^+$, the *weight valuation* of ξ is the valuation wt_ξ of $\mathbb{C}(C)$ [LWX21, (2.6)] such that

$$\mathrm{wt}_\xi\left(\sum_\chi f_\chi\right) = \min\{\langle \chi, \xi \rangle \mid f_\chi \neq 0\} \quad \text{for } f_\chi \in S_\chi.$$

A primitive $\zeta \in \mathfrak{t}_{\mathbb{Z}} \cap \mathfrak{t}^+$ defines the grading $S = \bigoplus_{j \geq 0} S_j$ with $S_j := \bigoplus_{\langle \chi, \zeta \rangle = j} S_\chi$, and the corresponding $\mathbb{G}_m$-action on C.

For a finitely generated graded $\mathbb{C}$-algebra $D = \bigoplus_{j \geq 0} D_j$ with $D_0 = \mathbb{C}$, the *vertex* of $\operatorname{Spec} D$ is the closed point defined by the ideal $\bigoplus_{j>0} D_j$. The *Hilbert series* of such an algebra D is $\mathrm{Hilb}_D(t) := \sum_{j \geq 0} (\dim_{\mathbb{C}} D_j) t^j$.

Our convention for the eigenspaces S_χ, and for the action of $\mathbb{T}$ on differential forms in Lemma 2.9, is the one of [OAI26a, Section 3]. For log Fano cones we refer to [LX18, Definition 2.23] and [LWX21, Definition 2.12]. K-semistable log Fano cones are defined in [LX18, Definition 2.28] and [LWX21, Definition 2.23], and K-polystable log Fano cones are defined in [LWX21, Definition 2.23].

Lemma 2.9. *Let $C = \operatorname{Spec} S$ be a normal affine variety of dimension d with a faithful action of an algebraic torus $\mathbb{T}$ satisfying the assumptions of Definition 2.8, with vertex o and Reeb cone $\mathfrak{t}^+$, and let $\xi \in \mathfrak{t}^+$. Assume that C is klt and that for some integer $I \geq 1$ the reflexive power $(\omega_S^{\otimes I})^{**}$ is generated by a $\mathbb{T}$-homogeneous element of character χ_0, where ω_S carries the $\mathbb{T}$-action induced by the action of $\mathbb{T}$ on differential forms, so that df has character χ for any $f \in S_\chi$. Then the following hold.*

(1) The valuation wt_ξ is quasi-monomial and centered at o, and if $\mathbb{R}_{>0}\xi \cap \mathfrak{t}_\mathbb{Z} \neq \emptyset$, then wt_ξ is a positive multiple of a divisorial valuation [LX18, Definition 2.8 and Lemma 2.16].

(2) $a(\mathrm{wt}_\xi, C) = \langle \chi_0, \xi \rangle / I$ [LX18, Lemma 2.18 and Remark 2.19].

(3) If ξ is a primitive element of $\mathfrak{t}_\mathbb{Z}$ and $S = \bigoplus_{j \geq 0} S_j$ is the grading defined by ξ, then the stabilizer of a closed point $p \in C \backslash \{o\}$ for the corresponding $\mathbb{G}_m$-action is μ_m, where m is the greatest common divisor of the degrees of the homogeneous elements of S which do not vanish at p, and

$$\mathrm{vol}(\mathrm{wt}_\xi) = \lim_{s \to 0^+} s^d \sum_{j \geq 0} (\dim_\mathbb{C} S_j) e^{-sj}.$$

(4) For any real number $c > 0$, we have $c\xi \in \mathfrak{t}^+$, $\mathrm{wt}_{c\xi} = c\,\mathrm{wt}_\xi$, $\mathrm{vol}(\mathrm{wt}_{c\xi}) = c^{-d}\,\mathrm{vol}(\mathrm{wt}_\xi)$, and $a(\mathrm{wt}_{c\xi}, C) = c a(\mathrm{wt}_\xi, C)$.

Proof. (1) By [LX18, Lemma 2.16], wt_ξ is a quasi-monomial valuation [LX18, Definition 2.8] centered at o. Suppose that $\xi = c\zeta$ for some real number $c > 0$ and some $\zeta \in \mathfrak{t}_\mathbb{Z}$. Then $\langle \chi, \xi \rangle = c\langle \chi, \zeta \rangle \in c\mathbb{Z}$ for any character χ of $\mathbb{T}$, so wt_ξ takes values in $c\mathbb{Z}$. Therefore the rational rank of wt_ξ is 1, and [LX18, Definition 2.8] with $r = 1$ implies that $\mathrm{wt}_\xi = \alpha_1 \operatorname{ord}_{E_1}$ for some prime divisor E_1 over C and some real number $\alpha_1 > 0$.

(2) This is [LX18, Lemma 2.18 and Remark 2.19], applied to the generator of $(\omega_S^{\otimes I})^{**}$ of character χ_0.

(3) Let $g \in \mathbb{G}_m$. Since C is affine and S is spanned by homogeneous elements, $g \cdot p = p$ if and only if $f(g \cdot p) = f(p)$ for any homogeneous $f \in S$. By Definition 2.8, $f(g \cdot p) = g^j f(p)$ for any $f \in S_j$. Therefore $g \cdot p = p$ if and only if $g^j = 1$ for any $j \geq 0$ such that some $f \in S_j$ satisfies $f(p) \neq 0$. Since $p \neq o$ and o is the common zero locus of $\bigoplus_{j>0} S_j$, at least one such j is positive, so m is a positive integer. The subgroup of $\mathbb{Z}$ generated by all such j is $m\mathbb{Z}$. Thus the stabilizer of p is $\{g \in \mathbb{G}_m \mid g^m = 1\} = \mu_m$.

We prove the volume formula. Let $\mathfrak{m}_o := \bigoplus_{j>0} S_j$ be the ideal of o, and for any integer $N \geq 1$ let $\mathfrak{b}_N := \bigoplus_{j \geq N} S_j$. Any $u \in S \backslash \mathfrak{m}_o$ has a non-zero component in $S_0 = \mathbb{C}$, so $\mathrm{wt}_\xi(u) = 0$. Therefore $\mathfrak{a}_N(\mathrm{wt}_\xi) = \mathfrak{b}_N \mathcal{O}_{C,o}$. Since S is finitely generated and $S_0 = \mathbb{C}$, we have $\dim_\mathbb{C} S_j < +\infty$ for any j, and $\mathfrak{m}_o^N \subset \mathfrak{b}_N$. Thus $S/\mathfrak{b}_N$ is an Artinian local ring with maximal ideal $\mathfrak{m}_o/\mathfrak{b}_N$, and

$$\mathcal{O}_{C,o}/\mathfrak{a}_N(\mathrm{wt}_\xi) \cong S/\mathfrak{b}_N \cong \bigoplus_{j<N} S_j$$

as $\mathbb{C}$-vector spaces. For any integer $N \geq 0$ let

$$c_N := \sum_{j \leq N} \dim_\mathbb{C} S_j = \dim_\mathbb{C} \mathcal{O}_{C,o}/\mathfrak{a}_{N+1}(\mathrm{wt}_\xi).$$

Since wt_ξ takes integer values on $\mathcal{O}_{C,o}$, Definition 2.2 and [LWX21, Definition 2.10 and the paragraph after it] imply that the limit defining $\mathrm{vol}(\mathrm{wt}_\xi)$ exists and

$$c_N = \frac{\mathrm{vol}(\mathrm{wt}_\xi)}{d!} N^d + o(N^d) \quad \text{as } N \to +\infty.$$

Since $c_0 = \dim_{\mathbb{C}} S_0$ and $c_N - c_{N-1} = \dim_{\mathbb{C}} S_N$ for any $N \geq 1$, we have

$$\mathrm{Hilb}_S(e^{-s}) = \sum_{j\geq 0}(\dim_{\mathbb{C}} S_j)e^{-sj} = (1-e^{-s})\sum_{N\geq 0} c_N e^{-sN}$$

for any real number $s > 0$. Since $x \to x^d e^{-sx}$ increases on $[0, d/s]$ and decreases on $[d/s, +\infty)$, comparing with $\int_0^{+\infty} x^d e^{-sx}dx = d!s^{-d-1}$ implies that

$$\Big|\sum_{N\geq 0} N^d e^{-sN} - d!s^{-d-1}\Big| \leq 2(d/s)^d.$$

Thus $s^d(1-e^{-s})\sum_{N\geq 0} N^d e^{-sN} \to d!$ as $s \to 0^+$. Let $\epsilon > 0$. There exists $\lambda_\epsilon > 0$ such that $|c_N - \mathrm{vol}(\mathrm{wt}_\xi)N^d/d!| \leq \epsilon N^d + \lambda_\epsilon$ for any $N \geq 0$. Since $1-e^{-s} \leq s$ and $\sum_{N\geq 0} e^{-sN} = (1-e^{-s})^{-1}$, we have

$$\Big|s^d \,\mathrm{Hilb}_S(e^{-s}) - \frac{\mathrm{vol}(\mathrm{wt}_\xi)}{d!}s^d(1-e^{-s})\sum_{N\geq 0} N^d e^{-sN}\Big| \leq \epsilon s^d(1-e^{-s})\sum_{N\geq 0} N^d e^{-sN} + \lambda_\epsilon s^d.$$

As $s \to 0^+$, the left-hand side differs from $|s^d \,\mathrm{Hilb}_S(e^{-s}) - \mathrm{vol}(\mathrm{wt}_\xi)|$ by a quantity tending to 0, and the right-hand side tends to $\epsilon d!$, as $d \geq 1$. Since $\epsilon > 0$ is arbitrary, $s^d \,\mathrm{Hilb}_S(e^{-s}) \to \mathrm{vol}(\mathrm{wt}_\xi)$ as $s \to 0^+$.

(4) Since $\langle \chi, c\xi\rangle = c\langle \chi, \xi\rangle$ for any character χ of $\mathbb{T}$, we have $c\xi \in \mathfrak{t}^+$, and $\mathrm{wt}_{c\xi} = c\,\mathrm{wt}_\xi$ by Definition 2.8. Hence $\mathfrak{a}_t(\mathrm{wt}_{c\xi}) = \mathfrak{a}_{t/c}(\mathrm{wt}_\xi)$ for any $t > 0$, and Definition 2.2 implies $\mathrm{vol}(\mathrm{wt}_{c\xi}) = c^{-d}\,\mathrm{vol}(\mathrm{wt}_\xi)$. By (2) for ξ and for $c\xi$, $a(\mathrm{wt}_{c\xi}, C) = \langle \chi_0, c\xi\rangle/I = ca(\mathrm{wt}_\xi, C)$. □

3. Hypersurfaces, complete intersections and products

In this section, we bound by Q_d the normalized volume of klt germs of dimension d at least four that are hypersurfaces or local complete intersections and are neither smooth nor analytically ordinary double points (Propositions 3.1 and 3.4), and we show that the product of the ordinary double point of dimension $d-1$ with $\mathbb{A}^1$ has normalized volume exactly Q_d (Proposition 3.5). Propositions 3.1 and 3.4 are the last step of both routes toward Theorem 1.2(1): in dimension four we show that $(X \ni x)$ itself is a hypersurface germ, and in dimension at least five we derive a contradiction once the cone obtained by degeneration is a local complete intersection.

The main case of Proposition 3.1 below is a hypersurface germ of multiplicity two whose quadratic part is degenerate. There the blow-up of the point only bounds the normalized volume by the value of an ordinary double point. A weighted blow-up with a smaller weight on a kernel coordinate of the quadratic part bounds it by Q_d, and the choice of weight works only in dimension at least four. In dimension three the bound fails (Remark 1.4).

Proposition 3.1. *Let $d \geq 4$ be an integer and let $(X \ni x)$ be a klt germ of dimension d. Assume that $\mathrm{edim}(X \ni x) \leq d+1$ and that $(X \ni x)$ is neither smooth nor analytically an ordinary double point. Then $\widehat{\mathrm{vol}}(X \ni x) \leq Q_d$. If moreover $\mathrm{mult}(X \ni x) \geq 3$, then $\widehat{\mathrm{vol}}(X \ni x) \leq 3(d-2)^d < Q_d$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we write $(X \ni x)$ as a hypersurface $\{f = 0\}$ in a smooth variety of dimension $d+1$ and show that $2 \leq \mathrm{mult}(X \ni x) \leq d$.

Since $(X \ni x)$ is not smooth, $\mathrm{edim}(X \ni x) \neq d$ [Har77, I, Theorem 5.1], so $\mathrm{edim}(X \ni x) = d+1$. Possibly shrinking X around x, let $X \subset W'$ be a closed embedding into a smooth variety W'. The kernel of the surjection $\mathfrak{m}_{W',x}/\mathfrak{m}^2_{W',x} \to \mathfrak{m}_{X,x}/\mathfrak{m}^2_{X,x}$ is of dimension $s := \dim W' - d - 1$ and is spanned by the classes of elements of the ideal of X. Let $g_1, \ldots, g_s$ be elements of the ideal of X whose classes form a basis of the kernel. Possibly shrinking W' around x, the closed subscheme $W := \{g_1 = \cdots = g_s = 0\}$ of W' is a smooth variety of dimension $d+1$ which contains X [Mat89, Theorem 14.2]. Since X is normal, $\mathcal{O}_{X,x}$ is a domain. Since $\dim \mathcal{O}_{W,x} = d+1$ and $\dim \mathcal{O}_{X,x} = d$, the ideal of X in $\mathcal{O}_{W,x}$ is a prime of height one [Mat89, Theorems 17.8 and 17.4(i)], hence

principal [Mat89, Theorems 20.1 and 20.3]. Let $f \in \mathcal{O}_{W,x}$ be a generator. Possibly shrinking W around x, f is a regular function on W and $X = \{f = 0\}$.

Let $z_0, \dots, z_d$ be a regular system of parameters of $\mathcal{O}_{W,x}$. Then $\widehat{\mathcal{O}}_{W,x} = \mathbb{C}[[z_0, \dots, z_d]]$ [Har77, I, Theorems 5.4A and 5.5A], and hence

$$\widehat{\mathcal{O}}_{X,x} = \mathbb{C}[[z_0, \dots, z_d]]/(f)$$

[Mat89, Theorem 8.11], where f also denotes the Taylor expansion of f in $z_0, \dots, z_d$. Since $z_0, \dots, z_d$ are analytic coordinates of W at x, the Taylor expansion of f converges, and the analytic local ring of X at x is $\mathbb{C}\{z_0, \dots, z_d\}/(f)$.

Let $k := \operatorname{ord}_x f$ and let f_k be the homogeneous part of degree k of f. We omit the verification of $\operatorname{gr}_{\mathfrak{m}}(\mathcal{O}_{X,x}) \cong \mathbb{C}[z_0, \dots, z_d]/(f_k)$, which holds because $\operatorname{gr}_{\mathfrak{m}}(\mathcal{O}_{W,x}) = \mathbb{C}[z_0, \dots, z_d]$ [Mat89, Theorem 14.4] is a domain, so the initial forms of the elements of $f\mathcal{O}_{W,x}$ generate the ideal (f_k) [Eis95, Exercises 5.2 and 5.3]. The isomorphism implies $k = \operatorname{mult}(X \ni x)$ [Eis95, Exercise 12.7(a)]. Since $\operatorname{edim}(X \ni x) = d + 1 = \dim W$, we have $f \in \mathfrak{m}_{W,x}^2$, so $k \geq 2$.

Since X is a normal Cartier divisor on the smooth variety W and X is klt near x, the pair (W, X) is plt near x by inversion of adjunction [KM98, Theorem 5.50]. Let $h : \widetilde{W} \to W$ be the blow-up of x with exceptional divisor E, and let $\widetilde{X}$ be the strict transform of X. Since $K_{\widetilde{W}} = h^* K_W + dE$ and $h^* X = \widetilde{X} + kE$, we have

$$K_{\widetilde{W}} + \widetilde{X} + (k - d)E = h^*(K_W + X),$$

so $a(E, W, X) = d + 1 - k$. Since E is exceptional over W with center x and (W, X) is plt near x, $a(E, W, X) > 0$. Thus $k \leq d$.

Step 2. In this step, we recall the weighted bound of [Liu25, Proposition 4.1] and prove the proposition when $k \geq 3$.

For $\mathbf{w} = (w_0, \dots, w_d) \in \mathbb{R}_{>0}^{d+1}$, let $v_{\mathbf{w}}$ be the monomial valuation of $\mathbb{C}[[z_0, \dots, z_d]]$ with $v_{\mathbf{w}}(z_i) = w_i$, that is,

$$v_{\mathbf{w}}\Big(\sum_{\alpha} c_\alpha z^\alpha\Big) = \min\Big\{\sum_{i=0}^{d} w_i \alpha_i \;\Big|\; c_\alpha \neq 0\Big\}.$$

The result [Liu25, Proposition 4.1] is formulated for klt hypersurface germs of any dimension $n \geq 2$ whose completed local ring is the quotient of the formal power series ring in $n+1$ variables by one element. Applied to $\widehat{\mathcal{O}}_{X,x} = \mathbb{C}[[z_0, \dots, z_d]]/(f)$, it implies that

$$\widehat{\operatorname{vol}}(X \ni x) \leq \frac{\big(\sum_{i=0}^{d} w_i - v_{\mathbf{w}}(f)\big)^d v_{\mathbf{w}}(f)}{\prod_{i=0}^{d} w_i} \tag{3.1}$$

for any $\mathbf{w} \in \mathbb{R}_{>0}^{d+1}$.

Suppose that $k \geq 3$. For $w_0 = \cdots = w_d = 1$ we have $v_{\mathbf{w}}(f) = k$, and (3.1) implies $\widehat{\operatorname{vol}}(X \ni x) \leq k(d + 1 - k)^d$. The function $\beta(t) := t(d + 1 - t)^d$ satisfies

$$\beta'(t) = (d + 1)(1 - t)(d + 1 - t)^{d-1} < 0 \quad \text{for } t \in (1, d + 1),$$

and $3 \leq k \leq d$ by **Step 1**. Thus $k(d + 1 - k)^d \leq \beta(3) = 3(d - 2)^d$. By Lemma 2.7(2),

$$\frac{3(d-2)^d}{Q_d} = \frac{3}{2} \cdot \frac{d-2}{d} \cdot \Big(\frac{d-1}{d}\Big)^{d-1} \leq \frac{3(d-2)}{4d} < 1.$$

Thus $\widehat{\operatorname{vol}}(X \ni x) \leq 3(d - 2)^d < Q_d$.

Step 3. In this step, we prove the proposition when $k = 2$.

Suppose that $k = 2$. If the quadratic form f_2 in $z_0, \dots, z_d$ is non-degenerate, then by the holomorphic Morse lemma [GLS07, Chapter I, Theorem 2.46] there exist analytic coordinates $y_0, \dots, y_d$ of W at x such that $f = y_0^2 + \cdots + y_d^2$. Then $(X \ni x)$ is analytically an ordinary double point, a contradiction. Hence $f_2 \neq 0$ is degenerate. Since any invertible linear change of $z_0, \dots, z_d$ is again a regular system of parameters of $\mathcal{O}_{W,x}$, possibly replacing $z_0, \dots, z_d$ with such

a change that diagonalizes f_2, we may assume that $f_2 = z_0^2 + \cdots + z_m^2$ for some $0 \le m \le d-1$. In particular, z_d does not appear in f_2.

Let $b := (d-2)/(d-1)$ and $\mathbf{w} := (1, \dots, 1, b)$, so that $w_d = b$. Then $0 < b < 1$. The monomials of degree two of f are those of f_2, which do not involve z_d, so they have weight 2. Any monomial of degree at least 3 has weight at least $3b$, and $3b \ge 2$ if and only if $d \ge 4$. Thus $v_{\mathbf{w}}(f) = 2$. Since $\sum_{i=0}^{d} w_i = d + b$, $\prod_{i=0}^{d} w_i = b$, and $d - 2 + b = db$, (3.1) implies

$$\widehat{\mathrm{vol}}(X \ni x) \le \frac{2(d-2+b)^d}{b} = 2d^d b^{d-1} = 2d^d \Big(\frac{d-2}{d-1}\Big)^{d-1} = Q_d.$$

Since $k = \mathrm{mult}(X \ni x)$ by **Step 1**, **Steps 2 and 3** prove the proposition. □

For local complete intersections of embedding codimension at least two, a deformation to an intersection of two quadrics bounds the normalized volume strictly below Q_d.

Lemma 3.2. *Let $d \ge 4$ be an integer and let $(X \ni x)$ be a klt germ of dimension d. Assume that $(X \ni x)$ is a local complete intersection and $\mathrm{edim}(X \ni x) \ge d + 2$. Then $\widehat{\mathrm{vol}}(X \ni x) \le 4(d-2)^d < Q_d$.*

Proof. Let $e := \mathrm{edim}(X \ni x)$ and $r := e - d \ge 2$. Possibly shrinking X around x and cutting a smooth variety containing X by elements of the ideal of X whose classes form a basis of the kernel of the map of cotangent spaces at x, we may assume that X is a closed subvariety of a smooth variety W of dimension e [Mat89, Theorem 14.2]. Since $\dim \mathfrak{m}_{W,x}/\mathfrak{m}_{W,x}^2 = e = \dim \mathfrak{m}_{X,x}/\mathfrak{m}_{X,x}^2$, the ideal J of X in $\mathcal{O}_{W,x}$ is contained in $\mathfrak{m}_{W,x}^2$. Since $\mathcal{O}_{X,x} = \mathcal{O}_{W,x}/J$ is a complete intersection, J is generated by an $\mathcal{O}_{W,x}$-regular sequence $f_1, \dots, f_r$ [BH98, Theorem 2.3.3(c)], of length $r = e - d$ because $\dim \mathcal{O}_{X,x} = d$. Thus $f_1, \dots, f_r \in \mathfrak{m}_{W,x}^2$. Possibly shrinking W around x, the elements $f_1, \dots, f_r$ and a regular system of parameters $y_1, \dots, y_e$ of $\mathcal{O}_{W,x}$ are regular functions on W, and $X = \{f_1 = \cdots = f_r = 0\}$.

Let ϕ_1, ϕ_2 be general quadratic forms and $l_3, \dots, l_r$ general linear forms in $y_1, \dots, y_e$. Let t be the coordinate of $\mathbb{A}^1$ and let $\mathcal{X} \subset W \times \mathbb{A}^1$ be the closed subscheme defined by

$$f_1 + t\phi_1, \quad f_2 + t\phi_2, \quad f_j + tl_j \quad (3 \le j \le r).$$

For $t \in \mathbb{A}^1$, let X_t be the fiber of $\mathcal{X}$ over t and $x_t := (x, t)$. Then $X_0 = X$, and $x_t \in X_t$ for any t.

We show that $\mathcal{X}$ is a klt variety near x_0 which is flat over $\mathbb{A}^1$ with Cartier canonical divisor. Since $\mathcal{X}$ is defined by r equations in the smooth variety $W \times \mathbb{A}^1$ of dimension $e+1$, any irreducible component of $\mathcal{X}$ through x_0 is of dimension at least $e + 1 - r = d + 1$. Since $\mathcal{X} \cap \{t = 0\} = X$ is of dimension d, $\dim \mathcal{O}_{\mathcal{X},x_0} \le d + 1$. Thus $\mathcal{O}_{\mathcal{X},x_0}$ is a complete intersection of dimension $d + 1$, hence Cohen–Macaulay and Gorenstein. Since $\dim \mathcal{O}_{\mathcal{X},x_0}/(t) = d$, $\mathrm{ht}(t) = 1$, and hence t is a nonzerodivisor on $\mathcal{O}_{\mathcal{X},x_0}$ [Mat89, Theorem 17.4(i) and (iii)]. Possibly shrinking $\mathcal{X}$ around x_0, $\mathcal{X}$ is flat over $\mathbb{A}^1$ [Har77, III, Proposition 9.7] and $K_{\mathcal{X}}$ is Cartier.

We show that $\mathcal{X}$ is normal near x_0. Let Z be an irreducible component of $\mathrm{Sing}\,\mathcal{X}$ which contains x_0. Since X is a Cartier divisor on $\mathcal{X}$, $\mathcal{X}$ is smooth at any point at which X is smooth, so $Z \cap X \subset \mathrm{Sing}\, X$. Since $x_0 \in Z \cap X$, we have

$$\dim Z \le \dim(Z \cap X) + 1 \le \dim \mathrm{Sing}\, X + 1 \le d - 1,$$

where the last inequality holds because X is normal. Thus $\mathrm{Sing}\,\mathcal{X}$ is of codimension at least two in $\mathcal{X}$ near x_0, and Serre's criterion [Har77, II, Theorem 8.22A] implies that $\mathcal{X}$ is normal near x_0. Since X is klt and $K_{\mathcal{X}} + X$ is Cartier, inversion of adjunction [KM98, Theorem 5.50] implies that $(\mathcal{X}, X)$ is plt near x_0, hence $\mathcal{X}$ is klt near x_0. The fibers X_t are the members of the base point free linear system on $\mathcal{X}$ defined by t, so for general t, X_t is klt near x_t [KM98, Lemma 5.17(1)]. Since a non-empty open subset of $\mathbb{A}^1$ has finite complement, there exists an open subset $T^\circ \subset \mathbb{A}^1$ with $0 \in T^\circ$ such that X_t is klt near x_t for any $t \in T^\circ$. Possibly shrinking $\mathcal{X}$ around the image of the section $t \to x_t$, the morphism $\mathcal{X} \times_{\mathbb{A}^1} T^\circ \to T^\circ$ with the section $t \to x_t$ is a $\mathbb{Q}$-Gorenstein flat family of klt singularities.

We bound $\widehat{\mathrm{vol}}(X_t \ni x_t)$ for general $t \neq 0$. We regard linear and quadratic forms in $y_1, \dots, y_e$ as forms on the tangent space $T_xW \cong \mathbb{C}^e$. Let $L \subset T_xW$ be the common kernel of $l_3, \dots, l_r$, so that $\dim L = e - r + 2 = d + 2$, and let $f_i^{(2)}$ be the quadratic part of f_i in $y_1, \dots, y_e$. For $t \neq 0$, the linear parts at x of $f_j + tl_j$, $3 \leq j \leq r$, are the linearly independent forms tl_j. Thus, identifying $W \times \{t\}$ with W, the subscheme $W_t := \{f_j + tl_j = 0,\ 3 \leq j \leq r\}$ is smooth of dimension $d+2$ near x with tangent space L at x [Mat89, Theorem 14.2], and $X_t = H_1 \cap H_2$ near x, where $H_i := \{f_i + t\phi_i = 0\} \subset W_t$. The quadratic part at x of $(f_i + t\phi_i)|_{W_t}$ is

$$\bar{\phi}_i := (f_i^{(2)} + t\phi_i)|_L, \quad i = 1, 2.$$

Since ϕ_1, ϕ_2 are general and $t \neq 0$, $\bar{\phi}_1$ and $\bar{\phi}_2$ are general quadratic forms on L, so $\{\bar{\phi}_1 = \bar{\phi}_2 = 0\} \subset \mathbb{P}(L) \cong \mathbb{P}^{d+1}$ is a smooth complete intersection of dimension $d - 1 \geq 3$, which is irreducible because it is connected [Har77, III, Exercise 5.5(b)]. Let $h_t : \widetilde{W}_t \to W_t$ be the blow-up of x with exceptional divisor $F_t \cong \mathbb{P}(L)$, and let $\widetilde{H}_i$ be the strict transform of H_i. Then $h_t^* H_i = \widetilde{H}_i + 2F_t$ and $\widetilde{H}_i \cap F_t = \{\bar{\phi}_i = 0\}$. Since $\{\bar{\phi}_1 = \bar{\phi}_2 = 0\}$ is smooth of codimension two in F_t, the divisors $\widetilde{H}_1$, $\widetilde{H}_2$, and F_t meet transversally along $\widetilde{H}_1 \cap \widetilde{H}_2 \cap F_t$. Thus $\widetilde{H}_1 \cap \widetilde{H}_2$ is smooth of dimension d near F_t, and no irreducible component of $\widetilde{H}_1 \cap \widetilde{H}_2$ is contained in F_t, as $\dim(\widetilde{H}_1 \cap \widetilde{H}_2 \cap F_t) = d - 1$. Hence the strict transform $\widetilde{X}_t$ of X_t is $\widetilde{H}_1 \cap \widetilde{H}_2$ near F_t, it is smooth along

$$E_t := \widetilde{X}_t \cap F_t = \{\bar{\phi}_1 = \bar{\phi}_2 = 0\} \subset \mathbb{P}(L),$$

and $F_t|_{\widetilde{X}_t} = E_t$. Since $K_{\widetilde{W}_t} = h_t^* K_{W_t} + (d+1)F_t$, adjunction implies that near E_t,

$$K_{\widetilde{X}_t} = (K_{\widetilde{W}_t} + \widetilde{H}_1 + \widetilde{H}_2)|_{\widetilde{X}_t} = \big(h_t^*(K_{W_t} + H_1 + H_2) + (d-3)F_t\big)|_{\widetilde{X}_t} = (h_t|_{\widetilde{X}_t})^* K_{X_t} + (d-3)E_t.$$

Thus $a(E_t, X_t, 0) = d - 2$. Since $\widetilde{X}_t$ is the blow-up of X_t at x_t [Har77, II, Corollary 7.15] and is smooth along the reduced exceptional divisor E_t, the ideal $\mathfrak{a}_m(\mathrm{ord}_{E_t})$ is the integral closure of $\mathfrak{m}_{x_t}^m$ for any $m \in \mathbb{N}$, so $\mathrm{vol}(\mathrm{ord}_{E_t}) = \mathrm{mult}(X_t \ni x_t)$, which is the degree of $E_t \subset \mathbb{P}(L)$ (cf. Remark 3.3). Since E_t is a complete intersection of two quadrics in $\mathbb{P}(L)$, $(\mathcal{O}_{\mathbb{P}(L)}(1)^{d-1} \cdot E_t) = 4$. Therefore

$$\widehat{\mathrm{vol}}(X_t \ni x_t) \leq a(\mathrm{ord}_{E_t}, X_t)^d \,\mathrm{vol}(\mathrm{ord}_{E_t}) = 4(d-2)^d$$

for general $t \in T^\circ$.

By [BL21, Theorem 1], the function $t \to \widehat{\mathrm{vol}}(X_t \ni x_t)$ on T° is lower semicontinuous, so the subset $\{t \in T^\circ \mid \widehat{\mathrm{vol}}(X_t \ni x_t) > 4(d-2)^d\}$ of T° is open. It is disjoint from a non-empty open subset of T°, hence it is empty since T° is irreducible. In particular, $\widehat{\mathrm{vol}}(X \ni x) = \widehat{\mathrm{vol}}(X_0 \ni x_0) \leq 4(d-2)^d$. Finally, by Lemma 2.7(2),

$$\frac{4(d-2)^d}{Q_d} = 2 \cdot \frac{d-2}{d} \cdot \Big(\frac{d-1}{d}\Big)^{d-1} \leq \frac{d-2}{d} < 1. \qquad \square$$

Remark 3.3. The estimate $\widehat{\mathrm{vol}}(X \ni x) \leq k(d+1-k)^d$ for a hypersurface germ of multiplicity k, which goes back to [LX19, Lemma 3.1], the estimate $\widehat{\mathrm{vol}}(X \ni x) \leq Q_d$ in multiplicity two with degenerate quadratic part, and the estimate of Lemma 3.2 appear in the proof of [Liu22, Theorem 2.12].

Part (2) of Proposition 3.4 below is the form used in Construction-Proposition 4.1(4). There the germ $(X' \ni x')$ is the cone obtained by a degeneration of $(X \ni x)$, and Construction-Proposition 4.1(1) only says that $\widehat{\mathrm{vol}}(X' \ni x') = \widehat{\mathrm{vol}}(X \ni x)$ and $\mathrm{edim}(X' \ni x') \geq \mathrm{edim}(X \ni x)$. When the cone is an ordinary double point, the embedding dimension forces $(X \ni x)$ itself to be a hypersurface germ.

Proposition 3.4. *Let $d \geq 4$ be an integer and let $(X \ni x)$ be a klt germ of dimension d. Assume that $(X \ni x)$ is neither smooth nor analytically an ordinary double point. Then the following hold.*

(1) If $(X \ni x)$ is a local complete intersection, then $\widehat{\mathrm{vol}}(X \ni x) \leq Q_d$.

(2) *Let $(X' \ni x')$ be a klt germ of dimension d which is a local complete intersection. Assume that $\widehat{\mathrm{vol}}(X' \ni x') = \widehat{\mathrm{vol}}(X \ni x)$ and $\mathrm{edim}(X \ni x) \leq \mathrm{edim}(X' \ni x')$. Then $\widehat{\mathrm{vol}}(X \ni x) \leq Q_d$.*

Proof. (1) Since $(X \ni x)$ is not smooth, $\mathrm{edim}(X \ni x) \geq d+1$ [Har77, I, Theorem 5.1]. If $\mathrm{edim}(X \ni x) = d+1$, then $\widehat{\mathrm{vol}}(X \ni x) \leq Q_d$ by Proposition 3.1. If $\mathrm{edim}(X \ni x) \geq d+2$, then $\widehat{\mathrm{vol}}(X \ni x) \leq 4(d-2)^d < Q_d$ by Lemma 3.2.

(2) There are three cases.

Suppose that $(X' \ni x')$ is smooth. Then $\mathrm{edim}(X \ni x) \leq \mathrm{edim}(X' \ni x') = d$, so $(X \ni x)$ is smooth [Har77, I, Theorem 5.1], a contradiction. Thus this case does not occur.

Suppose that $(X' \ni x')$ is analytically an ordinary double point. Let $\mathcal{O}^{\mathrm{an}}_{X',x'}$ be the analytic local ring of X' at x'. The completions of $\mathcal{O}_{X',x'}$ and $\mathcal{O}^{\mathrm{an}}_{X',x'}$ are both isomorphic to $\widehat{\mathcal{O}}_{X',x'}$ [Ser56, Proposition 3], and $A/\mathfrak{m}_A^2 \cong \widehat{A}/\widehat{\mathfrak{m}}_A^2$ for any Noetherian local ring $(A, \mathfrak{m}_A)$ [Mat89, Theorem 8.11]. Thus $\mathrm{edim}(X' \ni x') = \dim_{\mathbb{C}} \mathfrak{m}/\mathfrak{m}^2$, where $\mathfrak{m}$ is the maximal ideal of $\mathcal{O}^{\mathrm{an}}_{X',x'}$. Since

$$\mathcal{O}^{\mathrm{an}}_{X',x'} \cong \mathbb{C}\{z_0, \dots, z_d\}/(z_0^2 + \cdots + z_d^2),$$

we have $\mathrm{edim}(X' \ni x') = d+1$. Thus $\mathrm{edim}(X \ni x) \leq d+1$, and Proposition 3.1 implies that $\widehat{\mathrm{vol}}(X \ni x) \leq Q_d$.

Suppose that $(X' \ni x')$ is neither smooth nor analytically an ordinary double point. Then (1) applied to $(X' \ni x')$ implies that $\widehat{\mathrm{vol}}(X \ni x) = \widehat{\mathrm{vol}}(X' \ni x') \leq Q_d$. □

A test valuation only bounds the normalized volume from above. For the equality in Proposition 3.5, we use both directions of [Liu22, Proposition 2.17] for the trivial family, and then a translation moves a general point of the base to the origin.

Proposition 3.5. *Let $d \geq 4$ be an integer, let $T := \{z_1^2 + \cdots + z_d^2 = 0\} \subset \mathbb{A}^d$ with origin o, and let $X := T \times \mathbb{A}^1$ and $x := (o, 0)$. Then $(X \ni x)$ is a klt germ of dimension d which is neither smooth nor analytically an ordinary double point, and $\widehat{\mathrm{vol}}(X \ni x) = Q_d$.*

Proof. Let $\phi := z_1^2 + \cdots + z_d^2$ and let s be the coordinate of $\mathbb{A}^1$. A reducible quadratic form is a product of two linear forms and has rank at most two, while ϕ has rank $d \geq 4$. Thus ϕ is irreducible, and T and $X = \operatorname{Spec} \mathbb{C}[z_1, \dots, z_d, s]/(\phi)$ are integral hypersurfaces. In particular, T and X are Cohen–Macaulay and Gorenstein, and K_T and K_X are Cartier. The partial derivatives of ϕ vanish simultaneously only at $z_1 = \cdots = z_d = 0$, so $\operatorname{Sing} T = \{o\}$ and $\operatorname{Sing} X = \{o\} \times \mathbb{A}^1$, which are of codimension $d-1 \geq 3$. By Serre's criterion [Har77, II, Theorem 8.22A], T and X are normal.

Let $g : \widetilde{\mathbb{A}}^d \to \mathbb{A}^d$ be the blow-up of o with exceptional divisor $F \cong \mathbb{P}^{d-1}$, let $\widetilde{T}$ be the strict transform of T, and let $\mu := g|_{\widetilde{T}} : \widetilde{T} \to T$. Then $E := \widetilde{T} \cap F$ is the smooth quadric $\{\phi = 0\} \subset \mathbb{P}^{d-1}$, and $\widetilde{T}$ is the total space of $\mathcal{O}_E(-1)$, so $\widetilde{T}$ is smooth and E is the exceptional divisor of μ. Since $K_{\widetilde{\mathbb{A}}^d} = g^* K_{\mathbb{A}^d} + (d-1)F$ and $g^* T = \widetilde{T} + 2F$, adjunction implies

$$K_{\widetilde{T}} = \mu^* K_T + (d-3)E.$$

Thus $a(E, T, 0) = d - 2 > 0$, and T is klt. Moreover, $\mu \times \mathrm{id}_{\mathbb{A}^1} : \widetilde{T} \times \mathbb{A}^1 \to X$ is a log resolution with exceptional divisor $E \times \mathbb{A}^1$, and

$$K_{\widetilde{T} \times \mathbb{A}^1} = (\mu \times \mathrm{id}_{\mathbb{A}^1})^* K_X + (d-3)(E \times \mathbb{A}^1),$$

so X is klt. Hence $(X \ni x)$ is a klt germ of dimension d.

Since $x \in \operatorname{Sing} X$, $(X \ni x)$ is not smooth. The singular locus of the analytic germ of X at x is the germ of $\{o\} \times \mathbb{A}^1$ at x, which is of dimension one, while the singular locus of the analytic germ of $\{z_0^2 + \cdots + z_d^2 = 0\}$ at 0 is the point 0. Since an isomorphism of analytic germs maps the singular locus onto the singular locus, $(X \ni x)$ is not analytically an ordinary double point.

Since T is the ordinary double point of dimension $d-1 \geq 3$ (Definition 2.3), Theorem 2.4(2) implies that $\widehat{\mathrm{vol}}(T \ni o) = 2(d-2)^{d-1}$. Let $p : X \to \mathbb{A}^1$ be the projection and $\sigma(b) := (o, b)$ for $b \in \mathbb{A}^1$. The morphism p is flat, any fiber of p is T, and the relative canonical divisor of p is the pullback of K_T, so p with the section σ is a $\mathbb{Q}$-Gorenstein flat family of klt singularities over the smooth curve $\mathbb{A}^1$. By [Liu22, Proposition 2.17], for a general $b \in \mathbb{A}^1$ we have

$$\frac{\widehat{\mathrm{vol}}(X \ni (o,b))}{d^d} = \frac{\widehat{\mathrm{vol}}(T \ni o)}{(d-1)^{d-1}} = 2\Big(\frac{d-2}{d-1}\Big)^{d-1} = q_d.$$

Thus $\widehat{\mathrm{vol}}(X \ni (o,b)) = d^d q_d = Q_d$ by Lemma 2.7(1). The automorphism $(y, s) \to (y, s-b)$ of X maps (o, b) to x. The induced isomorphism of germs $(X \ni (o,b)) \cong (X \ni x)$ induces a bijection between the real valuations centered at (o, b) and those centered at x which preserves log discrepancies and volumes, so it preserves the normalized volume. Therefore $\widehat{\mathrm{vol}}(X \ni x) = \widehat{\mathrm{vol}}(X \ni (o,b)) = Q_d$. □

4. Reduction to Gorenstein cones

In this section, we prove Construction-Proposition 4.1 and Lemma 4.3. Construction-Proposition 4.1 replaces a klt germ which is not smooth and whose normalized volume exceeds the bound of Theorem 1.2 by a cone with a torus action whose normalized volume equals that of the germ and whose embedding dimension is at least that of the germ. It is used both in dimension four and in dimensions at least five. Under the replacement, the volume assumption becomes three structural properties of the cone: the singularity of the cone is isolated, the cone is Gorenstein, and, unless the germ is analytically an ordinary double point, the cone is not a local complete intersection.

The isolatedness step uses the volume assumption without any margin: for germs whose singular locus has positive dimension, Lemma 2.5 and Theorem 2.4(2) imply exactly the bound of Theorem 1.2, and the product germ of Proposition 3.5, whose singularity is not isolated, attains it.

Construction-Proposition 4.1. Let $d \geq 4$ be an integer and let $(X \ni x)$ be a klt germ of dimension d which is not smooth and satisfies $\widehat{\mathrm{vol}}(X \ni x) > Q_d$. Then there exist a normal affine variety $C = \operatorname{Spec} S$ of dimension d with a faithful action of an algebraic torus $\mathbb{T}$ as in Definition 2.8, with vertex o, a Reeb vector $\xi \in \mathfrak{t}^+$, and a $\mathbb{T}$-homogeneous element $\eta \in \omega_S$ such that the following hold.

(1) The log Fano cone (C, ξ) is K-polystable, $a(\mathrm{wt}_\xi, C) = d$, and
$$\widehat{\mathrm{vol}}(C \ni o) = a(\mathrm{wt}_\xi, C)^d \,\mathrm{vol}(\mathrm{wt}_\xi) = \widehat{\mathrm{vol}}(X \ni x).$$
Moreover, $\operatorname{edim}(X \ni x) \leq \operatorname{edim}(C \ni o)$.
(2) The variety C is klt, $C \backslash \{o\}$ is smooth, $o \in \operatorname{Sing}(C)$, the ring S is Gorenstein, and $\omega_S = S\eta$.
(3) For any primitive $\zeta \in \mathfrak{t}_{\mathbb{Z}} \cap \mathfrak{t}^+$, the degree r_ζ of η for the grading defined by ζ is a positive integer, $r_\zeta = a(\mathrm{wt}_\zeta, C)$, and $r_\zeta^d \,\mathrm{vol}(\mathrm{wt}_\zeta) \geq \widehat{\mathrm{vol}}(C \ni o)$.
(4) If $(X \ni x)$ is not analytically an ordinary double point, then $(C \ni o)$ is not a local complete intersection.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we construct C, $\mathbb{T}$, and ξ, we show that C is klt, and we prove (1).

By [XZ25, Theorem 1.2(1)], there exists a quasi-monomial valuation v of $\mathbb{C}(X)$ centered at x such that $a(v, X)^d \,\mathrm{vol}(v) = \widehat{\mathrm{vol}}(X \ni x)$, the graded ring $S' := \mathrm{gr}_v \mathcal{O}_{X,x}$ is finitely generated, and $C_0 := \operatorname{Spec} S'$, with the induced action of an algebraic torus $\mathbb{T}$ and the induced Reeb vector ξ_0, is a K-semistable log Fano cone. The variety C_0 is klt. Let $\mathbb{T}_1 \subset \mathbb{T}$ be the kernel of the action of $\mathbb{T}$ on C_0. The torus $\mathbb{T}/\mathbb{T}_1$ acts faithfully on C_0, the eigenspaces of $\mathbb{T}/\mathbb{T}_1$ in S' are the

non-zero $\mathbb{T}$-eigenspaces S'_χ, and the image of ξ_0 in the Lie algebra of $\mathbb{T}/\mathbb{T}_1$ defines the weight valuation wt_{ξ_0}. Possibly replacing $\mathbb{T}$ with $\mathbb{T}/\mathbb{T}_1$, we may assume that $\mathbb{T}$ acts faithfully on C_0. Then $C_0 = \operatorname{Spec} S'$ and $\mathbb{T}$ satisfy the assumptions of Definition 2.8. Let o_0 be the vertex of C_0. By [LX18, Lemma 2.58 and Theorem 1.3], $a(\mathrm{wt}_{\xi_0}, C_0) = a(v, X)$, $\mathrm{vol}(\mathrm{wt}_{\xi_0}) = \mathrm{vol}(v)$, and wt_{ξ_0} computes $\widehat{\mathrm{vol}}(C_0 \ni o_0)$. Thus $\widehat{\mathrm{vol}}(C_0 \ni o_0) = \widehat{\mathrm{vol}}(X \ni x)$. By [OAI26b, Lemma 2.2],

$$\mathrm{edim}(X \ni x) \le \mathrm{edim}(C_0 \ni o_0). \tag{4.1}$$

By [LWX21, Theorem 1.2 and Definitions 2.14 and 2.16], there exist an affine variety $\mathcal{C} = \operatorname{Spec} \mathcal{S}$, a flat morphism $\mathcal{C} \to \mathbb{A}^1 = \operatorname{Spec} \mathbb{C}[t]$, and an action of $\mathbb{T} \times \mathbb{G}_m$ on $\mathcal{C}$ with the following properties.

(a) The torus $\mathbb{T}$ acts trivially on $\mathbb{A}^1$, and $\mathbb{G}_m$ acts on $\mathbb{A}^1$ by multiplication.
(b) The divisor $K_{\mathcal{C}}$ is $\mathbb{Q}$-Cartier.
(c) Over $\mathbb{A}^1\backslash\{0\}$, the variety $\mathcal{C}$ is $\mathbb{T}$-equivariantly isomorphic to $C_0 \times (\mathbb{A}^1\backslash\{0\})$.
(d) The fiber $C = \operatorname{Spec} S$ of $\mathcal{C}$ over 0 is normal and klt, and (C, ξ_0) is a K-polystable log Fano cone.

Let $\mathcal{C}_1 \cong C_0$ be the fiber of $\mathcal{C}$ over 1, and let $\mathcal{S} = \bigoplus_\chi \mathcal{S}_\chi$ be the decomposition into $\mathbb{T}$-eigenspaces. Since t is $\mathbb{T}$-invariant, $\mathcal{S}_\chi/t\mathcal{S}_\chi = S_\chi$ and $\mathcal{S}_\chi/(t-1)\mathcal{S}_\chi \cong S'_\chi$ for any character χ. Since $\mathcal{S}$ is flat over $\mathbb{C}[t]$, each $\mathcal{S}_\chi$ is a flat $\mathbb{C}[t]$-module [LWX21, p. 1183]. In particular, $\mathcal{S}_0$ is torsion free, and (c) implies

$$\mathbb{C}[t] \subset \mathcal{S}_0 \subset \mathcal{S}_0[t^{-1}] = S'_0[t, t^{-1}] = \mathbb{C}[t, t^{-1}].$$

If $\mathcal{S}_0$ contained an element $g \notin \mathbb{C}[t]$, then, multiplying g by a power of t and subtracting a polynomial, we would obtain $t^{-1} \in \mathcal{S}_0$, hence $S_0 = \mathcal{S}_0/t\mathcal{S}_0 = 0$, which is absurd since $1 \in S_0$. Thus $\mathcal{S}_0 = \mathbb{C}[t]$ and $S_0 = \mathbb{C}$. Since $\mathbb{T}$ is reductive and $\mathcal{S}$ is a finitely generated $\mathcal{S}_0$-algebra, each $\mathcal{S}_\chi$ is a finitely generated $\mathcal{S}_0$-module, hence a free $\mathbb{C}[t]$-module of finite rank. Therefore

$$\dim_{\mathbb{C}} S_\chi = \dim_{\mathbb{C}} S'_\chi \quad \text{for any character } \chi \text{ of } \mathbb{T}. \tag{4.2}$$

By (4.2), the characters χ with $S_\chi \neq 0$ are exactly those with $S'_\chi \neq 0$. A torus acts faithfully on an affine variety if and only if the characters of the non-zero eigenspaces generate the character group. Hence $\mathbb{T}$ acts faithfully on C, the Reeb cones of C and C_0 are equal, and $C = \operatorname{Spec} S$ and $\mathbb{T}$ satisfy the assumptions of Definition 2.8. Let o be the vertex of C.

We compare the volumes. By [LWX21, Definition 2.10 and the paragraph after it], for $\xi' \in \mathfrak{t}^+$,

$$\mathrm{vol}(\mathrm{wt}_{\xi'}) = \lim_{\lambda \to +\infty} \frac{d!}{\lambda^d} \sum_{\langle \chi, \xi' \rangle \le \lambda} \dim_{\mathbb{C}} S_\chi$$

on C, and on C_0 the right-hand side with S'_χ in place of S_χ computes $\mathrm{vol}(\mathrm{wt}_{\xi'})$. By (4.2), the value of $\mathrm{vol}(\mathrm{wt}_{\xi'})$ on C equals the value of $\mathrm{vol}(\mathrm{wt}_{\xi'})$ on C_0.

We compare the log discrepancies. By [LWX21, p. 1183], there exist an integer $k \ge 1$ and a $\mathbb{T} \times \mathbb{G}_m$-equivariant nowhere vanishing section s of $\mathcal{O}_{\mathcal{C}}(kK_{\mathcal{C}})$. Let χ_0 be the $\mathbb{T}$-character of s. The fibers C and $\mathcal{C}_1$ are the normal Cartier divisors $\{t = 0\}$ and $\{t = 1\}$, and the functions t and $t - 1$ are $\mathbb{T}$-invariant. Since $\mathcal{C}$ is regular at any codimension one point of C or $\mathcal{C}_1$, adjunction holds at the codimension one points of C and of $\mathcal{C}_1$. Since $(\omega_S^{\otimes k})^{**}$ and $(\omega_{S'}^{\otimes k})^{**}$ are reflexive, it follows that the restrictions of s to C and to $\mathcal{C}_1 \cong C_0$ are $\mathbb{T}$-homogeneous generators of $(\omega_S^{\otimes k})^{**}$ and $(\omega_{S'}^{\otimes k})^{**}$ of character χ_0. Thus the assumptions of Lemma 2.9 hold for C and for C_0, with k in place of I, and Lemma 2.9(2) implies

$$a(\mathrm{wt}_{\xi'}, C) = \frac{\langle \chi_0, \xi' \rangle}{k} = a(\mathrm{wt}_{\xi'}, C_0) \quad \text{for any } \xi' \in \mathfrak{t}^+. \tag{4.3}$$

Since (C, ξ_0) is K-polystable, it is K-semistable, and [LX18, Theorem 1.3] implies that wt_{ξ_0} computes $\widehat{\mathrm{vol}}(C \ni o)$. By (4.3) and the equality of volumes,

$$\widehat{\mathrm{vol}}(C \ni o) = a(\mathrm{wt}_{\xi_0}, C)^d \,\mathrm{vol}(\mathrm{wt}_{\xi_0}) = \widehat{\mathrm{vol}}(C_0 \ni o_0) = \widehat{\mathrm{vol}}(X \ni x). \tag{4.4}$$

We compare the embedding dimensions. Let $\mathcal{J} := \bigoplus_{\chi\neq 0}\mathcal{S}_\chi$. Then $\mathcal{S}/\mathcal{J} = \mathcal{S}_0 = \mathbb{C}[t]$, so $\mathcal{J}$ defines a section $\sigma : \mathbb{A}^1 \to \mathcal{C}$ with $\sigma(0) = o$ and with $\sigma(1)$ the vertex of $\mathcal{C}_1 \cong C_0$. By [Har77, II, Exercise 5.8(a), Propositions 8.7 and 8.10], the fiber of the coherent sheaf $\sigma^*\Omega_{\mathcal{C}/\mathbb{A}^1}$ at a closed point $c \in \mathbb{A}^1$ is the cotangent space of the fiber of $\mathcal{C}$ over c at $\sigma(c)$, and the dimension of the fiber at c is upper semicontinuous in c, and by (c) it equals $\operatorname{edim}(C_0 \ni o_0)$ for every $c \neq 0$, since for such c the fiber of $\mathcal{C}$ over c is $\mathbb{T}$-equivariantly isomorphic to C_0 with $\sigma(c)$ corresponding to o_0. Hence $\operatorname{edim}(C_0 \ni o_0) \leq \operatorname{edim}(C \ni o)$, and (4.1) implies

$$\operatorname{edim}(X \ni x) \leq \operatorname{edim}(C \ni o). \tag{4.5}$$

Finally, we normalize the Reeb vector. By Lemma 2.9(1), wt_{ξ_0} is centered at o, and since C is klt, $a(\mathrm{wt}_{\xi_0}, C) > 0$. Let

$$\xi := \frac{d}{a(\mathrm{wt}_{\xi_0}, C)}\xi_0 \in \mathfrak{t}^+.$$

Since the assumptions of Lemma 2.9 hold for C, Lemma 2.9(4), applied to ξ_0 and the positive real number $d/a(\mathrm{wt}_{\xi_0}, C)$, implies $a(\mathrm{wt}_\xi, C) = d$ and $a(\mathrm{wt}_\xi, C)^d \operatorname{vol}(\mathrm{wt}_\xi) = a(\mathrm{wt}_{\xi_0}, C)^d \operatorname{vol}(\mathrm{wt}_{\xi_0})$. By [LWX21, Definitions 2.14 and 2.16], test configurations, special test configurations and product test configurations of (C, ξ') for $\xi' \in \mathfrak{t}^+$ do not depend on ξ'. For $c > 0$ and $\xi' \in \mathfrak{t}^+$, [LWX21, (2.7)] and Lemma 2.9(4) show that the vector field used in [LWX21, Definition 2.22] for $c\xi'$ is c times the one for ξ', and, by the formula of [LWX21, Definition 2.10], the volume function on the Reeb cone of the central fiber is homogeneous of degree $-d$. Hence any $\mathbb{Q}$-Gorenstein test configuration has equal generalized Futaki invariants with respect to ξ' and $c\xi'$, and by [LWX21, Definition 2.23], $(C, c\xi')$ is K-polystable if and only if (C, ξ') is. Since ξ is a positive multiple of ξ_0, (C, ξ) is K-polystable. Together with (4.4) and (4.5), this proves (1). The variety C is klt by (d).

Step 2. In this step we show that $o \in \operatorname{Sing}(C)$ and that $C\backslash\{o\}$ is smooth.

Since x is a singular point of X and $\dim X = d$, we have $\operatorname{edim}(X \ni x) > d$. By (1), $\operatorname{edim}(C \ni o) > d = \dim C$, so $o \in \operatorname{Sing}(C)$.

Let $f_1, \ldots, f_m$ be non-constant $\mathbb{T}$-homogeneous generators of the $\mathbb{C}$-algebra S, with characters $\chi_1, \ldots, \chi_m$. Since $S_0 = \mathbb{C}$, each χ_i is non-zero, and any χ with $S_\chi \neq 0$ is a sum of $\chi_1, \ldots, \chi_m$ with non-negative integer coefficients. Hence

$$\mathfrak{t}^+ = \{\xi' \in \mathfrak{t} \mid \langle \chi_i, \xi' \rangle > 0 \text{ for } 1 \leq i \leq m\}$$

is a non-empty open cone in $\mathfrak{t}$, and there exists a primitive element of $\mathfrak{t}_\mathbb{Z} \cap \mathfrak{t}^+$. Fix one, and let $S = \bigoplus_{j\geq 0} S_j$ be the grading defined by it. Then $S_0 = \mathbb{C}$, and the ideal of o in S is $\mathfrak{m}_o := \bigoplus_{j>0} S_j$.

Suppose that there exists $p \in \operatorname{Sing}(C)$ with $p \neq o$. Since o is the common zero locus of the elements of $\mathfrak{m}_o$, there exist $j > 0$ and $f \in S_j$ with $f(p) \neq 0$. For $\lambda \in \mathbb{G}_m$, $f(\lambda \cdot p) = \lambda^j f(p)$, so the $\mathbb{G}_m$-orbit of p is a curve. Since any homogeneous element of positive degree vanishes at o, the orbit morphism $\mathbb{G}_m \to C$ of p extends to a morphism $\mathbb{A}^1 \to C$ sending 0 to o. Thus the closure of the orbit of p is an irreducible curve through o. Since $\operatorname{Sing}(C)$ is closed and $\mathbb{G}_m$-invariant, it contains the closure of the orbit of p. Let V be an irreducible component of $\operatorname{Sing}(C)$ which contains the closure of the orbit of p. Then $o \in V$ and $\dim V \geq 1$. By Lemma 2.5, there exist a closed point $x' \in V$ and a hyperplane section H of an affine neighborhood of o in C such that $(H \ni x')$ is a klt germ of dimension $d-1$, $x' \in \operatorname{Sing}(H)$, and

$$\frac{\widehat{\operatorname{vol}}(C \ni o)}{d^d} \leq \frac{\widehat{\operatorname{vol}}(H \ni x')}{(d-1)^{d-1}}.$$

By Theorem 2.4(2) in dimension $d-1$, $\widehat{\operatorname{vol}}(H \ni x') \leq 2(d-2)^{d-1}$. Hence

$$\widehat{\operatorname{vol}}(C \ni o) \leq 2d^d \left(\frac{d-2}{d-1}\right)^{d-1} = Q_d,$$

which contradicts (1) and the assumption $\widehat{\mathrm{vol}}(X \ni x) > Q_d$. Thus $C\backslash\{o\}$ is smooth. When $d = 4$, the bound $\widehat{\mathrm{vol}}(C \ni o) \le Q_4$ for germs with $\dim_o \mathrm{Sing}(C) \ge 1$ is [Liu22, Corollary 2.18].

Step 3. In this step we show that K_C is Cartier at o and we prove (2).

Since C is klt, K_C is $\mathbb{Q}$-Cartier. Let I be the smallest positive integer such that IK_C is Cartier at o, and let U be an affine neighborhood of o in C such that $\mathcal{O}_U(IK_U) \cong \mathcal{O}_U$. We fix such an isomorphism, and we let

$$f : Y := \mathrm{Spec} \bigoplus_{i=0}^{I-1} H^0(U, \mathcal{O}_U(iK_U)) \to U$$

be the index one cover [KM98, Definition 5.19, Proposition 5.20]. Then f is a finite Galois morphism of degree I which is étale in codimension one, $K_Y = f^*K_U$, and Y is klt.

We show that $f^{-1}(o)$ is one point. For $0 \le i < I$, let A_i be the stalk of $\mathcal{O}_U(iK_U)$ at o, and let $A := \bigoplus_{i=0}^{I-1} A_i$, with multiplication defined by the fixed isomorphism. Then $A_0 = \mathcal{O}_{C,o}$, and $f^{-1}(o)$ is the underlying set of $\mathrm{Spec}\, A/\mathfrak{m}_o A$. Let $0 < i < I$ and $s \in A_i$. Then $s^I \in A_0$. If $s^I \notin \mathfrak{m}_o\mathcal{O}_{C,o}$, then $s^{I-1}(s^I)^{-1} \in A_{I-i}$ is the inverse of s in A, so the multiplication by s is an isomorphism $A_0 \to A_i$ of $\mathcal{O}_{C,o}$-modules. Then iK_C is Cartier at o, which contradicts the minimality of I. Thus $s^I \in \mathfrak{m}_o\mathcal{O}_{C,o}$, and the image of s in $A/\mathfrak{m}_o A$ is nilpotent. Since sums of nilpotent elements are nilpotent, the reduced ring of $A/\mathfrak{m}_o A$ is a quotient of $A_0/\mathfrak{m}_o A_0 = \mathbb{C}$. Hence $f^{-1}(o)$ consists of one point y.

By [XZ21, Theorem 1.3] and Theorem 2.4(1),

$$I \cdot \widehat{\mathrm{vol}}(C \ni o) = \widehat{\mathrm{vol}}(Y \ni y) \le d^d.$$

By (1) and Lemma 2.7(1) and (3),

$$\widehat{\mathrm{vol}}(C \ni o) = \widehat{\mathrm{vol}}(X \ni x) > Q_d = d^d q_d \ge d^d q_4 = \frac{16}{27} d^d > \frac{d^d}{2}.$$

Hence $I = 1$, that is, K_C is Cartier at o.

By [Kol13, Corollary 2.88(1)], S is Cohen–Macaulay. Since K_C is Cartier at o, the stalk of ω_S at o is free of rank one, so $\dim_{\mathbb{C}} \omega_S/\mathfrak{m}_o\omega_S = 1$. The quotient map $\omega_S \to \omega_S/\mathfrak{m}_o\omega_S$ is $\mathbb{T}$-equivariant, so there exists a $\mathbb{T}$-homogeneous $\eta \in \omega_S$ whose image in $\omega_S/\mathfrak{m}_o\omega_S$ is non-zero. Then $\omega_S = S\eta + \mathfrak{m}_o\omega_S$. With the grading of **Step 2**, the ring S is a *local ring with *maximal ideal $\mathfrak{m}_o$, and ω_S is a finite graded S-module. By the graded Nakayama lemma [BH98, Exercise 1.5.24(a)], $\omega_S = S\eta$. Since S is a domain and ω_S is torsion free, the map $S \to \omega_S$ given by η is an isomorphism. Thus S is Gorenstein and $\omega_S = S\eta$. Together with **Steps 1 and 2**, this proves (2).

Step 4. In this step we prove (3) and (4).

Let χ_η be the $\mathbb{T}$-character of η, and let $\zeta \in \mathfrak{t}_{\mathbb{Z}} \cap \mathfrak{t}^+$ be primitive. Then η has degree $r_\zeta = \langle \chi_\eta, \zeta \rangle \in \mathbb{Z}$ for the grading defined by ζ. By (2), the assumptions of Lemma 2.9 hold for C with $I = 1$ and $\chi_0 = \chi_\eta$, so Lemma 2.9(2) implies

$$a(\mathrm{wt}_\zeta, C) = \langle \chi_\eta, \zeta \rangle = r_\zeta.$$

By Lemma 2.9(1), wt_ζ is a positive multiple of a divisorial valuation centered at o. Since C is klt, $r_\zeta > 0$. Since wt_ζ is centered at o, Definition 2.2 implies $r_\zeta^d \,\mathrm{vol}(\mathrm{wt}_\zeta) \ge \widehat{\mathrm{vol}}(C \ni o)$. This proves (3).

Assume that $(X \ni x)$ is not analytically an ordinary double point, and suppose that $(C \ni o)$ is a local complete intersection. By **Step 1**, $(C \ni o)$ is a klt germ of dimension d, and by (1), $\widehat{\mathrm{vol}}(C \ni o) = \widehat{\mathrm{vol}}(X \ni x)$ and $\mathrm{edim}(X \ni x) \le \mathrm{edim}(C \ni o)$. Since $(X \ni x)$ is neither smooth nor analytically an ordinary double point, Proposition 3.4(2), with $(C \ni o)$ in place of $(X' \ni x')$, implies $\widehat{\mathrm{vol}}(X \ni x) \le Q_d$, a contradiction. This proves (4). □

Remark 4.2. Construction-Proposition 4.1(1) asks for K-polystability, rather than for the K-semistable cone given by [XZ25, Theorem 1.2(1)], only because the assumptions of [OAI26b,

Theorem 3.1], which Proposition 6.4 uses, include K-polystability (see Remark 6.3). Elsewhere in the paper, including the argument for $d = 4$, K-semistability of (C, ξ) is used only through the fact that wt_ξ computes $\widehat{\mathrm{vol}}(C \ni o)$. The bound of Proposition 3.4(2) used in Construction-Proposition 4.1(4) is attained by the product germ of Proposition 3.5, which is a hypersurface germ. Thus Construction-Proposition 4.1(4), like the smoothness of $C\backslash\{o\}$ in Construction-Proposition 4.1(2), uses the assumption $\widehat{\mathrm{vol}}(X \ni x) > Q_d$ without any margin.

The minimizing Reeb vector ξ may be irrational, and later sections approximate ξ by integral gradings. Proposition 5.5 and Lemma 6.2 use Lemma 4.3 for the approximation.

Lemma 4.3. *Notation as in Construction-Proposition 4.1. The real-valued function*

$$a(\mathrm{wt}_{\xi'}, C)^d \,\mathrm{vol}(\mathrm{wt}_{\xi'})$$

of $\xi' \in \mathfrak{t}^+$ is invariant under multiplication of ξ' by positive real numbers, bounded from below by $\widehat{\mathrm{vol}}(C \ni o)$, equal to $\widehat{\mathrm{vol}}(C \ni o)$ at ξ, and continuous at ξ.

Proof. Let χ_η be the $\mathbb{T}$-character of η. By Construction-Proposition 4.1(2), the assumptions of Lemma 2.9 hold for C with $I = 1$ and $\chi_0 = \chi_\eta$, so Lemma 2.9(2) implies

$$a(\mathrm{wt}_{\xi'}, C) = \langle \chi_\eta, \xi' \rangle \quad \text{for any } \xi' \in \mathfrak{t}^+. \tag{4.6}$$

By Lemma 2.9(4), the function is invariant under multiplication of ξ' by positive real numbers. Let $\xi' \in \mathfrak{t}^+$. By Lemma 2.9(1), $\mathrm{wt}_{\xi'}$ is a real valuation centered at o, and $a(\mathrm{wt}_{\xi'}, C) < +\infty$ by (4.6). Thus Definition 2.2 implies $a(\mathrm{wt}_{\xi'}, C)^d \,\mathrm{vol}(\mathrm{wt}_{\xi'}) \geq \widehat{\mathrm{vol}}(C \ni o)$. Equality at ξ is Construction-Proposition 4.1(1).

We prove continuity at ξ. Let $f_1, \dots, f_m$ be non-constant $\mathbb{T}$-homogeneous generators of the $\mathbb{C}$-algebra S, with characters $\chi_1, \dots, \chi_m$. Since $S_0 = \mathbb{C}$, each f_i is in the ideal of o, and $\langle \chi_i, \xi \rangle > 0$ for $1 \leq i \leq m$. Let $0 < \epsilon < 1$. There exists a neighborhood W of ξ in $\mathfrak{t}$ such that for any $\xi' \in W$ and $1 \leq i \leq m$,

$$(1 - \epsilon)\langle \chi_i, \xi \rangle \leq \langle \chi_i, \xi' \rangle \leq (1 + \epsilon)\langle \chi_i, \xi \rangle.$$

Any χ with $S_\chi \neq 0$ is a sum of $\chi_1, \dots, \chi_m$ with non-negative integer coefficients, since S_χ is spanned by monomials in $f_1, \dots, f_m$. Hence

$$(1 - \epsilon)\langle \chi, \xi \rangle \leq \langle \chi, \xi' \rangle \leq (1 + \epsilon)\langle \chi, \xi \rangle \quad \text{for any } \xi' \in W \text{ and any such } \chi.$$

In particular, $W \subset \mathfrak{t}^+$, and by Definition 2.8,

$$(1 - \epsilon)\,\mathrm{wt}_\xi(g) \leq \mathrm{wt}_{\xi'}(g) \leq (1 + \epsilon)\,\mathrm{wt}_\xi(g) \quad \text{for any } \xi' \in W \text{ and } g \in S.$$

An element $g' \in S$ with $g'(o) \neq 0$ has a non-zero component in S_0, so $\mathrm{wt}_\xi(g') = \mathrm{wt}_{\xi'}(g') = 0$. Since any element of $\mathcal{O}_{C,o}$ is a quotient g/g' of such elements, $(1 - \epsilon)\,\mathrm{wt}_\xi(g) \leq \mathrm{wt}_{\xi'}(g) \leq (1 + \epsilon)\,\mathrm{wt}_\xi(g)$ for any $\xi' \in W$ and $g \in \mathcal{O}_{C,o}$. Hence, for any $\lambda > 0$ and $\xi' \in W$,

$$\mathfrak{a}_{\lambda/(1-\epsilon)}(\mathrm{wt}_\xi) \subset \mathfrak{a}_\lambda(\mathrm{wt}_{\xi'}) \subset \mathfrak{a}_{\lambda/(1+\epsilon)}(\mathrm{wt}_\xi),$$

and Definition 2.2 implies

$$(1 + \epsilon)^{-d} \,\mathrm{vol}(\mathrm{wt}_\xi) \leq \mathrm{vol}(\mathrm{wt}_{\xi'}) \leq (1 - \epsilon)^{-d} \,\mathrm{vol}(\mathrm{wt}_\xi).$$

Since ϵ is arbitrary, $\mathrm{vol}(\mathrm{wt}_{\xi'})$ is continuous at ξ. By (4.6), $a(\mathrm{wt}_{\xi'}, C)$ is continuous at ξ. Thus $a(\mathrm{wt}_{\xi'}, C)^d \,\mathrm{vol}(\mathrm{wt}_{\xi'})$ is continuous at ξ. □

5. Klt singularities of dimension four

In this section, we prove Proposition 5.1, the case of dimension four of Theorem 1.2(1).

Proposition 5.1. *Let $(X \ni x)$ be a klt germ of dimension 4 which is neither smooth nor analytically an ordinary double point. Then $\widehat{\mathrm{vol}}(X \ni x) \leq Q_4 = 4096/27$.*

The proof follows [OAI26a], with the constants α and D of Set-up 5.2 in place of those of [OAI26a] (see Remark 5.3). New arguments are needed only for Propositions 5.5 and 5.7. The proof of Proposition 5.1 is given in Subsection 5.4.

Set-up 5.2. Let $(X \ni x)$, $C = \operatorname{Spec} S$, o, $\mathbb{T}$, ξ, η, α, and D be as follows, and for a given primitive $\zeta \in \mathfrak{t}_{\mathbb{Z}} \cap \mathfrak{t}^+$ let r, λ, and h_ζ be as in (4).

(1) $(X \ni x)$ is a klt germ of dimension 4 which is not smooth and satisfies $\widehat{\mathrm{vol}}(X \ni x) > Q_4$.
(2) $C = \operatorname{Spec} S$, o, $\mathbb{T}$, ξ, and η are as in Construction-Proposition 4.1 for $(X \ni x)$.
(3) $\alpha := 64/27$ and $D := 128/81$, so that $Q_4 = 64\alpha$ and $3D = 2\alpha$.
(4) Whenever a primitive $\zeta \in \mathfrak{t}_{\mathbb{Z}} \cap \mathfrak{t}^+$ is given, $S = \bigoplus_{j \ge 0} S_j$ denotes the grading defined by ζ, $r := r_\zeta$, $\lambda := \alpha/r$, $h_\zeta := \mathrm{vol}(\mathrm{wt}_\zeta)$, and every homogeneous element of degree j of a graded S-module has weight λj.
(5) The modules of differential forms and the canonical module ω_S carry the grading induced by the natural action of $\mathbb{G}_m$ on differential forms, so that df has degree j for $f \in S_j$, and η has degree r and weight $\lambda r = \alpha$.
(6) A homogeneous homomorphism of graded modules lowers weights by e if it maps elements of weight y to elements of weight $y - e$.

Remark 5.3. The results of [OAI26a, Section 3] and [OAI26a, Proposition 4.3(a)] are formulated for the cone of [OAI26a, Proposition 2.2], which is constructed under the assumption $\widehat{\mathrm{vol}}(X \ni x) \ge 162$. The proofs of [OAI26a, Lemmas 3.1 and 3.3] use only that C is a Gorenstein klt affine variety of dimension 4 with smooth $C\backslash\{o\}$, a faithful positive grading with $S_0 = \mathbb{C}$, and a homogeneous generator of ω_S of degree r. The computation of [OAI26a, Remark 3.2] is valid for every normal $\mathbb{Q}$-Gorenstein positively graded affine cone with a homogeneous generator of a reflexive pluricanonical power, and we use it in Subsection 5.3 for the cones $\operatorname{Spec} S/P$ of dimension two, where $P \subset S$ is a homogeneous prime ideal. The results [OAI26a, Lemmas 4.1 and 4.2] are formulated for an arbitrary finitely generated positively graded normal Cohen–Macaulay domain whose degree-zero part is $\mathbb{C}$ and whose grading action need not be faithful, and an arbitrary real number $\lambda > 0$ such that $\lambda m < 1$ for the order m of the stabilizer of every closed point other than the vertex. Propositions 5.5, 5.7, 5.9, and 5.10 adapt the arguments of [OAI26a, Propositions 3.4, 4.3, 5.1, and 6.1] to the constants α and D. We prove Proposition 5.7 with the test functions of the proof of [OAI26a, Proposition 4.3] and the new constants, except that in dimension 3 the test function is centered at $2/3$, which is the center used in dimension 2 in that proof, instead of the midpoint $D/2$ of $(0, D)$. With the center $D/2 = 64/81$, the lower bound obtained in the step for dimension 3 of the proof of [OAI26a, Proposition 4.3] is the negative number $-2811713/172186884$.

Proposition 5.5(1) bounds the stabilizers by means of [OAI26a, Lemma 3.3]. A stabilizer of order $m \ge r/\alpha$ satisfies $mh_\zeta r^3 \ge \widehat{\mathrm{vol}}(C \ni o)/\alpha$, and for $mh_\zeta r^3 > 64$ the bound of [OAI26a, Lemma 3.3] decreases near the parameter 0. Thus Proposition 5.5(1) needs $\widehat{\mathrm{vol}}(C \ni o)/\alpha > 64$, and $\alpha = Q_4/64$ is the largest constant for which $\widehat{\mathrm{vol}}(C \ni o) > Q_4$ implies $\widehat{\mathrm{vol}}(C \ni o)/\alpha > 64$. The constant D is determined by the threshold $3/2$ of Proposition 5.9, which is the lower bound for $\mathrm{lct}(C \ni o; \mathfrak{m}_o)$ needed in Proposition 5.10. If $\mathrm{lct}(C \ni o; \mathfrak{m}_o) \le 3/2$, then the boundary built in Proposition 5.9 from homogeneous elements of weight in $(0, D)$ has weight less than $(3/2)D = \alpha$, which is the weight of η, and hence the vertex is not an lc center of the boundary. This is why $D = 2\alpha/3$, and a larger α yields a longer interval $(0, D)$ in Proposition 5.7.

5.1. **Stabilizers and vertex discrepancies.** In this subsection, we prove Proposition 5.5, which bounds the stabilizers of the gradings near the Reeb ray $\mathbb{R}_{>0}\xi$ and the log discrepancies of the divisors over C centered at the vertex. We first recall the extraction of the vertex constructed in [OAI26a, Lemma 3.1], together with the charts of the extraction.

Construction-Proposition 5.4 ([OAI26a, Lemma 3.1])**.** Notation and conditions as in Set-up 5.2, and let $\zeta \in \mathfrak{t}_{\mathbb{Z}} \cap \mathfrak{t}^+$ be primitive. Let $\widetilde{C}$ be the quotient of $(C\backslash\{o\}) \times \mathbb{A}^1$ by the action of

$\mathbb{G}_m$ given by the grading on the first factor and by weight -1 on the coordinate u of $\mathbb{A}^1$, and let $E \subset \widetilde{C}$ be the image of $\{u = 0\}$. Then the following hold.

(1) The morphism $(C\backslash\{o\}) \times \mathbb{A}^1 \to C$ which pulls back every $f \in S_j$ to $u^j f$ induces a proper birational morphism $\pi : \widetilde{C} \to C$ which is an isomorphism over $C\backslash\{o\}$, $\pi^{-1}(o) = E$ as sets, E is a prime divisor isomorphic to $\operatorname{Proj} S$, $(\widetilde{C}, E)$ is lc, and
$$K_{\widetilde{C}} = \pi^* K_C + (r-1)E.$$

(2) Let $k > 0$ be an integer, let $g \in S_k$ be non-zero, and let $W := \{g = 1\} \subset C$. Then the following hold.
   (a) W is smooth and stable under $\mu_k \subset \mathbb{G}_m$.
   (b) The image of $\{g \neq 0\} \times \mathbb{A}^1$ in $\widetilde{C}$ is an open subset isomorphic to $(W \times \mathbb{A}^1)/\mu_k$.
   (c) The quotient morphism $W \times \mathbb{A}^1 \to (W \times \mathbb{A}^1)/\mu_k$ is finite and étale in codimension one.
   (d) The composite $W \times \mathbb{G}_m \to \widetilde{C} \to C$ is a finite étale morphism onto the open subset $\{g \neq 0\}$ of C.

(3) Let $p \in C\backslash\{o\}$ be a closed point whose stabilizer for ζ is μ_m, and let $\tilde{p} \in E$ be the point corresponding to the orbit of p. Then the following hold.
   (a) mE is Cartier at $\tilde{p}$.
   (b) The completion of $\mathcal{O}_{\widetilde{C},\tilde{p}}$ is the ring of μ_m-invariants of $\mathbb{C}[[u, z_1, z_2, z_3]]$ for a μ_m-action which is faithful on u and on the span of z_1, z_2, z_3 and whose quotient map is étale in codimension one, such that E corresponds to $\{u = 0\}$ and every $f \in S_j$ is the product of u^j and a power series in z_1, z_2, z_3.

Proof. (1) Since $\zeta \in \mathfrak{t}^+$ is integral, $\langle \chi, \zeta \rangle$ is a positive integer for any character $\chi \neq 0$ with $S_\chi \neq 0$. Thus the grading defined by ζ is positive, and S_0 is the eigenspace of the trivial character, which is $\mathbb{C}$ by Definition 2.8. Since ζ is primitive, the cocharacter $\zeta : \mathbb{G}_m \to \mathbb{T}$ is injective, so the action of $\mathbb{G}_m$ on C defined by ζ is faithful. By Construction-Proposition 4.1(2) and (3), C is a Gorenstein klt affine variety of dimension 4, $C\backslash\{o\}$ is smooth, and $\omega_S = S\eta$, where η is homogeneous of degree r. Thus the conditions of [OAI26a, Lemma 3.1] listed in Remark 5.3 hold, and $\widetilde{C}$, E, and π are the objects constructed in [OAI26a, Lemma 3.1]. Thus (1) follows from [OAI26a, Lemma 3.1].

(2) Since $k > 0$, g vanishes at o, so $W \subset C\backslash\{o\}$. For $p \in W$ and $t \in \mathbb{G}_m$ we have $g(t \cdot p) = t^k g(p) = t^k$, so the derivative of g along the $\mathbb{G}_m$-orbit of p at $t = 1$ is equal to $k \neq 0$. Thus dg does not vanish at any point of W, and W is smooth because $C\backslash\{o\}$ is smooth. For $\epsilon \in \mu_k$ and $p \in W$ we have $g(\epsilon \cdot p) = \epsilon^k = 1$, so W is stable under μ_k. The group μ_k acts on $W \times \mathbb{A}^1$ by $\epsilon \cdot (p, u) = (\epsilon \cdot p, \epsilon^{-1} u)$.

For $(q, u) \in \{g \neq 0\} \times \mathbb{A}^1$ and $t \in \mathbb{G}_m$, we have $g(t \cdot q) = t^k g(q)$, so $(t \cdot q, t^{-1}u) \in W \times \mathbb{A}^1$ if and only if $t^k = g(q)^{-1}$. The solutions t of $t^k = g(q)^{-1}$ form one coset of μ_k in $\mathbb{G}_m$. Thus any $\mathbb{G}_m$-orbit in $\{g \neq 0\} \times \mathbb{A}^1$ intersects $W \times \mathbb{A}^1$ in exactly one μ_k-orbit, and the image of $\{g \neq 0\} \times \mathbb{A}^1$ in $\widetilde{C}$, which is open because $\{g \neq 0\} \times \mathbb{A}^1$ is open and $\mathbb{G}_m$-stable, is $(W \times \mathbb{A}^1)/\mu_k$, since the morphism $\mathbb{G}_m \times W \times \mathbb{A}^1 \to \{g \neq 0\} \times \mathbb{A}^1$, $(t, p, u) \to (t \cdot p, t^{-1}u)$, is the quotient by the free μ_k-action $\epsilon \cdot (t, p, u) = (t\epsilon^{-1}, \epsilon \cdot p, \epsilon^{-1}u)$ and is $\mathbb{G}_m$-equivariant for the action on the factor t, so $(\{g \neq 0\} \times \mathbb{A}^1)/\mathbb{G}_m = (W \times \mathbb{A}^1)/\mu_k$.

If $\epsilon \in \mu_k$ fixes a point (p, u) with $u \neq 0$, then $\epsilon^{-1}u = u$, so $\epsilon = 1$. Hence μ_k acts freely on $W \times \mathbb{G}_m$, and $W \times \mathbb{G}_m \to (W \times \mathbb{G}_m)/\mu_k$ is finite and étale. By (1), the composite $W \times \mathbb{G}_m \to C$ sends (p, u) to $u \cdot p$, because $f(u \cdot p) = u^j f(p)$ for $f \in S_j$. The image of $\{g \neq 0\} \times \mathbb{G}_m$ in $\widetilde{C}$ is $(W \times \mathbb{G}_m)/\mu_k$, it is contained in $\widetilde{C}\backslash E$, and π maps it isomorphically onto $\{g \neq 0\}$ by (1). Thus $W \times \mathbb{G}_m \to C$ is a finite étale morphism onto $\{g \neq 0\}$.

Let $1 \neq \epsilon \in \mu_k$. By the previous paragraph, the fixed points of ϵ in $W \times \mathbb{A}^1$ are contained in $W \times \{0\}$. Let W' be an irreducible component of W, and suppose that ϵ fixes W' pointwise. Since the action of ϵ on C commutes with the action of $\mathbb{G}_m$, ϵ fixes the image of $W' \times \mathbb{G}_m$ in C pointwise.

The image of $W' \times \mathbb{G}_m$ in C is a non-empty open subset of C because $W \times \mathbb{G}_m \to \{g \neq 0\}$ is étale, so ϵ acts trivially on the variety C, which contradicts the faithfulness of the action of $\mathbb{G}_m$. Thus the fixed locus of ϵ in $W \times \mathbb{A}^1$ contains no irreducible component of $W \times \{0\}$, so it has codimension at least 2 in $W \times \mathbb{A}^1$. Hence the finite quotient morphism $W \times \mathbb{A}^1 \to (W \times \mathbb{A}^1)/\mu_k$ is étale in codimension one.

(3) Since $p \neq o$, there exists a homogeneous $g \in S_k$ with $k > 0$ and $g(p) \neq 0$. Replacing p with a point of the $\mathbb{G}_m$-orbit of p changes neither the stabilizer nor $\tilde{p}$, so we may assume that $g(p) = 1$, that is, $p \in W$ in the notation of (2). Then $\tilde{p}$ is the image of $(p, 0)$ in $(W \times \mathbb{A}^1)/\mu_k$, and the stabilizer of $(p, 0)$ in μ_k is μ_m. Since μ_m is finite, there exist formal coordinates z_1, z_2, z_3 of W at p on which μ_m acts by characters, and the completion of $\mathcal{O}_{\widetilde{C},\tilde{p}}$ is the ring of μ_m-invariants of $\mathbb{C}[[u, z_1, z_2, z_3]]$, the completion of $\mathcal{O}_{W\times\mathbb{A}^1,(p,0)}$. This is the completed local model of [OAI26a, Lemma 3.1]. In $\mathbb{C}[[u, z_1, z_2, z_3]]$, E corresponds to $\{u = 0\}$, and the quotient map is étale in codimension one by (2). The action on u is $\epsilon \cdot u = \epsilon^{-1}u$, which is faithful. By [OAI26a, Lemma 3.1], mE is Cartier at $\tilde{p}$.

Suppose that $\epsilon \in \mu_m$ acts trivially on the span of z_1, z_2, z_3, that is, on the tangent space T_pW. Since the action of ϵ commutes with the action of $\mathbb{G}_m$ and fixes p, ϵ fixes the $\mathbb{G}_m$-orbit of p pointwise. By (2), dg vanishes on T_pW and not on $T_p(\mathbb{G}_m \cdot p)$, so

$$T_pC = T_pW \oplus T_p(\mathbb{G}_m \cdot p),$$

and ϵ acts trivially on T_pC. Since μ_m is finite and p is a smooth point of C, there exist formal coordinates of C at p on which ϵ acts linearly, so ϵ acts trivially on $\widehat{\mathcal{O}}_{C,p}$. Thus ϵ acts trivially on $\mathcal{O}_{C,p} \subset \widehat{\mathcal{O}}_{C,p}$, hence on $\mathbb{C}(C)$ and on C, so $\epsilon = 1$ by the faithfulness of the action of $\mathbb{G}_m$. This shows that μ_m acts faithfully on the span of z_1, z_2, z_3.

Finally, let $f \in S_j$. By (1), the pullback of f to $W \times \mathbb{A}^1$ is $u^j \cdot f|_W$, and $f|_W$ is a power series in z_1, z_2, z_3 in the completion of $\mathcal{O}_{W,p}$. This proves (3). □

Since the margin $\widehat{\mathrm{vol}}(C \ni o)/\alpha - 64$ can be arbitrarily small, the open subset $\mathcal{U}$ in Proposition 5.5(1) depends on C. Proposition 5.1 uses only one primitive grading in $\mathcal{U}$.

Proposition 5.5. *Notation and conditions as in Set-up 5.2.*

(1) There exists an open subset $\mathcal{U} \subset \mathfrak{t}^+$ with $\xi \in \mathcal{U}$ and $\mathbb{R}_{>0}\mathcal{U} = \mathcal{U}$ such that for every primitive $\zeta \in \mathfrak{t}_{\mathbb{Z}} \cap \mathcal{U}$ and every closed point $p \in C\backslash\{o\}$, the stabilizer of p for ζ has order m with $\alpha m < r_\zeta$.

(2) Every prime divisor F over C with $\mathrm{center}_C F = \{o\}$ satisfies $a(F, C, 0) \geq 3$. In particular C is terminal.

Proof. (1) Let $\beta_0 := \widehat{\mathrm{vol}}(C \ni o)/\alpha$. By Construction-Proposition 4.1(1) and Set-up 5.2(1) and (3),

$$\widehat{\mathrm{vol}}(C \ni o) = \widehat{\mathrm{vol}}(X \ni x) > Q_4 = 64\alpha,$$

so $\beta_0 > 64$. For real numbers $\beta > 0$ and $\delta \geq 0$, let

$$f_\beta(\delta) := \frac{(1 + 3\delta)^4}{(1 + \delta\beta^{1/3})^3}.$$

Then $f_{\beta_0}(0) = 1$ and

$$f'_{\beta_0}(0) = 12 - 3\beta_0^{1/3} < 0,$$

because $\beta_0^{1/3} > 4$. Thus there exists a rational number $\delta > 0$ such that $\kappa := f_{\beta_0}(\delta) < 1$, and we fix δ and κ. Since $\delta > 0$, for any $\beta \geq \beta_0$ we have $1 + \delta\beta^{1/3} \geq 1 + \delta\beta_0^{1/3}$, so

$$f_\beta(\delta) \leq f_{\beta_0}(\delta) = \kappa.$$

Since $\kappa < 1$, we have $\widehat{\mathrm{vol}}(C \ni o) < \widehat{\mathrm{vol}}(C \ni o)/\kappa$. By Lemma 4.3 (with $d = 4$), the function of $\xi' \in \mathfrak{t}^+$ in Lemma 4.3 is equal to $\widehat{\mathrm{vol}}(C \ni o)$ at ξ and continuous at ξ, so there exists an open

subset $\mathcal{U}_0 \subset \mathfrak{t}^+$ with $\xi \in \mathcal{U}_0$ such that

$$a(\mathrm{wt}_{\xi'}, C)^4 \operatorname{vol}(\mathrm{wt}_{\xi'}) < \frac{\widehat{\mathrm{vol}}(C \ni o)}{\kappa} \quad \text{for any } \xi' \in \mathcal{U}_0. \tag{5.1}$$

Let $\mathcal{U} := \mathbb{R}_{>0}\mathcal{U}_0$. Then $\mathcal{U}$ is open, $\xi \in \mathcal{U}$, $\mathbb{R}_{>0}\mathcal{U} = \mathcal{U}$, and $\mathcal{U} \subset \mathfrak{t}^+$ because $\mathbb{R}_{>0}\mathfrak{t}^+ = \mathfrak{t}^+$. Since the function $a(\mathrm{wt}_{\xi'}, C)^4 \operatorname{vol}(\mathrm{wt}_{\xi'})$ is invariant under multiplication of ξ' by positive real numbers (Lemma 4.3), (5.1) holds for any $\xi' \in \mathcal{U}$.

Let $\zeta \in \mathfrak{t}_{\mathbb{Z}} \cap \mathcal{U}$ be primitive, let $r := r_\zeta$, and let $p \in C\backslash\{o\}$ be a closed point. By Lemma 2.9(3), the stabilizer of p for ζ is μ_m for a positive integer m. By Construction-Proposition 4.1(3), $r = a(\mathrm{wt}_\zeta, C)$, so

$$r^4 h_\zeta < \frac{\widehat{\mathrm{vol}}(C \ni o)}{\kappa}. \tag{5.2}$$

Suppose that $\alpha m \geq r$. Let $\beta := m h_\zeta r^3$. By Construction-Proposition 4.1(3), $r^4 h_\zeta \geq \widehat{\mathrm{vol}}(C \ni o)$, so

$$\beta = \frac{m}{r} \cdot r^4 h_\zeta \geq \frac{1}{\alpha} \widehat{\mathrm{vol}}(C \ni o) = \beta_0.$$

By Remark 5.3 and Construction-Proposition 4.1(2) and (3), the hypotheses of [OAI26a, Lemma 3.3] hold for the grading defined by ζ and the point p. Thus by [OAI26a, Lemma 3.3], $\widehat{\mathrm{vol}}(C \ni o) \leq (r + 3s)^4 h_\zeta / (1 + s(m h_\zeta)^{1/3})^3$ for every rational number $s > 0$. For the positive rational number $s = r\delta$ we obtain

$$\widehat{\mathrm{vol}}(C \ni o) \leq \frac{(r + 3r\delta)^4 h_\zeta}{\left(1 + r\delta (m h_\zeta)^{1/3}\right)^3} = r^4 h_\zeta \cdot \frac{(1 + 3\delta)^4}{(1 + \delta \beta^{1/3})^3} = r^4 h_\zeta f_\beta(\delta) \leq \kappa r^4 h_\zeta.$$

This contradicts (5.2). Thus $\alpha m < r$.

(2) Since $\mathcal{U}$ is a non-empty open subset of $\mathfrak{t}$ with $\mathbb{R}_{>0}\mathcal{U} = \mathcal{U}$ and $\mathfrak{t}_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q}$ is dense in $\mathfrak{t}$, there exists a primitive $\zeta \in \mathfrak{t}_{\mathbb{Z}} \cap \mathcal{U}$. Let $r := r_\zeta$, and let $\pi : \widetilde{C} \to C$ and E be as in Construction-Proposition 5.4 for ζ. Let F be a prime divisor over C with $\operatorname{center}_C F = \{o\}$. Since π is proper, $\operatorname{center}_{\widetilde{C}} F$ exists and is contained in $\pi^{-1}(o) = E$. Let $\tilde{p}$ be a closed point of $\operatorname{center}_{\widetilde{C}} F$. Then $\tilde{p} \in E$ corresponds to the orbit of a closed point $p \in C\backslash\{o\}$, and the stabilizer of p for ζ is μ_m with $\alpha m < r$ by (1). By Construction-Proposition 5.4(3), mE is Cartier at $\tilde{p}$. Since $\tilde{p} \in \operatorname{center}_{\widetilde{C}} F \subset \operatorname{Supp} E$, $\operatorname{mult}_F(mE)$ is a positive integer, so

$$\operatorname{mult}_F E \geq \frac{1}{m}.$$

By Construction-Proposition 5.4(1), $K_{\widetilde{C}} + E = \pi^* K_C + rE$, and $(\widetilde{C}, E)$ is lc. Thus

$$a(F, C, 0) = a(F, \widetilde{C}, E) + r \operatorname{mult}_F E \geq \frac{r}{m} > \alpha = \frac{64}{27} > 2.$$

By Construction-Proposition 4.1(2), $\omega_S = S\eta$, so K_C is Cartier. Since K_C is Cartier, $a(F, C, 0) \in \mathbb{Z}$ (see [KM98, Definition 2.25]). Thus $a(F, C, 0) \geq 3$.

Let F be an exceptional prime divisor over C with $\operatorname{center}_C F \neq \{o\}$. Then $\operatorname{center}_C F$ intersects the smooth open subset $C\backslash\{o\}$, so $a(F, C, 0) \geq 2$ because smooth varieties are terminal (see [KM98, Corollary 2.31(2)]). Thus C is terminal. □

5.2. **Homogeneous elements of small weight.** In this subsection, we prove Proposition 5.7, which provides the homogeneous elements of weight in $(0, D)$ used in Proposition 5.9: on the cone C (case (a)) and on the possible minimal lc centers of dimension two and three (case (b)). In the case $d = 4$ we use the test function $g(x) = g_3(x - 1) + g_3(x + 1)$ that [OAI26a] uses for rings of dimension four (see Remark 5.3), and the case $d = 4$ needs two inequalities between $M_2(g)$, $M_4(g)$, $M_6(g)$, and $g(\alpha/2)$, with the constants α and D of Set-up 5.2(3). We prove them in the following lemma.

Lemma 5.6. *Let $g_3 : \mathbb{R} \to \mathbb{R}$ be the function*

$$g_3(v) := \int_{-1/2}^{1/2} \cos^3(\pi\tau) e^{2\pi i v\tau} d\tau,$$

let $g(x) := g_3(x-1)+g_3(x+1)$, and let $M_k(g) := \int_{\mathbb{R}} x^k |g(x)|^2 dx$ for $k \in \{2,4,6\}$. Let $\alpha := 64/27$, $D := 128/81$, and $\ell := \alpha/2 + D$. Then the following hold.

(1) $\ell^2 M_2(g) - M_4(g) > 0$.

(2) $\dfrac{6561}{8192}\Big(\ell^2\big(M_4(g) - M_2(g)\big) - M_6(g) + M_4(g)\Big) > \alpha\Big(\ell^2 - \dfrac{\alpha^2}{4}\Big)|g(\alpha/2)|^2$.

Proof. Since $\cos^3(\pi\tau)$ is even, $g_3(v) = \int_{-1/2}^{1/2} \cos^3(\pi\tau)\cos(2\pi v\tau)d\tau$ is real. For any real number $v \notin \{\pm 1/2, \pm 3/2\}$, a direct calculation (write $\cos(k\pi\tau)\cos(2\pi v\tau) = \frac{1}{2}\big(\cos((k+2v)\pi\tau) + \cos((k-2v)\pi\tau)\big)$ for $k = 1, 3$ and integrate over $[-1/2, 1/2]$) shows

$$\int_{-1/2}^{1/2} \cos(\pi\tau) e^{2\pi i v\tau} d\tau = \frac{2\cos(\pi v)}{\pi(1-4v^2)}, \qquad \int_{-1/2}^{1/2} \cos(3\pi\tau) e^{2\pi i v\tau} d\tau = -\frac{6\cos(\pi v)}{\pi(9-4v^2)}.$$

Since $\cos^3(\pi\tau) = \frac{1}{4}\big(3\cos(\pi\tau) + \cos(3\pi\tau)\big)$ and

$$\frac{3}{4}\cdot\frac{2}{1-4v^2} - \frac{1}{4}\cdot\frac{6}{9-4v^2} = \frac{12}{(1-4v^2)(9-4v^2)},$$

we have

$$g_3(v) = \frac{12\cos(\pi v)}{\pi(1-4v^2)(9-4v^2)} \quad \text{for } v \notin \{\pm 1/2, \pm 3/2\}.$$

Let $f(\tau) := 2\cos(2\pi\tau)\cos^3(\pi\tau)$ for $|\tau| \le 1/2$ and $f(\tau) := 0$ otherwise. Since

$$e^{2\pi i(x\mp 1)\tau} = e^{\mp 2\pi i\tau} e^{2\pi i x\tau},$$

we have

$$g(x) = \int_{\mathbb{R}} f(\tau) e^{2\pi i x\tau} d\tau, \qquad f(\tau) = \cos(\pi\tau) + \frac{3}{4}\cos(3\pi\tau) + \frac{1}{4}\cos(5\pi\tau) \quad \text{for } |\tau| \le 1/2.$$

The functions f, f', and f'' vanish at $\tau = \pm 1/2$, because f is the product of $\cos^3(\pi\tau)$ and a smooth function on $[-1/2, 1/2]$. Thus $f \in C^2(\mathbb{R})$, f''' is piecewise continuous and bounded, and hence $f \in H^3(\mathbb{R})$. Let $M_0(g) := \int_{\mathbb{R}} |g(x)|^2 dx$. By [Fol99, Theorems 8.22 and 8.29],

$$M_{2j}(g) = (2\pi)^{-2j} \int_{-1/2}^{1/2} |f^{(j)}(\tau)|^2 d\tau \quad \text{for } 0 \le j \le 3.$$

For $k, k' \in \{1,3,5\}$ with $k \ne k'$, the functions $\cos(k\pi\tau)$ and $\cos(k'\pi\tau)$ are orthogonal in $L^2([-1/2, 1/2])$, so are $\sin(k\pi\tau)$ and $\sin(k'\pi\tau)$, and

$$\int_{-1/2}^{1/2} \cos^2(k\pi\tau) d\tau = \int_{-1/2}^{1/2} \sin^2(k\pi\tau) d\tau = \frac{1}{2}.$$

Since $f^{(j)}$ is the sum of $\pm(k\pi)^j$ times the coefficient of $\cos(k\pi\tau)$ in f times $\cos(k\pi\tau)$ or $\sin(k\pi\tau)$ over $k \in \{1,3,5\}$, we obtain

$$M_{2j}(g) = \frac{1}{2}\left(\left(\frac{1}{2}\right)^{2j} + \frac{9}{16}\left(\frac{3}{2}\right)^{2j} + \frac{1}{16}\left(\frac{5}{2}\right)^{2j}\right) \quad \text{for } 0 \le j \le 3. \tag{5.3}$$

Thus

$$M_0(g) = \frac{13}{16}, \qquad M_2(g) = \frac{61}{64}, \qquad M_4(g) = \frac{685}{256}, \qquad M_6(g) = \frac{11101}{1024},$$

which are also listed in Table 1.

We have $\ell = 224/81$ and $\ell^2 = 50176/6561$. Let

$$\kappa_1 := \ell^2 M_2(g) - M_4(g), \qquad \kappa_2 := \ell^2\big(M_4(g) - M_2(g)\big) - M_6(g) + M_4(g).$$

Substituting the values of $M_{2j}(g)$, we obtain

$$\kappa_1 = \frac{7748659}{1679616} > 0, \qquad \kappa_2 = \frac{3739327}{746496}, \qquad \frac{6561}{8192}\kappa_2 = \frac{33653943}{8388608}. \tag{5.4}$$

In particular, (1) holds.

We prove (2). The constant 6561/8192 in (2) is $Q_4/(6\alpha^4)$ (Lemma 2.7(1)). We have $\alpha/2-1 = 5/27$ and $\alpha/2+1 = 59/27$. For $v = 5/27$ and $v = 59/27$,

$$(1-4v^2, 9-4v^2) = \Big(\frac{629}{729}, \frac{6461}{729}\Big) \quad \text{and} \quad (1-4v^2, 9-4v^2) = \Big(-\frac{13195}{729}, -\frac{7363}{729}\Big),$$

respectively, so $(1-4v^2)(9-4v^2) > 0$ for both values of v. Since $\cos(59\pi/27) = \cos(5\pi/27)$, we have

$$g(\alpha/2) = \frac{\kappa_3 \cos(5\pi/27)}{\pi}, \qquad \kappa_3 := 12 \cdot 729^2 \left(\frac{1}{629 \cdot 6461} + \frac{1}{13195 \cdot 7363}\right) = \frac{191714152104}{117265825495} > 0.$$

In particular, $0 < \kappa_3 < 16349/10000$. Next we show $\pi > 157/50$. By Machin's formula,

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239).$$

By the tangent addition formula, $\tan(2\arctan(1/5)) = 5/12$, $\tan(4\arctan(1/5)) = 120/119$, and $\tan(4\arctan(1/5) - \arctan(1/239)) = 1$. Since $x - x^3/3 < \arctan x < x$ for any $x > 0$, we have

$$0 < 4\Big(\frac{1}{5} - \frac{1}{3 \cdot 5^3}\Big) - \frac{1}{239} < 4\arctan(1/5) - \arctan(1/239) < \frac{4}{5} < \frac{\pi}{2},$$

which implies Machin's formula, and

$$\pi > 16\Big(\frac{1}{5} - \frac{1}{3 \cdot 5^3}\Big) - \frac{4}{239} = \frac{281476}{89625} > \frac{157}{50}.$$

We have $\ell^2 - \alpha^2/4 = 40960/6561$ and $\alpha(\ell^2 - \alpha^2/4) = 2621440/177147$. Since $|\cos(5\pi/27)| \le 1$, $\pi > 157/50$, and $0 < \kappa_3 < 16349/10000$,

$$\alpha\Big(\ell^2 - \frac{\alpha^2}{4}\Big)|g(\alpha/2)|^2 < \frac{2621440}{177147}\Big(\frac{50}{157} \cdot \frac{16349}{10000}\Big)^2 < 4.0118 < \frac{33653943}{8388608} = \frac{6561}{8192}\kappa_2$$

by (5.4). Thus (2) holds. □

Remark 5.3 explains why [OAI26a, Lemmas 4.1 and 4.2] are available for the rings and the real numbers λ in Proposition 5.7.

Proposition 5.7. *Let $T = \bigoplus_{k\ge 0} T_k$ be a finitely generated graded $\mathbb{C}$-algebra with $T_0 = \mathbb{C}$ which is a normal Cohen–Macaulay domain of dimension $d \in \{2,3,4\}$. Let $\lambda > 0$ be a real number, let ω_T carry the grading induced by the natural action of $\mathbb{G}_m$ on differential forms, and let every homogeneous element of degree k of T, of ω_T, or of $(\omega_T^{\otimes q})^{**}$ for an integer $q \ge 1$ have weight λk. Assume that every closed point of $\operatorname{Spec} T$ other than the vertex has finite stabilizer of order m with $\lambda m < 1$. Let $\alpha := 64/27$ and $D := 128/81$, and assume moreover the following.*

(a) If $d = 4$, then ω_T is generated by a homogeneous element of weight α, and

$$\alpha^4 \lim_{s\to 0^+} s^4 \sum_{k\ge 0} (\dim_{\mathbb{C}} T_k) e^{-s\lambda k} > Q_4.$$

(b) If $d \in \{2,3\}$, then there exist an integer $q \ge 1$, a real number $e > 0$, and an injective homogeneous T-linear map $(\omega_T^{\otimes q})^{**} \to T$ which maps elements of weight y to elements of weight $y - e$.

Then there exists an integer $k > 0$ such that $\lambda k < D$ and $T_k \ne 0$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we set up the identity of [OAI26a, Lemma 4.1] and the positivity of [OAI26a, Lemma 4.2].

Let

$$H_T(s) := \sum_{k\geq 0}(\dim_{\mathbb{C}} T_k)e^{-s\lambda k} \quad \text{for } s>0.$$

By [BH98, Proposition 4.4.1], there exist positive integers $a_1,\dots,a_d$ and $B\in\mathbb{Z}[z,z^{-1}]$ with $B(1)>0$ such that

$$\sum_{k\geq 0}(\dim_{\mathbb{C}} T_k)z^k = \frac{B(z)}{\prod_{i=1}^d(1-z^{a_i})}.$$

Thus

$$H_T(s) = \frac{B(e^{-\lambda s})}{\prod_{i=1}^d(1-e^{-\lambda a_i s})}$$

extends to a meromorphic function near $s=0$ with a pole of order d at $s=0$. Write the polar part of H_T at $s=0$ as $\sum_{j=0}^{d-1}\gamma_j j!/s^{j+1}$ with $\gamma_j\in\mathbb{R}$, and let

$$P_T(y) := \sum_{j=0}^{d-1}\gamma_j y^j.$$

Then $(d-1)!\gamma_{d-1} = \lim_{s\to 0^+} s^d H_T(s) = B(1)/(\lambda^d a_1\cdots a_d) > 0$. For $y\in\mathbb{R}$ let δ_y be the unit point mass at y, and let

$$\mu_T := \sum_{k\geq 0}(\dim_{\mathbb{C}} T_k)\delta_{\lambda k} + (-1)^{d-1}\sum_{k\in\mathbb{Z}}\big(\dim_{\mathbb{C}}(\omega_T)_k\big)\delta_{-\lambda k}.$$

The dimensions $\dim_{\mathbb{C}} T_k$ and $\dim_{\mathbb{C}}(\omega_T)_k$ vanish for k sufficiently negative and are bounded by a polynomial in $|k|$, by [BH98, Proposition 4.4.1] for the finite graded T-modules T and ω_T. Thus the integral $\int F d\mu_T$ converges absolutely for any Schwartz function F on $\mathbb{R}$. For such F let $\hat{F}(\tau) := \int_{\mathbb{R}} F(y)e^{-2\pi i y\tau}dy$. Since T is a finitely generated graded normal Cohen–Macaulay domain with $T_0=\mathbb{C}$ and $\lambda m<1$ for the order m of the stabilizer of every closed point of $\operatorname{Spec} T$ other than the vertex, the hypotheses of [OAI26a, Lemma 4.1] hold (Remark 5.3), and

$$\int F d\mu_T = \int_{\mathbb{R}} F(y)P_T(y)dy \tag{5.5}$$

for any Schwartz function F such that $\operatorname{Supp}\hat{F}\subset(-1,1)$. If $d\in\{2,3\}$, then $d\geq 2$, and the map in (b) satisfies the hypothesis of [OAI26a, Lemma 4.2], so $(\omega_T)_k=0$ for any integer $k\leq 0$, and $\gamma_{d-2}>0$.

Suppose, for a contradiction, that $T_k=0$ for any integer $k>0$ with $\lambda k<D$. Then

$$T_k\neq 0 \ \text{ implies } \ k=0 \ \text{ or } \ \lambda k\geq D. \tag{5.6}$$

Step 2. In this step we introduce the test functions and justify the limits taken below.

For $b\in\{1,2,3\}$ let $f_b(\tau) := \cos^b(\pi\tau)$ for $|\tau|\leq 1/2$ and $f_b(\tau):=0$ otherwise, and let

$$g_b(x) := \int_{\mathbb{R}} f_b(\tau)e^{2\pi i x\tau}d\tau.$$

By definition, g_3 is the function g_3 of Lemma 5.6. Let $g(x) := g_3(x-1)+g_3(x+1)$. Then $g(x)=\int_{\mathbb{R}} f(\tau)e^{2\pi i x\tau}d\tau$, where $f(\tau) := 2\cos(2\pi\tau)f_3(\tau)$. For a function h on $\mathbb{R}$ and an integer $k\geq 0$ let

$$M_k(h) := \int_{\mathbb{R}} x^k|h(x)|^2dx.$$

Let $b\in\{1,2,3\}$, let $\varphi\in H^b(\mathbb{R})$ be real and even with $\operatorname{Supp}\varphi\subset[-1/2,1/2]$, let $\varphi_n\in C_c^\infty((-1/2,1/2))$, $n\geq 1$, be real and even with $\varphi_n\to\varphi$ in $H^b(\mathbb{R})$, and let $\psi(x) := \int\varphi(\tau)e^{2\pi i x\tau}d\tau$ and $\psi_n(x) := \int\varphi_n(\tau)e^{2\pi i x\tau}d\tau$. We show the following.

(i) $M_{2k}(\psi) = (2\pi)^{-2k}\int|\varphi^{(k)}|^2 d\tau$ for $0 \le k \le b$, and

$$\int_{\mathbb{R}} x^i|\psi_n(x-x_0)|^2dx \to \int_{\mathbb{R}} x^i|\psi(x-x_0)|^2dx$$

as $n \to +\infty$ for any $x_0 \in \mathbb{R}$ and any integer $0 \le i \le 2b$.

(ii) $\psi_n(x) \to \psi(x)$ as $n \to +\infty$ for any $x \in \mathbb{R}$.

(iii) For any polynomial w, any $x_0 \in \mathbb{R}$, and any n, the function $F(x) := w(x)|\psi_n(x-x_0)|^2$ is a Schwartz function with $\operatorname{Supp}\hat{F} \subset (-1,1)$. In particular, (5.5) holds for F.

Since $\int \varphi^{(k)}(\tau)e^{2\pi i x\tau}d\tau = (-2\pi i x)^k\psi(x)$ for $0 \le k \le b$, the Plancherel theorem [Fol99, Theorems 8.22 and 8.29] implies the formula for $M_{2k}(\psi)$ and

$$\big\|x^k(\psi_n-\psi)\big\|_{L^2} = (2\pi)^{-k}\big\|\varphi_n^{(k)}-\varphi^{(k)}\big\|_{L^2} \to 0 \quad \text{for } 0 \le k \le b.$$

For $0 \le i \le 2b$ we have $|x|^i \le (1+|x|^b)^2$ and $\big||\psi_n|^2-|\psi|^2\big| \le |\psi_n-\psi|(|\psi_n|+|\psi|)$, so by the Cauchy–Schwarz inequality

$$\Big|\int_{\mathbb{R}} x^i\big(|\psi_n|^2-|\psi|^2\big)dx\Big| \le \big\|(1+|x|^b)(\psi_n-\psi)\big\|_{L^2}\big\|(1+|x|^b)(|\psi_n|+|\psi|)\big\|_{L^2} \to 0.$$

Since $\int x^i|\psi_n(x-x_0)|^2dx = \int(x+x_0)^i|\psi_n(x)|^2dx$ is a linear combination of $\int x^{i'}|\psi_n(x)|^2dx$, $0 \le i' \le i$, with coefficients independent of n, (i) holds. Since $\operatorname{Supp}\varphi_n$ and $\operatorname{Supp}\varphi$ are contained in $[-1/2,1/2]$, $|\psi_n(x)-\psi(x)| \le \|\varphi_n-\varphi\|_{L^1} \le \|\varphi_n-\varphi\|_{L^2}$ for any x, which implies (ii). For (iii), ψ_n is a Schwartz function since $\varphi_n \in C_c^\infty(\mathbb{R})$. By the definition and Fubini's theorem, the Fourier transform of $|\psi_n|^2$ is $\tau \to \int \varphi_n(\sigma)\overline{\varphi_n(\sigma-\tau)}d\sigma$, which vanishes for $|\tau| > 2c$ if $\operatorname{Supp}\varphi_n \subset [-c,c]$ with $c < 1/2$. The translation by x_0 multiplies the Fourier transform by $e^{-2\pi i x_0\tau}$, and the multiplication by w replaces a Fourier transform by a linear combination of the derivatives of that Fourier transform. Thus $\hat{F}$ is supported in $[-2c,2c] \subset (-1,1)$, and (iii) holds.

We now construct approximants. Let $b \in \{1,2,3\}$. The derivatives $f_b^{(j)}$, $0 \le j \le b-1$, vanish at $\tau = \pm 1/2$, so $f_b \in C^{b-1}(\mathbb{R})$, and the b-th derivative of f_b is piecewise continuous and bounded. Thus $f_b \in H^b(\mathbb{R})$. As $\epsilon \to 0^+$, the dilations $f_b(\tau/(1-\epsilon))$ converge to f_b in $H^b(\mathbb{R})$, since their derivatives of order at most b converge pointwise almost everywhere, by the dominated convergence theorem. The convolution of $f_b(\tau/(1-\epsilon))$ with an even smooth mollifier supported in $(-\epsilon/4,\epsilon/4)$ belongs to $C_c^\infty((-1/2,1/2))$, is real and even, and converges to $f_b(\tau/(1-\epsilon))$ in $H^b(\mathbb{R})$ as the support of the mollifier shrinks to 0 [Eva10, Section 5.3, Theorem 1]. Hence there exist real even $f_{b,n} \in C_c^\infty((-1/2,1/2))$, $n \ge 1$, with $f_{b,n} \to f_b$ in $H^b(\mathbb{R})$. Let $G_{b,n}(x) := \int f_{b,n}(\tau)e^{2\pi i x\tau}d\tau$. Then (i)–(iii) hold for $(\varphi,\varphi_n,\psi,\psi_n) = (f_b,f_{b,n},g_b,G_{b,n})$.

Let $G_n(x) := G_{3,n}(x-1) + G_{3,n}(x+1)$. Then G_n is real and even, and

$$G_n(x) = \int_{\mathbb{R}} 2\cos(2\pi\tau)f_{3,n}(\tau)e^{2\pi i x\tau}d\tau.$$

By the Leibniz rule, $2\cos(2\pi\tau)f_{3,n} \to f$ in $H^3(\mathbb{R})$, and f is real and even with $\operatorname{Supp} f \subset [-1/2,1/2]$. Thus (i)–(iii) hold for $(\varphi,\varphi_n,\psi,\psi_n) = (f,2\cos(2\pi\tau)f_{3,n},g,G_n)$.

By (i), $M_{2k}(g_b) = (2\pi)^{-2k}\int|f_b^{(k)}|^2d\tau$ for $0 \le k \le b$. We have $\int_{-1/2}^{1/2}\cos^2(\pi\tau)d\tau = 1/2$ and $\int_{-1/2}^{1/2}\cos^4(\pi\tau)d\tau = 3/8$. Moreover, $|f_1'(\tau)|^2 = \pi^2\sin^2(\pi\tau)$ for $|\tau| < 1/2$, and $f_2(\tau) = (1+\cos(2\pi\tau))/2$ for $|\tau| \le 1/2$, so

$$\int|f_1'|^2d\tau = \frac{\pi^2}{2}, \qquad \int|f_2'|^2d\tau = \frac{\pi^2}{2}, \qquad \int|f_2''|^2d\tau = 2\pi^4.$$

Since g_1 and g_2 are even, $M_1(g_2) = M_3(g_2) = 0$. Expanding $(x+2/3)^k$, we obtain the values in Table 1.

Step 3. In this step we treat the case $d = 4$.

h	$M_0(h)$	$M_2(h)$	$M_4(h)$	$M_6(h)$
g_1	1/2	1/8		
g_2	3/8	1/8	1/8	
g	13/16	61/64	685/256	11101/1024

k	$\frac{8}{3}\int_{\mathbb{R}} y^k \lvert g_2(y-2/3)\rvert^2 dy$
0	1
1	2/3
2	7/9
3	26/27
4	115/81

TABLE 1. Moments of g_1, g_2, and g, and moments of the translate of g_2 by $2/3$ normalized by $M_0(g_2) = 3/8$.

By (a), ω_T is generated by a homogeneous element of degree α/λ, so $\dim_{\mathbb{C}}(\omega_T)_k = \dim_{\mathbb{C}} T_{k-\alpha/\lambda}$ for any $k \in \mathbb{Z}$. Since $(-1)^{d-1} = -1$,

$$\mu_T = \sum_{k\geq 0} (\dim_{\mathbb{C}} T_k)(\delta_{\lambda k} - \delta_{-\lambda k-\alpha}).$$

Let ν be the image of μ_T under the translation $y \to y + \alpha/2$. Then

$$\nu = \sum_{k\geq 0} (\dim_{\mathbb{C}} T_k)(\delta_{\lambda k+\alpha/2} - \delta_{-\lambda k-\alpha/2}),$$

and the reflection $x \to -x$ maps ν to $-\nu$. If F is a Schwartz function with $\operatorname{Supp}\hat{F} \subset (-1,1)$, then so is $F(y+\alpha/2)$, and (5.5) implies

$$\int F d\nu = \int_{\mathbb{R}} F(x) P_T(x-\alpha/2) dx. \tag{5.7}$$

We show that $P_T(x-\alpha/2)$ is an odd polynomial. Let

$$\Phi(x) := P_T(x-\alpha/2) + P_T(-x-\alpha/2) = \sum_{j=0}^{3} \phi_j x^j.$$

For any Schwartz function F with $\operatorname{Supp}\hat{F} \subset (-1,1)$, applying (5.7) to $F(x)$ and $F(-x)$ and using $\int F(-x)d\nu = -\int F d\nu$, we obtain $\int F(x)\Phi(x)dx = 0$. Let $\chi \in C_c^\infty((-1,1))$ with $\chi = 1$ near 0. For $0 \leq j \leq 3$, the inverse Fourier transform F_j of $\tau^j \chi(\tau)$ is a Schwartz function with $\operatorname{Supp}\hat{F}_j \subset (-1,1)$, and

$$\int_{\mathbb{R}} x^{j'} F_j(x) dx = (-2\pi i)^{-j'} \frac{d^{j'}}{d\tau^{j'}} \left(\tau^j \chi(\tau)\right)\Big|_{\tau=0}$$

is non-zero for $j' = j$ and zero for $j' \neq j$, $0 \leq j' \leq 3$. Since $\int F_j \Phi dx = 0$, we obtain $\phi_j = 0$ for $0 \leq j \leq 3$, and $P_T(x-\alpha/2)$ is odd. The coefficient of x^3 in $P_T(x-\alpha/2)$ is $\gamma_3 = \frac{1}{6}\lim_{s\to 0^+} s^4 H_T(s)$. By (a) and $Q_4 = 4096/27$ (Lemma 2.7), $\gamma_3 > Q_4/(6\alpha^4) = 6561/8192$. Thus

$$P_T(x-\alpha/2) = \gamma(x^3 + cx), \qquad \gamma := \gamma_3 > \frac{6561}{8192}, \qquad c \in \mathbb{R}.$$

We show $c \geq -1$. Let $n \geq 1$ and $F(x) := x|G_{2,n}(x)|^2$. The coefficients of ν are non-negative at positive positions and non-positive at negative positions, and $x|G_{2,n}(x)|^2$ has the sign of x. Thus $\int F d\nu \geq 0$. By (iii) and (5.7),

$$\gamma \int_{\mathbb{R}} (x^4 + cx^2)|G_{2,n}(x)|^2 dx \geq 0.$$

Letting $n \to +\infty$, (i) and Table 1 imply $\gamma(M_4(g_2) + cM_2(g_2)) = \gamma(1+c)/8 \geq 0$. Hence $c \geq -1$.

Let $\ell := \alpha/2 + D = 224/81$. By (5.6) and $T_0 = \mathbb{C}$, the positions where the coefficient of ν is non-zero are $\pm\alpha/2$, where the coefficients are ± 1, and positions x with $|x| \geq \ell$, where the coefficient is positive if $x \geq \ell$ and negative if $x \leq -\ell$. Let $n \geq 1$ and

$$F_n(x) := x(\ell^2 - x^2)|G_n(x)|^2.$$

Then $F_n(x) \le 0$ for $x \ge \ell$, and $F_n(x) \ge 0$ for $x \le -\ell$. Hence the contribution of any position x with $|x| \ge \ell$ to $\int F_n d\nu$ is non-positive. Since G_n is even, F_n is odd, and

$$\int F_n d\nu \le F_n(\alpha/2) - F_n(-\alpha/2) = \alpha\Big(\ell^2 - \frac{\alpha^2}{4}\Big)|G_n(\alpha/2)|^2.$$

By (iii) and (5.7), for any $n \ge 1$,

$$\gamma \int_{\mathbb{R}} (x^4 + cx^2)(\ell^2 - x^2)|G_n(x)|^2 dx \le \alpha\Big(\ell^2 - \frac{\alpha^2}{4}\Big)|G_n(\alpha/2)|^2.$$

The polynomial $(x^4 + cx^2)(\ell^2 - x^2)$ has degree $6 = 2 \cdot 3$. Letting $n \to +\infty$, (i) and (ii) for $(f, 2\cos(2\pi\tau) f_{3,n}, g, G_n)$ imply

$$\gamma\Big(\ell^2 M_4(g) - M_6(g) + c\big(\ell^2 M_2(g) - M_4(g)\big)\Big) \le \alpha\Big(\ell^2 - \frac{\alpha^2}{4}\Big)|g(\alpha/2)|^2. \tag{5.8}$$

By Lemma 5.6(1), $\ell^2 M_2(g) - M_4(g) > 0$. Since $c \ge -1$, the left-hand side of (5.8) is at least

$$\gamma\Big(\ell^2\big(M_4(g) - M_2(g)\big) - M_6(g) + M_4(g)\Big).$$

Since the right-hand side of Lemma 5.6(2) is non-negative, Lemma 5.6(2) shows that the expression in parentheses in the last display is positive. Since $\gamma > 6561/8192$, Lemma 5.6(2) then shows that the left-hand side of (5.8) is strictly larger than $\alpha(\ell^2 - \alpha^2/4)|g(\alpha/2)|^2$. This contradicts (5.8).

Step 4. In this step we treat the case $d = 3$.

By **Step 1**, $\gamma_2 > 0$, $\gamma_1 > 0$, and $(\omega_T)_k = 0$ for any $k \le 0$. Let $\gamma := \gamma_2$, $u := \gamma_1/\gamma_2$, and $c := \gamma_0/\gamma_2$. Then

$$P_T(y) = \gamma(y^2 + uy + c), \qquad \gamma > 0, \qquad u > 0.$$

Since $(-1)^{d-1} = 1$, μ_T is a positive measure.

We show $c \ge u^2/4 - 1/4$. For $n \ge 1$, the function $|G_{1,n}(y + u/2)|^2$ is non-negative, so by (iii) and (5.5),

$$\gamma \int_{\mathbb{R}} (y^2 + uy + c)|G_{1,n}(y + u/2)|^2 dy = \gamma \int_{\mathbb{R}} \Big(x^2 + c - \frac{u^2}{4}\Big)|G_{1,n}(x)|^2 dx \ge 0.$$

Letting $n \to +\infty$, (i) and Table 1 imply

$$\gamma\Big(M_2(g_1) + \Big(c - \frac{u^2}{4}\Big)M_0(g_1)\Big) = \frac{\gamma}{2}\Big(c - \frac{u^2}{4} + \frac{1}{4}\Big) \ge 0.$$

Hence $c \ge u^2/4 - 1/4$.

By (5.6) and $(\omega_T)_k = 0$ for $k \le 0$, the positions where the coefficient of μ_T is non-zero are negative, equal to 0, or at least D. For $n \ge 1$ let

$$F_n(y) := y(D - y)|G_{2,n}(y - 2/3)|^2.$$

Then $F_n(y) \le 0$ for $y < 0$ and for $y \ge D$, and $F_n(0) = 0$. Since μ_T is positive, $\int F_n d\mu_T \le 0$. By (iii) and (5.5),

$$\gamma \int_{\mathbb{R}} y(D - y)(y^2 + uy + c)|G_{2,n}(y - 2/3)|^2 dy \le 0.$$

The polynomial $y(D-y)(y^2+uy+c) = Dy^3 - y^4 + u(Dy^2 - y^3) + c(Dy - y^2)$ has degree $4 = 2 \cdot 2$. Letting $n \to +\infty$ and dividing by $\frac{3}{8}\gamma > 0$, (i) and Table 1 imply

$$D \cdot \frac{26}{27} - \frac{115}{81} + u\Big(D \cdot \frac{7}{9} - \frac{26}{27}\Big) + c\Big(D \cdot \frac{2}{3} - \frac{7}{9}\Big) = \frac{223}{2187} + \frac{194}{729}u + \frac{67}{243}c \le 0.$$

Since $67/243 > 0$ and $c \ge u^2/4 - 1/4$, the left-hand side is at least

$$\frac{67}{972}u^2 + \frac{194}{729}u + \frac{289}{8748},$$

which is positive since $u > 0$. This is a contradiction.

Step 5. In this step we treat the case $d = 2$.

By **Step 1**, $\gamma_1 > 0$, $\gamma_0 > 0$, and $(\omega_T)_k = 0$ for any $k \leq 0$. Let $\gamma := \gamma_1$ and $c := \gamma_0/\gamma_1$. Then

$$P_T(y) = \gamma(y + c), \qquad \gamma > 0, \qquad c > 0.$$

Since $(-1)^{d-1} = -1$, the coefficients of μ_T at the positions $-\lambda k$, $k > 0$, are non-positive, and by (5.6) the coefficients at non-negative positions are non-negative and non-zero only at 0 and at positions at least D. For $n \geq 1$ let

$$F_n(y) := y^2(D - y)|G_{2,n}(y - 2/3)|^2.$$

Then $F_n(y) \geq 0$ for $y < 0$, $F_n(0) = 0$, and $F_n(y) \leq 0$ for $y \geq D$. Thus $\int F_n d\mu_T \leq 0$. By (iii) and (5.5),

$$\gamma \int_{\mathbb{R}} y^2(D - y)(y + c)|G_{2,n}(y - 2/3)|^2 dy \leq 0.$$

The polynomial $y^2(D-y)(y+c) = Dy^3 - y^4 + c(Dy^2 - y^3)$ has degree $4 = 2 \cdot 2$. Letting $n \to +\infty$ and dividing by $\frac{3}{8}\gamma > 0$, (i) and Table 1 imply

$$D \cdot \frac{26}{27} - \frac{115}{81} + c\Big(D \cdot \frac{7}{9} - \frac{26}{27}\Big) = \frac{223}{2187} + \frac{194}{729}c \leq 0,$$

which contradicts $c > 0$.

Step 6. In this step we finish the proof.

By **Steps 3, 4, and 5**, the assumption of **Step 1** leads to a contradiction for any $d \in \{2, 3, 4\}$. Thus there exists an integer $k > 0$ such that $\lambda k < D$ and $T_k \neq 0$. □

5.3. **A log canonical threshold.** In this subsection, we bound from below the log canonical threshold of the ideal of the vertex of C, and we deduce that $(C \ni o)$ has embedding dimension at most 5.

Definition 5.8. Notation and conditions as in Set-up 5.2, and let $\mathfrak{m}_o \subset S$ be the ideal of o. For a non-zero ideal $\mathfrak{a} \subset S$ and a prime divisor F over C, let $\operatorname{mult}_F \mathfrak{a} := \min\{\operatorname{mult}_F f \mid f \in \mathfrak{a}\backslash\{0\}\}$, where $\operatorname{mult}_F f$ is the coefficient of F in the divisor of the pullback of f to a normal variety on which F is a divisor. The *log canonical threshold* of $\mathfrak{m}_o$ is

$$\operatorname{lct}(C \ni o; \mathfrak{m}_o) := \inf_F \frac{a(F, C, 0)}{\operatorname{mult}_F \mathfrak{m}_o},$$

where F runs through the prime divisors over C with $\operatorname{center}_C F = \{o\}$.

Since the common zero locus of $\mathfrak{m}_o$ is $\{o\}$, a prime divisor F over C satisfies $\operatorname{mult}_F \mathfrak{m}_o > 0$ if and only if $\operatorname{center}_C F = \{o\}$, so $\operatorname{lct}(C \ni o; \mathfrak{m}_o)$ is the infimum of $a(F, C, 0)/\operatorname{mult}_F \mathfrak{m}_o$ over all prime divisors F over C with $\operatorname{mult}_F \mathfrak{m}_o > 0$.

The bound $\operatorname{lct}(C \ni o; \mathfrak{m}_o) > 3/2$ of Proposition 5.9, together with the lower bound 3 for the log discrepancies of the divisors centered at o in Proposition 5.5(2), implies that any exceptional divisor over a general hyperplane section of C through o has positive discrepancy (Proposition 5.10). Thus the section is terminal Gorenstein, and the embedding dimension of the section at o is at most 4.

Conditions (2) and (3) of Proposition 5.9 are the conclusions of Proposition 5.7 for $T = S$ under (a) and for $T = S/P$ under (b).

Proposition 5.9. *Notation and conditions as in Set-up 5.2 and Definition 5.8, and let $\zeta \in \mathfrak{t}_{\mathbb{Z}} \cap \mathfrak{t}^+$ be primitive. Assume the following.*

(1) Any closed point of $C\backslash\{o\}$ has stabilizer for ζ of order m with $\lambda m < 1$.
(2) There exists an integer $j > 0$ such that $\lambda j < D$ and $S_j \neq 0$.

(3) *For any homogeneous prime ideal $P \subset S$ such that S/P is normal and Cohen–Macaulay of dimension 2 or 3, if there exist an integer $q \geq 1$, a real number $e > 0$, and an injective homogeneous S/P-linear map $(\omega_{S/P}^{\otimes q})^{**} \to S/P$ which lowers weights by e, where $\omega_{S/P}$ carries the grading induced by the natural action of $\mathbb{G}_m$ on differential forms, then $(S/P)_k \neq 0$ for some integer $k > 0$ with $\lambda k < D$.*

Then $\operatorname{lct}(C \ni o; \mathfrak{m}_o) > 3/2$.

Proof. We use the grading $S = \bigoplus_{j \geq 0} S_j$ defined by ζ. Since $S_0 = \mathbb{C}$ (Definition 2.8), $\mathfrak{m}_o = \bigoplus_{j>0} S_j$. The proof is long, so we divide it into several steps.

Step 1. In this step, assuming $\operatorname{lct}(C \ni o; \mathfrak{m}_o) \leq 3/2$, we construct a $\mathbb{G}_m$-invariant $\mathbb{Q}$-divisor $B \geq 0$ on C such that (C, B) is lc and any lc center of (C, B) contains o and is contained in the common zero locus of the homogeneous elements of S of weight in $(0, D)$.

Suppose that $\operatorname{lct}(C \ni o; \mathfrak{m}_o) \leq 3/2$. Since S is finitely generated and $S_0 = \mathbb{C}$, $\dim_{\mathbb{C}} S_j < +\infty$ for any j, so $\dim_{\mathbb{C}} \bigoplus_{0<\lambda j<D} S_j < +\infty$. By (2), $\bigoplus_{0<\lambda j<D} S_j \neq 0$. Let $f_1, \dots, f_s$ be a basis of $\bigoplus_{0<\lambda j<D} S_j$ consisting of homogeneous elements, let w_i be the weight of f_i, and let $w_0 := \max_i w_i$, so that $w_0 < D$. Choose $w \in \lambda \mathbb{N}^+$ such that $w/w_i \in \mathbb{N}^+$ for any i, and let

$$\mathfrak{b} := \left(f_1^{w/w_1}, \dots, f_s^{w/w_s}\right) \subset S.$$

Any f_i is contained in $\mathfrak{m}_o$ and $w_0/w_i \geq 1$. Thus for any prime divisor F over C whose center on C is $\{o\}$,

$$\frac{w_0}{w} \operatorname{mult}_F \mathfrak{b} = \min_i \frac{w_0}{w_i} \operatorname{mult}_F f_i \geq \min_i \operatorname{mult}_F f_i \geq \operatorname{mult}_F \mathfrak{m}_o > 0. \tag{5.9}$$

Let

$$t := \inf_F \frac{a(F, C, 0)}{\frac{w_0}{w} \operatorname{mult}_F \mathfrak{b}},$$

where F runs through the prime divisors over C with $o \in \operatorname{center}_C F$ and $\operatorname{mult}_F \mathfrak{b} > 0$. By (5.9), any prime divisor F over C with $\operatorname{center}_C F = \{o\}$ is in the range of the infimum defining t. Thus (5.9) and the assumption $\operatorname{lct}(C \ni o; \mathfrak{m}_o) \leq 3/2$ imply that $t \leq 3/2$.

Let $\pi_Y : Y \to C$ be a log resolution of $\mathfrak{b}$, write $\mathfrak{b}\mathcal{O}_Y = \mathcal{O}_Y(-F_{\mathfrak{b}})$, and let Δ_Y be the divisor with $K_Y + \Delta_Y = \pi_Y^* K_C$. Here K_C is Cartier, because S is Gorenstein (Construction-Proposition 4.1(2)). Since C is klt (Construction-Proposition 4.1(2)), $a(F, C, 0) > 0$ for any prime divisor F over C. Since $o \in V(\mathfrak{b})$, $\pi_Y^{-1}(o) \subset \operatorname{Supp} F_{\mathfrak{b}}$. Let

$$c := \min_F \frac{a(F, C, 0)}{\operatorname{mult}_F \mathfrak{b}},$$

where F runs through the prime divisors on Y with $o \in \pi_Y(F)$ and $\operatorname{mult}_F \mathfrak{b} > 0$, and let F_0 be a prime divisor on Y at which the minimum defining c is attained. Then $c \in \mathbb{Q}_{>0}$. For a prime divisor F on Y, the coefficient of F in $\Delta_Y + cF_{\mathfrak{b}}$ is $1 - a(F, C, 0) + c \operatorname{mult}_F \mathfrak{b}$. Thus over a neighborhood of o, all coefficients of $\Delta_Y + cF_{\mathfrak{b}}$ are at most 1, the coefficient of F_0 equals 1, where $o \in \pi_Y(F_0)$, and the support of $\Delta_Y + cF_{\mathfrak{b}}$ is contained in the simple normal crossing divisor $\operatorname{Exc}(\pi_Y) \cup \operatorname{Supp} F_{\mathfrak{b}}$. Thus for any prime divisor F over C with $o \in \operatorname{center}_C F$,

$$a(F, C, 0) - c \operatorname{mult}_F \mathfrak{b} = a(F, Y, \Delta_Y + cF_{\mathfrak{b}}) \geq 0$$

by [KM98, Lemma 2.30] and [Kol13, Claim 2.10.3] (log discrepancies are discrepancies plus 1). Thus the infimum defining t is attained at F_0, and $c = tw_0/w$. Since $a(F_0, C, 0) > 0$, $0 < t \leq 3/2$, and since $w_0 < D$,

$$tw_0 < \frac{3}{2} D = \alpha.$$

Choose an integer $N > c$, let $\phi_1, \dots, \phi_N$ be general elements of the linear span of $f_1^{w/w_1}, \dots, f_s^{w/w_s}$, let $H_\nu \subset C$ be the divisor of ϕ_ν, and let

$$B := \frac{c}{N} \sum_{\nu=1}^{N} H_\nu.$$

Any ϕ_ν is homogeneous of weight w, so B is $\mathbb{G}_m$-invariant, and $K_C + B$ is $\mathbb{Q}$-Cartier. Since the pullbacks of $f_1^{w/w_1}, \dots, f_s^{w/w_s}$ generate $\mathcal{O}_Y(-F_{\mathfrak{b}})$, we have $\pi_Y^* H_\nu = F_{\mathfrak{b}} + M_\nu$, where M_ν is a general member of a base-point-free linear system on Y given by s sections of $\mathcal{O}_Y(-F_{\mathfrak{b}})$, and M_ν may be zero. By [Har77, III, Theorem 10.8] applied to the morphism $Y \to \mathbb{P}^{s-1}$ defined by $f_1^{w/w_1}, \dots, f_s^{w/w_s}$, restricted to Y and to each stratum of the simple normal crossing divisor $\operatorname{Exc}(\pi_Y) \cup \operatorname{Supp} F_{\mathfrak{b}} \cup M_1 \cup \cdots \cup M_{\nu-1}$ (all of which are smooth), with hyperplanes of $\mathbb{P}^{s-1}$ (see also [Har77, III, Remark 10.9.2]), we may assume that $M_1, \dots, M_N$ are smooth and that $\operatorname{Exc}(\pi_Y) \cup \operatorname{Supp} F_{\mathfrak{b}} \cup M_1 \cup \cdots \cup M_N$ is a simple normal crossing divisor. We have

$$K_Y + \Delta_Y + cF_{\mathfrak{b}} + \frac{c}{N} \sum_{\nu=1}^{N} M_\nu = \pi_Y^*(K_C + B),$$

and the coefficients of the M_ν are $c/N < 1$. Thus (C, B) is lc over a neighborhood of o, $\pi_Y(F_0)$ is an lc center of (C, B) containing o, and any lc center of (C, B) containing o is the image of a stratum of the components of $\Delta_Y + cF_{\mathfrak{b}}$ with coefficient 1 [KM98, Lemma 2.30], [Kol13, (2.10.1)]. If the coefficient of a prime divisor F on Y in $\Delta_Y + cF_{\mathfrak{b}}$ equals 1, then $c \operatorname{mult}_F \mathfrak{b} = a(F, C, 0) > 0$, so $\pi_Y(F)$ is contained in the zero locus $V(\mathfrak{b}) = V(f_1, \dots, f_s)$ of $\mathfrak{b}$. Thus any lc center of (C, B) containing o is contained in $V(f_1, \dots, f_s)$.

The group $\mathbb{G}_m$ acts on C through the grading, so that $f(\epsilon \cdot p) = \epsilon^j f(p)$ for any $f \in S_j$, $\epsilon \in \mathbb{G}_m$, and closed point $p \in C$. For $j > 0$, $\epsilon^j f(p)$ tends to 0 as ϵ tends to 0, so the closure of any orbit contains o. Thus any non-empty closed $\mathbb{G}_m$-invariant subset of C contains o. Since Y is a log resolution of (C, B) over all of C, by [KM98, Lemma 2.30, Corollary 2.31(3)] the non-lc locus of (C, B) is the image under π_Y of the union of the components of $\Delta_Y + cF_{\mathfrak{b}} + \frac{c}{N} \sum_{\nu=1}^{N} M_\nu$ with coefficient > 1, which is closed since π_Y is proper. The non-lc locus of (C, B) is $\mathbb{G}_m$-invariant and does not contain o, so it is empty. Thus (C, B) is lc. Since (C, B) is lc, all coefficients of $\Delta_Y + cF_{\mathfrak{b}} + \frac{c}{N} \sum_{\nu=1}^{N} M_\nu$ are at most 1, so by [KM98, Lemma 2.30] and [Kol13, (2.10.1)–(2.10.2)] any lc center of (C, B) is the image of a stratum of the components with coefficient 1. Hence (C, B) has finitely many lc centers, and the connected group $\mathbb{G}_m$ permutes them, so any lc center of (C, B) is $\mathbb{G}_m$-invariant and contains o. Thus any lc center of (C, B) is contained in $V(f_1, \dots, f_s)$.

Step 2. In this step we show that o is not an lc center of (C, B).

Let $\pi : \widetilde{C} \to C$ and $E \subset \widetilde{C}$ be as in Construction-Proposition 5.4 for ζ, and let $\widetilde{B}$ be the strict transform of B on $\widetilde{C}$. Let $n > 0$ be an integer, let $g \in S_n$ be non-zero, let $W := \{g = 1\} \subset C$, and let $\tau : W \times \mathbb{A}^1 \to (W \times \mathbb{A}^1)/\mu_n$ be the quotient morphism. By Construction-Proposition 5.4(2), W is smooth, $(W \times \mathbb{A}^1)/\mu_n$ is an open subset of $\widetilde{C}$, and τ is finite and étale in codimension one. By Construction-Proposition 5.4(1), the pullback of a non-zero $f \in S_j$ to $W \times \mathbb{A}^1$ is $u^j f|_W$, where $f|_W \neq 0$ because $W \times \mathbb{G}_m \to C$ is finite étale onto the dense open subset $\{g \neq 0\}$. Since $\tau^{-1}(E) = \{u = 0\}$ and τ is étale in codimension one, $\tau^* E = \{u = 0\}$ and $\operatorname{mult}_E \pi^* f = j$. Each ϕ_ν has degree w/λ, so

$$\pi^* B = \widetilde{B} + \frac{cw}{\lambda} E = \widetilde{B} + \frac{tw_0}{\lambda} E, \qquad \tau^* \widetilde{B} = \operatorname{pr}_1^* B_W,$$

where $B_W := \frac{c}{N} \sum_\nu (\phi_\nu|_W = 0)$ is a $\mathbb{Q}$-divisor on W and $\operatorname{pr}_1 : W \times \mathbb{A}^1 \to W$ is the projection.

The pullback of (C, B) under the finite étale morphism $W \times \mathbb{G}_m \to \{g \neq 0\}$ is $(W \times \mathbb{G}_m, \operatorname{pr}_1^* B_W)$, which is lc by **Step 1** and [KM98, Proposition 5.20(4)]. The product of a log resolution of (W, B_W) with $\mathbb{G}_m$ (respectively with $\mathbb{A}^1$) is a log resolution of $(W \times \mathbb{G}_m, \operatorname{pr}_1^* B_W)$ (respectively

of $(W \times \mathbb{A}^1, \mathrm{pr}_1^* B_W + \{u = 0\})$, and the coefficients of the log pullbacks are those of (W, B_W) together with the coefficient 1 of $\{u = 0\}$. Thus (W, B_W) is lc, and $(W \times \mathbb{A}^1, \mathrm{pr}_1^* B_W + \{u = 0\})$ is lc. Since τ is finite and étale in codimension one,

$$K_{W\times\mathbb{A}^1} + \{u = 0\} + \mathrm{pr}_1^* B_W = \tau^*(K_{\widetilde{C}} + E + \widetilde{B}),$$

so $(\widetilde{C}, E + \widetilde{B})$ is lc over $(W \times \mathbb{A}^1)/\mu_n$ [KM98, Proposition 5.20(4)]. Since $E \cong \operatorname{Proj} S$, any point of E is contained in $(W \times \mathbb{A}^1)/\mu_n$ for some homogeneous g of positive degree. Moreover, $\widetilde{C}\backslash E \cong C\backslash\{o\}$, and (C, B) is lc. Thus $(\widetilde{C}, E + \widetilde{B})$ is lc.

By Construction-Proposition 5.4(1),

$$K_{\widetilde{C}} + E + \widetilde{B} = \pi^*(K_C + B) + \Big(r - \frac{tw_0}{\lambda}\Big)E,$$

and since $r = \alpha/\lambda$ and $tw_0 < \alpha$ (**Step 1**),

$$r - \frac{tw_0}{\lambda} = \frac{\alpha - tw_0}{\lambda} > 0.$$

Let F' be a prime divisor over C with $\mathrm{center}_C F' = \{o\}$. Then $\mathrm{center}_{\widetilde{C}} F' \subset \pi^{-1}(o) = E$. Since a positive multiple of E is Cartier (Construction-Proposition 5.4(3)) and $\operatorname{Supp} E = \pi^{-1}(o)$, $\mathrm{mult}_{F'} E > 0$. Since $(\widetilde{C}, E + \widetilde{B})$ is lc,

$$a(F', C, B) = a(F', \widetilde{C}, E + \widetilde{B}) + \Big(r - \frac{tw_0}{\lambda}\Big)\,\mathrm{mult}_{F'} E > 0.$$

Thus o is not an lc center of (C, B).

Step 3. In this step we choose a minimal lc center Z of (C, B) and show that the coordinate ring of Z has no non-zero homogeneous element of weight in $(0, D)$.

By **Step 1**, (C, B) has an lc center, and any lc center of (C, B) contains o and is contained in $V(f_1, \dots, f_s) \neq C$. Let Z be an lc center of (C, B) which is minimal with respect to inclusion. By **Step 2**, $Z \neq \{o\}$, so $1 \leq \dim Z \leq 3$. Since C is a normal affine variety, (C, B) is lc, and $K_C + B$ is $\mathbb{Q}$-Cartier, [FG12, Theorem 7.2] implies that Z is normal and that there exists a $\mathbb{Q}$-divisor $B_Z \geq 0$ on Z such that (Z, B_Z) is klt. Thus Z is Cohen–Macaulay [Kol13, Corollary 2.88(1)].

Let $P \subset S$ be the ideal of Z, and let $T := S/P$. By **Step 1**, Z is $\mathbb{G}_m$-invariant, so P is homogeneous, $T = \bigoplus_{k\geq 0} T_k$ is graded with T_k the image of S_k, and T is a normal Cohen–Macaulay domain of dimension 1, 2, or 3. The action of $\mathbb{G}_m$ on $Z = \operatorname{Spec} T$ is the restriction of the action on C, so for any closed point p of $Z\backslash\{o\}$, the stabilizer of p for the action on Z equals the stabilizer of p for the action on C. By (1), the order m of the stabilizer of a closed point of $Z\backslash\{o\}$ satisfies $\lambda m < 1$. For an integer $k > 0$ with $\lambda k < D$, S_k is spanned by some of the f_i, and the f_i vanish on Z. Thus

$$T_k = 0 \quad \text{for any integer } k > 0 \text{ with } \lambda k < D. \tag{5.10}$$

Step 4. In this step we treat the case $\dim Z = 1$.

Let $\mathfrak{m}_T := \bigoplus_{k>0} T_k$. Since T is a normal domain of dimension 1, the localization of T at $\mathfrak{m}_T$ is a discrete valuation ring [Har77, I, Theorem 6.2A], so $\dim_{\mathbb{C}} \mathfrak{m}_T/\mathfrak{m}_T^2 = 1$. Let $z \in \mathfrak{m}_T$ be a homogeneous element whose class spans $\mathfrak{m}_T/\mathfrak{m}_T^2$. By the graded Nakayama lemma [BH98, Exercise 1.5.24(a)], $\mathfrak{m}_T = zT$. By induction on the degree, any homogeneous element of T is a scalar multiple of a power of z. Since $T_0 = \mathbb{C}$ and $\dim T = 1$, $T = \mathbb{C}[z]$ is a polynomial ring in z. Let $k > 0$ be the degree of z. A closed point $p \in Z\backslash\{o\}$ satisfies $z(p) \neq 0$, and $\epsilon \in \mathbb{G}_m$ fixes p if and only if $\epsilon^k = 1$. Thus the stabilizer of p is μ_k, and by **Step 3**, $\lambda k < 1 < D$. Since $z \in T_k$ is non-zero, this contradicts (5.10).

Step 5. In this step we treat the case $\dim Z = 2$.

Since (Z, B_Z) is klt, Z is $\mathbb{Q}$-factorial [dFH09, Proposition 7.2, Corollary 7.15]. In particular, K_Z and B_Z are $\mathbb{Q}$-Cartier, and $(Z, 0)$ is klt, because for any prime divisor F over Z, $a(F, Z, 0) = a(F, Z, B_Z) + \mathrm{mult}_F B_Z \geq a(F, Z, B_Z) > 0$.

Choose an integer $q \geq 1$ such that qK_Z is Cartier near the vertex o, and let $L := (\omega_T^{\otimes q})^{**}$ with the grading of (3). The localization of L at $\mathfrak{m}_T$ is free of rank one, so $\dim_{\mathbb{C}} L/\mathfrak{m}_T L = 1$. Let $\sigma \in L$ be a homogeneous element whose class spans $L/\mathfrak{m}_T L$. By the graded Nakayama lemma [BH98, Exercise 1.5.24(a)], $L = T\sigma$. Since L is torsion-free of rank one, multiplication by σ is an isomorphism $T \to L$. Let k be the degree of σ.

Let ℓ be the greatest common divisor of the integers $k' > 0$ with $T_{k'} \neq 0$. Then $\mu_\ell \subset \mathbb{G}_m$ acts trivially on T, hence on the smooth locus Z° of Z and on $\omega_{Z^\circ}^{\otimes q}$. Since L is the module of sections of $\omega_{Z^\circ}^{\otimes q}$ over Z° [Sta, Tag 0EBJ], μ_ℓ acts trivially on L, so ℓ divides k. Let $T = \bigoplus_{k' \geq 0} T'_{k'}$ be the grading with $T'_{k'} := T_{\ell k'}$. It is defined by a faithful action of $\mathbb{G}_m$, and σ has degree k/ℓ for the grading of L induced by the action on differential forms. Let v be the valuation of $\mathbb{C}(Z)$ defined by $T = \bigoplus_{k'} T'_{k'}$, that is, $v(f)$ is the smallest degree of a non-zero homogeneous component of $f \in T\backslash\{0\}$. Since $T'_0 = \mathbb{C}$, $v \geq 0$ on T and $v > 0$ exactly on $\mathfrak{m}_T\backslash\{0\}$, so the center of v on Z is the vertex o. Since the greatest common divisor of the integers k' with $T'_{k'} \neq 0$ is 1, v has value group $\mathbb{Z}$. If $f, g \in T\backslash\{0\}$ satisfy $v(f/g) = 0$, then $\deg f_a = \deg g_a$ for the lowest-degree homogeneous components f_a of f and g_a of g, and the residue class of f/g is that of f_a/g_a. Thus the residue field of v is the field of ratios f/g with $f, g \in T$ homogeneous, $g \neq 0$, and $\deg f = \deg g$, which has transcendence degree $\dim T - 1$ (since the degrees of the non-zero homogeneous elements of T' generate $\mathbb{Z}$, $\mathbb{C}(Z)$ is generated over the residue field of v by one ratio of two homogeneous elements of T' whose degrees differ by 1, which is transcendental over the residue field of v), so v is a divisorial valuation [KM98, Lemma 2.45]. By [OAI26a, Remark 3.2] (see Remark 5.3),

$$a(v, Z) = \frac{k}{\ell q}.$$

Since $(Z, 0)$ is klt, $a(v, Z) > 0$, so $k > 0$ and the weight of σ is $\lambda k > 0$. The isomorphism $L \to T$ which maps σ to 1 is homogeneous and lowers weights by $\lambda k > 0$. By (3), applied to P, there exists an integer $k' > 0$ with $\lambda k' < D$ and $T_{k'} \neq 0$. This contradicts (5.10).

Step 6. In this step we treat the case $\dim Z = 3$.

Let $U := Z\backslash\{o\}$. Since $C\backslash\{o\}$ is smooth (Construction-Proposition 4.1(2)) and U is a prime divisor on $C\backslash\{o\}$, U is a Cartier divisor on $C\backslash\{o\}$. Thus ω_U is invertible and

$$\omega_U \cong \big(\omega_{C\backslash\{o\}} \otimes \mathcal{O}_{C\backslash\{o\}}(U)\big)|_U$$

[Sta, Tag 0B4A]. The isomorphism is the adjunction isomorphism defined by the exact sequence

$$0 \to \mathcal{O}_{C\backslash\{o\}}(-U) \to \mathcal{O}_{C\backslash\{o\}} \to \mathcal{O}_U \to 0,$$

so it is $\mathbb{G}_m$-equivariant for the natural actions on differential forms.

Let $w_1 < D$ be the weight of f_1, and let $q := \operatorname{mult}_Z f_1$. Since f_1 vanishes on Z, $q \geq 1$. On $C\backslash\{o\}$, $\operatorname{div}(f_1) - qU \geq 0$, so multiplication by f_1 is a homomorphism $\mathcal{O}_{C\backslash\{o\}}(qU) \to \mathcal{O}_{C\backslash\{o\}}$ which raises weights by w_1. The restriction of multiplication by f_1 to U is non-zero: near the generic point of U, U is defined by an equation h and $f_1 = h^q\psi$ with ψ not divisible by h, and the restriction maps the class of h^{-q} to the class of ψ modulo h. By Construction-Proposition 4.1(2), $\omega_{C\backslash\{o\}}$ is generated by η, which has weight α (Set-up 5.2(5)). We obtain the composite

$$\omega_U^{\otimes q} \cong \big(\omega_{C\backslash\{o\}}^{\otimes q} \otimes \mathcal{O}_{C\backslash\{o\}}(qU)\big)|_U \to \mathcal{O}_{C\backslash\{o\}}(qU)|_U \to \mathcal{O}_U,$$

where the first arrow divides by η^q and lowers weights by $q\alpha$, and the second arrow is multiplication by f_1. The composite is a non-zero homogeneous homomorphism which lowers weights by

$$q\alpha - w_1 > 0,$$

because $q \geq 1$ and $w_1 < D < \alpha$.

Since Z is normal of dimension 3, $\operatorname{depth} \mathcal{O}_{Z,o} \geq 2$. By [Sta, Tag 0EBJ], the modules of sections over U of $\mathcal{O}_U$ and of $\omega_U^{\otimes q}$ are T and $(\omega_T^{\otimes q})^{**}$ respectively. Taking sections over U, we obtain a homogeneous T-linear map $(\omega_T^{\otimes q})^{**} \to T$ which lowers weights by $q\alpha - w_1 > 0$. The map is injective, since it is non-zero and $(\omega_T^{\otimes q})^{**}$ is torsion-free of rank one. By **Step 3**, T is normal

and Cohen–Macaulay of dimension 3. By (3), there exists an integer $k > 0$ with $\lambda k < D$ and $T_k \neq 0$. This contradicts (5.10).

By **Steps 3–6**, the assumption $\operatorname{lct}(C \ni o; \mathfrak{m}_o) \leq 3/2$ leads to a contradiction. Thus $\operatorname{lct}(C \ni o; \mathfrak{m}_o) > 3/2$. □

Proposition 5.10. *Notation and conditions as in Set-up 5.2 and Definition 5.8, and assume that* $\operatorname{lct}(C \ni o; \mathfrak{m}_o) > 3/2$. *Let* $H \subset C$ *be the zero locus of a general linear combination of a finite set of generators of* $\mathfrak{m}_o$. *Then* $(H \ni o)$ *is a normal terminal Gorenstein germ of dimension* 3, *and* $\operatorname{edim}(C \ni o) \leq 5$.

Proof. Let $f : Y \to C$ be a log resolution of $\mathfrak{m}_o$ which is an isomorphism over $C\backslash\{o\}$, let $E_1, \dots, E_n$ be the exceptional prime divisors of f, and write

$$\mathfrak{m}_o \mathcal{O}_Y = \mathcal{O}_Y\Big(-\sum_i b_i E_i\Big), \qquad K_Y = f^* K_C + \sum_i (a_i - 1) E_i,$$

where $a_i := a(E_i, C, 0)$. Here K_C is Cartier because S is Gorenstein (Construction-Proposition 4.1(2)). Any E_i satisfies $\operatorname{center}_C E_i = \{o\}$, so $b_i = \operatorname{mult}_{E_i} \mathfrak{m}_o$ is a positive integer, and $a_i \geq 3$ by Proposition 5.5(2).

Let $g_1, \dots, g_l$ be the generators of $\mathfrak{m}_o$, and let $h := \sum_k c_k g_k$ with general $c_k \in \mathbb{C}$, so that $H = \{h = 0\}$. Since the pullbacks of $g_1, \dots, g_l$ generate $\mathcal{O}_Y(-\sum_i b_i E_i)$, we have $f^*H = H_Y + \sum_i b_i E_i$, where H_Y is a general member of a base-point-free linear system on Y given by l sections of $\mathcal{O}_Y(-\sum_i b_i E_i)$, and H_Y is the strict transform of H. By [Har77, III, Theorem 10.8] applied to the morphism $Y \to \mathbb{P}^{l-1}$ defined by $g_1, \dots, g_l$, restricted to Y and to each stratum of the simple normal crossing divisor $\sum_i E_i$, with hyperplanes of $\mathbb{P}^{l-1}$ (see also [Har77, III, Remark 10.9.2]), we may assume that H_Y is smooth and that $H_Y + \sum_i E_i$ is a simple normal crossing divisor. In particular, $H\backslash\{o\} \cong H_Y \setminus \operatorname{Exc}(f)$ is smooth.

The local ring $\mathcal{O}_{C,o}$ is a Gorenstein domain and $h \neq 0$, so h is a nonzerodivisor of $\mathcal{O}_{C,o}$. Thus $\mathcal{O}_{H,o} = \mathcal{O}_{C,o}/(h)$ is Gorenstein and Cohen–Macaulay of dimension 3 [BH98, Proposition 3.1.20, Theorem 2.1.3(a), and Proposition 3.1.19(b)]. Since $\mathcal{O}_{H,o}$ satisfies S_2 and $H\backslash\{o\}$ is smooth, $\mathcal{O}_{H,o}$ is normal by Serre's criterion [Har77, II, Theorem 8.22A]. Since H is a Cartier divisor on the Gorenstein variety C, $K_H = (K_C + H)|_H$ is Cartier near o [Sta, Tag 0B4A].

By Definition 5.8, $a_i/b_i \geq \operatorname{lct}(C \ni o; \mathfrak{m}_o) > 3/2$ for any i. If $b_i \geq 2$, then $a_i - b_i > b_i/2 \geq 1$. If $b_i = 1$, then $a_i - b_i \geq 3 - 1 = 2$. Thus

$$a_i - 1 - b_i > 0 \quad \text{for any } i.$$

Let $f_H : H_Y \to H$ be the restriction of f. Since $K_Y + H_Y = f^*(K_C + H) + \sum_i (a_i - 1 - b_i) E_i$, adjunction implies

$$K_{H_Y} = f_H^* K_H + \sum_i (a_i - 1 - b_i) E_i|_{H_Y}.$$

By the choice of H_Y, $\sum_i E_i|_{H_Y}$ is a reduced simple normal crossing divisor on H_Y, so any component of $E_i|_{H_Y}$ is an exceptional prime divisor of f_H with discrepancy $a_i - 1 - b_i > 0$. Since f_H is an isomorphism over $H\backslash\{o\}$, any exceptional prime divisor of f_H is a component of some $E_i|_{H_Y}$. Let G be a prime divisor over H which is exceptional over H_Y. Since H_Y is smooth of dimension 3, $a(G, H_Y, 0) \geq 2$ [KM98, Corollary 2.31(2)], and so

$$a(G, H, 0) = a(G, H_Y, 0) + \sum_i (a_i - 1 - b_i) \operatorname{mult}_G(E_i|_{H_Y}) \geq 2.$$

Thus any exceptional prime divisor over H has positive discrepancy, and $(H \ni o)$ is terminal. Since K_H is Cartier, $(H \ni o)$ is a terminal germ of index one.

If o is a smooth point of H, then $\operatorname{edim}(H \ni o) = 3$. Otherwise, $(H \ni o)$ is an isolated compound Du Val point [Rei83, Theorem (1.1)], so the completion of $\mathcal{O}_{H,o}$ is the quotient of a power series ring in four variables by one power series [Rei83, (0.5)]. Since the embedding dimension does not

change under completion, $\operatorname{edim}(H \ni o) \le 4$. Finally, $\mathfrak{m}_{H,o}/\mathfrak{m}_{H,o}^2 \cong \mathfrak{m}_{C,o}/(\mathfrak{m}_{C,o}^2 + (h))$, so

$$\operatorname{edim}(C \ni o) \le \operatorname{edim}(H \ni o) + 1 \le 5. \quad \square$$

5.4. **The bound in dimension four.** In this subsection, we prove Proposition 5.1.

Proof of Proposition 5.1. Suppose that $\widehat{\mathrm{vol}}(X \ni x) > Q_4$. Then $(X \ni x)$ satisfies Set-up 5.2(1), and we use the notation of Set-up 5.2. By Proposition 5.5(1), there exists an open subset $\mathcal{U} \subset \mathfrak{t}^+$ with $\xi \in \mathcal{U}$ and $\mathbb{R}_{>0}\mathcal{U} = \mathcal{U}$ such that for any primitive $\zeta \in \mathfrak{t}_{\mathbb{Z}} \cap \mathcal{U}$ and any closed point $p \in C\backslash\{o\}$, the stabilizer of p for ζ has order m with $\alpha m < r_\zeta$. The rays spanned by elements of $\mathfrak{t}_{\mathbb{Z}}$ are dense in $\mathfrak{t}^+$, and $\mathcal{U}$ is a non-empty open cone, so there exists a primitive $\zeta \in \mathfrak{t}_{\mathbb{Z}} \cap \mathcal{U}$. Let $S = \bigoplus_{j \ge 0} S_j$, $r = r_\zeta$, $\lambda = \alpha/r$, and h_ζ be as in Set-up 5.2(4) for ζ. We check the conditions (1), (2), and (3) of Proposition 5.9.

Condition (1). By the choice of ζ, the order m of the stabilizer of any closed point of $C\backslash\{o\}$ satisfies $\alpha m < r$, that is, $\lambda m < 1$.

Condition (2). We verify the hypotheses of Proposition 5.7 for $T := S$ and $d = 4$. The ring S is a finitely generated graded $\mathbb{C}$-algebra with $S_0 = \mathbb{C}$ (Definition 2.8), and S is a normal domain of dimension 4. Since S is Gorenstein (Construction-Proposition 4.1(2)), S is Cohen–Macaulay [BH98, Proposition 3.1.20]. By Condition (1), any closed point of $C\backslash\{o\}$ has stabilizer of order m with $\lambda m < 1$. By Construction-Proposition 4.1(2) and (3), $\omega_S = S\eta$, where η is homogeneous of degree $r \in \mathbb{N}^+$, and $\lambda = \alpha/r$. Since C is klt and $\omega_S = S\eta$, Lemma 2.9(3), applied to ζ, and Construction-Proposition 4.1(3) and (1) imply

$$r^4 \lim_{t \to 0^+} t^4 \sum_{j \ge 0} (\dim_{\mathbb{C}} S_j) e^{-tj} = r^4 h_\zeta \ge \widehat{\mathrm{vol}}(C \ni o) = \widehat{\mathrm{vol}}(X \ni x) > Q_4.$$

Since $\alpha/\lambda = r$, the substitution $t = \lambda s$ shows that the left-hand side of the last display is equal to

$$\alpha^4 \lim_{s \to 0^+} s^4 \sum_{j \ge 0} (\dim_{\mathbb{C}} S_j) e^{-s\lambda j},$$

and η has weight $\lambda r = \alpha$ (Set-up 5.2(5)). Thus condition (a) of Proposition 5.7 holds, and Proposition 5.7 implies that there exists an integer $j > 0$ with $\lambda j < D$ and $S_j \ne 0$.

Condition (3). Let $P \subset S$ be a homogeneous prime ideal such that $T := S/P$ is normal and Cohen–Macaulay of dimension 2 or 3, and assume that there exist an integer $q \ge 1$, a real number $e > 0$, and an injective homogeneous T-linear map $(\omega_T^{\otimes q})^{**} \to T$ which lowers weights by e. Since P is a proper homogeneous ideal and $S_0 = \mathbb{C}$, $P \subset \mathfrak{m}_o$, so T is a finitely generated graded $\mathbb{C}$-algebra with $T_0 = \mathbb{C}$. The vertex of $\operatorname{Spec} T$ is o, and the action of $\mathbb{G}_m$ on $\operatorname{Spec} T$ is the restriction of the action on C. Thus a closed point p of $\operatorname{Spec} T$ other than the vertex is a closed point of $C\backslash\{o\}$, the stabilizer of p for the action on $\operatorname{Spec} T$ equals the stabilizer of p for the action on C, and the order m of the stabilizer satisfies $\lambda m < 1$ by Condition (1). The homogeneous elements of degree k of T, of ω_T, and of $(\omega_T^{\otimes q})^{**}$ have weight λk (Set-up 5.2(4) and (5)). Thus T and λ satisfy the hypotheses of Proposition 5.7 with $d = \dim T \in \{2, 3\}$, and condition (b) of Proposition 5.7 holds by assumption. By Proposition 5.7, there exists an integer $k > 0$ with $\lambda k < D$ and $T_k \ne 0$.

By Proposition 5.9, $\operatorname{lct}(C \ni o; \mathfrak{m}_o) > 3/2$. By Proposition 5.10, $\operatorname{edim}(C \ni o) \le 5$. By Construction-Proposition 4.1(1), $\operatorname{edim}(X \ni x) \le \operatorname{edim}(C \ni o) \le 5$. Since $(X \ni x)$ is a klt germ of dimension 4 which is neither smooth nor analytically an ordinary double point, Proposition 3.1 with $d = 4$ implies that $\widehat{\mathrm{vol}}(X \ni x) \le Q_4$. This contradicts the assumption $\widehat{\mathrm{vol}}(X \ni x) > Q_4$. Thus $\widehat{\mathrm{vol}}(X \ni x) \le Q_4 = 4096/27$. $\square$

6. Large stabilizers and jet semigroups

In this section, we prove two results on the cone of Construction-Proposition 4.1 when $d \geq 5$, following the route of [OAI26b]. For integral gradings approximating the Reeb vector, we find points with large stabilizers (Proposition 6.4), and at the threshold Q_d, we show that the semigroup of Taylor leading terms of homogeneous elements at such a point is saturated (Proposition 6.10).

6.1. Fast primitive approximations.

In this subsection, we define fast primitive approximations of the Reeb vector ξ and show that, when $d \geq 5$, the corresponding gradings have points with large stabilizers.

Definition 6.1. Notation as in Construction-Proposition 4.1, and fix a norm $\|\cdot\|$ on $\mathfrak{t}$. For a primitive $\zeta \in \mathfrak{t}_{\mathbb{Z}} \cap \mathfrak{t}^+$, let $m_{\max}(\zeta)$ be the supremum of the orders of the stabilizers of the closed points of $C\backslash\{o\}$ for the $\mathbb{G}_m$-action defined by ζ. A sequence $(\zeta_i)_{i\geq 1}$ of primitive elements of $\mathfrak{t}_{\mathbb{Z}} \cap \mathfrak{t}^+$ is a *fast primitive approximation* of ξ if

$$\lim_{i\to+\infty} \left\| \zeta_i - \frac{r_{\zeta_i}}{d}\xi \right\| = 0.$$

Definition 6.1 asks for the absolute error to tend to zero, not only for the rays $\mathbb{R}_{>0}\zeta_i$ to converge to $\mathbb{R}_{>0}\xi$. The hypothesis of [OAI26b, Theorem 3.1] on the gradings is also a condition on the absolute error, and the absolute error also controls the volumes of the weight valuations and the degrees of the generators of S along the sequence (Lemma 6.2(3)(4)). Here $(r_{\zeta_i}/d)\xi$ is the multiple of ξ whose weight valuation has log discrepancy $a(\mathrm{wt}_{\zeta_i}, C) = r_{\zeta_i}$ (Lemma 2.9(4) and Construction-Proposition 4.1(1)–(3)).

For the sequence constructed in [OAI26b, Lemma 2.6], (1), (3), and (4) of the following lemma are proved there, for the cone of [OAI26b, Proposition 2.5], which is constructed under a lower bound on the normalized volume. Here the cone is that of Construction-Proposition 4.1, and we apply (2)–(4) to an arbitrary fast primitive approximation, so we include the short proofs of (1)–(4).

Lemma 6.2. *Notation as in Definition 6.1. For a sequence $(\zeta_i)_{i\geq 1}$ of primitive elements of $\mathfrak{t}_{\mathbb{Z}} \cap \mathfrak{t}^+$, write $r_i := r_{\zeta_i}$ and $h_i := \mathrm{vol}(\mathrm{wt}_{\zeta_i})$, and for a non-zero $f \in S_\chi$ write $\deg_{\zeta_i}(f) := \langle \chi, \zeta_i \rangle$, the degree of f for the grading defined by ζ_i.*

(1) There exists a fast primitive approximation of ξ.

(2) If $\mathbb{R}_{>0}\xi$ contains a primitive element ζ_0 of $\mathfrak{t}_{\mathbb{Z}}$, then the constant sequence $\zeta_i = \zeta_0$ is a fast primitive approximation of ξ, and any fast primitive approximation (ζ_i) of ξ satisfies $\zeta_i = \zeta_0$ for $i \gg 0$. Otherwise $r_i \to +\infty$ for any fast primitive approximation (ζ_i) of ξ.

(3) If (ζ_i) is a fast primitive approximation of ξ, then $\lim_{i\to+\infty} r_i^d h_i = \widehat{\mathrm{vol}}(C \ni o)$.

(4) Let (ζ_i) be a fast primitive approximation of ξ and let $\mathcal{G}$ be a finite set of non-constant $\mathbb{T}$-homogeneous generators of the $\mathbb{C}$-algebra S. Then there exist real numbers $0 < c' < c''$ such that $c' r_i \leq \deg_{\zeta_i}(f) \leq c'' r_i$ for any $f \in \mathcal{G}$ and any $i \gg 0$. In particular $m_{\max}(\zeta_i) \leq c'' r_i < +\infty$ for $i \gg 0$.

Proof. Let χ_η be the character of η. By Construction-Proposition 4.1(3) and the grading of Definition 2.8, $r_\zeta = \langle \chi_\eta, \zeta \rangle$ for any primitive $\zeta \in \mathfrak{t}_{\mathbb{Z}} \cap \mathfrak{t}^+$. Since C is klt and $\omega_S = S\eta$ (Construction-Proposition 4.1(2)), Lemma 2.9(2) with $I = 1$ and $\chi_0 = \chi_\eta$ implies

$$a(\mathrm{wt}_{\xi'}, C) = \langle \chi_\eta, \xi' \rangle \quad \text{for any } \xi' \in \mathfrak{t}^+.$$

Thus $\langle \chi_\eta, \xi \rangle = a(\mathrm{wt}_\xi, C) = d$ by Construction-Proposition 4.1(1). For a fast primitive approximation (ζ_i) of ξ, we write $\epsilon_i := \zeta_i - (r_i/d)\xi$, so that $\|\epsilon_i\| \to 0$.

(1) We first show that $\mathfrak{t}^+$ is an open convex cone in $\mathfrak{t}$. Let $f_1, \ldots, f_s$ be non-constant $\mathbb{T}$-homogeneous generators of the $\mathbb{C}$-algebra S, with $f_j \in S_{\chi_j}$. Since $S_0 = \mathbb{C}$, $\chi_j \neq 0$ for any j. Any character $\chi \neq 0$ with $S_\chi \neq 0$ is a sum $\sum_j n_j \chi_j$ with integers $n_j \geq 0$, not all zero. Hence

$$\mathfrak{t}^+ = \{\xi' \in \mathfrak{t} \mid \langle \chi_j, \xi' \rangle > 0 \text{ for } 1 \leq j \leq s\},$$

which is an open convex cone.

Let $u_1, \dots, u_n$ be a basis of $\mathfrak{t}_{\mathbb{Z}}$, and write $\xi = \sum_j \theta_j u_j$ with $\theta_j \in \mathbb{R}$. By Dirichlet's theorem on simultaneous approximation, for real numbers $\theta_1, \dots, \theta_n$ there exist infinitely many integers $t > 0$ and integers $p_1, \dots, p_n$ with $|t\theta_j - p_j| < t^{-1/n}$ for $1 \le j \le n$ [Cas57, Chapter I, Theorem VI]. Thus there exist integers $0 < t_1 < t_2 < \cdots$ and integers p_{ij} such that $|t_i\theta_j - p_{ij}| < t_i^{-1/n}$ for $1 \le j \le n$. Let $q_i := \sum_j p_{ij} u_j \in \mathfrak{t}_{\mathbb{Z}}$ and $e_i := q_i - t_i\xi$. The coordinates of e_i are bounded by $t_i^{-1/n}$, and all norms on $\mathfrak{t}$ are equivalent, so $\|e_i\| \to 0$. Since $q_i/t_i = \xi + e_i/t_i \to \xi \in \mathfrak{t}^+$ and $\mathfrak{t}^+$ is an open cone, $q_i \in \mathfrak{t}^+$ for $i \gg 0$. Possibly discarding finitely many indices, we may assume that $q_i \in \mathfrak{t}^+$ for any i. In particular $q_i \neq 0$. Let $b_i \ge 1$ be the largest integer such that $q_i \in b_i\mathfrak{t}_{\mathbb{Z}}$, and let $\zeta_i := q_i/b_i$. Then ζ_i is a primitive element of $\mathfrak{t}_{\mathbb{Z}} \cap \mathfrak{t}^+$, and

$$r_i = \frac{\langle \chi_\eta, q_i \rangle}{b_i} = \frac{t_i d + \langle \chi_\eta, e_i \rangle}{b_i}.$$

Therefore

$$\zeta_i - \frac{r_i}{d}\xi = \frac{1}{b_i}\Big(e_i - \frac{\langle \chi_\eta, e_i \rangle}{d}\xi\Big).$$

Since $b_i \ge 1$, $\|e_i\| \to 0$, and $\langle \chi_\eta, \cdot \rangle$ is a linear function on $\mathfrak{t}$, the right-hand side tends to 0. Thus (ζ_i) is a fast primitive approximation of ξ.

(2) Assume that $\zeta_0 \in \mathbb{R}_{>0}\xi$ is a primitive element of $\mathfrak{t}_{\mathbb{Z}}$, and write $\zeta_0 = \lambda\xi$ with $\lambda > 0$. Then $\zeta_0 \in \mathfrak{t}^+$, and $r_{\zeta_0} = \langle \chi_\eta, \zeta_0 \rangle = \lambda d$. Hence $\zeta_0 = (r_{\zeta_0}/d)\xi$, and the constant sequence $\zeta_i = \zeta_0$ is a fast primitive approximation of ξ. Let (ζ_i) be any fast primitive approximation of ξ. Then

$$r_{\zeta_0}\zeta_i - r_i\zeta_0 = r_{\zeta_0}\zeta_i - \frac{r_i r_{\zeta_0}}{d}\xi = r_{\zeta_0}\epsilon_i.$$

The left-hand side is an element of the lattice $\mathfrak{t}_{\mathbb{Z}}$, and the right-hand side tends to 0. Hence $r_{\zeta_0}\zeta_i = r_i\zeta_0$ for $i \gg 0$. Since ζ_i and ζ_0 are primitive and $r_i, r_{\zeta_0} > 0$, $\zeta_i = \zeta_0$ for $i \gg 0$.

Now assume that $\mathbb{R}_{>0}\xi$ contains no primitive element of $\mathfrak{t}_{\mathbb{Z}}$, and let (ζ_i) be a fast primitive approximation of ξ. Suppose that $r_i \not\to +\infty$. Then there exist a real number $R > 0$ and infinitely many indices i with $r_i \le R$. For any such i, $\|\zeta_i\| \le R\|\xi\|/d + \|\epsilon_i\|$ is bounded, so ζ_i takes only finitely many values in $\mathfrak{t}_{\mathbb{Z}}$. Thus there exists a primitive $\zeta \in \mathfrak{t}_{\mathbb{Z}} \cap \mathfrak{t}^+$ with $\zeta_i = \zeta$ for infinitely many i. For any i with $\zeta_i = \zeta$, $\epsilon_i = \zeta - (r_\zeta/d)\xi$ is constant and tends to 0, so $\zeta = (r_\zeta/d)\xi \in \mathbb{R}_{>0}\xi$, a contradiction. Hence $r_i \to +\infty$.

(3) Since $r_i \ge 1$,

$$\frac{d}{r_i}\zeta_i = \xi + \frac{d}{r_i}\epsilon_i \to \xi.$$

By Construction-Proposition 4.1(3), $a(\mathrm{wt}_{\zeta_i}, C) = r_i$. The function $a(\mathrm{wt}_{\xi'}, C)^d \operatorname{vol}(\mathrm{wt}_{\xi'})$ of $\xi' \in \mathfrak{t}^+$ is invariant under multiplication by positive real numbers, continuous at ξ, and equal to $\widehat{\mathrm{vol}}(C \ni o)$ at ξ (Lemma 4.3). Therefore

$$r_i^d h_i = a(\mathrm{wt}_{\zeta_i}, C)^d \operatorname{vol}(\mathrm{wt}_{\zeta_i}) = a(\mathrm{wt}_{(d/r_i)\zeta_i}, C)^d \operatorname{vol}(\mathrm{wt}_{(d/r_i)\zeta_i}) \to \widehat{\mathrm{vol}}(C \ni o).$$

(4) Since $\dim C = d > 0$, $S \neq \mathbb{C}$, so $\mathcal{G} \neq \emptyset$. For $f \in \mathcal{G}$, let χ_f be the character with $f \in S_{\chi_f}$. Since f is non-constant and $S_0 = \mathbb{C}$, $\chi_f \neq 0$, so $\langle \chi_f, \xi \rangle > 0$. Let

$$c' := \min_{f \in \mathcal{G}} \frac{\langle \chi_f, \xi \rangle}{2d} \quad \text{and} \quad c'' := \max_{f \in \mathcal{G}} \frac{\langle \chi_f, \xi \rangle}{d} + c'.$$

Then $0 < c' < c''$. For $f \in \mathcal{G}$,

$$\deg_{\zeta_i}(f) = \langle \chi_f, \zeta_i \rangle = \frac{r_i}{d}\langle \chi_f, \xi \rangle + \langle \chi_f, \epsilon_i \rangle.$$

Since $\mathcal{G}$ is finite and $\|\epsilon_i\| \to 0$, $|\langle \chi_f, \epsilon_i \rangle| \le c'$ for any $f \in \mathcal{G}$ and any $i \gg 0$. Since $r_i \ge 1$, $c' \le c' r_i$, hence $c' r_i \le \deg_{\zeta_i}(f) \le c'' r_i$ for any $f \in \mathcal{G}$ and any $i \gg 0$.

Let $i \gg 0$ be such that $c' r_i \le \deg_{\zeta_i}(f) \le c'' r_i$ for any $f \in \mathcal{G}$, and let $p \in C \setminus \{o\}$ be a closed point. The elements of $\mathcal{G}$ are $\mathbb{T}$-homogeneous of non-zero characters, so they are contained

in the ideal $\bigoplus_{\chi\neq 0} S_\chi$ of o. Since $\mathcal{G}$ generates the $\mathbb{C}$-algebra S, any $\mathbb{T}$-homogeneous element of non-zero character is a polynomial in the elements of $\mathcal{G}$ without constant term. Hence $\mathcal{G}$ generates the ideal of o. Since $p \neq o$, there exists $f \in \mathcal{G}$ with $f(p) \neq 0$. The element f is homogeneous of degree $\deg_{\zeta_i}(f) > 0$ for the grading defined by ζ_i. By Lemma 2.9(3) applied to ζ_i, the stabilizer of p is μ_m, where m divides $\deg_{\zeta_i}(f)$. Thus $m \leq \deg_{\zeta_i}(f) \leq c''r_i$. Since p is arbitrary, $m_{\max}(\zeta_i) \leq c''r_i$. □

Remark 6.3. The standing assumptions of [OAI26b, Section 3] are the conclusions of [OAI26b, Proposition 2.5], which is proved there under a lower bound on the normalized volume. The proof of [OAI26b, Theorem 3.1] uses only the following properties of the cone: $d \geq 5$, the ring S is a normal Gorenstein domain, $C\backslash\{o\}$ is smooth, the torus action is faithful, the log Fano cone (C, ξ) is K-polystable, wt_ξ minimizes the normalized volume, and $a(\mathrm{wt}_\xi, C) = d$. K-polystability enters there through the Ricci-flat Kähler cone metric on $C\backslash\{o\}$ with Reeb vector ξ, which is given by the Yau–Tian–Donaldson theorem for Fano cones and which [OAI26b, Lemma 3.5] uses to bound from below the weights of homogeneous symmetric tensors [OAI26b, Section 3.3]. By Construction-Proposition 4.1, the cone in Proposition 6.4 has the properties listed above.

Proposition 6.4. *Let $d \geq 5$ be an integer and let $(X \ni x)$ be a klt germ of dimension d which is neither smooth nor analytically an ordinary double point and satisfies $\widehat{\mathrm{vol}}(X \ni x) > Q_d$. Let C, $\mathbb{T}$, and ξ be as in Construction-Proposition 4.1. Then any fast primitive approximation (ζ_i) of ξ satisfies $m_{\max}(\zeta_i) > r_{\zeta_i}/(d-1)$ for $i \gg 0$. If $\mathbb{R}_{>0}\xi$ contains a primitive element ζ_0 of $\mathfrak{t}_\mathbb{Z}$, then $m_{\max}(\zeta_0) > r_{\zeta_0}/(d-1)$.*

Proof. Let (ζ_i) be a fast primitive approximation of ξ. Suppose that there exist infinitely many indices i with $m_{\max}(\zeta_i) \leq r_{\zeta_i}/(d-1)$. Possibly replacing (ζ_i) with the subsequence consisting of such indices, which is again a fast primitive approximation of ξ, we may assume that $m_{\max}(\zeta_i) \leq r_{\zeta_i}/(d-1)$ for any i. The conditions of [OAI26b, Theorem 3.1] on the gradings are that (ζ_i) is a fast primitive approximation of ξ and $m_{\max}(\zeta_i) \leq r_{\zeta_i}/(d-1)$ for any i, where (ζ_i) may be the constant sequence $\zeta_i = \zeta_0$ when $\mathbb{R}_{>0}\xi$ contains a primitive element ζ_0 of $\mathfrak{t}_\mathbb{Z}$. Hence, by Remark 6.3, the cone C with the Reeb vector ξ and the sequence (ζ_i) satisfy the hypotheses of [OAI26b, Theorem 3.1]. Thus $(C \ni o)$ is smooth or an ordinary double point [OAI26b, Theorem 3.1]. In particular, $(C \ni o)$ is a local complete intersection. Since $(X \ni x)$ is not analytically an ordinary double point, this contradicts Construction-Proposition 4.1(4). Hence $m_{\max}(\zeta_i) > r_{\zeta_i}/(d-1)$ for $i \gg 0$.

If $\mathbb{R}_{>0}\xi$ contains a primitive element ζ_0 of $\mathfrak{t}_\mathbb{Z}$, then the constant sequence $\zeta_i = \zeta_0$ is a fast primitive approximation of ξ (Lemma 6.2(2)). Applying the first part to the constant sequence, we obtain $m_{\max}(\zeta_0) > r_{\zeta_0}/(d-1)$. □

6.2. **Jet data.** In this subsection, we fix the set-up for the rest of Section 6 and recall from [OAI26b] the semigroup Γ, the lattice Λ, and the cone $\mathcal{K}$ attached to a point of $C\backslash\{o\}$ with a large stabilizer.

Set-up 6.5. Let d, N, $(X \ni x)$, $C = \operatorname{Spec} S$, o, $\mathbb{T}$, ξ, η, (ζ_i), r_i, x_i, and m_i be as follows.

(1) $d \geq 5$ is an integer and $N := d - 1$.
(2) $(X \ni x)$ is a klt germ of dimension d which is neither smooth nor analytically an ordinary double point and satisfies $\widehat{\mathrm{vol}}(X \ni x) > Q_d$.
(3) $C = \operatorname{Spec} S$, o, $\mathbb{T}$, ξ, and η are as in Construction-Proposition 4.1 for $(X \ni x)$.
(4) $(\zeta_i)_{i\geq 1}$ is a fast primitive approximation of ξ, and $r_i := r_{\zeta_i}$.
(5) If $\mathbb{R}_{>0}\xi$ contains a primitive element ζ_0 of $\mathfrak{t}_\mathbb{Z}$, then $\zeta_i = \zeta_0$ for any i.
(6) For any i, the point $x_i \in C\backslash\{o\}$ is a closed point whose stabilizer for the $\mathbb{G}_m$-action defined by ζ_i is μ_{m_i} with $m_i \geq r_i/N$.

By Lemma 6.2(2), Set-up 6.5(5) holds after removing finitely many terms of (ζ_i), and it is used only to obtain Lemma 6.8 and Proposition 6.10 for every i when $\mathbb{R}_{>0}\xi$ contains a primitive

element of $\mathfrak{t}_{\mathbb{Z}}$. By Proposition 6.4, points x_i as in Set-up 6.5(6) exist after removing finitely many terms of (ζ_i).

Remark 6.6. In [OAI26b, Sections 4–7] the cone C is assumed to satisfy $\widehat{\mathrm{vol}}(C \ni o) \geq 2N^d$, which is stronger than $\widehat{\mathrm{vol}}(C \ni o) > Q_d$ since $Q_d < 2N^d$ (Lemma 2.7(5)), and [OAI26b, Section 7] assumes in addition the bound $\widehat{\mathrm{vol}} \leq 2(k-1)^k$ for singular klt germs of any dimension $k < d$, which is Theorem 2.4(2). In Sections 6 and 7 and in Appendix A we use [OAI26b, Lemma 2.7, Proposition 4.1, Lemma 4.2, Theorem 4.4, Lemmas 4.5 and 4.7, Lemma 5.2, Theorem 5.3, Lemmas 6.1 and 6.2, Proposition 6.4, Corollary 6.6, Lemmas 7.1, 7.3, and 7.4, and Proposition 7.5]. In [OAI26b], the assumption $\widehat{\mathrm{vol}}(C \ni o) \geq 2N^d$ enters the proofs of the results listed above only through the first inequality of [OAI26b, Lemma 4.3] and through [OAI26b, Theorem 4.8, Lemma 6.3, and Lemma 7.2]. Under the assumption $\widehat{\mathrm{vol}}(C \ni o) > Q_d$, we replace them by Lemma 6.8, Proposition 6.10, Lemma 7.3, and Proposition 7.9, respectively, and we apply the results listed above with the hypotheses given in the corresponding statements below. Of the standing data of [OAI26b, Section 6], the proof of [OAI26b, Lemma 6.2] uses only that S is a positively graded normal Gorenstein domain whose canonical module has a homogeneous generator.

Construction-Proposition 6.7 is [OAI26b, Lemma 2.7 and Proposition 4.1] written in the notation of this paper (see Remark 6.6), and we do not reprove it.

Construction-Proposition 6.7 ([OAI26b, Lemma 2.7 and Proposition 4.1])**.** Notation and conditions as in Set-up 6.5. For any i, write ζ, r, and m for ζ_i, r_i, and m_i. The objects $\mathcal{B}$, $\mathcal{L}$, $z_1, \dots, z_N$, $b_1, \dots, b_N$, Γ, Λ, c, κ, $\mathcal{K}$, and φ below depend on i, which we suppress from the notation.

(1) (a) The quotient stack $\mathcal{B} := [(C\backslash\{o\})/\mathbb{G}_m]$ for the $\mathbb{G}_m$-action defined by ζ is a smooth proper Deligne–Mumford stack of dimension N with trivial generic stabilizer and projective coarse moduli space $\operatorname{Proj} S$.
(b) There exists an ample line bundle $\mathcal{L}$ on $\mathcal{B}$ such that $S_j = H^0(\mathcal{B}, j\mathcal{L})$ for any $j \geq 0$ and $-K_{\mathcal{B}} = r\mathcal{L}$.

(2) (a) Choose μ_m-equivariant algebraic parameters $z_1, \dots, z_N$ of a smooth chart of $\mathcal{B}$ at the image of x_i, with characters $b_1, \dots, b_N \in \mathbb{Z}/m\mathbb{Z}$, and a local generator of $\mathcal{L}$ of character 1, so that any $s \in S_j$ is a power series in $z_1, \dots, z_N$.
(b) Order the monomials in $z_1, \dots, z_N$ first by total degree and then lexicographically, and call the smallest monomial occurring in $s \neq 0$ the *leading monomial* of s.
(c) The set

$$\Gamma := \{(j, \gamma) \in \mathbb{Z} \times \mathbb{Z}^N_{\geq 0} \mid z^\gamma \text{ is the leading monomial of some } s \in S_j\backslash\{0\}\}$$

is a semigroup which generates the group

$$\Lambda := \{(j, \gamma) \in \mathbb{Z} \times \mathbb{Z}^N \mid j \equiv b_1\gamma_1 + \cdots + b_N\gamma_N \bmod m\}.$$

(d) $c := (r, 1, \dots, 1) \in \Lambda$.
(e) $\kappa := r/(Nm)$ satisfies $0 < \kappa \leq 1$.

(3) Let $\mathcal{K} \subset \mathbb{R} \times \mathbb{R}^N$ be the closure of the convex cone spanned by Γ. Then there exists a convex positively homogeneous function $\varphi : \mathbb{R}^N_{\geq 0} \to \mathbb{R}$ such that

$$\mathcal{K} = \{(t, y) \in \mathbb{R} \times \mathbb{R}^N_{\geq 0} \mid Nt \geq r\varphi(y)\}.$$

(4) There exists a real number $\epsilon > 0$ such that for any i, the function φ for i satisfies $\varphi(y) \geq \epsilon(y_1 + \cdots + y_N)$ for any $y \in \mathbb{R}^N_{\geq 0}$ (by [OAI26b, Proposition 4.1(ii)], such a constant exists for $i \gg 0$, and for each of the finitely many remaining i such a constant exists for ζ_i alone; we take the minimum).

6.3. **The canonical vector.** In this subsection, we prove that $c \in \mathrm{Int}(\mathcal{K})$ for $i \gg 0$ and that $\Gamma = \mathcal{K} \cap \Lambda$ (Proposition 6.10).

Lemma 6.8 turns the lower bound on the normalized volume in Set-up 6.5(2) and the minimality of $\widehat{\mathrm{vol}}(C \ni o)$ into two inequalities on the function φ of Construction-Proposition 6.7(3), and Proposition 6.10 combines them.

Lemma 6.8. *Notation and conditions as in Set-up 6.5 and Construction-Proposition 6.7. Let $\mathsf{X} = (\mathsf{X}_1, \dots, \mathsf{X}_N)$ consist of independent exponential random variables with mean 1. For any i, let $\mathcal{H} := \int_{\mathbb{R}^N_{\geq 0}} e^{-\varphi(y)} dy$ and $\beta(y) := \varphi(y) - y_1 - \cdots - y_N$, where φ is the function of Construction-Proposition 6.7(3) for i. Then $\mathcal{H} > 2\rho_d/\kappa$ for any i, and there exist real numbers $\tau_i \geq 0$ with $\lim_{i \to +\infty} \tau_i = 0$ such that*

$$\mathbb{E}\big[(1 + \beta(\mathsf{X}))e^{-\beta(\mathsf{X})}\big] \geq -\tau_i \mathcal{H}$$

for any i. If $\zeta_i = \zeta_0$ for every i, then the last inequality holds with $\tau_i = 0$ for any i.

Proof. Fix i and suppress it from the notation. For $y \in \mathbb{R}^N_{\geq 0}$ write $|y| := y_1 + \cdots + y_N$. By Construction-Proposition 6.7(4), $\varphi(y) \geq \epsilon|y|$ for any $y \in \mathbb{R}^N_{\geq 0}$. Thus $\mathcal{H} < +\infty$, and

$$\mu := \mathcal{H}^{-1} e^{-\varphi(y)} dy$$

is a probability measure on $\mathbb{R}^N_{\geq 0}$ under which all moments of $|y|$ are finite. We write $\mathbb{E}_\mu$ for the expectation with respect to μ. For a real number $\delta \geq 0$ let

$$F(\delta) := N^d (1 + \delta)^d \kappa \int_{\mathbb{R}^N_{\geq 0}} e^{-\varphi(y) - \delta|y|} dy.$$

By [OAI26b, Lemma 4.2] (see Remark 6.6), for any rational number $\delta \geq 0$ there exists a valuation v_δ of $\mathbb{C}(C)$ centered at o such that $a(v_\delta, C) = N(1 + \delta)$ and $a(v_\delta, C)^d \,\mathrm{vol}(v_\delta) = F(\delta)$, and $v_0 = (N/r)\,\mathrm{wt}_\zeta$. By the definition of the normalized volume,

$$F(\delta) \geq \widehat{\mathrm{vol}}(C \ni o) \quad \text{for any rational number } \delta \geq 0. \tag{6.1}$$

Since $a(\lambda v, C)^d \,\mathrm{vol}(\lambda v) = a(v, C)^d \,\mathrm{vol}(v)$ for any valuation v centered at o and any real number $\lambda > 0$, and since $a(\mathrm{wt}_\zeta, C) = r$ (Construction-Proposition 4.1(3)), we have

$$F(0) = r^d \,\mathrm{vol}(\mathrm{wt}_\zeta). \tag{6.2}$$

We prove the first inequality. By (6.2), Construction-Proposition 4.1(3) and (1), and Set-up 6.5(2),

$$N^d \kappa \mathcal{H} = F(0) = r^d \,\mathrm{vol}(\mathrm{wt}_\zeta) \geq \widehat{\mathrm{vol}}(C \ni o) = \widehat{\mathrm{vol}}(X \ni x) > Q_d.$$

Since $Q_d / N^d = 2\rho_d$ (Lemma 2.7(1)), we have $\mathcal{H} > 2\rho_d/\kappa$.

We prove the second inequality. Since φ is positively homogeneous, the substitution $y \to y/t$ shows that

$$\int_{\mathbb{R}^N_{\geq 0}} e^{-t\varphi(y)} dy = t^{-N} \mathcal{H} \quad \text{for any real number } t > 0.$$

Differentiating once and twice at $t = 1$, which is allowed by dominated convergence since $\varphi(y) \geq \epsilon|y|$, we obtain $\mathbb{E}_\mu[\varphi] = N$ and $\mathbb{E}_\mu[\varphi^2] = N(N+1)$. Since $|y| \leq \varphi(y)/\epsilon$, we have

$$\mathbb{E}_\mu[|y|] \leq \frac{N}{\epsilon} \quad \text{and} \quad \mathbb{E}_\mu[|y|^2] \leq \frac{N(N+1)}{\epsilon^2}. \tag{6.3}$$

Let

$$\Phi(\delta) := \frac{F(\delta)}{F(0)} = (1 + \delta)^d \mathbb{E}_\mu\big[e^{-\delta|y|}\big] \quad \text{for } \delta \geq 0.$$

Since all moments of $|y|$ under μ are finite, Φ is twice differentiable on $[0, +\infty)$, and Φ' and Φ'' are computed under the integral sign:

$$\Phi'(\delta) = d(1 + \delta)^{d-1} \mathbb{E}_\mu\big[e^{-\delta|y|}\big] - (1 + \delta)^d \mathbb{E}_\mu\big[|y| e^{-\delta|y|}\big],$$

$$\Phi''(\delta) = d(d-1)(1+\delta)^{d-2}\mathbb{E}_\mu\big[e^{-\delta|y|}\big] - 2d(1+\delta)^{d-1}\mathbb{E}_\mu\big[|y|e^{-\delta|y|}\big] + (1+\delta)^d\mathbb{E}_\mu\big[|y|^2e^{-\delta|y|}\big].$$

By (6.3), for $0 \le \delta \le 1$ we have

$$|\Phi''(\delta)| \le \ell := 2^d\Big(d^2 + \frac{dN}{\epsilon} + \frac{N(N+1)}{\epsilon^2}\Big), \tag{6.4}$$

and ℓ does not depend on i since ϵ does not depend on i. Moreover, $\Phi'(0) = d - \mathbb{E}_\mu[|y|]$. Since X has density $e^{-|y|}$ on $\mathbb{R}^N_{\ge 0}$, $\beta(y) = \varphi(y) - |y|$, $\mathbb{E}_\mu[\varphi] = N$, and $d = N+1$, we obtain

$$\mathbb{E}\big[(1+\beta(\mathsf{X}))e^{-\beta(\mathsf{X})}\big] = \int_{\mathbb{R}^N_{\ge 0}} \big(1+\varphi(y)-|y|\big)e^{-\varphi(y)}dy = \mathcal{H}\big(1+N-\mathbb{E}_\mu[|y|]\big) = \mathcal{H}\Phi'(0). \tag{6.5}$$

Now we write F_i and Φ_i for the functions F and Φ for i, and let $\tau_i := \max\{0, -\Phi_i'(0)\}$. By (6.5), the second inequality holds for any i. It remains to show that $\lim_{i\to+\infty}\tau_i = 0$, and that $\tau_i = 0$ for any i if $\zeta_i = \zeta_0$ for every i.

Suppose that τ_i does not converge to 0. Then there exist a real number $\theta > 0$ and an infinite set I of indices such that $\Phi_i'(0) \le -\theta$ for any $i \in I$. Choose a rational number δ with $0 < \delta \le \min\{1, \theta/\ell\}$. By Taylor's formula and (6.4), for any $i \in I$ we have

$$\Phi_i(\delta) \le 1 + \Phi_i'(0)\delta + \frac{\ell\delta^2}{2} \le 1 - \frac{\theta\delta}{2},$$

so (6.1) implies

$$\widehat{\mathrm{vol}}(C \ni o) \le F_i(\delta) \le \Big(1 - \frac{\theta\delta}{2}\Big)F_i(0).$$

By (6.2) and Lemma 6.2(3), $F_i(0) = r_i^d\,\mathrm{vol}(\mathrm{wt}_{\zeta_i})$ converges to $\widehat{\mathrm{vol}}(C \ni o)$. Letting $i \to +\infty$ in I, we obtain $\widehat{\mathrm{vol}}(C \ni o) \le (1 - \theta\delta/2)\widehat{\mathrm{vol}}(C \ni o)$, which is impossible since $\widehat{\mathrm{vol}}(C \ni o) > Q_d > 0$. Thus $\lim_{i\to+\infty}\tau_i = 0$.

Assume that $\zeta_i = \zeta_0$ for every i. Then the sequence $(r_i^d\,\mathrm{vol}(\mathrm{wt}_{\zeta_i}))$ is constant, so Lemma 6.2(3) and (6.2) imply $F_i(0) = \widehat{\mathrm{vol}}(C \ni o)$ for any i. By (6.1), $\Phi_i(\delta) \ge 1 = \Phi_i(0)$ for any rational number $\delta > 0$. Letting $\delta \to 0^+$ through rational numbers, we obtain $\Phi_i'(0) \ge 0$, that is, $\tau_i = 0$. □

For $d = 5$, the constant $C_* = \frac{3}{4}e^{4/3} - \frac{e}{3}$ of [OAI26b, Theorem 4.4] is larger than $2\rho_5$. The constant C_* is a supremum over all numbers N of variables, and by [OAI26b, Theorem 4.4] it is approached by the vectors $(2, -2/(N-1), \dots, -2/(N-1))$ as $N \to +\infty$. For $N = 4$, Lemma 6.9 replaces C_* by $137/72$, which is smaller than $2\rho_5$.

Lemma 6.9. *Let $\mathsf{X}_1, \dots, \mathsf{X}_4$ be independent exponential random variables with mean 1, and let $g : \mathbb{R} \to \mathbb{R}$ be the function with $g(y) = 1$ for $y \le 0$ and $g(y) = (1+y)e^{-y}$ for $y > 0$. Then*

$$\frac{3}{4}e^{4/3}\mathbb{E}\Big[g\Big(\frac{4}{3} + a_1\mathsf{X}_1 + a_2\mathsf{X}_2 + a_3\mathsf{X}_3 + a_4\mathsf{X}_4\Big)\Big] \le \frac{137}{72}$$

for all real numbers $a_1, \dots, a_4$ with $a_1 + a_2 + a_3 + a_4 = 0$.

We prove Lemma 6.9 in Appendix A.

Proposition 6.10. *Notation and conditions as in Set-up 6.5 and Construction-Proposition 6.7. Then for $i \gg 0$, and for any i if $\zeta_i = \zeta_0$ for every i, the objects Γ, Λ, $\mathcal{K}$, and c of Construction-Proposition 6.7 for i satisfy the following.*

1. *$c \in \mathrm{Int}(\mathcal{K})$.*
2. *$\mathcal{K}$ is a rational polyhedral cone, and for any facet of $\mathcal{K}$, the primitive element of $\mathrm{Hom}(\Lambda, \mathbb{Z})$ which is non-negative on $\mathcal{K}$ and vanishes on the facet takes the value 1 at c.*
3. *$\Gamma = \mathcal{K} \cap \Lambda$ and $\mathrm{Int}(\mathcal{K}) \cap \Lambda = c + \Gamma$. In particular Γ is a finitely generated semigroup.*

Proof. The proof is long, so we divide it into several steps. In the proof, we write $\mathbf{1} := (1, \dots, 1) \in \mathbb{R}^N$ and $|y| := y_1 + \cdots + y_N$ for $y \in \mathbb{R}^N_{\ge 0}$.

Step 1. In this step we show that, for any i, $c \in \mathrm{Int}(\mathcal{K})$ if and only if $\varphi(\mathbf{1}) < N$, where φ is the function of Construction-Proposition 6.7(3) for i.

By Construction-Proposition 6.7(3), a point $(t,y) \in \mathbb{R}\times\mathbb{R}^N$ belongs to $\mathcal{K}$ if and only if $y \in \mathbb{R}^N_{\geq 0}$ and $Nt \geq r\varphi(y)$, and $c = (r, \mathbf{1})$. If $\varphi(\mathbf{1}) \geq N$, then for any real number $s > 0$ we have $N(r-s) < Nr \leq r\varphi(\mathbf{1})$, so $(r-s,\mathbf{1}) \notin \mathcal{K}$. Since $(r-s,\mathbf{1}) \to c$ as $s \to 0^+$, $c \notin \mathrm{Int}(\mathcal{K})$. Conversely, φ is a finite convex function on $\mathbb{R}^N_{>0}$, so it is continuous on $\mathbb{R}^N_{>0}$ [Roc70, Theorem 10.1]. Thus, if $\varphi(\mathbf{1}) < N$, then

$$\{(t,y) \in \mathbb{R}\times\mathbb{R}^N_{>0} \mid Nt > r\varphi(y)\}$$

is an open subset of $\mathbb{R}\times\mathbb{R}^N$ which is contained in $\mathcal{K}$ and contains c. Hence $c \in \mathrm{Int}(\mathcal{K})$.

Step 2. In this step we show that, if $\varphi(\mathbf{1}) \geq N$ for some i, then the inequality (6.9) below holds for i and for some real numbers $a_1,\dots,a_N$ with $a_1+\cdots+a_N = 0$.

Let X, $\mathcal{H}$, and β be as in Lemma 6.8, and let g be as in Lemma 6.9. Since $\frac{d}{dy}\big((1+y)e^{-y}\big) = -ye^{-y}$, we have $0 < g \leq 1$. By Lemma 6.8, there exist real numbers $\tau_i \geq 0$ with $\lim_{i\to+\infty}\tau_i = 0$, and with $\tau_i = 0$ for any i if $\zeta_i = \zeta_0$ for every i, such that

$$\mathcal{H} > \frac{2\rho_d}{\kappa} \quad\text{and}\quad \mathbb{E}\big[(1+\beta(\mathsf{X}))e^{-\beta(\mathsf{X})}\big] \geq -\tau_i\mathcal{H} \quad\text{for any } i. \tag{6.6}$$

Fix i with $\varphi(\mathbf{1}) \geq N$. Since $\mathbf{1}$ is an interior point of $\mathbb{R}^N_{\geq 0}$, φ has a subgradient $w = (w_1,\dots,w_N)$ at $\mathbf{1}$ [Roc70, Theorem 23.4], that is,

$$\varphi(y) \geq \varphi(\mathbf{1}) + \sum_{k=1}^N w_k(y_k - 1) \quad\text{for any } y \in \mathbb{R}^N_{\geq 0}.$$

Since φ is positively homogeneous, $\varphi(0) = 0$ and $\varphi(2\cdot\mathbf{1}) = 2\varphi(\mathbf{1})$. Taking $y = 0$ and $y = 2\cdot\mathbf{1}$, we obtain $w_1+\cdots+w_N \geq \varphi(\mathbf{1})$ and $w_1+\cdots+w_N \leq \varphi(\mathbf{1})$. Thus $w_1+\cdots+w_N = \varphi(\mathbf{1})$, and

$$\varphi(y) \geq w_1y_1+\cdots+w_Ny_N \quad\text{for any } y \in \mathbb{R}^N_{\geq 0}.$$

For $1 \leq k \leq N$ let

$$a_k := w_k - 1 - \frac{\varphi(\mathbf{1}) - N}{N}.$$

Then $a_1+\cdots+a_N = 0$, and $a_k \leq w_k - 1$ since $\varphi(\mathbf{1}) \geq N$. Thus, for any $y \in \mathbb{R}^N_{\geq 0}$,

$$\beta(y) = \varphi(y) - |y| \geq \sum_{k=1}^N (w_k - 1)y_k \geq \sum_{k=1}^N a_k y_k. \tag{6.7}$$

Next we show that, for any real numbers $\psi \leq \sigma$,

$$\Big(1 + \frac{3}{4}(1+\sigma)\Big)e^{-\sigma} \leq \frac{3}{4}e^{4/3}g\Big(\frac{4}{3}+\psi\Big). \tag{6.8}$$

Let $h(\sigma) := (1+\frac{3}{4}(1+\sigma))e^{-\sigma}$ for $\sigma \in \mathbb{R}$. Then $h'(\sigma) = -(1+\frac{3}{4}\sigma)e^{-\sigma}$. Thus h is increasing on $(-\infty, -\frac{4}{3}]$ and decreasing on $[-\frac{4}{3}, +\infty)$, and $h(\sigma) \leq h(-\frac{4}{3}) = \frac{3}{4}e^{4/3}$ for any $\sigma \in \mathbb{R}$. If $\psi \leq -\frac{4}{3}$, then $g(\frac{4}{3}+\psi) = 1$, and (6.8) follows. If $\psi > -\frac{4}{3}$, then

$$h(\sigma) \leq h(\psi) = \Big(1+\frac{3}{4}(1+\psi)\Big)e^{-\psi} = \frac{3}{4}e^{4/3}\Big(1+\frac{4}{3}+\psi\Big)e^{-4/3-\psi} = \frac{3}{4}e^{4/3}g\Big(\frac{4}{3}+\psi\Big).$$

By (6.7), we may apply (6.8) with $\sigma = \beta(\mathsf{X})$ and $\psi = a_1\mathsf{X}_1+\cdots+a_N\mathsf{X}_N$. Taking expectations, which are finite since $\varphi(y) \geq \epsilon|y|$ and $0 < g \leq 1$, we obtain

$$\mathbb{E}\big[e^{-\beta(\mathsf{X})}\big] + \frac{3}{4}\mathbb{E}\big[(1+\beta(\mathsf{X}))e^{-\beta(\mathsf{X})}\big] \leq \frac{3}{4}e^{4/3}\mathbb{E}\Big[g\Big(\frac{4}{3}+a_1\mathsf{X}_1+\cdots+a_N\mathsf{X}_N\Big)\Big].$$

Since X has density $e^{-|y|}$ on $\mathbb{R}^N_{\geq 0}$,

$$\mathbb{E}\big[e^{-\beta(\mathsf{X})}\big] = \int_{\mathbb{R}^N_{\geq 0}} e^{-\varphi(y)}dy = \mathcal{H}.$$

By (6.6), we obtain

$$\Big(1-\frac{3}{4}\tau_i\Big)\mathcal{H} \le \frac{3}{4}e^{4/3}\mathbb{E}\Big[g\Big(\frac{4}{3}+a_1\mathsf{X}_1+\cdots+a_N\mathsf{X}_N\Big)\Big]. \tag{6.9}$$

Step 3. In this step we prove (1).

By **Step 1**, it suffices to show that $\varphi(\mathbf{1}) < N$ for $i \gg 0$, and for any i if $\zeta_i = \zeta_0$ for every i. Assume that $\varphi(\mathbf{1}) \ge N$ for some i, and let $a_1, \dots, a_N$ be the real numbers of **Step 2** for i. Denote by J_i the right-hand side of (6.9). Then $J_i > 0$ since $g > 0$. Since $0 < \kappa \le 1$ (Construction-Proposition 6.7(2)), (6.6) implies $\mathcal{H} > 2\rho_d$. If $1 - \frac{3}{4}\tau_i > 0$, then (6.9) implies $(1-\frac{3}{4}\tau_i)2\rho_d < (1-\frac{3}{4}\tau_i)\mathcal{H} \le J_i$, and if $1-\frac{3}{4}\tau_i \le 0$, then $(1-\frac{3}{4}\tau_i)2\rho_d \le 0 < J_i$. In both cases,

$$\Big(1-\frac{3}{4}\tau_i\Big)2\rho_d < J_i. \tag{6.10}$$

Suppose first that $d = 5$. Then $N = 4$ and $a_1+a_2+a_3+a_4 = 0$, so $J_i \le 137/72$ by Lemma 6.9. By Lemma 2.7(3), $2\rho_5 = 253125/131072$, so

$$2\rho_5 - \frac{137}{72} = \frac{33517}{1179648} > 0.$$

Since $\lim_{i\to+\infty}\tau_i = 0$, (6.10) fails for $i \gg 0$, and it fails for any i if $\tau_i = 0$ for every i.

Suppose that $d \ge 6$. By [OAI26b, Theorem 4.4] (see Remark 6.6), $J_i \le C_* := \frac{3}{4}e^{4/3} - \frac{e}{3}$. For any integer $k \ge 9$, the ratio of the $(k+1)$-th term to the k-th term of the exponential series of $e^{4/3}$ is $\frac{4/3}{k+1} \le \frac{4/3}{10}$, so

$$e^{4/3} < \sum_{k=0}^{8}\frac{(4/3)^k}{k!} + \frac{(4/3)^9}{9!}\cdot\frac{1}{1-(4/3)/10} = \frac{917330371}{241805655} < \frac{19}{5}.$$

Moreover,

$$e > \sum_{k=0}^{6}\frac{1}{k!} = \frac{1957}{720} > \frac{1359}{500}.$$

Thus

$$C_* < \frac{3}{4}\cdot\frac{19}{5} - \frac{1}{3}\cdot\frac{1359}{500} = \frac{243}{125}.$$

By Lemma 2.7(3), $(\rho_d)_{d\ge4}$ is increasing and $2\rho_6 = 95551488/48828125$. Thus

$$2\rho_d \ge 2\rho_6 = \frac{95551488}{48828125} > \frac{94921875}{48828125} = \frac{243}{125} > C_* \ge J_i.$$

Since $\lim_{i\to+\infty}\tau_i = 0$, (6.10) fails for $i \gg 0$, and it fails for any i if $\tau_i = 0$ for every i.

In both cases, $\varphi(\mathbf{1}) < N$ for $i \gg 0$. If $\zeta_i = \zeta_0$ for every i, then $\tau_i = 0$ for any i, so $\varphi(\mathbf{1}) < N$ for any i. By **Step 1**, (1) holds.

Step 4. In this step we prove (2) and (3) for any i such that $c \in \mathrm{Int}(\mathcal{K})$.

The standing data at the beginning of [OAI26b, Section 5] are given by Construction-Proposition 6.7, as follows. By Construction-Proposition 6.7(1), $\mathcal{B}$ is a smooth proper Deligne–Mumford stack with trivial generic stabilizer and projective coarse moduli space, $\mathcal{L}$ is ample, and $-K_{\mathcal{B}} = r\mathcal{L}$. By Construction-Proposition 6.7(2), the local generator of $\mathcal{L}$ has character 1, which is faithful on μ_m, Γ generates Λ, and $c \in \Lambda$. By Construction-Proposition 6.7(2)–(4), $\mathcal{K}$ spans $\mathbb{R}\times\mathbb{R}^N$, $\mathcal{K}$ contains the ray $\mathbb{R}_{\ge0}(1,0)$, the projection of $\mathcal{K}$ to $\mathbb{R}^N$ is $\mathbb{R}^N_{\ge0}$, and $y_1+\cdots+y_N \le Nt/(r\epsilon)$ for any $(t,y)\in\mathcal{K}$, so $y_1+\cdots+y_N$ is bounded on $\mathcal{K}$ when t is bounded. For the assumption $\widehat{\mathrm{vol}}(C \ni o) \ge 2N^d$ of [OAI26b, Section 4], whose data are also used in [OAI26b, Section 5], see Remark 6.6. Here (1) takes the place of [OAI26b, Theorem 4.8].

By [OAI26b, Lemma 5.2], any $u \in \Lambda$ with $u + c \in \mathrm{Int}(\mathcal{K})$ belongs to Γ. Since $c \in \mathrm{Int}(\mathcal{K})$, [OAI26b, Theorem 5.3] shows that $\mathcal{K}$ is a rational polyhedral cone, that for any facet of $\mathcal{K}$ the primitive element of $\mathrm{Hom}(\Lambda,\mathbb{Z})$ which is non-negative on $\mathcal{K}$ and vanishes on the facet takes the

value 1 at c, that $\Gamma = \mathcal{K} \cap \Lambda$ and $\mathrm{Int}(\mathcal{K}) \cap \Lambda = c + \Gamma$, and that Γ is a finitely generated saturated affine semigroup with group Λ. This proves (2) and (3). □

7. The Taylor degeneration and order-two slices

In this section, we degenerate the cone C by two filtrations, to $\operatorname{Spec} \bar{S}$ and to $\operatorname{Spec} D[v_*] = \operatorname{Spec} D \times \mathbb{A}^1$, where $D = \bar{S}/(\bar{v})$ is a graded ring whose Hilbert series is $(1-t)^{-N}$, the Hilbert series of a polynomial ring in N variables, and we take slices at the points of $\operatorname{Spec} D$ with non-trivial stabilizer. The total space $\mathcal{X}$ of the second degeneration (Construction-Proposition 7.4) connects $\operatorname{Spec} D \times \mathbb{A}^1$ with C. From Subsection 7.3 on, we study D, $\bar{S}$, and $\mathcal{X}$, and Lemma 7.6 carries the complete intersection property from D back to C.

7.1. The Taylor ring.

In this subsection, we construct the ring $\bar{S}$ associated with Taylor orders and the quotient D of $\bar{S}$, and we prove that D is reduced.

Construction-Proposition 7.1 ([OAI26b, Lemma 6.1])**.** Notation and conditions as in Set-up 6.5 and Construction-Proposition 6.7, and fix i such that the conclusions of Proposition 6.10 hold for i. Write ζ, r, and m for ζ_i, r_i, and m_i. For $s \in S_j \backslash \{0\}$, let $T(s)$ be the total degree of the leading monomial of s. We call $T(s)$ the *Taylor order* of s. For $q \in \mathbb{Z}$, let $T_q S \subset S$ be the graded subspace spanned by the homogeneous elements s with $s = 0$ or $T(s) \geq q$, and let

$$\bar{S} := \bigoplus_{q \geq 0} T_q S / T_{q+1} S,$$

bigraded by the degree and the T-degree q. Then the following hold.

(1) $\bar{S}$ is a finitely generated normal Gorenstein domain, and $\omega_{\bar{S}}$ is generated by an element of bidegree (r, N).
(2) There exists $v \in S_m$ with $T(v) = 0$ such that the class $\bar{v} \in \bar{S}$ of v satisfies $\bar{S}[\bar{v}^{-1}] = \mathbb{C}[\bar{v}, \bar{v}^{-1}, u_1, \dots, u_N]$ for some elements $u_1, \dots, u_N$ of T-degree 1.
(3) The ring $D := \bar{S}/(\bar{v})$ is bigraded by the degree and the T-degree, and the following hold for the T-grading of D.
  (a) D is a finitely generated equidimensional Gorenstein $\mathbb{C}$-algebra of dimension N.
  (b) $D_0 = \mathbb{C}$ and $\mathrm{Hilb}_D(t) = (1-t)^{-N}$.
  (c) ω_D is generated by an element of bidegree $(r - m, N)$.
(4) There exist homogeneous elements $e_l \in S$ whose images in D form a bihomogeneous basis of D, such that the following hold.
  (a) The elements $v^n e_l$ with $n \geq 0$ form a basis of the $\mathbb{C}$-vector space S, and the element 1 is the only $v^n e_l$ with $n + T(e_l) = 0$.
  (b) For any $q \in \mathbb{Z}$, the elements $v^n e_l$ with $n \geq 0$ and $T(e_l) \geq q$ form a basis of $T_q S$.

Proof. We first show that $\bar{S}$ is a ring and a domain. Let $s \in S_j \backslash \{0\}$ and $s' \in S_{j'} \backslash \{0\}$. The order of monomials in Construction-Proposition 6.7(2) is compatible with multiplication, so the leading monomial of ss' is the product of the leading monomials of s and s'. Thus

$$T(ss') = T(s) + T(s').$$

Hence $T_q S \cdot T_{q'} S \subset T_{q+q'} S$ for any $q, q' \in \mathbb{Z}$, the multiplication of S induces a bigraded ring structure on $\bar{S}$, and the product of two non-zero bihomogeneous elements of $\bar{S}$ is non-zero. Comparing the components of largest bidegree for the lexicographic order on $\mathbb{Z}^2$, we see that the product of two non-zero elements of $\bar{S}$ is non-zero. Thus $\bar{S}$ is a domain.

By Remark 6.6, the assumption $\widehat{\mathrm{vol}}(C \ni o) \geq 2N^d$ of [OAI26b, Section 6] enters [OAI26b, Lemma 6.1] only through the identities

$$\Gamma = \mathcal{K} \cap \Lambda \quad \text{and} \quad \mathrm{Int}(\mathcal{K}) \cap \Lambda = c + \Gamma,$$

together with the finite generation of Γ and the rational polyhedrality of $\mathcal{K}$, which hold by Proposition 6.10(2) and (3). The other inputs of [OAI26b, Lemma 6.1] are the finiteness of Taylor orders in any degree and the realization of the monomials z^γ with $(j, \gamma) \in \Lambda$ as leading

monomials after multiplication by powers of v. They come from [OAI26b, Proposition 4.1], and we deduce them directly from Construction-Proposition 6.7.

For the finiteness, let $s \in S_j \backslash \{0\}$ with leading monomial z^γ. Then $(j,\gamma) \in \Gamma \subset \mathcal{K}$, so Construction-Proposition 6.7(3) and (4) imply

$$r\epsilon T(s) = r\epsilon(\gamma_1 + \cdots + \gamma_N) \le r\varphi(\gamma) \le Nj.$$

Thus $T(s) \le Nj/(r\epsilon)$, and $S_j \cap T_q S = 0$ for any $q > Nj/(r\epsilon)$.

For the realization, we have $(m,0) \in \Lambda$, and $(m,0) \in \mathcal{K}$ since $\varphi(0) = 0$. By Proposition 6.10(3), $(m,0) \in \Gamma$, that is, there exists $v \in S_m$ whose leading monomial is 1, so $T(v) = 0$. Let $(j,\gamma) \in \Lambda$ with $\gamma \in \mathbb{Z}^N_{\ge 0}$, and let $n \ge 0$ be an integer such that $N(j+nm) \ge r\varphi(\gamma)$. Then $(j+nm,\gamma) \in \mathcal{K} \cap \Lambda = \Gamma$ by Proposition 6.10(3), so z^γ is the leading monomial of an element of S_{j+nm}.

Thus [OAI26b, Lemma 6.1] holds for our data, where the dimension n and the log discrepancy $A_X(v)$ of [OAI26b] are written d and $a(v,X)$ here, and it implies (1)–(4), except that $\bar{S}$ is a domain, which we proved above. □

We use the following lemma for the filtration $(T_q S)_{q\in\mathbb{Z}}$ with $k = N$ (Lemma 7.3) and for the filtration $(W_p S)_{p\in\mathbb{Z}}$ of Construction-Proposition 7.4 with $k = d$.

Lemma 7.2 ([OAI26b, Lemma 6.2])**.** *Let $C = \operatorname{Spec} S$, o, and η be as in Construction-Proposition 4.1, let $S = \bigoplus_{j\ge 0} S_j$ be the grading defined by a primitive $\zeta \in \mathfrak{t}_\mathbb{Z} \cap \mathfrak{t}^+$, and let $(\mathcal{F}_p S)_{p\in\mathbb{Z}}$ be a decreasing multiplicative filtration of S by graded subspaces with $\mathcal{F}_p S = S$ for $p \le 0$ such that for any j we have $S_j \cap \mathcal{F}_p S = 0$ for all sufficiently large p. Assume that $\operatorname{gr}_\mathcal{F} S$ is a finitely generated Gorenstein ring whose canonical module is generated by an element of bidegree (r_ζ, k), and that*

$$R_\mathcal{F} := \bigoplus_{p\in\mathbb{Z}} \lambda^{-p} \mathcal{F}_p S \subset S[\lambda,\lambda^{-1}]$$

is normal and Gorenstein. Then the canonical module of $R_\mathcal{F}$ is generated by an element which on $\{\lambda \ne 0\}$ is a non-zero constant multiple of $\lambda^{-k}\eta \wedge d\lambda$. Moreover, if $\tilde{w}$ is a quasi-monomial valuation of $\mathbb{C}(\operatorname{Spec} R_\mathcal{F})$ centered on $\{\lambda = 0\}$ whose restriction w to $\mathbb{C}(C)$ is centered at o, then

$$a(w,C) \le a(\tilde{w}, \operatorname{Spec} R_\mathcal{F}) + (k-1)\tilde{w}(\lambda).$$

Proof. This is [OAI26b, Lemma 6.2]. The conditions above on $(\mathcal{F}_p S)_{p\in\mathbb{Z}}$, $\operatorname{gr}_\mathcal{F} S$, and $R_\mathcal{F}$ are the hypotheses in the statement of [OAI26b, Lemma 6.2], with $Q = k$, and, for the grading defined by ζ, S is a positively graded normal Gorenstein domain and $\omega_S = S\eta$ with η homogeneous of degree r_ζ (Definition 2.8 and Construction-Proposition 4.1(2) and (3)). By Remark 6.6, [OAI26b, Lemma 6.2] holds for S with the grading defined by ζ. □

The reducedness of D is used in the proof of Construction-Proposition 7.4 and is a hypothesis of Theorem 9.1.

Lemma 7.3. *Notation as in Construction-Proposition 7.1. Then D is reduced.*

Proof. The proof is long, so we divide it into several steps. Since Taylor orders are non-negative, $T_q S = S$ for $q \le 0$. Let

$$R_T := \bigoplus_{q\in\mathbb{Z}} \lambda^{-q} T_q S \subset S[\lambda,\lambda^{-1}].$$

Step 1. In this step, we show that R_T is a finitely generated normal Gorenstein domain, λ is a nonzerodivisor of R_T, and $R_T/(\lambda) \cong \bar{S}$.

Since the leading monomial of a product is the product of the leading monomials, $(T_q S)_{q\in\mathbb{Z}}$ is a decreasing multiplicative filtration of S by ideals with $T_0 S = S$. By [BH98, Lemma 4.5.2], λ is a nonzerodivisor of R_T, $R_T/(\lambda) \cong \bar{S}$, and $R_T[\lambda^{-1}] = S[\lambda,\lambda^{-1}]$. Under the isomorphism $R_T/(\lambda) \cong \bar{S}$, the class of $\lambda^{-q}s$ with $s \in S_j \cap T_q S$ is the class of s in $T_q S/T_{q+1}S$.

The elements $v^n e_l$ of Construction-Proposition 7.1(4) are homogeneous and $\dim_{\mathbb{C}} S_j < +\infty$, so only finitely many of them belong to S_j. By Construction-Proposition 7.1(4), $S_j \cap T_q S$ is spanned by those with $T(e_l) \geq q$. Thus

$$S_j \cap T_q S = 0 \quad \text{for any } q \gg 0. \tag{7.1}$$

Let $s_1, \dots, s_{k'} \in S$ be homogeneous elements whose classes generate $\bar{S}$ (Construction-Proposition 7.1(1)), let $q_l := T(s_l)$, and for $\alpha \in \mathbb{Z}^{k'}_{\geq 0}$ write $s^\alpha := s_1^{\alpha_1} \cdots s_{k'}^{\alpha_{k'}}$. Let $s \in S_j \cap T_q S$ with $q \geq 0$. Then there exist $c_\alpha \in \mathbb{C}$, non-zero only for the α with $s^\alpha \in S_j$ and $\alpha_1 q_1 + \cdots + \alpha_{k'} q_{k'} = q$, such that $s' := s - \sum_\alpha c_\alpha s^\alpha \in T_{q+1} S$. Hence

$$\lambda^{-q} s = \sum_\alpha c_\alpha \prod_{l=1}^{k'} (\lambda^{-q_l} s_l)^{\alpha_l} + \lambda \cdot \lambda^{-q-1} s'.$$

By descending induction on q, which starts by (7.1), R_T is generated by $\lambda, \lambda^{-q_1} s_1, \dots, \lambda^{-q_{k'}} s_{k'}$. In particular, R_T is finitely generated.

Let P be a prime ideal of R_T. If $\lambda \in P$, then $(R_T)_P/(\lambda)$ is a localization of $\bar{S}$, hence Gorenstein, so $(R_T)_P$ is Gorenstein [BH98, Proposition 3.1.19(b)]. If $\lambda \notin P$, then $(R_T)_P$ is a localization of $S[\lambda, \lambda^{-1}]$, the coordinate ring of $C \times \mathbb{G}_m$, which is Gorenstein: S is Gorenstein (Construction-Proposition 4.1(2)), so $S[\lambda]$ is Gorenstein [BH98, Corollary 3.3.21], and $S[\lambda, \lambda^{-1}]$ is a localization of $S[\lambda]$ [BH98, Proposition 3.1.19(a)]. Thus R_T is Gorenstein, hence Cohen–Macaulay [BH98, Proposition 3.1.20]. The ring R_T is a domain since $R_T \subset S[\lambda, \lambda^{-1}]$. We check Serre's criterion [Mat89, Theorem 23.8]. The condition (S_2) holds since R_T is Cohen–Macaulay. For (R_1), let P be a prime ideal of height one. If $\lambda \notin P$, then $(R_T)_P$ is a localization of the normal domain $S[\lambda, \lambda^{-1}]$, hence regular. If $\lambda \in P$, then $P/(\lambda)$ is a minimal prime of $R_T/(\lambda) \cong \bar{S}$, which is zero since $\bar{S}$ is a domain, so $P = (\lambda)$ and the maximal ideal of $(R_T)_P$ is generated by λ. Thus $(R_T)_P$ is a discrete valuation ring. Hence R_T is normal.

Step 2. In this step, we fix an irreducible component H of $\{\bar{v} = 0\} \subset \operatorname{Spec} \bar{S}$, let $k_H := \operatorname{ord}_H(\bar{v})$, and construct a valuation w of $\mathbb{C}(C)$ centered at o such that $\operatorname{vol}(w) \leq 1$ and $a(w, C) \leq N + 1/k_H$.

Let $\mathfrak{p} \subset \bar{S}$ be the prime ideal of H. Since $\bar{S}$ is a normal domain and $\mathfrak{p}$ is a minimal prime of $(\bar{v})$ with $\bar{v} \neq 0$, $\mathfrak{p}$ has height one and $\bar{S}_\mathfrak{p}$ is a discrete valuation ring, so $k_H \geq 1$ is defined. Let $P \subset R_T$ be the preimage of $\mathfrak{p}$ under $R_T \to R_T/(\lambda) \cong \bar{S}$. Since $\lambda \in P$ is a nonzerodivisor of $(R_T)_P$ and $(R_T)_P/(\lambda) \cong \bar{S}_\mathfrak{p}$, we have $\dim (R_T)_P = 2$, and the maximal ideal of $(R_T)_P$ is generated by λ and an element ϖ whose image in $\bar{S}_\mathfrak{p}$ is a uniformizer. Thus $(R_T)_P$ is a regular local ring with regular system of parameters λ, ϖ. Since $T(v) = 0$, the image of $v = \lambda^0 v \in R_T$ in $\bar{S}$ is $\bar{v}$, which is a unit times the k_H-th power of the image of ϖ in $\bar{S}_\mathfrak{p}$. Hence

$$v = g_0 \varpi^{k_H} + \lambda g_1$$

for a unit g_0 and an element g_1 of $(R_T)_P$.

Let $\tilde{w}$ be the monomial valuation of $\mathbb{C}(\operatorname{Spec} R_T)$ in the coordinates λ, ϖ of $(R_T)_P$ with $\tilde{w}(\lambda) = 1$ and $\tilde{w}(\varpi) = 1/k_H$. Then $\tilde{w}$ is quasi-monomial, and the center of $\tilde{w}$ on $\operatorname{Spec} R_T$ is the closure of P, which is contained in $\{\lambda = 0\}$. Near the generic point of the closure of P, the loci $\{\lambda = 0\}$ and $\{\varpi = 0\}$ are prime divisors with simple normal crossings, and each of them has log discrepancy 1 on $\operatorname{Spec} R_T$. Thus [Li18, Section 2.1, (9)] implies

$$a(\tilde{w}, \operatorname{Spec} R_T) = 1 + \frac{1}{k_H}.$$

Let w be the restriction of $\tilde{w}$ to $\mathbb{C}(C) \subset \mathbb{C}(C)(\lambda) = \mathbb{C}(\operatorname{Spec} R_T)$. Then $w(v) = \tilde{w}(g_0 \varpi^{k_H} + \lambda g_1) \geq 1$. Let e_l be the elements of Construction-Proposition 7.1(4). Since $e_l \in T_{T(e_l)} S$, we have $\lambda^{-T(e_l)} e_l \in R_T$, so $\tilde{w}(e_l) \geq T(e_l) \tilde{w}(\lambda) = T(e_l)$. Thus

$$w(v^n e_l) \geq n + T(e_l) \quad \text{for any } n \geq 0 \text{ and any } l.$$

We show that w is centered at o. Since $T(1) = 0$ and $S_0 = \mathbb{C}$, any e_l with $T(e_l) > 0$ has positive degree, and v has degree $m > 0$. Hence, by Construction-Proposition 7.1(4), the elements

$v^n e_l$ other than 1 have positive degree and span a subspace of codimension one in S. Thus they span the ideal $\mathfrak{m}_o = \bigoplus_{j>0} S_j$ of o. Since $w \geq 0$ on the basis $(v^n e_l)$ and $w \geq 1$ on the basis elements other than 1, we have $w \geq 0$ on S and $w \geq 1$ on $\mathfrak{m}_o$. Thus w is centered at o.

For $p > 0$, let $\mathfrak{a}_p := \{f \in S \mid w(f) \geq p\}$. Then $\mathfrak{a}_p$ contains the elements $v^n e_l$ with $n + T(e_l) \geq p$, so $\dim_{\mathbb{C}} S/\mathfrak{a}_p$ is at most the number of pairs (n, l) with $n + T(e_l) < p$. Since the images of the e_l in D form a bihomogeneous basis of D and the image of e_l has T-degree $T(e_l)$, Construction-Proposition 7.1(3) implies

$$\sum_{n \geq 0} \sum_{l} t^{n+T(e_l)} = \frac{\mathrm{Hilb}_D(t)}{1-t} = \frac{1}{(1-t)^d}.$$

Hence $\dim_{\mathbb{C}} S/\mathfrak{a}_p \leq \binom{\lceil p \rceil - 1 + d}{d}$. Since $w \geq 1$ on $\mathfrak{m}_o$, we have $\mathfrak{m}_o^{\lceil p \rceil} \subset \mathfrak{a}_p$, so $S/\mathfrak{a}_p$ is supported at o and $\dim_{\mathbb{C}} S/\mathfrak{a}_p = \dim_{\mathbb{C}} \mathcal{O}_{C,o}/\mathfrak{a}_p(w)$. Thus

$$\mathrm{vol}(w) = \lim_{p \to +\infty} \frac{d! \dim_{\mathbb{C}} S/\mathfrak{a}_p}{p^d} \leq 1.$$

The hypotheses of Lemma 7.2 hold for the filtration $(T_q S)_{q \in \mathbb{Z}}$ with $k = N$: $\zeta = \zeta_i$ is a primitive element of $\mathfrak{t}_{\mathbb{Z}} \cap \mathfrak{t}^+$ (Set-up 6.5(4)), the filtration is decreasing and multiplicative with $T_q S = S$ for $q \leq 0$ (**Step 1**), the finiteness condition is (7.1), $\mathrm{gr}_T S = \bar{S}$ is a finitely generated Gorenstein ring whose canonical module is generated by an element of bidegree (r, N) with $r = r_\zeta$ (Construction-Proposition 7.1(1)), and R_T is normal and Gorenstein (**Step 1**). Thus, by Lemma 7.2,

$$a(w, C) \leq a(\tilde{w}, \mathrm{Spec}\, R_T) + (N-1)\tilde{w}(\lambda) = N + \frac{1}{k_H}.$$

Step 3. In this step, we show that $k_H = 1$ for any irreducible component H of $\{\bar{v} = 0\}$, and we conclude the proof.

Since C is klt (Construction-Proposition 4.1(2)), $a(w, C) > 0$. By **Step 2**,

$$\widehat{\mathrm{vol}}(C \ni o) \leq a(w, C)^d \,\mathrm{vol}(w) \leq \left(N + \frac{1}{k_H}\right)^d.$$

Suppose that $k_H \geq 2$. Since $d = N + 1$, we have $\widehat{\mathrm{vol}}(C \ni o) \leq (N + 1/2)^d = N^d (1 + 1/(2N))^{N+1}$. Let $\psi(x) := (x+1)\log(1 + 1/(2x))$ for $x > 0$. Since $\log(1+y) < y$ for $y > 0$,

$$\psi'(x) = \log\left(1 + \frac{1}{2x}\right) - \frac{x+1}{x(2x+1)} < \frac{1}{2x} - \frac{x+1}{x(2x+1)} = -\frac{1}{2x(2x+1)} < 0.$$

Thus ψ is decreasing on $(0, +\infty)$, and since $N \geq 4$,

$$\left(1 + \frac{1}{2N}\right)^{N+1} = e^{\psi(N)} \leq e^{\psi(4)} = \left(\frac{9}{8}\right)^5 = \frac{59049}{32768}.$$

By Lemma 2.7(3), $\frac{59049}{32768} < \frac{253125}{131072} = 2\rho_5 \leq 2\rho_d$. By Lemma 2.7(1), $Q_d = 2\rho_d N^d$. Thus $\widehat{\mathrm{vol}}(C \ni o) < Q_d$. However, $\widehat{\mathrm{vol}}(C \ni o) = \widehat{\mathrm{vol}}(X \ni x) > Q_d$ by Construction-Proposition 4.1(1) and Set-up 6.5(2), a contradiction. Hence $k_H = 1$.

The minimal primes of $D = \bar{S}/(\bar{v})$ are the ideals $\mathfrak{p}D$, where $\mathfrak{p}$ runs over the prime ideals of the irreducible components H of $\{\bar{v} = 0\}$. For such $\mathfrak{p}$, the localization of D at $\mathfrak{p}D$ is $\bar{S}_{\mathfrak{p}}/(\bar{v})$, a field since $k_H = 1$. Thus D satisfies (R_0). Since D is Gorenstein (Construction-Proposition 7.1(3)), D is Cohen–Macaulay [BH98, Proposition 3.1.20], so D satisfies (S_1). By Serre's criterion for reducedness [Mat89, §23, p. 183], D is reduced. □

7.2. **The second filtration.** In this subsection, we construct a second filtration $(W_pS)_{p\in\mathbb{Z}}$ of S and the extended Rees algebra R of $(W_pS)_{p\in\mathbb{Z}}$, and we prove that $(C \ni o)$ is a local complete intersection if D is a complete intersection at the vertex.

For $q \geq 1$, the quotient S/T_qS is infinite-dimensional: $T(v) = 0$, so S/T_qS has as a basis the classes of the elements $v^n e_l$ with $n \geq 0$ and $T(e_l) < q$, among them the classes of all powers of v (Construction-Proposition 7.1(4)). For the filtration $(W_pS)_{p\in\mathbb{Z}}$, we have $v \in W_1S\backslash W_2S$, the ideal of o is W_1S, and the Hilbert series of $\mathrm{gr}_W S \cong D[v_*]$ for the W-degree is $\mathrm{Hilb}_D(t)/(1-t) = (1-t)^{-d}$ (Construction-Propositions 7.1(3) and 7.4(1)), so S/W_pS is finite-dimensional for any p.

Construction-Proposition 7.4 ([OAI26b, Proposition 6.4])**.** Notation as in Construction-Proposition 7.1. For $p \in \mathbb{Z}$, let $W_pS := \sum_{n=0}^{p} v^n T_{p-n}S$ if $p \geq 0$ and $W_pS := S$ if $p < 0$. The *extended Rees algebra* of $(W_pS)_{p\in\mathbb{Z}}$ is

$$R := \bigoplus_{p\in\mathbb{Z}} \lambda^{-p} W_pS \subset S[\lambda, \lambda^{-1}],$$

where $\lambda^{-p}s$ has W-degree p for $s \in W_pS$. Let $\mathcal{X} := \operatorname{Spec} R$. Let ι denote both the automorphism of R which multiplies the elements of W-degree p by $(-1)^p$ and the automorphism of $\bar{S}$ which multiplies the elements of T-degree q by $(-1)^q$. Then the following hold.

(1) The filtration $(W_pS)_{p\in\mathbb{Z}}$ and the ring $R/(\lambda)$ satisfy the following.
   (a) $(W_pS)_{p\in\mathbb{Z}}$ is a decreasing multiplicative filtration.
   (b) W_1S is the ideal of o.
   (c) $R/(\lambda) \cong D[v_*]$, where v_* is the class of $\lambda^{-1}v$, a variable of W-degree 1, and the W-degree on D is the T-degree.
   (d) The canonical module of $D[v_*]$ is generated by an element of bidegree (r, d) for the degree and the W-degree.

(2) R is a finitely generated normal Gorenstein domain, λ is a nonzerodivisor of R, $R/(\lambda-1) \cong S$, and the canonical module of R is generated by an element which on $\{\lambda \neq 0\}$ is a non-zero constant multiple of $\lambda^{-d}\eta \wedge d\lambda$.

(3) The automorphisms ι of R and of $\bar{S}$ satisfy the following.
   (a) The automorphism ι of R induces on $R/(\lambda) \cong D[v_*]$ the automorphism which multiplies the elements of D of T-degree q by $(-1)^q$ and maps v_* to $-v_*$.
   (b) The automorphism ι of $\bar{S}$ fixes $\bar{v}$ and induces on $D = \bar{S}/(\bar{v})$ the automorphism which multiplies the elements of T-degree q by $(-1)^q$.

Proof. By Remark 6.6, [OAI26b, Proposition 6.4] holds for our data: the assumption $\widehat{\mathrm{vol}}(C \ni o) \geq 2N^d$ of [OAI26b, Section 6] enters only through the reducedness of D [OAI26b, Lemma 6.3], which is Lemma 7.3 here. We write out (1)–(3) in our notation. Let e_l be the elements of Construction-Proposition 7.1(4), and let $\bar{e}_l \in D$ be the image of e_l.

(1) By Construction-Proposition 7.1(4), for any $q \in \mathbb{Z}$,

$$T_qS = \mathrm{Span}_{\mathbb{C}}\{v^n e_l \mid n \geq 0,\ T(e_l) \geq q\}. \tag{7.2}$$

By (7.2), for any $p \geq 0$,

$$W_pS = \mathrm{Span}_{\mathbb{C}}\{v^n e_l \mid n \geq 0,\ n + T(e_l) \geq p\}. \tag{7.3}$$

For the inclusion $\subset$ in (7.3), by (7.2), $v^{n'}T_{p-n'}S$ is spanned by the elements $v^{n'+n}e_l$ with $T(e_l) \geq p - n'$, and $(n' + n) + T(e_l) \geq p$. Conversely, if $n + T(e_l) \geq p$, then for $n' := \min\{n, p\}$ we have $v^{n-n'}e_l \in T_{p-n'}S$, so $v^n e_l = v^{n'} \cdot v^{n-n'}e_l \in W_pS$. By (7.3), $W_{p+1}S \subset W_pS$. Since $T_qS \cdot T_{q'}S \subset T_{q+q'}S$, we have

$$v^n T_{p-n}S \cdot v^{n'}T_{p'-n'}S \subset v^{n+n'}T_{p+p'-n-n'}S \subset W_{p+p'}S$$

for $0 \leq n \leq p$ and $0 \leq n' \leq p'$, so $(W_pS)_{p\in\mathbb{Z}}$ is multiplicative. By Construction-Proposition 7.1(4), the elements $v^n e_l$ with $n + T(e_l) \geq 1$ are the basis elements other than 1. Since $S_0 = \mathbb{C}$ and

$T(1) = 0$, they have positive degree, so they span the ideal $\bigoplus_{j>0} S_j$ of o. Thus W_1S is the ideal of o by (7.3).

By [BH98, Lemma 4.5.2], λ is a nonzerodivisor of R, $R/(\lambda) \cong \operatorname{gr}_W S := \bigoplus_{p\geq 0} W_pS/W_{p+1}S$, and $R[\lambda^{-1}] = S[\lambda, \lambda^{-1}]$. For $q \geq 0$, the kernel of the surjection $T_qS \to D_q$ to the elements of D of T-degree q is $vT_qS + T_{q+1}S$, which is contained in $W_{q+1}S$. Hence $T_qS \subset W_qS$ induces a map $D_q \to W_qS/W_{q+1}S$, and together with $v_* \to$ (class of v in W_1S/W_2S) we obtain a homomorphism of bigraded rings $D[v_*] \to \operatorname{gr}_W S$, where v_* has bidegree $(m, 1)$. It maps the basis $(\bar{e}_l v_*^n)$ of $D[v_*]$ to the classes of the elements $v^n e_l$ in $W_{n+T(e_l)}S/W_{n+T(e_l)+1}S$, which form a basis of $\operatorname{gr}_W S$ by (7.3). Thus $R/(\lambda) \cong \operatorname{gr}_W S \cong D[v_*]$, under which the class of $\lambda^{-1}v$ is v_* and the W-degree on D is the T-degree. Since D is Gorenstein, so is $D[v_*]$ [BH98, Corollary 3.3.21]. For the degree grading and for the W-grading, $D[v_*]$ is positively graded with degree-zero part $\mathbb{C}$, and v_* is a homogeneous $D[v_*]$-regular element. Thus, in each grading, [BH98, Corollary 3.6.14] implies

$$\omega_D \cong (\omega_{D[v_*]}/v_*\omega_{D[v_*]})(\deg v_*).$$

Since ω_D is generated by an element of bidegree $(r - m, N)$ (Construction-Proposition 7.1(3)) and v_* has bidegree $(m, 1)$, the canonical module of $D[v_*]$ is generated by an element of bidegree $(r - m, N) + (m, 1) = (r, d)$.

(2) Since $\dim_{\mathbb{C}} S_j < +\infty$, only finitely many basis elements $v^n e_l$ belong to S_j, so (7.3) implies

$$S_j \cap W_pS = 0 \quad \text{for any } j \text{ and any } p \gg 0. \tag{7.4}$$

Since $\operatorname{gr}_W S \cong D[v_*]$ is finitely generated, lifting finitely many bihomogeneous generators of $\operatorname{gr}_W S$ to elements $\lambda^{-p}s$ of R and using descending induction on p in any degree j, which starts by (7.4), we see that R is generated by λ and finitely many elements $\lambda^{-p}s$.

The ring R is a domain since $R \subset S[\lambda, \lambda^{-1}]$. Let P be a prime ideal of R. If $\lambda \in P$, then $R_P/(\lambda)$ is a localization of $D[v_*]$, which is Gorenstein by (1), so R_P is Gorenstein [BH98, Proposition 3.1.19(b)]. If $\lambda \notin P$, then R_P is a localization of $S[\lambda, \lambda^{-1}]$, the coordinate ring of $C \times \mathbb{G}_m$, which is Gorenstein: S is Gorenstein (Construction-Proposition 4.1(2)), so $S[\lambda]$ is Gorenstein [BH98, Corollary 3.3.21], and $S[\lambda, \lambda^{-1}]$ is a localization of $S[\lambda]$ [BH98, Proposition 3.1.19(a)]. Thus R is Gorenstein, hence Cohen–Macaulay [BH98, Proposition 3.1.20], and R satisfies (S_2). For (R_1), let P be a prime ideal of height one. If $\lambda \notin P$, then R_P is a localization of the normal domain $S[\lambda, \lambda^{-1}]$, hence regular. If $\lambda \in P$, then $P/(\lambda)$ is a minimal prime of $D[v_*]$. Since D is reduced (Lemma 7.3), so is $D[v_*]$, and $R_P/(\lambda)$ is a field. Thus the maximal ideal of R_P is generated by λ, and R_P is a discrete valuation ring. By Serre's criterion [Mat89, Theorem 23.8], R is normal.

Since λ is a unit in $R/(\lambda - 1)$, we have

$$R/(\lambda - 1) \cong R[\lambda^{-1}]/(\lambda - 1) = S[\lambda, \lambda^{-1}]/(\lambda - 1) \cong S.$$

Finally, the hypotheses of Lemma 7.2 hold for $(W_pS)_{p\in\mathbb{Z}}$ with $k = d$: $W_pS = S$ for $p \leq 0$, the filtration is decreasing and multiplicative by (1), the finiteness condition is (7.4), $\operatorname{gr}_W S \cong D[v_*]$ is a finitely generated Gorenstein ring whose canonical module is generated by an element of bidegree (r, d) by (1), and R is normal and Gorenstein. Thus, by Lemma 7.2, the canonical module of R is generated by an element which on $\{\lambda \neq 0\}$ is a non-zero constant multiple of $\lambda^{-d}\eta \wedge d\lambda$.

(3) The element λ has W-degree -1, so $\iota(\lambda) = -\lambda$ and ι preserves (λ). Under $R/(\lambda) \cong D[v_*]$, an element of D of T-degree q times v_*^n has W-degree $q + n$. Thus ι induces on $D[v_*]$ the automorphism which multiplies the elements of D of T-degree q by $(-1)^q$ and maps v_* to $-v_*$. On $\bar{S}$, the element $\bar{v}$ has T-degree $T(v) = 0$, so $\iota(\bar{v}) = \bar{v}$, and ι preserves $(\bar{v})$ and multiplies the elements of D of T-degree q by $(-1)^q$. □

If all closed points of $\operatorname{Spec} D$ other than the vertex had trivial stabilizer for the $\mathbb{G}_m$-action defined by the T-grading, then D would be a polynomial ring [OAI26b, Lemma 6.5]. The following lemma shows that $\operatorname{Spec} D$ always has a closed point with stabilizer of order at least

2, so [OAI26b, Lemma 6.5] does not recognize D as a polynomial ring. From Subsection 7.3 to Section 9, we study the closed points of $\operatorname{Spec} D$ with non-trivial stabilizer.

Lemma 7.5 ([OAI26b, Corollary 6.6])**.** *Notation as in Construction-Proposition 7.1. Then there exists a closed point of* $\operatorname{Spec} D$ *other than the vertex whose stabilizer for the* $\mathbb{G}_m$*-action defined by the* T*-grading is non-trivial.*

Proof. This is [OAI26b, Corollary 6.6]. By Remark 6.6, [OAI26b, Corollary 6.6] holds for our data: the assumption $\widehat{\mathrm{vol}}(C \ni o) \geq 2N^d$ of [OAI26b, Section 6] enters only through the reducedness of D [OAI26b, Lemma 6.3], which is Lemma 7.3 here, and C is singular at o by Construction-Proposition 4.1(2). □

The argument for $d \geq 5$ ends with Lemma 7.6: if D is a complete intersection at the vertex, then $(C \ni o)$ is a local complete intersection, which contradicts Construction-Proposition 4.1(4) since $(X \ni x)$ is not analytically an ordinary double point (Set-up 6.5(2)). Thus the goal of the following sections is to prove that D is a complete intersection at the vertex (Theorem 9.1).

Lemma 7.6. *Notation as in Construction-Proposition 7.4. If the localization of* D *at the ideal* $\bigoplus_{q>0} D_q$ *is a complete intersection ring, then* $(C \ni o)$ *is a local complete intersection.*

Proof. The proof is long, so we divide it into several steps. Let $\mathfrak{m}_D := \bigoplus_{q>0} D_q$.

Step 1. In this step, we construct a positive grading of R.

The ring R is bigraded: the elements of degree j and W-degree p form $\lambda^{-p}(S_j \cap W_p S)$. By Construction-Proposition 7.4(2), R is generated by λ and finitely many bihomogeneous elements $\lambda^{-p_l} s_l$ with $s_l \in S_{j_l} \cap W_{p_l} S$. Since $W_1 S$ is the ideal of o (Construction-Proposition 7.4(1)), we have $S_0 \cap W_p S = 0$ for $p \geq 1$, and for $p \leq 0$ the elements of $\lambda^{-p} S_0$ are multiples of λ^{-p}. Thus we may assume that $j_l > 0$ for any l. Fix an integer $M > \max_l p_l/j_l$, and let $\deg_M$ be the grading of R for which a bihomogeneous element of degree j and W-degree p has $\deg_M$ equal to $Mj - p$. Then $\deg_M \lambda = 1$ and $\deg_M(\lambda^{-p_l} s_l) = Mj_l - p_l > 0$, so

$$R = \bigoplus_{e \geq 0} R_e \quad \text{with } R_0 = \mathbb{C},$$

where R_e is the space of elements with $\deg_M$ equal to e. Let $\mathfrak{M} := \bigoplus_{e>0} R_e$, a maximal ideal of R.

Step 2. In this step, we show that $R_{\mathfrak{M}}$ is a complete intersection.

The grading $\deg_M$ induces a grading of $R/(\lambda) \cong D[v_*]$ with degree-zero part $R_0 = \mathbb{C}$, spanned by 1, which has bidegree $(0,0)$. Thus any non-zero bihomogeneous element of $D[v_*]$ of bidegree different from $(0,0)$ has positive $\deg_M$, and the image of $\mathfrak{M}$ in $D[v_*]$ is the ideal $\mathfrak{n}$ generated by $\mathfrak{m}_D$ and v_*. Since v_* is a nonzerodivisor of $D[v_*]_{\mathfrak{n}}$ and $D[v_*]_{\mathfrak{n}}/(v_*) \cong D_{\mathfrak{m}_D}$ is a complete intersection, $D[v_*]_{\mathfrak{n}}$ is a complete intersection [BH98, Theorem 2.3.4(a)]. Since $\lambda \in \mathfrak{M}$ is a nonzerodivisor of $R_{\mathfrak{M}}$ (Construction-Proposition 7.4(2)) and $R_{\mathfrak{M}}/(\lambda) \cong D[v_*]_{\mathfrak{n}}$, the ring $R_{\mathfrak{M}}$ is a complete intersection [BH98, Theorem 2.3.4(a)].

Step 3. In this step, we write R as a polynomial ring modulo a regular sequence.

Let $y_1, \dots, y_n \in R$ be homogeneous generators of the $\mathbb{C}$-algebra R with positive $\deg_M$, for example λ and the elements $\lambda^{-p_l} s_l$ of **Step 1**. Let $A := \mathbb{C}[Y_1, \dots, Y_n]$ be the polynomial ring graded by $\deg Y_1 := \deg_M y_1, \dots, \deg Y_n := \deg_M y_n$, let $A \to R$ be the surjection with $Y_1, \dots, Y_n \to y_1, \dots, y_n$, let I be the kernel, a homogeneous ideal, and let $\mathfrak{M}_A := (Y_1, \dots, Y_n)$, the preimage of $\mathfrak{M}$. Then $R_{\mathfrak{M}} \cong A_{\mathfrak{M}_A}/IA_{\mathfrak{M}_A}$, and $A_{\mathfrak{M}_A}$ is a regular local ring, hence Cohen–Macaulay [Mat89, Theorem 17.8]. By **Step 2** and [BH98, Theorem 2.3.3(c)], $IA_{\mathfrak{M}_A}$ is generated by an $A_{\mathfrak{M}_A}$-regular sequence, whose length is $k := \operatorname{ht} IA_{\mathfrak{M}_A}$ [Mat89, Theorem 17.4(iii)]. By Krull's height theorem, any generating set of $IA_{\mathfrak{M}_A}$ has at least k elements, so $\dim_{\mathbb{C}} I/\mathfrak{M}_A I = k$ by Nakayama's lemma. By the graded Nakayama lemma [BH98, Exercise 1.5.24(a)], I is generated

by k homogeneous elements $f_1, \dots, f_k$. Since $f_1, \dots, f_k$ generate an ideal of height k in the Cohen–Macaulay local ring $A_{\mathfrak{M}_A}$, they form an $A_{\mathfrak{M}_A}$-regular sequence [Mat89, Theorem 17.4(iii)].

We show that $f_1, \dots, f_k$ is an A-regular sequence. Let $0 \le k' < k$. The associated primes of the graded A-module $A/(f_1, \dots, f_{k'})$ are homogeneous [BH98, Lemma 1.5.6(b)(ii)], hence contained in $\mathfrak{M}_A$. Since $f_{k'+1}$ is a nonzerodivisor on $A_{\mathfrak{M}_A}/(f_1, \dots, f_{k'})$, $f_{k'+1}$ belongs to no associated prime of $A/(f_1, \dots, f_{k'})$, so $f_{k'+1}$ is a nonzerodivisor on $A/(f_1, \dots, f_{k'})$. Thus

$$R \cong A/(f_1, \dots, f_k)$$

with $f_1, \dots, f_k$ an A-regular sequence.

Step 4. In this step, we conclude the proof.

The element $\lambda - 1$ of the domain R is non-zero, hence a nonzerodivisor of R. Let $\mathfrak{q} \subset R$ be the preimage of the ideal $\mathfrak{m}_o$ of o under $R \to R/(\lambda - 1) \cong S$ (Construction-Proposition 7.4(2)), and let $\mathfrak{q}_A \subset A$ be the preimage of $\mathfrak{q}$. Since $f_1, \dots, f_k \in \mathfrak{q}_A$ is an A-regular sequence and $A_{\mathfrak{q}_A}$ is flat over A, $f_1, \dots, f_k$ is an $A_{\mathfrak{q}_A}$-regular sequence, and $R_{\mathfrak{q}} \cong A_{\mathfrak{q}_A}/(f_1, \dots, f_k)$ with $A_{\mathfrak{q}_A}$ regular local. Thus $R_{\mathfrak{q}}$ is a complete intersection [BH98, Theorem 2.3.3(c)]. Since $\lambda - 1 \in \mathfrak{q}$ is a nonzerodivisor of $R_{\mathfrak{q}}$ and $R_{\mathfrak{q}}/(\lambda - 1) \cong S_{\mathfrak{m}_o}$, the local ring $S_{\mathfrak{m}_o}$ is a complete intersection [BH98, Theorem 2.3.4(a)]. Thus $(C \ni o)$ is a local complete intersection. □

7.3. **Order-two slices.** In this subsection, we show that the non-trivial stabilizers on $\operatorname{Spec} D$ for the T-grading, which exist by Lemma 7.5, have order two (Proposition 7.10).

Since $\dim \operatorname{Spec} D = d - 1$, the ring D does not directly provide valuations of $\mathbb{C}(C)$, so we pass to the slice $(Y \ni y)$ of Construction-Proposition 7.7, a germ of dimension d with a μ_h-action whose function field is a finite extension of $\mathbb{C}(C)$. The decomposition of $\mathcal{O}_{Y,y}$ into isotypic parts for μ_h gives the factor $1/h$ in Proposition 7.10(2).

Construction-Proposition 7.7 ([OAI26b, Lemma 7.1])**.** Notation as in Construction-Proposition 7.4. Let $y_D \in \operatorname{Spec} D$ be a closed point other than the vertex whose stabilizer for the T-grading is μ_h with $h \ge 2$, and let $y := (y_D, 0) \in \operatorname{Spec} D[v_*] = \{\lambda = 0\} \subset \mathcal{X}$. Let $f \in R$ be a W-homogeneous element of positive W-degree with $f(y) \ne 0$, let $\mathcal{X}_f := \{f \ne 0\} \subset \mathcal{X}$, let $Y_0 := \{f = f(y)\} \subset \mathcal{X}_f$, and let $(Y \ni y)$ be the germ of Y_0 at y. Then the following hold.

(1) The morphism $g : Y_0 \times \mathbb{G}_m \to \mathcal{X}_f$ induced by the $\mathbb{G}_m$-action of the W-grading is finite and étale.

(2) $(Y \ni y)$ is a normal Gorenstein germ of dimension d preserved by $\mu_h \subset \mathbb{G}_m$, the restriction λ_Y of λ to Y is a non-zero element of $\mathfrak{m}_{Y,y}$ on which μ_h acts through a primitive character, Y is klt outside $\{\lambda_Y = 0\}$, and $\mathbb{C}(Y)$ is a finite extension of $\mathbb{C}(C)$, where the inclusion $\mathbb{C}(C) \subset \mathbb{C}(Y)$ is induced by restricting the elements of $S \subset R$ to Y.

(3) For every quasi-monomial valuation u of $\mathbb{C}(Y)$ centered at y, the restriction w of u to $\mathbb{C}(C)$ is a quasi-monomial valuation centered at o, and $\ell := u(\lambda_Y) > 0$ satisfies

$$a(w, C) \le N\ell + a(u, Y) \quad \text{and} \quad \operatorname{vol}(w) \le \ell^{-d}.$$

Proof. Statements (1), (2), and (3) are [OAI26b, Lemma 7.1], which holds for the present data by Remark 6.6. We indicate the reasons in the present notation. Let $\mathbb{G}_m$ act on $\mathcal{X}$ through the W-grading, so that $f'(\gamma \cdot z) = \gamma^p f'(z)$ for $\gamma \in \mathbb{G}_m$, $z \in \mathcal{X}$, and $f' \in R$ of W-degree p. We also write γ for the coordinate of the factor $\mathbb{G}_m$ of $Y_0 \times \mathbb{G}_m$, so that $g(y', \gamma) = \gamma \cdot y'$.

(1) Let $\mathcal{X}_f \to \mathbb{G}_m$ be the morphism defined by $f/f(y)$. Since $f(\gamma \cdot z) = \gamma^{\deg_W f} f(z)$, the morphism

$$Y_0 \times \mathbb{G}_m \to \mathcal{X}_f \times_{\mathbb{G}_m} \mathbb{G}_m, \quad (y', \gamma) \to (\gamma \cdot y', \gamma),$$

where $\mathbb{G}_m \to \mathbb{G}_m$ is the $\deg_W f$-th power map, is an isomorphism with inverse $(z, \gamma) \to (\gamma^{-1} \cdot z, \gamma)$. Thus g is the base change of the $\deg_W f$-th power map of $\mathbb{G}_m$, that is, g adjoins a $\deg_W f$-th root of $f/f(y)$. Since $\deg_W f > 0$, g is finite and étale.

(2) Since $W_p S = S$ for $p \le 0$, we have $R[\lambda^{-1}] = S[\lambda, \lambda^{-1}]$, the ring S is the W-degree 0 part of R, and

$$\mathcal{X} \setminus \{\lambda = 0\} \cong C \times \mathbb{G}_m.$$

In particular, $\dim \mathcal{X} = d+1$. By Construction-Proposition 7.4(2), $\mathcal{X}$ is normal and Gorenstein. Since g is étale, $Y_0 \times \mathbb{G}_m$ is normal and Gorenstein of dimension $d+1$. Since the projection $Y_0 \times \mathbb{G}_m \to Y_0$ is smooth and surjective, the local homomorphisms $\mathcal{O}_{Y_0,y'} \to \mathcal{O}_{Y_0\times\mathbb{G}_m,(y',1)}$ for the closed points $y' \in Y_0$ are flat, so Y_0 is normal [Mat89, Corollary to Theorem 23.9] and Gorenstein [Mat89, Theorem 23.4], of dimension d.

Since $v_*(y) = 0$ and the W-grading on D is the T-grading, μ_h fixes y. For $\gamma \in \mu_h$ we obtain $f(y) = f(\gamma \cdot y) = \gamma^{\deg_W f} f(y)$. Since $f(y) \neq 0$, h divides $\deg_W f$, so $f(\gamma \cdot z) = f(z)$ for $z \in \mathcal{X}$. Thus μ_h preserves Y_0 and fixes y.

Since λ has W-degree -1, we have $g^*\lambda = \gamma^{-1}\lambda_Y$, and $\gamma \in \mu_h$ acts on λ_Y by multiplication by γ^{-1}, which is a primitive character of μ_h. Since $y \in \{\lambda = 0\}$, $\lambda_Y \in \mathfrak{m}_{Y,y}$. If $\lambda_Y = 0$ in $\mathcal{O}_{Y,y}$, then λ vanishes on $g(Y_1 \times \mathbb{G}_m)$ for an open neighborhood Y_1 of y in Y_0. Since g is étale, $g(Y_1 \times \mathbb{G}_m)$ is a non-empty open subset of $\mathcal{X}$, which contradicts $\lambda \neq 0$ in the domain R. Hence $\lambda_Y \neq 0$.

Since C is klt (Construction-Proposition 4.1(2)), so is $\mathcal{X}\backslash\{\lambda = 0\} \cong C \times \mathbb{G}_m$. Since $g^*\lambda = \gamma^{-1}\lambda_Y$, we have

$$g^{-1}(\mathcal{X}\backslash\{\lambda = 0\}) = (Y_0\backslash\{\lambda_Y = 0\}) \times \mathbb{G}_m.$$

As g is étale, $(Y_0\backslash\{\lambda_Y = 0\}) \times \mathbb{G}_m$ is klt [KM98, Proposition 5.20(4)], hence Y is klt outside $\{\lambda_Y = 0\}$ (if $Y_0' \to Y_0$ is a log resolution, then $Y_0' \times \mathbb{G}_m \to Y_0 \times \mathbb{G}_m$ is a log resolution, and $a(E \times \mathbb{G}_m, Y_0 \times \mathbb{G}_m) = a(E, Y_0)$ for any prime divisor E on Y_0').

For $s \in S \subset R$, let $s|_Y \in \mathcal{O}_{Y,y}$ be the restriction of s to Y. Write $f = \lambda^{-\deg_W f} s$ with $s \in W_{\deg_W f} S$. Then

$$\lambda_Y^{\deg_W f} = \frac{s|_Y}{f(y)}.$$

Since $\mathcal{O}(Y_0)$ is generated by the restrictions of the elements of R and any W-homogeneous element of R of W-degree p has the form $\lambda^{-p}s'$ with $s' \in S$, the field $\mathbb{C}(Y)$ is generated by λ_Y over the image of S, and λ_Y is algebraic over the image of S. As $\dim Y = d = \dim C$, the transcendence degree of the image of S over $\mathbb{C}$ is d. Since S is a domain of dimension d, the map $S \to \mathbb{C}(Y)$, $s \to s|_Y$, is injective. Thus it induces an inclusion $\mathbb{C}(C) \subset \mathbb{C}(Y)$, and $\mathbb{C}(Y) = \mathbb{C}(C)(\lambda_Y)$ is a finite extension of $\mathbb{C}(C)$.

(3) We first check the hypotheses of Lemma 7.2 for the grading defined by ζ, the filtration $(W_pS)_{p\in\mathbb{Z}}$, and $k = d$. The element $\zeta = \zeta_i$ is a primitive element of $\mathfrak{t}_\mathbb{Z} \cap \mathfrak{t}^+$ (Set-up 6.5(4)). By Construction-Proposition 7.4(1), $(W_pS)_{p\in\mathbb{Z}}$ is a decreasing multiplicative filtration of S by graded subspaces, and $W_pS = S$ for $p \leq 0$ by definition. Since $\dim_\mathbb{C} S_j < +\infty$ and any non-zero element of S_j has finite Taylor order, $T_qS \cap S_j = 0$ for $q \gg 0$. As $v \in S_m$, we have

$$S_j \cap W_pS = \sum_{0\leq b\leq \min\{p, j/m\}} v^b (T_{p-b}S)_{j-bm},$$

hence $S_j \cap W_pS = 0$ for $p \gg 0$. Moreover, by [BH98, Lemma 4.5.2] and Construction-Proposition 7.4(1),

$$\mathrm{gr}_W S \cong R/(\lambda) \cong D[v_*],$$

which is finitely generated and Gorenstein since D is finitely generated and Gorenstein (Construction-Proposition 7.1(3), [BH98, Corollary 3.3.21]), and the canonical module of $D[v_*]$ is generated by an element of bidegree (r, d) by Construction-Proposition 7.4(1). Finally, R is normal and Gorenstein by Construction-Proposition 7.4(2).

Let u be a quasi-monomial valuation of $\mathbb{C}(Y)$ centered at y. Let $\tilde{u}$ be the valuation of $\mathbb{C}(Y_0 \times \mathbb{G}_m) = \mathbb{C}(Y)(\gamma)$ which takes a polynomial in γ with coefficients in $\mathbb{C}(Y)$ to the minimum of the values of u on the coefficients. If u is monomial in local coordinates $z_1, \ldots, z_d$ on a log resolution $Y' \to Y$, then $\tilde{u}$ is monomial in the pullbacks of $z_1, \ldots, z_d$ to $Y' \times \mathbb{G}_m$, with the weights of u. Hence $\tilde{u}$ is quasi-monomial and $a(\tilde{u}, Y_0 \times \mathbb{G}_m) = a(u, Y)$. Let $u_\mathcal{X}$ be the restriction of $\tilde{u}$ to $\mathbb{C}(\mathcal{X}) \subset \mathbb{C}(Y_0 \times \mathbb{G}_m)$, where the inclusion is g^*. Since g is finite and étale, $u_\mathcal{X}$ is quasi-monomial [Bou72, Chapter VI, §8, no. 1, Proposition 1], [JM12, Proposition 3.7] and

$$a(u_\mathcal{X}, \mathcal{X}) = a(\tilde{u}, Y_0 \times \mathbb{G}_m) = a(u, Y)$$

(cf. [KM98, Proposition 5.20]). The center of $u_{\mathcal{X}}$ on $\mathcal{X}$ is the closure of the orbit $\mathbb{G}_m \cdot y \subset \{\lambda = 0\}$ (the center of $\tilde{u}$ is the generic point of $\{y\} \times \mathbb{G}_m$, and g is finite), and

$$u_{\mathcal{X}}(\lambda) = \tilde{u}(\gamma^{-1}\lambda_Y) = u(\lambda_Y) = \ell.$$

Since u is centered at y and $0 \neq \lambda_Y \in \mathfrak{m}_{Y,y}$, we have $\ell > 0$.

Any $s \in S$ has W-degree 0, so s is invariant under $\mathbb{G}_m$ and g^*s is the pullback of $s|_Y$ under the projection $Y_0 \times \mathbb{G}_m \to Y_0$. Hence $u_{\mathcal{X}}(s) = u(s|_Y) = w(s)$, that is, w is also the restriction of $u_{\mathcal{X}}$ to $\mathbb{C}(C)$. Since $\mathbb{C}(Y)$ is a finite extension of $\mathbb{C}(C)$, the rational rank of w and the transcendence degree of the residue field of w are equal to those of u [Bou72, Chapter VI, §8, no. 1, Proposition 1]. Thus w satisfies the Abhyankar equality, and w is quasi-monomial (a valuation is quasi-monomial if and only if it satisfies the Abhyankar equality, see the paragraph before [LX18, Definition 2.8] and [JM12, Proposition 3.7]).

For $p \geq 0$ and $s \in W_pS$, we have $\lambda^{-p}s \in R$, hence $u_{\mathcal{X}}(\lambda^{-p}s) \geq 0$ and

$$w(s) \geq p\ell. \tag{7.5}$$

Since W_1S is the ideal of o (Construction-Proposition 7.4(1)), w is centered at o. By Lemma 7.2,

$$a(w, C) \leq a(u_{\mathcal{X}}, \mathcal{X}) + (d-1)u_{\mathcal{X}}(\lambda) = N\ell + a(u, Y).$$

It remains to bound $\operatorname{vol}(w)$. By Construction-Proposition 7.1(3), $\dim_{\mathbb{C}} D_q = \binom{q+N-1}{N-1}$ for $q \geq 0$. Since $\operatorname{gr}_W S \cong D[v_*]$ and v_* has W-degree 1, we obtain

$$\dim_{\mathbb{C}} W_qS/W_{q+1}S = \sum_{b=0}^{q} \binom{q-b+N-1}{N-1} = \binom{q+d-1}{d-1}$$

for $q \geq 0$, hence

$$\dim_{\mathbb{C}} S/W_pS = \sum_{q=0}^{p-1} \binom{q+d-1}{d-1} = \binom{p+d-1}{d} \quad \text{for } p \geq 0. \tag{7.6}$$

For $\tau > 0$, let $\mathfrak{a}_\tau(w) \subset S$ be the ideal of the elements s with $w(s) \geq \tau$, and let $p := \lceil \tau/\ell \rceil$. By (7.5), $W_pS \subset \mathfrak{a}_\tau(w)$. Since $W_pS \supset (W_1S)^p$, the ideal $\mathfrak{a}_\tau(w)$ is primary to the ideal of o, so by Definition 2.2,

$$\operatorname{vol}(w) = \lim_{\tau \to +\infty} \frac{d! \dim_{\mathbb{C}} S/\mathfrak{a}_\tau(w)}{\tau^d}. \tag{7.7}$$

Since $W_pS \subset \mathfrak{a}_\tau(w)$, (7.6) gives

$$\dim_{\mathbb{C}} S/\mathfrak{a}_\tau(w) \leq \dim_{\mathbb{C}} S/W_pS = \binom{\lceil \tau/\ell \rceil + d - 1}{d},$$

and (7.7) implies $\operatorname{vol}(w) \leq \ell^{-d}$. □

Lemma 7.8. *Notation as in Construction-Proposition 7.4. Let $y_D \in \operatorname{Spec} D$ be a closed point other than the vertex. Then the stabilizer of y_D for the T-grading is μ_h for some integer $h \geq 1$, and there exists a W-homogeneous element $f \in R$ of positive W-degree such that $f(y_D, 0) \neq 0$, where $(y_D, 0) \in \operatorname{Spec} D[v_*] = \{\lambda = 0\} \subset \mathcal{X}$.*

Proof. Let $\mathbb{G}_m$ act on $\operatorname{Spec} D$ through the T-grading, so that $e(t \cdot z) = t^j e(z)$ for $t \in \mathbb{G}_m$, $z \in \operatorname{Spec} D$, and $e \in D$ homogeneous of T-degree j. By Construction-Proposition 7.1(3), D is positively graded by the T-degree with $D_0 = \mathbb{C}$, so the homogeneous elements of positive T-degree generate the maximal ideal of the vertex. Since y_D is not the vertex, there exists a homogeneous element $e \in D$ of T-degree $j > 0$ with $e(y_D) \neq 0$. If $t \in \mathbb{G}_m$ fixes y_D, then $e(y_D) = e(t \cdot y_D) = t^j e(y_D)$, so $t^j = 1$. Thus the stabilizer of y_D is a closed subgroup of μ_j, that is, it is μ_h for some integer $h \geq 1$. By Construction-Proposition 7.4(1), the surjection $R \to R/(\lambda) \cong D[v_*]$ preserves the W-degree, and the W-degree on D is the T-degree. Hence e lifts to a W-homogeneous element $f \in R$ of W-degree j, and $f(y_D, 0) = e(y_D) \neq 0$. □

Proposition 7.9. *Notation as in Construction-Proposition 7.7. Then $(Y \ni y)$ is a klt germ and $y \in \mathrm{Sing}(Y)$.*

Proof. By Construction-Proposition 7.7(2), Y is normal and K_Y is Cartier. Suppose that $(Y \ni y)$ is not klt. Then there exists a prime divisor E over Y such that $a(E, Y, 0) \leq 0$ and the center of E on Y contains y [KM98, Corollary 2.32]. Since Y is klt outside $\{\lambda_Y = 0\}$ (Construction-Proposition 7.7(2)), the center of E is contained in $\{\lambda_Y = 0\}$, so $\mathrm{ord}_E(\lambda_Y) > 0$. Since Y is normal and $a(E, Y, 0) \leq 0$, E is exceptional over Y.

Let $\pi : Y' \to Y$ be a log resolution of $(Y, \{\lambda_Y = 0\})$ on which E is a prime divisor [KM98, Lemma 2.45], and let $y' \in E$ be a closed point with $\pi(y') = y$. Choose local coordinates $z_1, \ldots, z_d$ at y' such that $E = \{z_1 = 0\}$ and any prime component of $\mathrm{Exc}(\pi) \cup \pi^{-1}\{\lambda_Y = 0\}$ through y' is of the form $\{z_i = 0\}$. Let $E_i := \{z_i = 0\}$ near y', so that $E_1 = E$. For $0 < \epsilon < 1$, let u_ϵ be the monomial valuation in $z_1, \ldots, z_d$ with weights $(1, \epsilon, \ldots, \epsilon)$. Then u_ϵ is quasi-monomial and centered at $\pi(y') = y$, and

$$a(u_\epsilon, Y) = a(E, Y, 0) + \epsilon \sum_{i=2}^{d} a(E_i, Y, 0), \qquad \ell_\epsilon := u_\epsilon(\lambda_Y) = \mathrm{ord}_E(\lambda_Y) + \epsilon \sum_{i=2}^{d} \mathrm{ord}_{E_i}(\lambda_Y),$$

where the formula for $a(u_\epsilon, Y)$ holds by [Li18, Section 2.1, (9)]. Let w_ϵ be the restriction of u_ϵ to $\mathbb{C}(C)$. By Construction-Proposition 7.7(3), w_ϵ is a quasi-monomial valuation centered at o, and

$$a(w_\epsilon, C) \leq N\ell_\epsilon + a(u_\epsilon, Y) \quad \text{and} \quad \mathrm{vol}(w_\epsilon) \leq \ell_\epsilon^{-d}.$$

Since C is klt (Construction-Proposition 4.1(2)), $a(w_\epsilon, C) > 0$. Thus

$$\widehat{\mathrm{vol}}(C \ni o) \leq a(w_\epsilon, C)^d \,\mathrm{vol}(w_\epsilon) \leq \left(N + \frac{a(u_\epsilon, Y)}{\ell_\epsilon}\right)^d,$$

where the base on the right is positive. As $\epsilon \to 0$,

$$\frac{a(u_\epsilon, Y)}{\ell_\epsilon} \to \frac{a(E, Y, 0)}{\mathrm{ord}_E(\lambda_Y)} \leq 0,$$

hence $\widehat{\mathrm{vol}}(C \ni o) \leq N^d$. On the other hand, $\widehat{\mathrm{vol}}(C \ni o) = \widehat{\mathrm{vol}}(X \ni x) > Q_d$ by Construction-Proposition 4.1(1) and Set-up 6.5(2). By Lemma 2.7(1) and (3),

$$\frac{Q_d}{N^d} = 2\rho_d \geq 2\rho_5 > 1,$$

which contradicts $\widehat{\mathrm{vol}}(C \ni o) \leq N^d$. Thus $(Y \ni y)$ is a klt germ.

By Remark 6.6, [OAI26b, Lemma 7.3] holds for the present data, in which $R/(\lambda) \cong D[v_*]$ with v_* of W-degree 1 (Construction-Proposition 7.4(1)), $\bar{v}$ is a nonzerodivisor of the domain $\bar{S}$ and $\bar{S}[\bar{v}^{-1}]$ is a polynomial ring over $\mathbb{C}[\bar{v}, \bar{v}^{-1}]$ in N variables of T-degree 1 (Construction-Proposition 7.1(1) and (2)), μ_h acts on λ_Y through a primitive character (Construction-Proposition 7.7(2)), and $h \geq 2$. Hence $y \in \mathrm{Sing}(Y)$ by [OAI26b, Lemma 7.3]. □

Proposition 7.10. *Notation as in Construction-Proposition 7.7. Then the following hold.*

(1) There exists a real valuation w_Y of $\mathbb{C}(Y)$ centered at y such that

$$a(w_Y, Y)^d \,\mathrm{vol}(w_Y) = \widehat{\mathrm{vol}}(Y \ni y).$$

(2) For every w_Y as in (1), w_Y is μ_h-invariant, and, writing w for the restriction of w_Y to $\mathbb{C}(C)$, $\ell := w_Y(\lambda_Y)$, and $\sigma := \mathrm{vol}(w_Y)$, we have

$$\mathrm{vol}(w) \leq \min_{0 \leq t \leq 1} \left(\frac{t^d}{\ell^d} + \frac{(1-t)^d \sigma}{h}\right) \quad \text{and} \quad \frac{\widehat{\mathrm{vol}}(C \ni o)}{d^d} \leq \frac{p_d}{2} + \frac{\widehat{\mathrm{vol}}(Y \ni y)}{h d^d}.$$

(3) $h = 2$. Moreover, every closed point of $\mathrm{Spec}\, D$ other than the vertex has stabilizer of order at most 2 for the T-grading.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove (1) and the μ_h-invariance of w_Y in (2).

By Proposition 7.9, $(Y \ni y)$ is a klt germ. Hence a valuation w_Y as in (1) exists by [Blu18, Main Theorem]. Let w_Y be any such valuation. By [Xu20, Theorem 1.2], w_Y is quasi-monomial, and by [XZ21, Theorem 1.1], w_Y is unique up to scaling. Since μ_h acts on $(Y \ni y)$ (Construction-Proposition 7.7(2)), w_Y is μ_h-invariant by [XZ21, Corollary 1.2]. By Construction-Proposition 7.7(3) applied to $u = w_Y$, the valuation w is quasi-monomial and centered at o, $\ell > 0$, and

$$a(w, C) \le N\ell + a(w_Y, Y). \tag{7.8}$$

Step 2. In this step we show that the dimension of each isotypic part of $\mathcal{O}_{Y,y}/\mathfrak{a}_\tau(w_Y)$ is $\sigma\tau^d/(hd!) + o(\tau^d)$. This is [OAI26b, Lemma 7.4], and we include the argument in the present notation.

For $\tau \ge 0$, let $\mathfrak{a}_\tau(w_Y) \subset \mathcal{O}_{Y,y}$ be the ideal of the elements φ with $w_Y(\varphi) \ge \tau$. Since w_Y is μ_h-invariant, $\mathfrak{a}_\tau(w_Y)$ is μ_h-stable, and μ_h acts on $\mathcal{O}_{Y,y}/\mathfrak{a}_\tau(w_Y)$. For a character $\chi : \mu_h \to \mathbb{C}^*$, let $L_\chi(\tau)$ be the dimension of the χ-isotypic part of $\mathcal{O}_{Y,y}/\mathfrak{a}_\tau(w_Y)$, and let

$$L(\tau) := \dim_{\mathbb{C}} \mathcal{O}_{Y,y}/\mathfrak{a}_\tau(w_Y) = \sum_\chi L_\chi(\tau).$$

By the definition of $\sigma = \mathrm{vol}(w_Y)$, $L(\tau) = \sigma\tau^d/d! + o(\tau^d)$. Let χ_λ be the character through which μ_h acts on λ_Y. By Construction-Proposition 7.7(2), χ_λ generates the group of characters of μ_h, which is cyclic of order h.

Since $\mathfrak{a}_{\tau'}(w_Y) \subset \mathfrak{a}_\tau(w_Y)$ for $\tau' \ge \tau$, L_χ is nondecreasing. For $j \ge 0$, multiplication by λ_Y^j induces a μ_h-equivariant map

$$\mathcal{O}_{Y,y}/\mathfrak{a}_\tau(w_Y) \to \mathcal{O}_{Y,y}/\mathfrak{a}_{\tau+j\ell}(w_Y)$$

which shifts the characters by χ_λ^j. Since $w_Y(\lambda_Y^j\varphi) = j\ell + w_Y(\varphi)$ for $\varphi \in \mathcal{O}_{Y,y}$, the map is injective. Hence $L_\chi(\tau) \le L_{\chi\chi_\lambda^j}(\tau + j\ell)$. For any two characters χ and χ' there exists $0 \le j \le h-1$ with $\chi' = \chi\chi_\lambda^j$. Thus

$$L_\chi(\tau) \le L_{\chi'}(\tau + (h-1)\ell) \quad \text{and} \quad L_{\chi'}(\tau - (h-1)\ell) \le L_\chi(\tau)$$

for any χ, χ', and $\tau \ge (h-1)\ell$. Summing over χ', we obtain

$$\frac{L(\tau - (h-1)\ell)}{h} \le L_\chi(\tau) \le \frac{L(\tau + (h-1)\ell)}{h},$$

and therefore

$$L_\chi(\tau) = \frac{\sigma\tau^d}{hd!} + o(\tau^d) \quad \text{for any character } \chi \text{ of } \mu_h. \tag{7.9}$$

Step 3. In this step we prove the first inequality of (2). This is [OAI26b, Proposition 7.5], and we include the argument in the present notation.

For $\tau > 0$, let $\mathfrak{a}_\tau(w) \subset S$ be the ideal of the elements s with $w(s) \ge \tau$. Since w is centered at o, $\mathfrak{a}_\tau(w)$ is primary to the ideal of o, and

$$\mathrm{vol}(w) = \lim_{\tau \to +\infty} d! \dim_{\mathbb{C}}(S/\mathfrak{a}_\tau(w))/\tau^d.$$

Fix $0 < t < 1$, and for $\tau > 0$ let $p := \lfloor \tau t/\ell \rfloor$. Then $0 \le p\ell \le \tau t$.

Let $s \in W_p S$. Then $\lambda^{-p}s \in R$, so

$$(\lambda^{-p}s)|_Y = \lambda_Y^{-p}(s|_Y) \in \mathcal{O}_{Y,y}.$$

Since s has W-degree 0, $s|_Y$ is μ_h-invariant, so $\gamma^*(\lambda_Y^{-p}(s|_Y)) = \gamma^p\lambda_Y^{-p}(s|_Y)$ for $\gamma \in \mu_h$, that is, μ_h acts on $\lambda_Y^{-p}(s|_Y)$ through χ_λ^{-p}. Moreover,

$$w_Y(\lambda_Y^{-p}(s|_Y)) = w(s) - p\ell,$$

so $\lambda_Y^{-p}(s|_Y) \in \mathfrak{a}_{\tau-p\ell}(w_Y)$ if and only if $s \in \mathfrak{a}_\tau(w)$. Hence $s \to \lambda_Y^{-p}(s|_Y)$ induces an injection of $W_pS/(W_pS \cap \mathfrak{a}_\tau(w))$ into the χ_λ^{-p}-isotypic part of $\mathcal{O}_{Y,y}/\mathfrak{a}_{\tau-p\ell}(w_Y)$. From the exact sequence

$$0 \to W_pS/(W_pS \cap \mathfrak{a}_\tau(w)) \to S/\mathfrak{a}_\tau(w) \to S/(W_pS + \mathfrak{a}_\tau(w)) \to 0$$

we obtain

$$\dim_{\mathbb{C}} S/\mathfrak{a}_\tau(w) \le \dim_{\mathbb{C}} S/W_pS + L_{\chi_\lambda^{-p}}(\tau - p\ell).$$

By Construction-Proposition 7.1(3), $\dim_{\mathbb{C}} D_q = \binom{q+N-1}{N-1}$ for $q \ge 0$. Since $\mathrm{gr}_W S \cong D[v_*]$ with v_* of W-degree 1 (Construction-Proposition 7.4(1)), we have

$$\dim_{\mathbb{C}} S/W_pS = \binom{p+d-1}{d} \le \frac{(p+d)^d}{d!} \le \frac{(\tau t/\ell + d)^d}{d!}.$$

Since $p > \tau t/\ell - 1$, we have $\tau - p\ell < \tau(1-t) + \ell$, and since L_χ is nondecreasing, (7.9) implies

$$L_{\chi_\lambda^{-p}}(\tau - p\ell) \le \max_\chi L_\chi(\tau(1-t)+\ell) = \frac{\sigma(1-t)^d\tau^d}{hd!} + o(\tau^d),$$

where the maximum is over the h characters of μ_h. Multiplying by $d!/\tau^d$ and letting $\tau \to +\infty$, we obtain

$$\mathrm{vol}(w) \le \frac{t^d}{\ell^d} + \frac{(1-t)^d\sigma}{h}$$

for $0 < t < 1$. The right-hand side is continuous in t, so the inequality holds for $0 \le t \le 1$, and the first inequality of (2) follows.

Step 4. In this step we prove the second inequality of (2).

Since $(Y \ni y)$ is klt, $a(w_Y, Y) > 0$, so

$$t_0 := \frac{N\ell}{N\ell + a(w_Y, Y)} \in (0,1).$$

Since C is klt, $a(w, C) > 0$. By the definition of $\widehat{\mathrm{vol}}(C \ni o)$, (7.8), and **Step 3** with $t = t_0$, we have

$$\widehat{\mathrm{vol}}(C \ni o) \le a(w,C)^d \,\mathrm{vol}(w) \le (N\ell + a(w_Y,Y))^d\Big(\frac{t_0^d}{\ell^d} + \frac{(1-t_0)^d\sigma}{h}\Big) = N^d + \frac{a(w_Y,Y)^d\sigma}{h}.$$

Since $a(w_Y,Y)^d\sigma = \widehat{\mathrm{vol}}(Y \ni y)$ and $N^d = (d-1)^d = d^dp_d/2$, dividing by d^d implies the second inequality of (2).

Step 5. In this step we prove (3).

By Proposition 7.9, $(Y \ni y)$ is a klt germ of dimension d with $y \in \mathrm{Sing}(Y)$. By Theorem 2.4(2),

$$\widehat{\mathrm{vol}}(Y \ni y) \le 2(d-1)^d = d^dp_d.$$

Suppose that $h \ge 3$. Then (2) implies

$$\frac{\widehat{\mathrm{vol}}(C \ni o)}{d^d} \le \frac{p_d}{2} + \frac{p_d}{3} = \frac{5p_d}{6}.$$

On the other hand, $\widehat{\mathrm{vol}}(C \ni o) = \widehat{\mathrm{vol}}(X \ni x) > Q_d$ by Construction-Proposition 4.1(1) and Set-up 6.5(2). Since $d \ge 5$, Lemma 2.7(1) and (3) imply

$$\frac{Q_d}{d^d} = q_d = \rho_dp_d \ge \rho_5p_d > \frac{5p_d}{6},$$

a contradiction. Since $h \ge 2$, we obtain $h = 2$.

Let y'_D be a closed point of $\operatorname{Spec} D$ other than the vertex. By Lemma 7.8, the stabilizer of y'_D is $\mu_{h'}$ for some integer $h' \geq 1$, and there exists a W-homogeneous element $f' \in R$ of positive W-degree with $f'(y'_D, 0) \neq 0$. If $h' \geq 2$, then Construction-Proposition 7.7 holds for y'_D, h', and f' in place of y_D, h, and f, and the equality $h = 2$ of (3), applied to y'_D, h', and f', reads $h' = 2$. Thus $h' \leq 2$. □

8. The fixed locus in the Taylor degeneration

In this section, we prove that the fixed subscheme Z of the involution ι of Construction-Proposition 7.4 is a curve (Theorem 8.6 in Subsection 8.2). Theorem 8.6 is the input for the recognition theorem of Section 9.

8.1. **The fixed subscheme.** In this subsection, we compute the fixed subscheme Z of ι, and we bound $\dim Z$ and the singular loci of the slices of Construction-Proposition 7.7 at the closed points of Z.

Lemma 8.1 turns the inequality of Proposition 7.10(2) into a bound on the singular locus of the slice. One hyperplane section, followed by Theorem 2.4(2) in dimension $d-1$, bounds $\widehat{\operatorname{vol}}(Y \ni y)$ by a quantity that is not small enough to force $\widehat{\operatorname{vol}}(X \ni x) < Q_d$, so we cut twice.

Lemma 8.1. *Let $d \geq 5$ be an integer, let $(X \ni x)$ and $(Y \ni y)$ be klt germs of dimension d, and let $h \geq 2$ be an integer. Assume that*

$$\frac{\widehat{\operatorname{vol}}(X \ni x)}{d^d} \leq \frac{p_d}{2} + \frac{\widehat{\operatorname{vol}}(Y \ni y)}{hd^d}$$

and $\dim_y \operatorname{Sing}(Y) \geq 2$. Then $\widehat{\operatorname{vol}}(Y \ni y) \leq d^d p_{d-2}$ and $\widehat{\operatorname{vol}}(X \ni x) < Q_d$.

Proof. Since $\dim_y \operatorname{Sing}(Y) \geq 2$, there exists an irreducible component V of $\operatorname{Sing}(Y)$ such that $y \in V$ and $\dim V \geq 2$. Since $d \geq 3$, Lemma 2.5 applied to $(Y \ni y)$ and V implies that there exist a closed point $y_1 \in V$ and a hyperplane section Y_1 of an affine neighborhood of y in Y such that $y_1 \in Y_1$, $(Y_1 \ni y_1)$ is a klt germ of dimension $d-1$,

$$\frac{\widehat{\operatorname{vol}}(Y \ni y)}{d^d} \leq \frac{\widehat{\operatorname{vol}}(Y_1 \ni y_1)}{(d-1)^{d-1}},$$

and $\dim_{y_1} \operatorname{Sing}(Y_1) \geq \dim V - 1 \geq 1$. Let V_1 be an irreducible component of $\operatorname{Sing}(Y_1)$ such that $y_1 \in V_1$ and $\dim V_1 \geq 1$. Since $d-1 \geq 3$, Lemma 2.5 applied to $(Y_1 \ni y_1)$ and V_1 implies that there exist a closed point $y_2 \in V_1$ and a hyperplane section Y_2 of an affine neighborhood of y_1 in Y_1 such that $y_2 \in \operatorname{Sing}(Y_2)$, $(Y_2 \ni y_2)$ is a klt germ of dimension $d-2$, and

$$\frac{\widehat{\operatorname{vol}}(Y_1 \ni y_1)}{(d-1)^{d-1}} \leq \frac{\widehat{\operatorname{vol}}(Y_2 \ni y_2)}{(d-2)^{d-2}}.$$

Since $y_2 \in \operatorname{Sing}(Y_2)$, Theorem 2.4(2) in dimension $d-2$ implies that $\widehat{\operatorname{vol}}(Y_2 \ni y_2) \leq 2(d-3)^{d-2}$. Thus

$$\frac{\widehat{\operatorname{vol}}(Y \ni y)}{d^d} \leq \frac{\widehat{\operatorname{vol}}(Y_2 \ni y_2)}{(d-2)^{d-2}} \leq 2\left(\frac{d-3}{d-2}\right)^{d-2} = p_{d-2},$$

which is the first inequality. Since $h \geq 2$ and $\widehat{\operatorname{vol}}(Y \ni y) \geq 0$, the assumption and Lemma 2.7(4) imply that

$$\frac{\widehat{\operatorname{vol}}(X \ni x)}{d^d} \leq \frac{p_d}{2} + \frac{\widehat{\operatorname{vol}}(Y \ni y)}{2d^d} \leq \frac{p_d + p_{d-2}}{2} < p_{d-1} = q_d.$$

By Lemma 2.7(1), $d^d q_d = Q_d$. Hence $\widehat{\operatorname{vol}}(X \ni x) < Q_d$. □

A singular point of the central fiber of a one-parameter deformation need not be a singular point of the total space. Lemma 8.2 shows that a point of embedding codimension at least 2 in the central fiber is still a singular point of the total space.

Lemma 8.2. *Let $N \geq 2$ be an integer, let D be a reduced equidimensional finitely generated $\mathbb{C}$-algebra of dimension N, let R be an equidimensional finitely generated $\mathbb{C}$-algebra of dimension $N+2$, and let $\lambda \in R$ be a nonzerodivisor such that $R/(\lambda) \cong D[v]$ for a variable v. Let q be a closed point of $\operatorname{Spec} D[v] \subset \operatorname{Spec} R$ with image $z \in \operatorname{Spec} D$, and let $\delta_q := 1$ if $\lambda \notin \mathfrak{m}_q^2$ and $\delta_q := 0$ if $\lambda \in \mathfrak{m}_q^2$, where $\mathfrak{m}_q$ is the maximal ideal of the local ring of $\operatorname{Spec} R$ at q. Then*

$$\operatorname{edim}(\operatorname{Spec} R \ni q) - N - 2 = \operatorname{edim}(\operatorname{Spec} D \ni z) - N - 1 + \delta_q.$$

In particular $\Sigma_2(D) \times \mathbb{A}^1 \subset \operatorname{Sing}(\operatorname{Spec} R)$, where $\Sigma_2(D) \subset \operatorname{Spec} D$ denotes the closed subset whose closed points z satisfy $\operatorname{edim}(\operatorname{Spec} D \ni z) \geq N+2$.

Proof. We first check that $\Sigma_2(D)$ is well defined. Let $\Sigma \subset \operatorname{Spec} D$ be the set of points p such that $\dim_{k(p)} \Omega_{D/\mathbb{C}} \otimes k(p) \geq N+2$. By [Har77, II, Exercise 5.8(a)], Σ is closed. By [Har77, II, Proposition 8.7], $\Omega_{D/\mathbb{C}} \otimes k(z) \cong \mathfrak{m}_z/\mathfrak{m}_z^2$ for any closed point z of $\operatorname{Spec} D$ with maximal ideal $\mathfrak{m}_z$, so the closed points of Σ are exactly the closed points z with $\operatorname{edim}(\operatorname{Spec} D \ni z) \geq N+2$. Since closed points are dense in any closed subset of $\operatorname{Spec} D$ [Har77, II, Exercise 3.14], a closed subset of $\operatorname{Spec} D$ is determined by the closed points it contains.

Let $B := \mathcal{O}_{\operatorname{Spec} R, q}$. Since $q \in \operatorname{Spec} D[v]$, we have $\lambda \in \mathfrak{m}_q$, and $B/\lambda B$ is the local ring of $\operatorname{Spec} D[v]$ at q. The maximal ideal of $B/\lambda B$ is $\mathfrak{m}_q/\lambda B$, so the cotangent space of $\operatorname{Spec} D[v]$ at q is

$$\mathfrak{m}_q/(\mathfrak{m}_q^2 + \lambda B).$$

The subspace $(\mathfrak{m}_q^2 + \lambda B)/\mathfrak{m}_q^2$ of $\mathfrak{m}_q/\mathfrak{m}_q^2$ is spanned by the class of λ, so $\dim_{\mathbb{C}}(\mathfrak{m}_q^2 + \lambda B)/\mathfrak{m}_q^2 = \delta_q$. Thus

$$\operatorname{edim}(\operatorname{Spec} R \ni q) = \operatorname{edim}(\operatorname{Spec} D[v] \ni q) + \delta_q.$$

Let $b \in \mathbb{C}$ be the value of v at q, so that $q = (z, b)$. Since

$$\Omega_{D[v]/\mathbb{C}} \cong (\Omega_{D/\mathbb{C}} \otimes_D D[v]) \oplus D[v]dv$$

by [Har77, II, Exercise 8.3(a)], [Har77, II, Proposition 8.7] implies that $\operatorname{edim}(\operatorname{Spec} D[v] \ni q) = \operatorname{edim}(\operatorname{Spec} D \ni z) + 1$. Hence

$$\operatorname{edim}(\operatorname{Spec} R \ni q) = \operatorname{edim}(\operatorname{Spec} D \ni z) + 1 + \delta_q,$$

and subtracting $N+2$ from both sides implies the formula of the lemma.

Now let z be a closed point of $\Sigma_2(D)$ and let $b \in \mathbb{C}$. By the formula for $q = (z, b)$,

$$\operatorname{edim}(\operatorname{Spec} R \ni q) \geq N + 2 + 1 + \delta_q \geq N + 3.$$

Since R is equidimensional of dimension $N+2$ and q is a closed point, $\dim \mathcal{O}_{\operatorname{Spec} R, q} = N+2$ [Har77, I, Theorem 1.8A(b)]. Thus $\mathcal{O}_{\operatorname{Spec} R, q}$ is not regular, that is, $q \in \operatorname{Sing}(\operatorname{Spec} R)$. The closed points of $\Sigma_2(D) \times \mathbb{A}^1$ are the points (z, b) with z a closed point of $\Sigma_2(D)$ and $b \in \mathbb{C}$, $\operatorname{Sing}(\operatorname{Spec} R)$ is closed [Mat89, Corollary to Theorem 30.5], and closed points are dense in $\Sigma_2(D) \times \mathbb{A}^1$ [Har77, II, Exercise 3.14]. Therefore $\Sigma_2(D) \times \mathbb{A}^1 \subset \operatorname{Sing}(\operatorname{Spec} R)$. □

Construction-Proposition 8.3. Notation as in Construction-Proposition 7.4. Let $(D_{\mathrm{odd}}) \subset D$ be the ideal generated by the homogeneous elements of odd T-degree, let $Z := \operatorname{Spec} D/(D_{\mathrm{odd}})$, and let $\Sigma_2(D) \subset \operatorname{Spec} D$ be the closed subset whose closed points z satisfy $\operatorname{edim}(\operatorname{Spec} D \ni z) \geq N+2$. We call the subscheme of $\mathcal{X}$ defined by the ideal generated by the elements $b - \iota(b)$ with $b \in R$ the *fixed subscheme* of ι. Then the following hold.

(1) The fixed subscheme of ι is Z, embedded in $\{\lambda = 0\} = \operatorname{Spec} D[v_*]$ by $v_* = 0$.
(2) Every closed point $z \in Z$ other than the vertex belongs to $\operatorname{Sing}(\mathcal{X})$. Moreover, the stabilizer of z for the T-grading is μ_2, and for $y_D := z$ and every f as in Construction-Proposition 7.7, the germ $(Y \ni y)$ of Construction-Proposition 7.7 satisfies $\dim_y \operatorname{Sing}(Y) \leq 1$ and $\dim_z Z \leq \dim_y \operatorname{Sing}(Y) + 1$.
(3) $1 \leq \dim Z \leq 2$.
(4) If a closed point $z \in Z$ other than the vertex belongs to $\Sigma_2(D)$, then $\dim_z \Sigma_2(D) \leq 1$.

Proof of Construction-Proposition 8.3. For $t \in \mathbb{G}_m$, let σ_t be the automorphism of D which multiplies the elements of T-degree j by t^j. A closed point z of $\operatorname{Spec} D$ with maximal ideal $\mathfrak{m}_z \subset D$ is fixed by t if and only if $\sigma_t(\mathfrak{m}_z) = \mathfrak{m}_z$. By Construction-Proposition 7.4(3), σ_{-1} is the automorphism ι of D. Since (D_{odd}) is a homogeneous ideal which contains no non-zero element of T-degree 0, the ring $D/(D_{\mathrm{odd}})$ is positively graded with degree-zero part $\mathbb{C}$, and the vertex of Z is the vertex of $\operatorname{Spec} D$. We identify a closed point z of Z with the image of z in $\operatorname{Spec} D$, and, after (1) is proved, with the closed point $(z, 0)$ of $\operatorname{Spec} D[v_*] \subset \mathcal{X}$. A closed point z of $\operatorname{Spec} D$ belongs to Z if and only if $\sigma_{-1}(\mathfrak{m}_z) = \mathfrak{m}_z$. First let $z \in Z$ and $e \in D$, and write $e = e_0 + e_1$, where e_0 and e_1 are the sums of the homogeneous components of e of even and odd T-degree, respectively. Then $\iota(e) = e - 2e_1$, and $e_1 \in \mathfrak{m}_z$ since $(D_{\mathrm{odd}}) \subset \mathfrak{m}_z$, so $\iota(e) \in \mathfrak{m}_z$ if and only if $e \in \mathfrak{m}_z$. Conversely, assume that $\sigma_{-1}(\mathfrak{m}_z) = \mathfrak{m}_z$, and let $e \in D$ be homogeneous of odd T-degree. Then $e - e(z)$ and $\iota(e - e(z)) = -e - e(z)$ belong to $\mathfrak{m}_z$, so $2e(z) \in \mathfrak{m}_z \cap \mathbb{C} = 0$. Thus every homogeneous element of odd T-degree vanishes at z, and $z \in Z$.

(1). Let $J \subset R$ be the ideal generated by the elements $b - \iota(b)$ with $b \in R$. If $b \in R$ is homogeneous of W-degree p, then $b - \iota(b) = (1 - (-1)^p)b$, which is $2b$ if p is odd and 0 if p is even. Since any element of R is a finite sum of homogeneous elements, J is generated by the homogeneous elements of odd W-degree. Since $\lambda = \lambda^{-p} \cdot 1$ with $p = -1$ and $1 \in W_{-1}S = S$, λ is homogeneous of W-degree -1, so $\lambda \in J$. By Construction-Proposition 7.4(1), the isomorphism $R/(\lambda) \cong D[v_*]$ respects the W-degree, where v_* has W-degree 1 and the W-degree on D is the T-degree. Since the quotient map $R \to D[v_*]$ is surjective and graded, any homogeneous element of $D[v_*]$ of odd degree is the image of a homogeneous element of R of odd W-degree. Thus the image $\bar{J}$ of J in $D[v_*]$ is the ideal generated by the homogeneous elements of odd degree. A homogeneous element of $D[v_*]$ of degree p is $\sum_{k \geq 0} a_k v_*^k$ with $a_k \in D$ of T-degree $p - k$. Since v_* has degree 1, $v_* \in \bar{J}$, and if p is odd, then a_0 has odd T-degree. Hence

$$\bar{J} = v_* D[v_*] + (D_{\mathrm{odd}})D[v_*],$$

so $R/J \cong D[v_*]/\bar{J} \cong D/(D_{\mathrm{odd}})$, and the fixed subscheme $\operatorname{Spec} R/J$ of ι is $Z \subset \operatorname{Spec} D[v_*]$, embedded by $v_* = 0$.

(2). Let $z \in Z$ be a closed point other than the vertex. By the characterization of the closed points of Z above, -1 belongs to the stabilizer of z. By Lemma 7.8, the stabilizer of z is μ_h for some integer $h \geq 1$, so $h \geq 2$, and by the last sentence of Proposition 7.10(3), $h \leq 2$. Hence $h = 2$.

In particular Construction-Proposition 7.7 is available for $y_D := z$ and $y = (z, 0)$, and elements f as in Construction-Proposition 7.7 exist by Lemma 7.8. Now let f be any element as in Construction-Proposition 7.7, and let $\mathcal{X}_f$, Y_0, $(Y \ni y)$, and $g : Y_0 \times \mathbb{G}_m \to \mathcal{X}_f$ be as there. Since g is étale (Construction-Proposition 7.7(1)), for any closed point (y', t) of $Y_0 \times \mathbb{G}_m$ the local homomorphism

$$\mathcal{O}_{\mathcal{X}_f, g(y',t)} \to \mathcal{O}_{Y_0 \times \mathbb{G}_m, (y',t)}$$

is flat with closed fiber a field. The local homomorphism $\mathcal{O}_{Y_0, y'} \to \mathcal{O}_{Y_0 \times \mathbb{G}_m, (y',t)}$ is flat with closed fiber the local ring of $\mathbb{G}_m$ at t, which is regular. By [Mat89, Theorem 23.7] applied to both local homomorphisms, $g(y', t)$ is a regular point of $\mathcal{X}_f$ if and only if y' is a regular point of Y_0. Since Sing is closed [Mat89, Corollary to Theorem 30.5] and closed points are dense in closed subsets [Har77, II, Exercise 3.14], we obtain the equality of closed subsets

$$g^{-1}(\operatorname{Sing}(\mathcal{X}_f)) = \operatorname{Sing}(Y_0) \times \mathbb{G}_m. \tag{8.1}$$

Since $g(y, 1) = y$ and $y \in \operatorname{Sing}(Y)$ (Proposition 7.9), (8.1) implies that $z = y \in \operatorname{Sing}(\mathcal{X})$. As z is an arbitrary closed point of Z other than the vertex, any closed point of Z other than the vertex belongs to $\operatorname{Sing}(\mathcal{X})$.

Next we compare dimensions in the chart g. A homogeneous element of $D[v_*]$ of positive degree is $\sum_k a_k v_*^k$ with $a_0 \in \bigoplus_{j>0} D_j$, the ideal of the vertex of $\operatorname{Spec} D$ (Definition 2.8), so it vanishes at $(o_D, 0)$, where o_D is the vertex of $\operatorname{Spec} D$. Thus f vanishes at the vertex of Z, and any closed point of the open subset $Z_f := Z \cap \mathcal{X}_f$ of Z is a closed point of Z other than the

vertex. By the previous paragraph and the density of closed points, $Z_f \subset \operatorname{Sing}(\mathcal{X}_f)$, and (8.1) implies that

$$g^{-1}(Z_f) \subset \operatorname{Sing}(Y_0) \times \mathbb{G}_m.$$

The morphism $g^{-1}(Z_f) \to Z_f$ induced by g is étale, so $\dim \mathcal{O}_{g^{-1}(Z_f),(y,1)} = \dim \mathcal{O}_{Z_f,y}$ [Mat89, Theorem 15.1]. Hence

$$\dim_z Z = \dim_y Z_f = \dim_{(y,1)} g^{-1}(Z_f) \leq \dim_{(y,1)}(\operatorname{Sing}(Y_0) \times \mathbb{G}_m) = \dim_y \operatorname{Sing}(Y) + 1.$$

Finally, suppose that $\dim_y \operatorname{Sing}(Y) \geq 2$. Then $d \geq 5$ by Set-up 6.5(1). By Construction-Proposition 4.1, C is a normal affine variety of dimension d, and by Construction-Proposition 4.1(2), C is klt. Thus $(C \ni o)$ is a klt germ of dimension d, and by Proposition 7.9 and Construction-Proposition 7.7(2), $(Y \ni y)$ is a klt germ of dimension d. By Proposition 7.10(1)(2) and $h = 2$,

$$\frac{\widehat{\operatorname{vol}}(C \ni o)}{d^d} \leq \frac{p_d}{2} + \frac{\widehat{\operatorname{vol}}(Y \ni y)}{hd^d}.$$

Thus Lemma 8.1, applied to the germs $(C \ni o)$ and $(Y \ni y)$ and the integer $h = 2$, implies that $\widehat{\operatorname{vol}}(C \ni o) < Q_d$. On the other hand, $\widehat{\operatorname{vol}}(C \ni o) = \widehat{\operatorname{vol}}(X \ni x)$ by Construction-Proposition 4.1(1), and $\widehat{\operatorname{vol}}(X \ni x) > Q_d$ by Set-up 6.5(2), a contradiction. Thus $\dim_y \operatorname{Sing}(Y) \leq 1$.

(3). Let Z' be an irreducible component of Z with $\dim Z' \geq 1$. Then Z' has infinitely many closed points, so Z' contains a closed point z other than the vertex. By (2), $\dim Z' \leq \dim_z Z \leq \dim_y \operatorname{Sing}(Y) + 1 \leq 2$ for the slice $(Y \ni y)$ at z. Thus $\dim Z \leq 2$.

By Lemma 7.5, there exists a closed point z_0 of $\operatorname{Spec} D$ other than the vertex whose stabilizer for the T-grading is non-trivial, and by the last sentence of Proposition 7.10(3), the stabilizer of z_0 has order at most 2, so it is μ_2. Since $-1 \in \mu_2$ fixes z_0, the characterization of the closed points of Z above implies $z_0 \in Z$. Since (D_{odd}) is homogeneous, Z is stable under $\mathbb{G}_m$, so the orbit $\mathbb{G}_m \cdot z_0$ is contained in Z. Since the stabilizer of z_0 is finite, the morphism $\mathbb{G}_m \to Z$ given by $t \to t \cdot z_0$ has finite fibers, and the closure of $\mathbb{G}_m \cdot z_0$ is an irreducible curve in Z. Hence $\dim Z \geq 1$.

(4). Let $z \in Z$ be a closed point other than the vertex such that $z \in \Sigma_2(D)$, and let f, Y_0, $(Y \ni y)$, and g be as in (2), with $y = (z, 0)$. We check the hypotheses of Lemma 8.2 for N, D, and the elements R and λ of Construction-Proposition 7.4, with $v = v_*$ and $q = (z, 0) \in \operatorname{Spec} D[v_*]$: D is reduced by Lemma 7.3, and D is an equidimensional finitely generated $\mathbb{C}$-algebra of dimension N by Construction-Proposition 7.1(3). By Construction-Proposition 7.4(1)(2), R is a finitely generated domain, λ is a nonzerodivisor of R, and $R/(\lambda) \cong D[v_*]$. Since R is a finitely generated domain and λ is a non-zero nonunit, $\dim R = \dim R/(\lambda) + 1 = \dim D[v_*] + 1 = N + 2$ by Krull's principal ideal theorem, and R is equidimensional because R is a domain. Thus Lemma 8.2 implies that $\Sigma_2(D) \times \mathbb{A}^1 \subset \operatorname{Sing}(\mathcal{X})$. Let $\Sigma_f := (\Sigma_2(D) \times \mathbb{A}^1) \cap \mathcal{X}_f$. Then $\Sigma_f \subset \operatorname{Sing}(\mathcal{X}_f)$, and (8.1) implies that $g^{-1}(\Sigma_f) \subset \operatorname{Sing}(Y_0) \times \mathbb{G}_m$. Since the morphism $g^{-1}(\Sigma_f) \to \Sigma_f$ induced by g is étale, [Mat89, Theorem 15.1] and (2) imply that

$$\dim_z \Sigma_2(D) + 1 = \dim_y \Sigma_f = \dim_{(y,1)} g^{-1}(\Sigma_f) \leq \dim_y \operatorname{Sing}(Y) + 1 \leq 2.$$

Hence $\dim_z \Sigma_2(D) \leq 1$. □

The scheme Z is the fixed subscheme of ι, not the spectrum of the ring of invariants of ι. By (2), every closed point of Z other than the vertex is a closed point of $\operatorname{Spec} D$ whose stabilizer for the T-grading is μ_2.

8.2. **The dimension of the fixed subscheme.** In this subsection, we prove Theorem 8.6, which shows that the fixed subscheme Z of ι is a curve.

By Construction-Proposition 7.1(2), the fibers of the morphism $\operatorname{Spec} \bar{S} \to \mathbb{A}^1$ given by $\bar{v}$ over the points other than 0 are affine spaces on which all the T-weights are one, and ι has only the origin as a fixed point on each of them. Thus near a closed point of Z other than the vertex,

the subscheme of $\operatorname{Spec}\bar{S}$ defined by the ideal generated by the elements $b-\iota(b)$ with $b\in\bar{S}$ is supported on the central fiber $\operatorname{Spec}D$. Lemma 8.4 gives the local form of the support condition on $\operatorname{Spec}D$, which is the starting point for the computation of the local structure of Z.

Lemma 8.4. *Notation as in Construction-Proposition 8.3. Let $z\in Z$ be a closed point other than the vertex, regarded as a closed point of $\operatorname{Spec}D=\{\bar{v}=0\}\subset\operatorname{Spec}\bar{S}$. Let A be the completion of the local ring of $\operatorname{Spec}\bar{S}$ at z, let $t\in A$ be the image of $\bar{v}$, and let $J_A\subset A$ be the ideal generated by the elements $b-\iota(b)$ with $b\in A$, where ι acts on A through the automorphism ι of $\bar{S}$ of Construction-Proposition 7.4. Then t is an ι-invariant nonzerodivisor of A, and $t^k\in J_A$ for some integer $k\geq 1$.*

Proof. Let $\mathfrak{m}_z\subset D$ be the maximal ideal of z, and let $\mathfrak{n}_z\subset\bar{S}$ be the preimage of $\mathfrak{m}_z$. For $b\in D$, write $b=\sum_q b_q$ with b_q of T-degree q. Since $z\in Z$,

$$b-\iota(b)=2\sum_{q\text{ odd}}b_q\in(D_{\mathrm{odd}})\subset\mathfrak{m}_z.$$

Hence $\iota(\mathfrak{m}_z)=\mathfrak{m}_z$. By Construction-Proposition 7.4(3), the automorphism ι of $\bar{S}$ fixes $\bar{v}$ and induces the automorphism ι of $D=\bar{S}/(\bar{v})$. Thus $\iota(\mathfrak{n}_z)=\mathfrak{n}_z$, ι acts on the local ring $\mathcal{O}_{\operatorname{Spec}\bar{S},z}$ and on the $\mathfrak{n}_z$-adic completion A of $\mathcal{O}_{\operatorname{Spec}\bar{S},z}$, and $\iota(t)=t$.

By Construction-Proposition 7.1(1), $\bar{S}$ is a domain, and $\bar{v}\neq 0$ because $\bar{S}[\bar{v}^{-1}]\neq 0$ (Construction-Proposition 7.1(2)). Thus $\bar{v}$ is a nonzerodivisor of $\mathcal{O}_{\operatorname{Spec}\bar{S},z}$. Since A is flat over $\mathcal{O}_{\operatorname{Spec}\bar{S},z}$ [Mat89, Theorem 8.8], t is a nonzerodivisor of A.

Let $J\subset\bar{S}$ be the ideal generated by the elements $b-\iota(b)$ with $b\in\bar{S}$. For $b\in\bar{S}$ and an integer $n\geq 0$, we have $\iota(b\bar{v}^{-n})=\iota(b)\bar{v}^{-n}$. Hence $J\bar{S}[\bar{v}^{-1}]$ is the ideal of $\bar{S}[\bar{v}^{-1}]$ generated by the elements $b-\iota(b)$ with $b\in\bar{S}[\bar{v}^{-1}]$. By Construction-Proposition 7.1(2),

$$\bar{S}[\bar{v}^{-1}]=\mathbb{C}[\bar{v},\bar{v}^{-1},u_1,\dots,u_N],$$

where each u_i has T-degree 1 and $\bar{v}$ has T-degree 0. Thus $\iota(u_i)=-u_i$, so

$$u_i=\tfrac{1}{2}\bigl(u_i-\iota(u_i)\bigr)\in J\bar{S}[\bar{v}^{-1}]\qquad(1\leq i\leq N).$$

Moreover, the monomials in $\bar{v},\bar{v}^{-1},u_1,\dots,u_N$ are T-homogeneous, so any T-homogeneous element of $\bar{S}[\bar{v}^{-1}]$ of positive T-degree belongs to $(u_1,\dots,u_N)\subset J\bar{S}[\bar{v}^{-1}]$.

The ideal of D generated by the elements of positive T-degree is the maximal ideal of the vertex. Since z is not the vertex, there exists a T-homogeneous element $s\in D$ of positive T-degree with $s(z)\neq 0$. Since $\bar{v}$ has T-degree 0, there exists a T-homogeneous element $\tilde{s}\in\bar{S}$ with T-degree equal to that of s whose image in D is s. Then $\tilde{s}\in J\bar{S}[\bar{v}^{-1}]$, so $\bar{v}^n\tilde{s}\in J$ for some integer $n\geq 0$. Since $\tilde{s}(z)=s(z)\neq 0$, $\tilde{s}$ is a unit of $\mathcal{O}_{\operatorname{Spec}\bar{S},z}$, and hence $\bar{v}^n\in J\mathcal{O}_{\operatorname{Spec}\bar{S},z}$. For $b\in\bar{S}$ with image $b'\in A$, the image of $b-\iota(b)$ in A is $b'-\iota(b')$, so $JA\subset J_A$. Therefore $t^n\in J_A$, and hence $t^k\in J_A$ for $k:=\max\{n,1\}$. □

Lemma 8.5 computes the local dimension of Z at a hypersurface point. At such a point Z is smooth, its dimension is the number of ι-invariant directions in the cotangent space of $\operatorname{Spec}D$, and the T-degree N of a generator of ω_D (Construction-Proposition 7.1(3)) forces that number to be odd. This is the key to excluding two-dimensional components of Z.

Lemma 8.5. *Notation as in Construction-Proposition 8.3. Let $z\in Z$ be a closed point other than the vertex such that $\operatorname{edim}(\operatorname{Spec}D\ni z)\leq N+1$. Then $(\operatorname{Spec}D\ni z)$ is a singular hypersurface germ, and the local ring of Z at z is a regular local ring of odd dimension.*

Proof. The proof is long, so we divide it into several steps. Let $\mathcal{O}_{D,z}$ and $\mathcal{O}_{Z,z}$ be the local rings of $\operatorname{Spec}D$ and Z at z, and let $\mathfrak{m}_z\subset D$ be the maximal ideal of z. Let A be the completion of the local ring of $\operatorname{Spec}\bar{S}$ at z, let $t\in A$ be the image of $\bar{v}$, and let $J_A\subset A$ be the ideal generated by the elements $b-\iota(b)$ with $b\in A$. By Lemma 8.4, ι acts on A, t is an ι-invariant nonzerodivisor of A, and $t^k\in J_A$ for some integer $k\geq 1$. Let $B:=A/tA$. Since $\mathcal{O}_{D,z}$ is the quotient of the local ring

of $\operatorname{Spec}\bar{S}$ at z by $\bar{v}$, B is the completion of $\mathcal{O}_{D,z}$ [Mat89, Theorem 8.11]. For $b \in D$, the element $b - \iota(b)$ is twice the sum of the components of b of odd T-degree, so $b - \iota(b) \in (D_{\mathrm{odd}}) \subset \mathfrak{m}_z$. Hence $\iota(\mathfrak{m}_z) = \mathfrak{m}_z$, and ι acts on $\mathcal{O}_{D,z}$ and on B.

Step 1. In this step we write $\mathcal{O}_{D,z}$, ι-equivariantly, as a quotient of a regular local ring of dimension $\operatorname{edim}(\operatorname{Spec} D \ni z)$.

Since D is finitely generated with $D_0 = \mathbb{C}$ (Construction-Proposition 7.1(3)), there exist T-homogeneous generators $\bar{y}_1, \dots, \bar{y}_M$ of D of positive T-degrees $a_1, \dots, a_M$. Let $Q := \mathbb{C}[y_1, \dots, y_M]$ be the polynomial ring graded by $\deg y_j = a_j$, and let $Q \to D$ be the graded surjection sending y_j to $\bar{y}_j$. Let ι act on Q by multiplying the elements of degree q by $(-1)^q$. Then $Q \to D$ is ι-equivariant. Let $\mathfrak{n} \subset Q$ be the preimage of $\mathfrak{m}_z$, and let $I_\mathfrak{n}$ be the kernel of $Q_\mathfrak{n} \to \mathcal{O}_{D,z}$. Then $\iota(\mathfrak{n}) = \mathfrak{n}$. If a_j is odd, then $\bar{y}_j \in (D_{\mathrm{odd}}) \subset \mathfrak{m}_z$, so $\bar{y}_j(z) = 0$. Thus the elements $y_j - \bar{y}_j(z)$ satisfy $\iota(y_j - \bar{y}_j(z)) = (-1)^{a_j}(y_j - \bar{y}_j(z))$, and their classes form a basis of $\mathfrak{n}Q_\mathfrak{n}/\mathfrak{n}^2Q_\mathfrak{n}$.

Let e_+ and e_- be the dimensions of the eigenspaces of ι on $\mathfrak{m}_z\mathcal{O}_{D,z}/\mathfrak{m}_z^2\mathcal{O}_{D,z}$ for the eigenvalues 1 and -1. Then $e_+ + e_- = \operatorname{edim}(\operatorname{Spec} D \ni z) \le N + 1$. The map

$$\mathfrak{n}Q_\mathfrak{n}/\mathfrak{n}^2Q_\mathfrak{n} \to \mathfrak{m}_z\mathcal{O}_{D,z}/\mathfrak{m}_z^2\mathcal{O}_{D,z}$$

is surjective and ι-equivariant, with kernel $L := (I_\mathfrak{n} + \mathfrak{n}^2Q_\mathfrak{n})/\mathfrak{n}^2Q_\mathfrak{n}$ of dimension $M - e_+ - e_-$. If $g \in I_\mathfrak{n}$ has class $\bar{g} \in L$ with $\iota(\bar{g}) = \epsilon\bar{g}$ for some $\epsilon \in \{1, -1\}$, then $\frac{1}{2}(g + \epsilon\iota(g)) \in I_\mathfrak{n}$ has class $\bar{g}$ and is an eigenvector of ι with eigenvalue ϵ. Thus there exist $g_1, \dots, g_{M-e_+-e_-} \in I_\mathfrak{n}$ and $\epsilon_1, \dots, \epsilon_{M-e_+-e_-} \in \{1, -1\}$ such that $\iota(g_i) = \epsilon_i g_i$ and the classes of the g_i form a basis of L. Since ι has order two, the cotangent space of $Q_\mathfrak{n}$ is the direct sum of L and a representation isomorphic to $\mathfrak{m}_z\mathcal{O}_{D,z}/\mathfrak{m}_z^2\mathcal{O}_{D,z}$. Comparing the determinants of ι on both sides, we obtain

$$\prod_{j=1}^{M}(-1)^{a_j} = (-1)^{e_-}\prod_{i=1}^{M-e_+-e_-}\epsilon_i. \tag{8.2}$$

The elements $g_1, \dots, g_{M-e_+-e_-}$ are part of a regular system of parameters of the regular local ring $Q_\mathfrak{n}$ of dimension M. Hence $Q' := Q_\mathfrak{n}/(g_1, \dots, g_{M-e_+-e_-})$ is a regular local ring of dimension $e_+ + e_-$ [Mat89, Theorem 14.2] with maximal ideal $\mathfrak{n}'$, ι acts on Q', and $\mathcal{O}_{D,z} = Q'/I'$ for the image I' of $I_\mathfrak{n}$ in Q'. The map from the cotangent space of Q' to that of $\mathcal{O}_{D,z}$ is surjective between spaces of dimension $e_+ + e_-$, hence an isomorphism, and $I' \subset \mathfrak{n}'^2$.

Since D is equidimensional of dimension N (Construction-Proposition 7.1(3)), $\dim \mathcal{O}_{D,z} = N$ [Har77, I, Theorem 1.8A(b)], so $e_+ + e_- \ge N$ [Mat89, Theorem 13.4]. Thus $e_+ + e_- \in \{N, N+1\}$. If $e_+ + e_- = N$, then $\dim Q' = \dim \mathcal{O}_{D,z}$ and Q' is a domain [Mat89, Theorem 14.3], so $I' = 0$.

Step 2. In this step we write A/J_A as a quotient of a power series ring, and we prove that $e_+ + e_- = N + 1$.

Let P_0 be the completion of Q'. Then ι acts on P_0, and $B = P_0/I'P_0$ [Mat89, Theorem 8.11]. Lifting a basis of eigenvectors of the cotangent space of P_0 through the elements $\frac{1}{2}(x + \iota(x))$ and $\frac{1}{2}(x - \iota(x))$, we obtain elements $x_1, \dots, x_{e_++e_-}$ of the maximal ideal of P_0 which form a regular system of parameters, with $\iota(x_i) = x_i$ for $i \le e_+$ and $\iota(x_i) = -x_i$ for $i > e_+$. By [Mat89, Theorem 8.4], they define a surjection from $\mathbb{C}[[x_1, \dots, x_{e_++e_-}]]$ onto P_0. Since both rings are regular local rings of dimension $e_+ + e_-$, the surjection is an isomorphism. Thus $P_0 = \mathbb{C}[[x_1, \dots, x_{e_++e_-}]]$, and ι multiplies x_i by 1 for $i \le e_+$ and by -1 for $i > e_+$.

Let $P := \mathbb{C}[[t, x_1, \dots, x_{e_++e_-}]]$ with $\iota(t) = t$ and with ι acting on $x_1, \dots, x_{e_++e_-}$ as on P_0. For each i, let $\sigma_i \in \{1, -1\}$ be given by $\iota(x_i) = \sigma_i x_i$. Lifting the image of x_i in B to an element x of the maximal ideal of A and replacing x with $\frac{1}{2}(x + \sigma_i\iota(x))$, we obtain an element x_i' of the maximal ideal of A with $\iota(x_i') = \sigma_i x_i'$ whose image in B is the image of x_i. The maximal ideal of A is generated by $t, x_1', \dots, x_{e_++e_-}'$, because the image of the maximal ideal of A in $B = A/tA$ is generated by the images of the x_i. By [Mat89, Theorem 8.4], the continuous $\mathbb{C}$-algebra homomorphism $\varphi : P \to A$ with $\varphi(t) = t$ and $\varphi(x_i) = x_i'$ is surjective. The continuous homomorphisms $\iota \circ \varphi$ and $\varphi \circ \iota$ coincide on $t, x_1, \dots, x_{e_++e_-}$, so φ is ι-equivariant. Let $I := \operatorname{Ker}\varphi$.

Then $\iota(I) = I$, and the reduction of φ modulo t is the surjection $P_0 \to B$ above. If $tp \in I$ for some $p \in P$, then $t\varphi(p) = 0$, so $\varphi(p) = 0$ since t is a nonzerodivisor of A. Thus $I \cap tP = tI$, and

$$I/tI \cong (I + tP)/tP = \operatorname{Ker}(P_0 \to B) = I'P_0. \tag{8.3}$$

Let $\rho : P \to \mathbb{C}[[t, x_1, \dots, x_{e_+}]]$ be the surjection setting $x_i = 0$ for $i > e_+$. Then $\rho \circ \iota = \rho$, so $\rho(p - \iota(p)) = 0$ for $p \in P$. Conversely, $x_i = \frac{1}{2}(x_i - \iota(x_i))$ for $i > e_+$. Hence the ideal $J_P \subset P$ generated by the elements $p - \iota(p)$ with $p \in P$ is $\operatorname{Ker}\rho$. Since φ is surjective and ι-equivariant, $\varphi(J_P)A = J_A$. Therefore

$$A/J_A \cong P/(I + J_P) \cong \mathbb{C}[[t, x_1, \dots, x_{e_+}]]/\rho(I). \tag{8.4}$$

Suppose that $e_+ + e_- = N$. Then $I' = 0$ by **Step 1**, so $I = tI$ by (8.3), and $I = 0$ by Nakayama's lemma. By (8.4), $A/J_A \cong \mathbb{C}[[t, x_1, \dots, x_{e_+}]]$, a domain in which $t^k \neq 0$. This contradicts $t^k \in J_A$. Thus $e_+ + e_- = N + 1 > \dim \mathcal{O}_{D,z}$, so $\mathcal{O}_{D,z}$ is not regular [Mat89, Section 14], and $(\operatorname{Spec} D \ni z)$ is singular.

Step 3. In this step we prove that $(\operatorname{Spec} D \ni z)$ is a hypersurface germ with an ι-invariant equation, and that $\mathcal{O}_{Z,z}$ is a regular local ring of dimension e_+.

By Lemma 7.3, D is reduced, so I' is a radical ideal of Q'. Let $\mathfrak{p}_1, \dots, \mathfrak{p}_r$ be the minimal primes of I'. Since D is equidimensional of dimension N, $\dim Q'/\mathfrak{p}_i = N$ for any i [Har77, I, Theorem 1.8A(b)]. Since Q' is a regular local ring of dimension $N + 1$, $\operatorname{ht}\mathfrak{p}_i = 1$ [Mat89, Theorems 17.8 and 17.4(i)]. The ring Q' is a UFD [Mat89, Theorem 20.3], so $\mathfrak{p}_i = (f_i)$ for prime elements $f_1, \dots, f_r$ which are pairwise non-associate [Mat89, Theorem 20.1]. Thus

$$I' = \mathfrak{p}_1 \cap \cdots \cap \mathfrak{p}_r = (f_1 \cdots f_r)$$

is principal, and $I' \neq 0$ because $\dim \mathcal{O}_{D,z} = N < \dim Q'$. Since $\iota(I') = I'$ and $\dim_{\mathbb{C}} I'/\mathfrak{n}'I' = 1$, ι acts on $I'/\mathfrak{n}'I'$ by some $\epsilon \in \{1, -1\}$. For a generator f_0 of I', the element $f := \frac{1}{2}(f_0 + \epsilon\iota(f_0))$ has the class of f_0 in $I'/\mathfrak{n}'I'$, so $I' = fQ'$ by Nakayama's lemma, and $\iota(f) = \epsilon f$. Hence $B = P_0/fP_0$ with $f \neq 0$ [Mat89, Theorem 8.11], and $(\operatorname{Spec} D \ni z)$ is a hypersurface germ.

By (8.3), there exists $F_0 \in I$ whose image in P_0 is f. Let $F := \frac{1}{2}(F_0 + \epsilon\iota(F_0))$. Then $F \in I$, $\iota(F) = \epsilon F$, and the image of F in P_0 is f. By (8.3), $I = FP + tI$, so $I = FP$ by Nakayama's lemma, and $\rho(I)$ is generated by $F_+ := \rho(F)$. Suppose that $\epsilon = -1$. Then $F_+ = \rho(\iota(F)) = -F_+$, so $F_+ = 0$, and $A/J_A \cong \mathbb{C}[[t, x_1, \dots, x_{e_+}]]$ by (8.4). This contradicts $t^k \in J_A$ as in **Step 2**. Thus $\epsilon = 1$, that is, $\iota(f) = f$.

By (8.4) and $t^k \in J_A$, we have $t^k \in F_+\mathbb{C}[[t, x_1, \dots, x_{e_+}]]$. In particular, $F_+ \neq 0$. The ring $\mathbb{C}[[t, x_1, \dots, x_{e_+}]]$ is a UFD [Mat89, Theorem 20.3] in which t is a prime element. Since F_+ divides t^k, $F_+ = ut^l$ for a unit u and an integer $0 \leq l \leq k$. Since $A \neq 0$, F belongs to the maximal ideal of P, and so does F_+. Hence $l \geq 1$, and

$$A/J_A \cong \mathbb{C}[[t, x_1, \dots, x_{e_+}]]/(t^l).$$

Let $J_D \subset D$ be the ideal generated by the elements $b - \iota(b)$ with $b \in D$. For $b \in D$, $b - \iota(b)$ is twice the sum of the components of b of odd T-degree, and $b = \frac{1}{2}(b - \iota(b))$ if b is homogeneous of odd T-degree. Hence $J_D = (D_{\mathrm{odd}})$, $\mathcal{O}_{Z,z} = \mathcal{O}_{D,z}/J_D\mathcal{O}_{D,z}$, and the completion of $\mathcal{O}_{Z,z}$ is B/J_DB [Mat89, Theorem 8.11]. For $b \in D$ and $s' \in D\backslash\mathfrak{m}_z$,

$$\iota(b/s') - b/s' = \big(\iota(b)s' - b\iota(s')\big)/\big(s'\iota(s')\big) \in J_D\mathcal{O}_{D,z},$$

so ι acts trivially on $\mathcal{O}_{Z,z}$, and by continuity also on B/J_DB. Hence the ideal $J_B \subset B$ generated by the elements $b - \iota(b)$ with $b \in B$ is J_DB. Since $A \to B$ is surjective and ι-equivariant, the image of J_A in B is J_B. Therefore

$$\widehat{\mathcal{O}_{Z,z}} \cong B/J_B \cong A/(J_A + tA) \cong \mathbb{C}[[x_1, \dots, x_{e_+}]].$$

Since $\dim \widehat{\mathcal{O}_{Z,z}} = \dim \mathcal{O}_{Z,z}$, and $\mathcal{O}_{Z,z}$ is regular if and only if $\widehat{\mathcal{O}_{Z,z}}$ is regular [Har77, I, Theorem 5.4A], $\mathcal{O}_{Z,z}$ is a regular local ring of dimension e_+.

Step 4. In this step we prove that e_+ is odd, by computing the action of ι on the fiber at z of the canonical module of D in two ways.

Let $\Omega_Q := Q dy_1 \wedge \cdots \wedge dy_M$, graded so that $dy_1 \wedge \cdots \wedge dy_M$ has degree $a_1 + \cdots + a_M$, and let ι act on Ω_Q by multiplying the elements of degree q by $(-1)^q$. Since $\iota(y_j) = (-1)^{a_j} y_j$, the action of ι on Ω_Q is the natural action on differential forms, that is,

$$\iota(dy_1 \wedge \cdots \wedge dy_M) = d\iota(y_1) \wedge \cdots \wedge d\iota(y_M).$$

By [BH98, Example 3.6.10 and Proposition 3.6.12(b)], Ω_Q is the *canonical module of Q, and the graded module $\omega_D := {}^* \operatorname{Ext}_Q^{M-N}(D, \Omega_Q)$ is the *canonical module of D in the sense of [BH98, Definition 3.6.8], without shift. By [BH98, Proposition 3.6.9(b)], the *canonical module is unique up to homogeneous isomorphism. By Construction-Proposition 7.1(3), ω_D is generated by a homogeneous element ω_0 of T-degree N.

The automorphism ι acts on $\operatorname{Ext}_Q^{M-N}(D, \Omega_Q)$ by functoriality. Namely, for an ι-equivariant free resolution $F_\bullet \to D$, ι acts on $\operatorname{Hom}_Q(F_\bullet, \Omega_Q)$ by $h \to \iota \circ h \circ \iota$. The induced action on Ext does not depend on $F_\bullet$, because a comparison map c' between two such resolutions may be replaced with the ι-equivariant comparison map $\frac{1}{2}(c' + \iota \circ c' \circ \iota)$. Taking $F_\bullet$ graded, with ι multiplying the elements of degree q by $(-1)^q$, we see that ι multiplies the elements of ω_D of T-degree q by $(-1)^q$. In particular, $\iota(\omega_0) = (-1)^N \omega_0$.

Since $Q_\mathfrak{n}$ is flat over Q, there exists an ι-equivariant isomorphism

$$\omega_D \otimes_Q Q_\mathfrak{n} \cong \operatorname{Ext}_{Q_\mathfrak{n}}^{M-N}(\mathcal{O}_{D,z}, \Omega_Q \otimes_Q Q_\mathfrak{n}),$$

and the left-hand side is a canonical module of $\mathcal{O}_{D,z}$ [BH98, Proposition 3.6.9(a)]. Since ω_0 generates ω_D and ι acts trivially on the residue field $\mathcal{O}_{D,z}/\mathfrak{m}_z\mathcal{O}_{D,z} = \mathbb{C}$, ι acts on the fiber $\omega_D \otimes_D \mathcal{O}_{D,z}/\mathfrak{m}_z\mathcal{O}_{D,z}$ by $(-1)^N$.

We now compute the action of ι on the fiber $\omega_D \otimes_D \mathcal{O}_{D,z}/\mathfrak{m}_z\mathcal{O}_{D,z}$ in a different way. By **Step 2**, $e_+ + e_- = N + 1$, so the elements g_i of **Step 1** are $g_1, \ldots, g_{M-N-1}$. By **Step 3**, $I' = fQ'$, so there exists $\tilde{f} \in I_\mathfrak{n}$ whose image in Q' is f, and replacing $\tilde{f}$ with $\frac{1}{2}(\tilde{f} + \iota(\tilde{f}))$, we may assume that $\iota(\tilde{f}) = \tilde{f}$. Then $I_\mathfrak{n} = (g_1, \ldots, g_{M-N-1}, \tilde{f})$. The sequence $g_1, \ldots, g_{M-N-1}, \tilde{f}$ is $Q_\mathfrak{n}$-regular, since $g_1, \ldots, g_{M-N-1}$ are part of a regular system of parameters and f is a non-zero element of the domain Q'. Let $K_\bullet$ be the Koszul complex of $g_1, \ldots, g_{M-N-1}, \tilde{f}$, with basis $\theta_1, \ldots, \theta_{M-N}$ of K_1 mapping to $g_1, \ldots, g_{M-N-1}, \tilde{f}$. Then $K_\bullet$ is a free resolution of $\mathcal{O}_{D,z}$ [BH98, Corollary 1.6.14(b)], and it is ι-equivariant for $\iota(\theta_i) = \epsilon_i \theta_i$ $(i \le M - N - 1)$ and $\iota(\theta_{M-N}) = \theta_{M-N}$. The map sending a homomorphism $h : K_{M-N} \to \Omega_Q \otimes_Q Q_\mathfrak{n}$ to $h(\theta_1 \wedge \cdots \wedge \theta_{M-N})$ induces an isomorphism [BH98, Proposition 1.6.10(d)]

$$\operatorname{Ext}_{Q_\mathfrak{n}}^{M-N}(\mathcal{O}_{D,z}, \Omega_Q \otimes_Q Q_\mathfrak{n}) \cong \Omega_Q \otimes_Q \mathcal{O}_{D,z}.$$

Thus the fiber at z is spanned by the class of the homomorphism h_0 with

$$h_0(\theta_1 \wedge \cdots \wedge \theta_{M-N}) = dy_1 \wedge \cdots \wedge dy_M.$$

Since

$$(\iota \circ h_0 \circ \iota)(\theta_1 \wedge \cdots \wedge \theta_{M-N}) = \prod_{i=1}^{M-N-1} \epsilon_i \prod_{j=1}^{M} (-1)^{a_j} dy_1 \wedge \cdots \wedge dy_M,$$

(8.2) implies that ι acts on the fiber by $(-1)^{e_-}$. For $M - N = 1$, the computation above is the hypersurface adjunction of [Sta, Tag 0B4A].

Comparing the actions of ι on the fiber computed above, $(-1)^{e_-} = (-1)^N$. Since $e_+ + e_- = N + 1$, $e_+ = N + 1 - e_-$ is odd. By **Step 3**, $\mathcal{O}_{Z,z}$ is a regular local ring of odd dimension e_+. □

We now exclude two-dimensional components of Z.

Theorem 8.6. *Notation as in Construction-Proposition 8.3. Then* $\dim Z = 1$.

Proof. By Construction-Proposition 8.3(3), $1 \le \dim Z \le 2$. Suppose that $\dim Z = 2$, and let Z_0 be an irreducible component of Z with $\dim Z_0 = 2$.

Suppose that $Z_0 \subset \Sigma_2(D)$. Let $z \in Z_0$ be a closed point other than the vertex. Then $\dim_z \Sigma_2(D) \ge \dim_z Z_0 = 2$, which contradicts Construction-Proposition 8.3(4). Thus $Z_0 \not\subset \Sigma_2(D)$.

Hence the complement in Z_0 of $\Sigma_2(D)$, of the vertex, and of the irreducible components of Z other than Z_0 is a non-empty open subset U of Z_0. Let z be a closed point of U. Then $z \in Z$ is not the vertex, and $\operatorname{edim}(\operatorname{Spec} D \ni z) \le N + 1$ since $z \notin \Sigma_2(D)$. By Lemma 8.5, the local ring of Z at z is a regular local ring of odd dimension. On the other hand, Z_0 is the only irreducible component of Z through z, so $\dim \mathcal{O}_{Z,z} = \dim Z_0 = 2$ [Har77, I, Theorem 1.8A(b)], where $\mathcal{O}_{Z,z}$ is the local ring of Z at z. This is a contradiction. Therefore $\dim Z \le 1$, and $\dim Z = 1$. □

9. Graded algebras with a small fixed locus

In this section, we prove Theorem 9.1, a standalone algebraic result. The vertex of $\operatorname{Spec} D$ and the Hilbert series Hilb_D are as in Definition 2.8. Theorem 9.1 is the order-two version of [OAI26b, Lemma 6.5], which shows that such an algebra is a polynomial ring when all stabilizers are trivial. In Section 10, Theorem 9.1, with N in place of d, is used for the ring D of Section 7.

Theorem 9.1. *Let $d \ge 2$ be an integer, let $D = \bigoplus_{j \ge 0} D_j$ be a finitely generated graded $\mathbb{C}$-algebra with $D_0 = \mathbb{C}$, and let $\mathfrak{m} := \bigoplus_{j>0} D_j$. Assume that D is reduced, equidimensional, and Cohen–Macaulay of dimension d, that $\operatorname{Hilb}_D(t) = (1-t)^{-d}$, that every closed point of $\operatorname{Spec} D$ other than the vertex has stabilizer of order at most 2 for the $\mathbb{G}_m$-action induced by the grading, and that $\dim \operatorname{Spec} D/(D_{\mathrm{odd}}) \le 1$, where (D_{odd}) is the ideal generated by the homogeneous elements of odd degree. Then either D is a polynomial ring in d variables of degree 1, or there exists an isomorphism of graded $\mathbb{C}$-algebras*

$$D \cong \mathbb{C}[x_1, \dots, x_d, b]/(q),$$

where $\deg x_i = 1$ for $1 \le i \le d$, $\deg b = 2$, and $q \in \mathbb{C}[x_1, \dots, x_d]$ is a quadratic form of rank at least two. In particular D is generated by D_1 and D_2, the localization $D_{\mathfrak{m}}$ is a complete intersection ring, and $\dim_{\mathbb{C}} \mathfrak{m}/\mathfrak{m}^2 \le d+1$.

We prove Theorem 9.1 at the end of this section. The first lemma describes the stabilizers of the $\mathbb{G}_m$-action induced by a grading, and the sheaf $\mathcal{O}(1)$ when the action is free outside the vertex.

Lemma 9.2. *Let $B = \bigoplus_{j \ge 0} B_j$ be a finitely generated graded $\mathbb{C}$-algebra with $B_0 = \mathbb{C}$, and let $Y := \operatorname{Proj} B$. Then the following hold.*

1. *The stabilizer of a closed point $p \in \operatorname{Spec} B$ other than the vertex for the $\mathbb{G}_m$-action induced by the grading is μ_m, where m is the greatest common divisor of the degrees of the homogeneous elements of B which do not vanish at p.*
2. *If every closed point of $\operatorname{Spec} B$ other than the vertex has trivial stabilizer for the $\mathbb{G}_m$-action induced by the grading, then $\mathcal{L} := \mathcal{O}_Y(1)$ is an ample invertible sheaf, $\mathcal{O}_Y(j) \cong \mathcal{L}^{\otimes j}$ for every integer j, and the natural map $B_j \to H^0(Y, \mathcal{L}^{\otimes j})$ is an isomorphism for every sufficiently large integer j.*
3. *There exists an integer $e \ge 1$ such that $B^{(e)} := \bigoplus_{j \ge 0} B_{ej}$ is generated by B_e. For every such e, there exists an isomorphism $Y \cong \operatorname{Proj} B^{(e)}$ under which $\mathcal{O}_Y(e)$ corresponds to $\mathcal{O}_{\operatorname{Proj} B^{(e)}}(1)$. If $\dim B \ge 1$, then $\dim Y = \dim B - 1$.*

Proof. (1) Recall that $t \in \mathbb{G}_m$ sends a closed point p of $\operatorname{Spec} B$ to the closed point $t \cdot p$ with $g(t \cdot p) = t^j g(p)$ for any j and any $g \in B_j$, where $g(p) \in \mathbb{C}$ is the value of g at p. Let $g_1, \dots, g_n$ be homogeneous generators of the $\mathbb{C}$-algebra B, and let m' be the greatest common divisor of the degrees $\deg g_i$ over the indices i with $g_i(p) \ne 0$. Such indices exist because p is not the vertex.

Then $t \cdot p = p$ if and only if $t^{\deg g_i} g_i(p) = g_i(p)$ for $1 \le i \le n$, that is, if and only if $t^{\deg g_i} = 1$ for any i with $g_i(p) \neq 0$. Thus the stabilizer of p is $\mu_{m'}$.

It remains to show that $m' = m$. Since each g_i with $g_i(p) \neq 0$ is a homogeneous element not vanishing at p, $m \mid m'$. Conversely, let g be a homogeneous element with $g(p) \neq 0$. Then g is a linear combination of monomials $g_1^{a_1} \cdots g_n^{a_n}$ with $\sum_i a_i \deg g_i = \deg g$, and some monomial $g_1^{a_1} \cdots g_n^{a_n}$ in the combination does not vanish at p. Then $a_i = 0$ whenever $g_i(p) = 0$, so $m' \mid \deg g$. Hence $m' \mid m$, and $m = m'$.

(3) There exists an integer $e \ge 1$ such that $B^{(e)}$ is generated by B_e [Bou72, Chapter III, §1, no. 3, Proposition 3]. Let e be such an integer. For any homogeneous $f \in B$ of positive degree, $D_+(f) = D_+(f^e)$, $(B_{f^e})_0 = (B^{(e)}_{f^e})_0$, and $(B_{f^e})_e = (B^{(e)}_{f^e})_1$. The identifications of $(B_{f^e})_0$ with $(B^{(e)}_{f^e})_0$ and of $(B_{f^e})_e$ with $(B^{(e)}_{f^e})_1$ are compatible on the intersections of the open sets $D_+(f^e)$, so they glue to an isomorphism $Y \cong \operatorname{Proj} B^{(e)}$ under which $\mathcal{O}_Y(e)$ corresponds to $\mathcal{O}_{\operatorname{Proj} B^{(e)}}(1)$. Assume that $\dim B \ge 1$. Since B is finite over $B^{(e)}$ [Bou72, Chapter III, §1, no. 3, Proposition 2(ii)], $\dim B = \dim B^{(e)}$. The underlying topological space of $\operatorname{Proj} B^{(e)}$ is equal to that of the reduced scheme $(\operatorname{Proj} B^{(e)})_{\mathrm{red}}$, which is an algebraic set in $\mathbb{P}(B_e^\vee)$, and [Har77, I, Exercise 2.6], applied to each irreducible component of $\operatorname{Proj} B^{(e)}$, implies $\dim \operatorname{Proj} B^{(e)} = \dim B^{(e)} - 1$. Hence $\dim Y = \dim B - 1$.

(2) Let p be a closed point of $\operatorname{Spec} B$ other than the vertex. By (1), there exist homogeneous elements $f_1, \dots, f_l \in B$ with $f_i(p) \neq 0$ for $1 \le i \le l$ and $\gcd(\deg f_1, \dots, \deg f_l) = 1$. Let $f_p := f_1 \cdots f_l$, choose integers $c_1, \dots, c_l$ with $\sum_i c_i \deg f_i = 1$, and let $w_p := \prod_i f_i^{c_i} \in B_{f_p}$. Then w_p is a homogeneous unit of degree 1 of B_{f_p}, so multiplication by w_p^j is a bijection $(B_{f_p})_0 \to (B_{f_p})_j$ for any integer j. Hence

$$B_{f_p} = (B_{f_p})_0[w_p, w_p^{-1}],$$

and $\mathcal{O}_Y(j)|_{D_+(f_p)}$ is free with basis w_p^j. The open sets $D_+(f_p)$ cover Y, since a point of Y is a homogeneous prime ideal Q with $B_+ \not\subset Q$, where $B_+ := \bigoplus_{j>0} B_j$, and by the Nullstellensatz, Q is the intersection of the maximal ideals containing Q, so one of them is the maximal ideal of a closed point p other than the vertex, and $f_p \notin Q$ because $f_p(p) \neq 0$, that is, $Q \in D_+(f_p)$. Therefore $\mathcal{L}$ is invertible, and the multiplication map $\mathcal{L}^{\otimes j} \to \mathcal{O}_Y(j)$ is an isomorphism for any integer j, since on $D_+(f_p)$ it sends $w_p^{\otimes j}$ to w_p^j.

Choose e as in (3). Under the isomorphism $Y \cong \operatorname{Proj} B^{(e)}$ of (3), $\mathcal{L}^{\otimes e} \cong \mathcal{O}_Y(e)$ corresponds to $\mathcal{O}_{\operatorname{Proj} B^{(e)}}(1)$, which is very ample since $B^{(e)}$ is generated by B_e. Thus $\mathcal{L}$ is ample. Finally, for $0 \le c < e$ the graded $B^{(e)}$-module $M_c := \bigoplus_{i \ge 0} B_{ie+c}$ is finitely generated [Bou72, Chapter III, §1, no. 3, Proposition 2(ii)]. On $D_+(f^e)$ for any homogeneous $f \in B$ of positive degree, the sheaf associated with M_c and $\mathcal{O}_Y(c)$ are both the sheaf associated with the $(B_{f^e})_0$-module $(B_{f^e})_c$, so the sheaf associated with M_c is $\mathcal{O}_Y(c)$. Hence [Har77, II, Exercise 5.9(b)], applied to $B^{(e)}$ and to M_c for $0 \le c < e$, implies that $B_j \to H^0(Y, \mathcal{O}_Y(j)) = H^0(Y, \mathcal{L}^{\otimes j})$ is an isomorphism for any sufficiently large j. □

When $\operatorname{Spec} D$ has closed points with stabilizer of order two, the sheaf $\mathcal{O}(1)$ on $\operatorname{Proj} D$ need not be invertible (for example, for the quadric hypersurface $\mathbb{C}[x_1, \dots, x_d, b]/(q)$ of Theorem 9.1, at the points where only b is non-zero), and the theory of polarized varieties cannot be used directly. A double cover obtained by adjoining a square root of odd degree of an element of even degree removes the stabilizers of order two. The condition on (D_{odd}) is used only in the next construction: it guarantees an element of even degree that does not vanish at any point with stabilizer of order two. The hypotheses of Construction-Proposition 9.3 are those of Theorem 9.1 except the condition on Hilb_D.

Construction-Proposition 9.3. Let $d \ge 2$ be an integer and let $D = \bigoplus_{j \ge 0} D_j$ be a finitely generated graded $\mathbb{C}$-algebra with $D_0 = \mathbb{C}$. Assume that D is reduced, equidimensional, and Cohen–Macaulay of dimension d, that every closed point of $\operatorname{Spec} D$ other than the vertex has stabilizer

of order at most 2 for the $\mathbb{G}_m$-action induced by the grading, and that $\dim \operatorname{Spec} D/(D_{\mathrm{odd}}) \leq 1$, where (D_{odd}) is the ideal generated by the homogeneous elements of odd degree. Then there exist an odd integer $k \geq 3$ and $s \in D_{2k}$ such that $B := D[u]/(u^2 - s)$, graded by $\deg u = k$, is a finitely generated reduced equidimensional Cohen–Macaulay $\mathbb{C}$-algebra of dimension d which is a free D-module with basis $1, u$, and every closed point of $\operatorname{Spec} B$ other than the vertex has trivial stabilizer for the $\mathbb{G}_m$-action induced by the grading.

Proof. For a closed point $p \in \operatorname{Spec} D$ other than the vertex, let h_p be the greatest common divisor of the degrees of the homogeneous elements of D not vanishing at p. By Lemma 9.2(1) and the assumption on stabilizers, $h_p \in \{1, 2\}$. Moreover, $h_p = 2$ if and only if every homogeneous element of odd degree vanishes at p, that is, if and only if $p \in \operatorname{Spec} D/(D_{\mathrm{odd}})$.

Choose closed points $p_1, \dots, p_r \in \operatorname{Spec} D$ other than the vertex as follows: one point on each irreducible component of $\operatorname{Spec} D$, and one point on each irreducible component of $\operatorname{Spec} D/(D_{\mathrm{odd}})$ with $\dim = 1$. The first choice is possible since D is equidimensional of dimension $d \geq 2$.

We show that for $1 \leq i \leq r$ and any sufficiently large integer k, some element of D_{2k} does not vanish at p_i. Since the greatest common divisor of a set of positive integers is the greatest common divisor of a finite subset, there exist homogeneous $f_1, \dots, f_l \in D$ not vanishing at p_i with $\gcd(\deg f_1, \dots, \deg f_l) = h_{p_i}$. Let $a_j := \deg f_j / h_{p_i}$, so that $\gcd(a_1, \dots, a_l) = 1$, and let $\Lambda \subset \mathbb{Z}_{\geq 0}$ be the additive semigroup generated by $a_1, \dots, a_l$. The image of Λ in $\mathbb{Z}/a_1\mathbb{Z}$ is a submonoid of a finite group, hence a subgroup, and it contains the images of $a_1, \dots, a_l$, which generate $\mathbb{Z}/a_1\mathbb{Z}$. Thus for any residue class c modulo a_1 there exists $n_c \in \Lambda$ with $n_c \equiv c \pmod{a_1}$, and any integer $n \geq \max_c n_c$ belongs to $n_c + a_1\mathbb{Z}_{\geq 0} \subset \Lambda$ for $c \equiv n$. Hence any sufficiently large multiple of h_{p_i} is the degree of a monomial in $f_1, \dots, f_l$, which does not vanish at p_i. Since $h_{p_i} \mid 2$, the claim follows.

Fix an odd integer $k \geq 3$ such that for $1 \leq i \leq r$ some element of D_{2k} does not vanish at p_i. Then for $1 \leq i \leq r$, the elements of D_{2k} vanishing at p_i form a proper linear subspace of D_{2k}. Since $\mathbb{C}$ is infinite, a finite union of proper linear subspaces is not equal to D_{2k}. Choose $s \in D_{2k}$ with $s(p_i) \neq 0$ for $1 \leq i \leq r$. In particular s is not contained in any minimal prime of D.

We show that $s(q) \neq 0$ for any closed point $q \in \operatorname{Spec} D/(D_{\mathrm{odd}})$ other than the vertex. Let V be an irreducible component of $\operatorname{Spec} D/(D_{\mathrm{odd}})$ containing q. Since $\mathbb{G}_m$ is connected and permutes the finitely many irreducible components of the $\mathbb{G}_m$-stable closed subset $\operatorname{Spec} D/(D_{\mathrm{odd}})$, V is $\mathbb{G}_m$-stable. Any finite $\mathbb{G}_m$-stable set of closed points consists of points fixed by $\mathbb{G}_m$, since the orbit of a point is the image of the irreducible group $\mathbb{G}_m$. By Lemma 9.2(1), the only closed point fixed by $\mathbb{G}_m$ is the vertex. Hence $V \neq \{q\}$, so $\dim V = 1$ and V contains some p_i. Then $V \cap V(s)$ is a $\mathbb{G}_m$-stable proper closed subset of the irreducible curve V, hence a finite $\mathbb{G}_m$-stable set, so $V \cap V(s)$ consists of the vertex. Thus $s(q) \neq 0$.

Let $B := D[u]/(u^2 - s)$ with $\deg u = k$. Since $u^2 - s$ is homogeneous of degree $2k$ and monic in u, $B = D \oplus Du$ is a free D-module with basis $1, u$, and $B_j = D_j \oplus D_{j-k}u$. Thus B is finitely generated, $B_0 = \mathbb{C}$, B is integral over D, and $\dim B = d$. The polynomial ring $D[u]$ is Cohen–Macaulay [Mat89, Theorem 17.7], and the monic polynomial $u^2 - s$ is a nonzerodivisor of $D[u]$, so B is Cohen–Macaulay [Mat89, Theorem 17.3(ii)]. Localizing at the homogeneous maximal ideal of B, which contains all minimal primes of B since they are homogeneous (a minimal prime of a homogeneous ideal I is an associated prime of B/I, hence homogeneous [BH98, Lemma 1.5.6(b)(ii)]), [Mat89, Theorem 17.3(i)] implies that B is equidimensional.

We show that B is reduced. For a minimal prime P of D, let K_P be the fraction field of D/P and s_P the image of s in K_P. Since D is reduced, $D \to \prod_P K_P$ is injective, and since B is a free D-module with basis $1, u$, the induced map

$$B \to \prod_P K_P[u]/(u^2 - s_P)$$

is injective. Since $s \notin P$, $s_P \neq 0$, and since the characteristic is zero, $u^2 - s_P$ is a separable polynomial. Hence $K_P[u]/(u^2 - s_P)$ is a product of fields, and B is reduced.

Finally, let p' be a closed point of $\operatorname{Spec} B$ other than the vertex, and let p be the image of p' in $\operatorname{Spec} D$, which is a closed point since B is finite over D. If p is the vertex, then any element of $\bigoplus_{j>0} D_j$ vanishes at p', and $u(p')^2 = s(p) = 0$, so the generators of $\bigoplus_{j>0} B_j$ vanish at p' and p' is the vertex, a contradiction. Thus p is not the vertex. Let h' be the greatest common divisor of the degrees of the homogeneous elements of B not vanishing at p'. Any homogeneous element of D not vanishing at p does not vanish at p', so $h' \mid h_p$. If $h_p = 1$, then $h' = 1$. If $h_p = 2$, then $p \in \operatorname{Spec} D/(D_{\mathrm{odd}})$, so $u(p')^2 = s(p) \neq 0$, and $h' \mid \deg u = k$. Since k is odd and $h' \mid 2$, $h' = 1$. By Lemma 9.2(1), the stabilizer of p' is trivial. □

Under the hypotheses of Theorem 9.1, after the double cover, $\mathcal{O}_{\operatorname{Proj} B}(1)$ is invertible with $(\mathcal{O}_{\operatorname{Proj} B}(1)^{d-1}) = 2$, and the bound of Fujita on the base locus of a linear system shows that the linear system given by D_1 has finite base locus. If the grading of D is free outside the vertex, then $\mathcal{O}_{\operatorname{Proj} D}(1)$ is invertible with $(\mathcal{O}_{\operatorname{Proj} D}(1)^{d-1}) = 1$, and the bound of Fujita shows that the base locus is empty. Thus, after the double cover, the base locus is finite instead of empty.

Lemma 9.4. *Notation and conditions as in Construction-Proposition 9.3, and assume moreover that* $\operatorname{Hilb}_D(t) = (1-t)^{-d}$. *Then* $\dim D/(D_1) \leq 1$, *where* (D_1) *is the ideal of* D *generated by* D_1.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we show that $\operatorname{Proj} B$ is a union of integral components $Y_1, \dots, Y_c$ with $\sum_{i=1}^{c} ((\mathcal{O}_{\operatorname{Proj} B}(1)|_{Y_i})^{d-1}) = 2$.

Let $Y := \operatorname{Proj} B$ and $\mathcal{L} := \mathcal{O}_Y(1)$. By Construction-Proposition 9.3 and Lemma 9.2(2), $\mathcal{L}$ is an ample invertible sheaf, $\mathcal{O}_Y(j) \cong \mathcal{L}^{\otimes j}$ for any integer j, and $B_j \to H^0(Y, \mathcal{L}^{\otimes j})$ is an isomorphism for any sufficiently large j. Since B is reduced, Y is reduced. The minimal primes $\mathfrak{p}_1, \dots, \mathfrak{p}_c$ of B are homogeneous [BH98, Lemma 1.5.6(b)(ii)], and the irreducible components of Y are $Y_i := \operatorname{Proj}(B/\mathfrak{p}_i)$ for $1 \leq i \leq c$. Since B is equidimensional of dimension d, Lemma 9.2(3), applied to $B/\mathfrak{p}_i$, implies that each Y_i is an integral projective scheme with $\dim Y_i = d-1$.

Since $\operatorname{Hilb}_B(t) = (1+t^k)\operatorname{Hilb}_D(t) = (1+t^k)(1-t)^{-d}$,

$$\dim_{\mathbb{C}} B_j = \binom{j+d-1}{d-1} + \binom{j-k+d-1}{d-1} \quad \text{for } j \geq k,$$

which is a polynomial in j of degree $d-1$ with leading coefficient $2/(d-1)!$.

For an integral projective scheme Z over $\mathbb{C}$ with $\dim Z = d-1$ and an ample invertible sheaf $\mathcal{M}$ on Z, let $(\mathcal{M}^{d-1})$ be the number $d(Z, \mathcal{M})$ of [Fuj82, (1.2)], that is, $(d-1)!$ times the coefficient of j^{d-1} in the polynomial $j \to \chi(Z, \mathcal{M}^{\otimes j})$. It is an integer [Fuj82, (1.2)]. For $1 \leq i \leq c$, let $\mathcal{L}_i := \mathcal{L}|_{Y_i}$, which is ample [Har77, III, Exercise 5.7(a)].

Let $e \geq 1$ be an integer such that $B^{(e)}$ is generated by B_e (Lemma 9.2(3)). By Lemma 9.2(2) and (3), $Y \cong \operatorname{Proj} B^{(e)} \subset \mathbb{P}(B_e^\vee)$ and

$$\mathcal{O}_{\mathbb{P}(B_e^\vee)}(1)|_Y \cong \mathcal{O}_Y(e) \cong \mathcal{L}^{\otimes e}.$$

Since $B^{(e)}$ is reduced, $B^{(e)}$ is the homogeneous coordinate ring of the algebraic set $Y \subset \mathbb{P}(B_e^\vee)$, and $B^{(e)}/(\mathfrak{p}_i \cap B^{(e)})$ is the homogeneous coordinate ring of Y_i for $1 \leq i \leq c$. By [Har77, II, Exercise 5.9(b)] and [Har77, III, Exercise 5.2(b)], the Hilbert polynomial of Y_i in the sense of [Har77, I, §7] is $j \to \chi(Y_i, \mathcal{L}_i^{\otimes ej})$, so

$$\deg Y_i = e^{d-1}(\mathcal{L}_i^{d-1}).$$

The Hilbert polynomial of Y is equal to $\dim_{\mathbb{C}} B_{ej}$ for any sufficiently large j, so the formula for $\dim_{\mathbb{C}} B_j$ above implies $\deg Y = 2e^{d-1}$. Since $\dim(Y_i \cap Y_{i'}) < d-1$ for $i \neq i'$, induction on c with [Har77, I, Proposition 7.6(b)] implies $\deg Y = \sum_{i=1}^{c} \deg Y_i$, and [Har77, I, Proposition 7.6(a)] implies $\deg Y_i > 0$ for $1 \leq i \leq c$. Thus

$$\sum_{i=1}^{c} (\mathcal{L}_i^{d-1}) = 2,$$

where $(\mathcal{L}_i^{d-1})$ is a positive integer for $1 \le i \le c$. Hence one of the following holds.

- (**Case 1**) $c = 1$, that is, Y is integral, and $(\mathcal{L}^{d-1}) = 2$.
- (**Case 2**) $c = 2$, $Y = Y_1 \cup Y_2$, and $(\mathcal{L}_1^{d-1}) = (\mathcal{L}_2^{d-1}) = 1$.

Step 2. In this step we introduce the linear system $\mathcal{W}$ defined by D_1 and show that $\dim_{\mathbb{C}} \mathcal{W} = d$.

Since $k \ge 3$, $B_1 = D_1$. Let $\mathcal{W} \subset H^0(Y, \mathcal{L})$ be the image of D_1. Since $\mathrm{Hilb}_D(t) = (1-t)^{-d}$, $\dim_{\mathbb{C}} D_1 = d$. We show that $D_1 \to H^0(Y, \mathcal{L})$ is injective. Let $g \in D_1$ with image zero, and let p be a closed point of $\operatorname{Spec} B$ other than the vertex. Choose a homogeneous $f \in B$ of positive degree with $f(p) \ne 0$. The restriction of g to $D_+(f)$ is zero, so $f^n g = 0$ for some $n \ge 0$, and hence $g(p) = 0$. Thus g vanishes at any closed point of $\operatorname{Spec} B$ other than the vertex. Since $\dim V(\mathfrak{p}_i) = d \ge 2$, the closed points of $V(\mathfrak{p}_i)$ other than the vertex are dense in $V(\mathfrak{p}_i)$, so $g \in \mathfrak{p}_i$ for $1 \le i \le c$. Since B is reduced, $g = 0$. Hence $\dim_{\mathbb{C}} \mathcal{W} = d$.

Moreover, as closed subsets of Y,

$$\mathrm{Bs}(\mathcal{W}) = \mathrm{Proj}\big(B/(D_1)B\big),$$

since an element of D_1 vanishes at a point $y \in Y$ if and only if it is contained in the homogeneous prime ideal of y, the stalk of $\mathcal{L}$ at y being free of rank one.

Step 3. In this step we show that $\mathrm{Bs}(\mathcal{W})$ is a finite set.

We use [Fuj82, Theorem (2.1)] in the following form, see [Fuj82, (1.1), (1.4)] for the definitions: for an integral projective scheme X over $\mathbb{C}$, an ample invertible sheaf $\mathcal{L}'$ on X, and a non-zero linear subspace $\mathcal{W}' \subset H^0(X, \mathcal{L}')$,

$$\dim \mathrm{Bs}(\mathcal{W}') < \dim X + (\mathcal{L}'^{\dim X}) - \dim_{\mathbb{C}} \mathcal{W}', \tag{9.1}$$

where $\dim \emptyset = -1$. The right-hand side is the Δ-genus $\Delta(X, \mathcal{W}')$ of [Fuj82, (1.4)].

In **Case 1**, **Step 2** and (9.1) with $X = Y$, $\mathcal{L}' = \mathcal{L}$, $\mathcal{W}' = \mathcal{W}$ imply

$$\dim \mathrm{Bs}(\mathcal{W}) < (d-1) + 2 - d = 1,$$

so $\mathrm{Bs}(\mathcal{W})$ is finite.

In **Case 2**, we first show that $Y_1 \cap Y_2$ contains a prime divisor P of both Y_1 and Y_2. Since B is reduced, $\mathfrak{p}_1 \cap \mathfrak{p}_2 = 0$. The ideal $\mathfrak{p}_1 + \mathfrak{p}_2$ is homogeneous and contained in $B_+ := \bigoplus_{j>0} B_j$. Let Q be a minimal prime of $\mathfrak{p}_1 + \mathfrak{p}_2$, which is homogeneous [BH98, Lemma 1.5.6(b)(ii)]. Since $\mathfrak{p}_1 \not\subset \mathfrak{p}_2$ and $\mathfrak{p}_2 \not\subset \mathfrak{p}_1$, $\mathfrak{p}_i \subsetneq Q$ for $i = 1, 2$, so $\operatorname{ht} Q \ge 1$. Suppose that $\operatorname{ht} Q \ge 2$. Localizing the exact sequence

$$0 \to B \to B/\mathfrak{p}_1 \oplus B/\mathfrak{p}_2 \to B/(\mathfrak{p}_1 + \mathfrak{p}_2) \to 0$$

at Q and taking local cohomology H^i_Q supported at the maximal ideal of B_Q, we obtain an exact sequence

$$H^0_Q\big((B/\mathfrak{p}_1)_Q \oplus (B/\mathfrak{p}_2)_Q\big) \to H^0_Q\big((B/(\mathfrak{p}_1 + \mathfrak{p}_2))_Q\big) \to H^1_Q(B_Q).$$

The ring $(B/(\mathfrak{p}_1 + \mathfrak{p}_2))_Q$ is a non-zero Artinian local ring since Q is minimal over $\mathfrak{p}_1 + \mathfrak{p}_2$, so the middle term is non-zero. For $i = 1, 2$, $(B/\mathfrak{p}_i)_Q$ is a local domain of positive dimension since $\mathfrak{p}_i \subsetneq Q$, so the depth of $(B/\mathfrak{p}_i)_Q$ is at least one and the left term vanishes [BH98, Theorem 3.5.7(a)]. Hence $H^1_Q(B_Q) \ne 0$. But B_Q is a Cohen–Macaulay local ring of dimension $\operatorname{ht} Q \ge 2$, so $H^1_Q(B_Q) = 0$ [BH98, Theorem 3.5.7(a)], a contradiction. Thus $\operatorname{ht} Q = 1$ (compare [Har62]). Since $Q \subset B_+$ and B_{B_+} is a Cohen–Macaulay local ring of dimension d, $\dim B/Q = d - 1$ by [Mat89, Theorem 17.4(i)] for B_{B_+}. Hence $P := \mathrm{Proj}(B/Q) \subset Y_1 \cap Y_2$ is an irreducible closed subset with $\dim P = d - 2$ (Lemma 9.2(3)), that is, a prime divisor of both Y_1 and Y_2.

Next, we show that $\dim_{\mathbb{C}} \operatorname{Ker}\big(\mathcal{W} \to H^0(Y_1, \mathcal{L}_1)\big) \le 1$, where the map is the restriction. Let $\nu : Y_2^\nu \to Y_2$ be the normalization. Since ν is finite and birational, $\nu^*\mathcal{L}_2$ is ample [Har77, III, Exercise 5.7(d)] and $((\nu^*\mathcal{L}_2)^{d-1}) = (\mathcal{L}_2^{d-1}) = 1$ [KM98, Proposition 1.35(6)]. Fix a prime divisor P' of Y_2^ν with $\nu(P') = P$, which exists since ν is finite and surjective. Let $\sigma \ne 0$ be an element of the kernel. Since $Y = Y_1 \cup Y_2$ is reduced and $\sigma|_{Y_1} = 0$, $\sigma|_{Y_2} \ne 0$, and $\sigma|_{Y_2}$ vanishes on $Y_1 \cap Y_2 \supset P$. Since Y_2 is integral and ν is birational, $\nu^*(\sigma|_{Y_2}) \ne 0$. Let $F_\sigma \ge 0$ be the divisor of zeros of $\nu^*(\sigma|_{Y_2})$

on the normal variety Y_2^ν. Then $F_\sigma \geq P'$. Since $\nu^*\mathcal{L}_2$ is ample, $((\nu^*\mathcal{L}_2)^{d-2} \cdot E) := ((\nu^*\mathcal{L}_2|_E)^{d-2})$ is a positive integer for every prime divisor E of Y_2^ν [KM98, Theorem 1.37]. Since F_σ is the divisor of a section of $\nu^*\mathcal{L}_2$, $((\nu^*\mathcal{L}_2)^{d-1}) = ((\nu^*\mathcal{L}_2)^{d-2} \cdot F_\sigma)$ [KM98, Proposition 1.35(5)]. Since intersection numbers are additive over the components of F_σ counted with multiplicity [Laz04, Section 1.1.C] and $F_\sigma - P' \geq 0$,

$$1 = ((\nu^*\mathcal{L}_2)^{d-1}) = ((\nu^*\mathcal{L}_2)^{d-2} \cdot F_\sigma) \geq ((\nu^*\mathcal{L}_2)^{d-2} \cdot P') \geq 1.$$

Hence $F_\sigma = P'$. Let $\sigma' \neq 0$ be another element of the kernel. Then $\nu^*(\sigma'|_{Y_2})/\nu^*(\sigma|_{Y_2})$ is a rational function $\phi \in \mathbb{C}(Y_2^\nu)$ whose divisor is $F_{\sigma'} - F_\sigma = 0$. Since Y_2^ν is normal, ϕ and ϕ^{-1} are regular on any affine open subset of Y_2^ν [Har77, II, Proposition 6.3A]. Since Y_2^ν is a proper integral scheme over $\mathbb{C}$, $\Gamma(Y_2^\nu, \mathcal{O}_{Y_2^\nu}) = \mathbb{C}$ [Har77, II, Exercise 4.5(d)], so ϕ is a constant $a \in \mathbb{C}^*$. Thus $\nu^*((\sigma' - a\sigma)|_{Y_2}) = 0$, so $(\sigma' - a\sigma)|_{Y_2} = 0$. Since also $(\sigma' - a\sigma)|_{Y_1} = 0$ and Y is reduced, $\sigma' = a\sigma$. Since the situation is symmetric in Y_1 and Y_2, also $\dim_{\mathbb{C}} \operatorname{Ker}\big(\mathcal{W} \to H^0(Y_2, \mathcal{L}_2)\big) \leq 1$.

For $i = 1, 2$, let $\mathcal{W}_i \subset H^0(Y_i, \mathcal{L}_i)$ be the image of $\mathcal{W}$. Then $\dim_{\mathbb{C}} \mathcal{W}_i \geq d - 1 \geq 1$, and (9.1), applied to Y_i, $\mathcal{L}_i$, and $\mathcal{W}_i$, implies

$$\dim \operatorname{Bs}(\mathcal{W}_i) < (d-1) + (\mathcal{L}_i^{d-1}) - \dim_{\mathbb{C}} \mathcal{W}_i \leq (d-1) + 1 - (d-1) = 1.$$

Since $\operatorname{Bs}(\mathcal{W}) \cap Y_i = \operatorname{Bs}(\mathcal{W}_i)$ for $i = 1, 2$, $\operatorname{Bs}(\mathcal{W}) = \operatorname{Bs}(\mathcal{W}_1) \cup \operatorname{Bs}(\mathcal{W}_2)$ is finite.

Step 4. In this step we conclude the proof.

Recall from **Step 2** that $\operatorname{Bs}(\mathcal{W}) = \operatorname{Proj}(B/(D_1)B)$ as sets. By **Step 3**, $\operatorname{Proj}(B/(D_1)B)$ is a finite set. If $\dim B/(D_1)B \geq 2$, then some minimal prime $\mathfrak{q}$ of $(D_1)B$, which is homogeneous [BH98, Lemma 1.5.6(b)(ii)], satisfies $\dim B/\mathfrak{q} \geq 2$, and $\operatorname{Proj}(B/\mathfrak{q}) \subset \operatorname{Proj}(B/(D_1)B)$ satisfies, by Lemma 9.2(3), $\dim \operatorname{Proj}(B/\mathfrak{q}) = \dim B/\mathfrak{q} - 1 \geq 1$, a contradiction. Thus $\dim B/(D_1)B \leq 1$. Since

$$B/(D_1)B = \big(D/(D_1)\big)[u]/(u^2 - \bar{s}),$$

where $\bar{s}$ is the image of s in $D/(D_1)$, is a free $D/(D_1)$-module with basis $1, u$, it is integral over $D/(D_1)$ and contains $D/(D_1)$. Hence $\dim D/(D_1) = \dim B/(D_1)B \leq 1$. □

The next lemma is used in the proof of Theorem 9.1 for the one-dimensional ring $D/(x_1, \dots, x_{d-1})$, where $x_1, \dots, x_{d-1} \in D_1$ form a regular sequence on D.

Lemma 9.5. *Let $R = \bigoplus_{j \geq 0} R_j$ be a finitely generated graded $\mathbb{C}$-algebra with $R_0 = \mathbb{C}$. Assume that R is Cohen–Macaulay of dimension 1, that $\operatorname{Hilb}_R(t) = (1-t)^{-1}$, and that every closed point of $\operatorname{Spec} R$ other than the vertex has stabilizer of order at most 2 for the $\mathbb{G}_m$-action induced by the grading. Then for all non-zero $x \in R_1$ and $y \in R_2$, the subalgebra $\mathbb{C}[y] \subset R$ is a polynomial ring and $R = \mathbb{C}[y] \oplus \mathbb{C}[y]x$. In particular R is generated by R_1 and R_2.*

Proof. Let $R_+ := \bigoplus_{j>0} R_j$. We use twice that a graded R-module M with $M_{R_+} = 0$ is zero, since the annihilator of a non-zero homogeneous element of M is a proper homogeneous ideal of R, hence is contained in R_+ because $R_0 = \mathbb{C}$.

Let $R^{(2)} := \bigoplus_{j \geq 0} R_{2j}$ be the second Veronese subring, graded by $R_j^{(2)} := R_{2j}$. Let $r_1, \dots, r_l$ be homogeneous generators of the $\mathbb{C}$-algebra R of positive degree. Since any monomial $r_1^{a_1} \cdots r_l^{a_l}$ is the product of $\prod_i (r_i^2)^{\lfloor a_i/2 \rfloor}$ and $\prod_{a_i \text{ odd}} r_i$, the algebra $R^{(2)}$ is generated by $r_1^2, \dots, r_l^2$ and the products $\prod_{i \in I} r_i$ of even degree for $I \subset \{1, \dots, l\}$, and R is generated as an $R^{(2)}$-module by the products $\prod_{i \in I} r_i$. Thus $R^{(2)}$ is a finitely generated graded $\mathbb{C}$-algebra with $R_0^{(2)} = \mathbb{C}$, R is finite over $R^{(2)}$, $\dim R^{(2)} = \dim R = 1$, and $\operatorname{Hilb}_{R^{(2)}}(t) = (1-t)^{-1}$.

We show that any closed point of $\operatorname{Spec} R^{(2)}$ other than the vertex has trivial stabilizer. Let $\bar{p}$ be such a point. Since R is finite over $R^{(2)}$, there exists a closed point $p \in \operatorname{Spec} R$ over $\bar{p}$, and p is not the vertex because some homogeneous element of $R^{(2)}$ of positive degree does not vanish at $\bar{p}$. Let h be the greatest common divisor of the degrees of the homogeneous elements of R not vanishing at p, and let h' be the greatest common divisor of the degrees in $R^{(2)}$ of the homogeneous elements of $R^{(2)}$ not vanishing at $\bar{p}$. By Lemma 9.2(1), $h \in \{1, 2\}$. For any

homogeneous $g \in R$ with $g(p) \neq 0$, the element $g^2 \in R^{(2)}$ does not vanish at $\bar{p}$ and has degree $\deg g$ in $R^{(2)}$, so $h' \mid h$. If $h = 1$, then $h' = 1$. If $h = 2$, then any homogeneous $g \in R$ with $g(p) \neq 0$ has even degree, so $g \in R^{(2)}$ has degree $\deg g/2$ in $R^{(2)}$, and the greatest common divisor of the degrees $\deg g/2$ is $h/2 = 1$. Hence $h' = 1$. By Lemma 9.2(1), the stabilizer of $\bar{p}$ is trivial.

Let $Y' := \operatorname{Proj} R^{(2)}$ and $\mathcal{L}' := \mathcal{O}_{Y'}(1)$. By Lemma 9.2(2), $\mathcal{L}'$ is an ample invertible sheaf, and there exists an integer j_0 such that $R^{(2)}_j \to H^0(Y', \mathcal{L}'^{\otimes j})$ is an isomorphism for $j \geq j_0$. Since $\dim \operatorname{Proj} R^{(2)} = \dim R^{(2)} - 1 = 0$ (Lemma 9.2(3)), $Y' = \operatorname{Spec} G$ for an Artinian $\mathbb{C}$-algebra G with $\dim_{\mathbb{C}} G < +\infty$. An invertible module over the semilocal ring G is free of rank one, so $h^0(Y', \mathcal{L}'^{\otimes j}) = \dim_{\mathbb{C}} G$. For $j \geq j_0$, $\dim_{\mathbb{C}} G = h^0(Y', \mathcal{L}'^{\otimes j}) = \dim_{\mathbb{C}} R^{(2)}_j = 1$. Hence $G = \mathbb{C}$, that is, Y' is a single reduced point, and the multiplication maps

$$H^0(Y', \mathcal{L}'^{\otimes a}) \otimes H^0(Y', \mathcal{L}'^{\otimes b}) \to H^0(Y', \mathcal{L}'^{\otimes (a+b)})$$

are isomorphisms of one-dimensional vector spaces for any $a, b \geq 0$.

We show that $R^{(2)}$ is reduced. Since the minimal primes of $R^{(2)}$ are homogeneous [BH98, Lemma 1.5.6(b)(ii)], the nilradical of $R^{(2)}$ is homogeneous, so it suffices to show that any homogeneous nilpotent $g \in R^{(2)}$ is zero. Since $R^{(2)}_0 = \mathbb{C}$, we may assume that $\deg g = e > 0$ in $R^{(2)}$. The image $\bar{g} \in H^0(Y', \mathcal{L}'^{\otimes e}) \cong \mathbb{C}$ of g satisfies $\bar{g}^n = 0$ for some n, so $\bar{g} = 0$. For $j \geq j_0$ and $a \in R^{(2)}_j$, the image of ga in $H^0(Y', \mathcal{L}'^{\otimes (e+j)})$ is $\bar{g}\bar{a} = 0$, and since $e + j \geq j_0$, $ga = 0$. Thus $g \cdot \bigoplus_{j \geq j_0} R^{(2)}_j = 0$. Any product of $l+1$ of the generators $r_1, \dots, r_l$ contains some r_i twice, so $(R_+)^{l+1} \subset (r_1^2, \dots, r_l^2)R$. Since $r_i^2 \in R^{(2)}$ has positive degree in $R^{(2)}$,

$$(R_+)^{(l+1)j_0} \subset \Big(\bigoplus_{j \geq j_0} R^{(2)}_j \Big) R,$$

and hence $g(R_+)^{(l+1)j_0} = 0$. Thus the image of g in the one-dimensional Cohen–Macaulay local ring R_{R_+} is annihilated by a power of the maximal ideal, so it belongs to $H^0_{R_+}(R_{R_+}) = 0$ [BH98, Theorem 3.5.7(a)]. Applying the remark at the beginning of the proof to the graded module gR, we obtain $g = 0$.

Let $y \in R_2 = R^{(2)}_1$ be non-zero. Since $R^{(2)}$ is reduced, $y^j \neq 0$ for any $j \geq 0$, and since $\dim_{\mathbb{C}} R^{(2)}_j = 1$, $R^{(2)}_j = \mathbb{C}y^j$. Thus $R^{(2)} = \mathbb{C}[y]$, and $\mathbb{C}[y]$ is a polynomial ring since the non-zero elements y^j have distinct degrees.

Since R is finite over $\mathbb{C}[y]$, R/yR is finite over $\mathbb{C}$, so y is a system of parameters of the one-dimensional Cohen–Macaulay local ring R_{R_+}. Hence y is a nonzerodivisor on R_{R_+} [Mat89, Theorem 17.4(iii)], and by the remark at the beginning of the proof applied to the annihilator of y, y is a nonzerodivisor on R. Thus

$$\operatorname{Hilb}_{R/yR}(t) = (1 - t^2) \operatorname{Hilb}_R(t) = 1 + t.$$

Let $x \in R_1$ be non-zero. Since $(yR)_1 = 0$, the image of x spans $(R/yR)_1$, so R/yR is spanned by the images of 1 and x, that is, $R = (\mathbb{C}[y] + \mathbb{C}[y]x) + yR$. The ring $\mathbb{C}[y]$, graded by $\deg y = 2$, is positively graded with degree-zero part $\mathbb{C}$, so it is a *local ring with *maximal ideal $y\mathbb{C}[y]$ [BH98, Example 1.5.14(b)]. Since R is a finite graded $\mathbb{C}[y]$-module, the graded Nakayama lemma [BH98, Exercise 1.5.24(a)] implies $R = \mathbb{C}[y] + \mathbb{C}[y]x$. The natural surjection $\mathbb{C}[y] \oplus \mathbb{C}[y](-1) \to R$ of graded $\mathbb{C}[y]$-modules, sending the basis elements $(1, 0)$ and $(0, 1)$ to 1 and x, is an isomorphism since

$$\operatorname{Hilb}_{\mathbb{C}[y] \oplus \mathbb{C}[y](-1)}(t) = (1 + t)/(1 - t^2) = (1 - t)^{-1} = \operatorname{Hilb}_R(t).$$

Hence $R = \mathbb{C}[y] \oplus \mathbb{C}[y]x$. □

Proof of Theorem 9.1. The proof is long, so we divide it into several steps.

Step 1. In this step we cut D down to a one-dimensional ring by a regular sequence in D_1.

By Construction-Proposition 9.3 and Lemma 9.4, $\dim D/(D_1) \leq 1$. The ideal (D_1), and the associated primes of D/I for any ideal I generated by elements of D_1, are homogeneous [BH98, Lemma 1.5.6(b)(ii)], hence contained in $\mathfrak{m}$. Thus heights and dimensions of (D_1) and of the associated primes above may be computed in the Cohen–Macaulay local ring $D_{\mathfrak{m}}$ with $\dim D_{\mathfrak{m}} = d$, and [Mat89, Theorem 17.4(i)] implies

$$\operatorname{ht}(D_1) = d - \dim D/(D_1) \geq d - 1.$$

We construct $x_1, \ldots, x_{d-1} \in D_1$ forming a regular sequence on D by induction. Suppose that $0 \leq j < d-1$ and that $x_1, \ldots, x_j \in D_1$ form a regular sequence on D. By [Mat89, Theorems 17.4(i) and 17.6], any associated prime $\mathfrak{p}$ of $D/(x_1, \ldots, x_j)$ satisfies $\operatorname{ht}\mathfrak{p} = j < d-1 \leq \operatorname{ht}(D_1)$, so $D_1 \not\subset \mathfrak{p}$ and $D_1 \cap \mathfrak{p}$ is a proper linear subspace of D_1. Since $\mathbb{C}$ is infinite and there exist only finitely many associated primes, there exists $x_{j+1} \in D_1$ not contained in any of them, and x_{j+1} is a nonzerodivisor on $D/(x_1, \ldots, x_j)$.

Let $R := D/(x_1, \ldots, x_{d-1})$. Then R is Cohen–Macaulay [Mat89, Theorem 17.3(ii)], and $\dim R = d - (d-1) = 1$ by [Mat89, Theorem 17.4(i)]. For $0 \leq j < d-1$, the exact sequence of graded modules

$$0 \to \big(D/(x_1, \ldots, x_j)\big)(-1) \xrightarrow{\cdot x_{j+1}} D/(x_1, \ldots, x_j) \to D/(x_1, \ldots, x_{j+1}) \to 0$$

implies that $\operatorname{Hilb}_{D/(x_1,\ldots,x_{j+1})}(t) = (1-t)\operatorname{Hilb}_{D/(x_1,\ldots,x_j)}(t)$. Hence

$$\operatorname{Hilb}_R(t) = (1-t)^{d-1}\operatorname{Hilb}_D(t) = (1-t)^{-1}.$$

Since $x_1, \ldots, x_{d-1}$ are homogeneous, $\operatorname{Spec} R$ is a $\mathbb{G}_m$-stable closed subscheme of $\operatorname{Spec} D$ containing the vertex, so any closed point of $\operatorname{Spec} R$ other than the vertex has stabilizer of order at most 2.

Step 2. In this step we show that D is generated by D_1 and one element of D_2.

Since $D_0 = \mathbb{C}$, the degree-one part of $(x_1, \ldots, x_{d-1})$ is the span of $x_1, \ldots, x_{d-1}$. Since $\dim_{\mathbb{C}} D_1 = d$ and $\dim_{\mathbb{C}} R_1 = 1$, the elements $x_1, \ldots, x_{d-1}$ are linearly independent, and there exists $x_d \in D_1$ such that $x_1, \ldots, x_d$ is a basis of D_1 and the image $\bar{x}_d$ of x_d in R_1 is non-zero. Since $\dim_{\mathbb{C}} R_2 = 1$, there exists $b \in D_2$ whose image $\bar{b}$ in R_2 is non-zero. By Lemma 9.5,

$$R = \mathbb{C}[\bar{b}] \oplus \mathbb{C}[\bar{b}]\bar{x}_d.$$

Let $D' \subset D$ be the $\mathbb{C}$-subalgebra generated by $x_1, \ldots, x_d, b$. We show $D_j \subset D'$ by induction on j, the case $j = 0$ holding since $D_0 = \mathbb{C}$. Let $j > 0$ and $f \in D_j$. The image of f in R_j is $a\bar{b}^{j/2}$ if j is even and $a\bar{b}^{(j-1)/2}\bar{x}_d$ if j is odd, for some $a \in \mathbb{C}$. Let $g := ab^{j/2}$ or $g := ab^{(j-1)/2}x_d$, respectively. Then $g \in D'_j$ and $f - g \in (x_1, \ldots, x_{d-1}) \cap D_j = \sum_{i=1}^{d-1} x_i D_{j-1}$. By induction, $D_{j-1} \subset D'$, so $f \in D'$. Hence $D = D'$.

Step 3. In this step we identify the relation.

Suppose first that b is contained in the span of the products $x_i x_{i'}$ for $1 \leq i \leq i' \leq d$. Then D is generated by $x_1, \ldots, x_d$ by **Step 2**, so the graded surjection $\mathbb{C}[x_1, \ldots, x_d] \to D$ from the polynomial ring with $\deg x_i = 1$ is an isomorphism since both sides have Hilbert series $(1-t)^{-d}$. Thus D is a polynomial ring in d variables of degree 1.

Suppose next that b is not contained in the span of the products $x_i x_{i'}$. Let $P := \mathbb{C}[x_1, \ldots, x_d, b]$ be the polynomial ring with $\deg x_i = 1$ for $1 \leq i \leq d$ and $\deg b = 2$, and let $\pi : P \to D$ be the graded surjection with $\pi(x_i) = x_i$ for $1 \leq i \leq d$ and $\pi(b) = b$. Since $\operatorname{Hilb}_P(t) = (1-t)^{-d}(1-t^2)^{-1}$,

$$\dim_{\mathbb{C}} P_1 = d = \dim_{\mathbb{C}} D_1, \qquad \dim_{\mathbb{C}} P_2 = \binom{d+1}{2} + 1 = \dim_{\mathbb{C}} D_2 + 1.$$

Thus $\pi_1 : P_1 \to D_1$ is an isomorphism, and the kernel of π in degree 2 is one-dimensional. Let q be a non-zero element of $\operatorname{Ker}(\pi) \cap P_2$, and write $q = q_0 + cb$ with $q_0 \in \mathbb{C}[x_1, \ldots, x_d]$ a quadratic form and $c \in \mathbb{C}$. If $c \neq 0$, then $\pi(b) = -c^{-1}\pi(q_0)$ is contained in the span of the products $x_i x_{i'}$ in

D, contrary to the assumption. Thus $c = 0$, and $q = q_0 \in \mathbb{C}[x_1, \dots, x_d]$ is a non-zero quadratic form. Since P is a domain, q is a nonzerodivisor of P, so

$$\mathrm{Hilb}_{P/(q)}(t) = \frac{1-t^2}{(1-t)^d(1-t^2)} = (1-t)^{-d} = \mathrm{Hilb}_D(t).$$

Hence the induced graded surjection $P/(q) \to D$ is an isomorphism in any degree, and

$$D \cong \mathbb{C}[x_1, \dots, x_d, b]/(q).$$

We show that $\operatorname{rank} q \geq 2$. Suppose that $\operatorname{rank} q \leq 1$. Since $q \neq 0$, $q = a\ell^2$ for some $a \in \mathbb{C}^*$ and some non-zero $\ell \in \mathbb{C}[x_1, \dots, x_d]$ of degree 1. Since π_1 is an isomorphism, $\pi(\ell) \neq 0$, while $\pi(\ell)^2 = a^{-1}\pi(q) = 0$. This contradicts the assumption that D is reduced.

Step 4. In this step we show that $D_{\mathfrak{m}}$ is a complete intersection ring and $\dim_{\mathbb{C}} \mathfrak{m}/\mathfrak{m}^2 \leq d+1$.

By **Step 2**, D is generated by D_1 and D_2, and the ideal $\mathfrak{m}$ is generated by $x_1, \dots, x_d, b$. Thus $\mathfrak{m}/\mathfrak{m}^2$ is spanned by the images of $x_1, \dots, x_d, b$, and $\dim_{\mathbb{C}} \mathfrak{m}/\mathfrak{m}^2 \leq d+1$. If D is a polynomial ring in d variables, then $D_{\mathfrak{m}}$ is a regular local ring, hence a complete intersection ring [BH98, Proposition 3.1.20]. Otherwise, let $\mathfrak{n} := (x_1, \dots, x_d, b) \subset P$. Then $D_{\mathfrak{m}} \cong P_{\mathfrak{n}}/(q)$, where $P_{\mathfrak{n}}$ is a regular local ring and $q \neq 0$ is a nonzerodivisor of $P_{\mathfrak{n}}$, so $D_{\mathfrak{m}}$ is a complete intersection ring [BH98, Theorem 2.3.3(c)]. □

10. Proof of the main theorem

In this section, we prove Theorem 1.2 and Corollary 1.3.

Proof of Theorem 1.2. Part (2) is Proposition 3.5.

We prove (1). Let $(X \ni x)$ be a klt germ of dimension d which is neither smooth nor analytically an ordinary double point. If $d = 4$, then $\widehat{\mathrm{vol}}(X \ni x) \leq Q_4$ by Proposition 5.1.

Assume that $d \geq 5$, and suppose that $\widehat{\mathrm{vol}}(X \ni x) > Q_d$. Let $N := d-1$, and let $C = \operatorname{Spec} S$, o, $\mathbb{T}$, ξ, and η be as in Construction-Proposition 4.1 for $(X \ni x)$. Since $(X \ni x)$ is not analytically an ordinary double point, $(C \ni o)$ is not a local complete intersection by Construction-Proposition 4.1(4).

If $\mathbb{R}_{>0}\xi$ contains a primitive element ζ_0 of $\mathfrak{t}_{\mathbb{Z}}$, then the constant sequence $\zeta_i = \zeta_0$ is a fast primitive approximation of ξ by Lemma 6.2(2), and Proposition 6.4, the definition of $m_{\max}$ in Definition 6.1, and Lemma 2.9(3) imply that $C\backslash\{o\}$ contains a closed point whose stabilizer for the $\mathbb{G}_m$-action defined by ζ_0 is μ_m with $m > r_{\zeta_0}/N$. Otherwise, let (ζ_i) be a fast primitive approximation of ξ (Lemma 6.2(1)). By Proposition 6.4, the definition of $m_{\max}$, and Lemma 2.9(3), for any $i \gg 0$, $C\backslash\{o\}$ contains a closed point whose stabilizer for the $\mathbb{G}_m$-action defined by ζ_i is μ_m with $m > r_{\zeta_i}/N$. Since removing finitely many terms of (ζ_i) yields a fast primitive approximation of ξ (Definition 6.1), possibly replacing $(\zeta_i)_{i\geq 1}$ with $(\zeta_{i+i_0})_{i\geq 1}$ for some $i_0 \geq 0$, we may assume that such a closed point exists for any i. Since $m > r_i/N$ implies $m \geq r_i/N$, there exist (ζ_i), r_i, x_i, and m_i as in Set-up 6.5.

By Proposition 6.10, we fix i such that the conclusions of Proposition 6.10 hold for i, and we take the data of Construction-Proposition 6.7 for this i. Let $\bar{S}$, v, and D be as in Construction-Proposition 7.1, and let R, λ, and v_* be as in Construction-Proposition 7.4. We consider the T-grading $D = \bigoplus_{q\geq 0} D_q$ of Construction-Proposition 7.1(3), and let $\mathfrak{m} := \bigoplus_{q>0} D_q$. Since $d \geq 5$, we have $N \geq 4$. We check the hypotheses of Theorem 9.1, applied to D with N in place of d.

By Construction-Proposition 7.1(3), D is a finitely generated equidimensional Gorenstein $\mathbb{C}$-algebra of dimension N with $D_0 = \mathbb{C}$ and $\mathrm{Hilb}_D(t) = (1-t)^{-N}$. Since D is Gorenstein, D is Cohen–Macaulay [BH98, Proposition 3.1.20]. By Lemma 7.3, D is reduced.

To use Proposition 7.10(3), we first construct data as in Construction-Proposition 7.7. By Lemma 7.5 and Lemma 7.8, $\operatorname{Spec} D$ has a closed point y_D other than the vertex whose stabilizer for the T-grading is μ_h with $h \geq 2$. Let $y := (y_D, 0) \in \operatorname{Spec} D[v_*]$. By Lemma 7.8, there exists a W-homogeneous element $f \in R$ of positive W-degree with $f(y) \neq 0$. Thus y_D, h, and f satisfy

the conditions of Construction-Proposition 7.7, and Proposition 7.10(3) implies that every closed point of $\operatorname{Spec} D$ other than the vertex has stabilizer of order at most 2 for the T-grading.

By Theorem 8.6, the subscheme $Z = \operatorname{Spec} D/(D_{\mathrm{odd}})$ of Construction-Proposition 8.3 satisfies $\dim Z = 1$.

Thus all hypotheses of Theorem 9.1, applied to D with N in place of d, hold. In particular, $D_{\mathfrak{m}}$ is a complete intersection ring. By Lemma 7.6, $(C \ni o)$ is a local complete intersection, a contradiction. Therefore $\widehat{\mathrm{vol}}(X \ni x) \le Q_d$. □

Proof of Corollary 1.3. By Lemma 2.7(5), $Q_d < 2(d-1)^d$. By Lemma 2.7(2),

$$\frac{2(d-1)^d}{d^d} = 2\left(\frac{d-1}{d}\right)^{d-1} \cdot \frac{d-1}{d} \le \frac{d-1}{d} < 1,$$

hence $2(d-1)^d < d^d$.

Let $(X \ni x)$ be a klt germ of dimension d such that $\widehat{\mathrm{vol}}(X \ni x) \ge Q_d$. If x is a smooth point of X, then $\widehat{\mathrm{vol}}(X \ni x) = d^d$ by Theorem 2.4(1). If $(X \ni x)$ is analytically an ordinary double point, then $x \in \operatorname{Sing}(X)$ (the completion of $\mathcal{O}_{X,x}$ is $\mathbb{C}[[z_0, \dots, z_d]]/(z_0^2 + \cdots + z_d^2)$, which is not regular), and $\widehat{\mathrm{vol}}(X \ni x) = 2(d-1)^d$ by Theorem 2.4(2). Otherwise, $\widehat{\mathrm{vol}}(X \ni x) \le Q_d$ by Theorem 1.2(1), hence $\widehat{\mathrm{vol}}(X \ni x) = Q_d$.

It remains to show that Q_d, $2(d-1)^d$, and d^d are normalized volumes of klt germs of dimension d. By Theorem 2.4(1), $\widehat{\mathrm{vol}}(\mathbb{A}^d \ni 0) = d^d$. By Theorem 1.2(2), Q_d is the normalized volume of a klt germ of dimension d. Let $\mathcal{Q} \subset \mathbb{P}^d$ be a smooth quadric hypersurface, and let $(X' \ni x')$ be the germ at the vertex of the affine cone over $\mathcal{Q}$ with respect to $\mathcal{O}_{\mathcal{Q}}(1)$. Then $(X' \ni x')$ is an ordinary double point of dimension d (by [Har77, III, Exercise 5.5(a)] the section ring is $\mathbb{C}[z_0, \dots, z_d]/(q)$ for an equation q of $\mathcal{Q}$, and $q = z_0^2 + \cdots + z_d^2$ after a linear change of coordinates since $\mathcal{Q}$ is smooth). Since $\mathcal{O}_{\mathcal{Q}}(-K_{\mathcal{Q}}) \cong \mathcal{O}_{\mathcal{Q}}(d-1)$ and $d - 1 > 0$, $(X' \ni x')$ is a klt germ [Kol13, Lemma 3.1(3)]. Since $x' \in \operatorname{Sing}(X')$ (as for $(X \ni x)$ in the second paragraph of this proof), the equality case of Theorem 2.4(2) implies that $\widehat{\mathrm{vol}}(X' \ni x') = 2(d-1)^d$. □

Appendix A. An exponential bound in four variables

We prove Lemma 6.9 from Section 6.

Proof of Lemma 6.9. The proof is long, so we divide it into several steps.

Step 1. In this step we reduce the lemma to an upper bound for a function Φ of one real variable, which is defined at the end of the step.

Recall that $\mathsf{X}_1, \dots, \mathsf{X}_4$ are independent exponential random variables with mean 1, and that $g(y) = 1$ for $y \le 0$ and $g(y) = (1+y)e^{-y}$ for $y > 0$. For $a = (a_1, \dots, a_4) \in \mathbb{R}^4$, let $a \cdot \mathsf{X} := a_1\mathsf{X}_1 + \cdots + a_4\mathsf{X}_4$, let $\|a\|$ be the Euclidean norm of a, and let

$$F(a) := \frac{3}{4} e^{4/3} \mathbb{E}\Big[g\Big(\frac{4}{3} + a \cdot \mathsf{X}\Big)\Big].$$

We need to show that $F(a) \le \frac{137}{72}$ for any $a \in \mathbb{R}^4$ with $a_1 + \cdots + a_4 = 0$. For $y > 0$,

$$\frac{d}{dy}\big((1+y)e^{-y}\big) = -ye^{-y} < 0,$$

so $0 \le g \le 1$. Since g is continuous, the dominated convergence theorem implies that F is continuous. Moreover,

$$F(0) = \frac{3}{4} e^{4/3} \cdot \frac{7}{3} e^{-4/3} = \frac{7}{4}.$$

We bound F near infinity. Let $a^{(1)}, a^{(2)}, \dots$ be vectors in $\mathbb{R}^4$ with $a_1^{(i)} + \cdots + a_4^{(i)} = 0$ and $\|a^{(i)}\| \to +\infty$, such that $F(a^{(i)})$ converges. Possibly passing to a subsequence, we may assume

that $a^{(i)}/\|a^{(i)}\|$ converges to a vector $b=(b_1,\dots,b_4)$. Then $\|b\|=1$ and $b_1+\cdots+b_4=0$. Since $b\neq 0$, the random variable $b_1\mathsf{X}_1+\cdots+b_4\mathsf{X}_4$ has a density, so

$$\mathbb{P}(b_1\mathsf{X}_1+\cdots+b_4\mathsf{X}_4=0)=0.$$

If $b_1\mathsf{X}_1+\cdots+b_4\mathsf{X}_4<0$, then $a^{(i)}\cdot\mathsf{X}\to-\infty$, so $g(\frac{4}{3}+a^{(i)}\cdot\mathsf{X})=1$ for $i\gg 0$. If $b_1\mathsf{X}_1+\cdots+b_4\mathsf{X}_4>0$, then $a^{(i)}\cdot\mathsf{X}\to+\infty$, so $g(\frac{4}{3}+a^{(i)}\cdot\mathsf{X})\to 0$, as $(1+y)e^{-y}\to 0$ when $y\to+\infty$. By the dominated convergence theorem,

$$\lim_{i\to+\infty}F(a^{(i)})=\frac{3}{4}e^{4/3}\mathbb{P}(b_1\mathsf{X}_1+\cdots+b_4\mathsf{X}_4<0).$$

The mean of $b_1\mathsf{X}_1+\cdots+b_4\mathsf{X}_4$ is $b_1+\cdots+b_4=0$, so [OAI26b, Lemma 4.5] implies that $\mathbb{P}(b_1\mathsf{X}_1+\cdots+b_4\mathsf{X}_4<0)\le 1-e^{-1}$. Thus

$$\lim_{i\to+\infty}F(a^{(i)})\le\frac{3}{4}e^{4/3}(1-e^{-1})=\frac{3}{4}(e^{4/3}-e^{1/3})<\frac{3}{4}\Big(\frac{19}{5}-\frac{4}{3}\Big)=\frac{37}{20}.$$

Here $e^{1/3}>1+\frac{1}{3}=\frac{4}{3}$, since all terms of the exponential series at $\frac{1}{3}$ are positive. For $j\ge 9$ we have $\frac{(4/3)^{j+1}}{(j+1)!}\le\frac{2}{15}\cdot\frac{(4/3)^j}{j!}$, so bounding the terms of degree at least 9 of the exponential series by a geometric series, we obtain

$$e^{4/3}\le\sum_{j=0}^{8}\frac{(4/3)^j}{j!}+\frac{(4/3)^9}{9!}\cdot\frac{1}{1-\frac{2}{15}}=\frac{917330371}{241805655}<\frac{19}{5}.$$

Both [OAI26b, Lemma 4.5] and [OAI26b, Lemma 4.7] concern only independent exponential random variables, and they have no geometric hypothesis (see Remark 6.6). By [OAI26b, Lemma 4.7] in four variables, any local maximum point $a\neq 0$ of the function $a\to\mathbb{E}g(\frac{4}{3}+a\cdot\mathsf{X})$ on the hyperplane $a_1+\cdots+a_4=0$ is, after a permutation of the coordinates, equal to $(t,-\frac{t}{3},-\frac{t}{3},-\frac{t}{3})$ for some $t>0$.

Let s be the supremum of F on the hyperplane $a_1+\cdots+a_4=0$. Then $s\le\frac{3}{4}e^{4/3}$ since $g\le 1$. Assume that $s>\frac{137}{72}$, and choose $a^{(1)},a^{(2)},\dots$ on the hyperplane with $F(a^{(i)})\to s$. If the sequence $a^{(i)}$ is unbounded, then, passing to a subsequence with $\|a^{(i)}\|\to+\infty$, the previous paragraphs imply that $s<\frac{37}{20}<\frac{137}{72}$, a contradiction. Thus the sequence $a^{(i)}$ is bounded. Passing to a subsequence, we may assume that $a^{(i)}$ converges to a point a^* of the hyperplane, and $F(a^*)=s$ since F is continuous. In particular, a^* is a local maximum point of F on the hyperplane, and $a^*\neq 0$ since $F(0)=\frac{7}{4}<\frac{137}{72}<s$. By [OAI26b, Lemma 4.7] in four variables, after a permutation of the coordinates, $a^*=(t,-\frac{t}{3},-\frac{t}{3},-\frac{t}{3})$ for some $t>0$. Since $\mathsf{X}_1,\dots,\mathsf{X}_4$ are independent and identically distributed, $s=F(a^*)=\Phi(t)$, where

$$\mathsf{Z}:=\mathsf{X}_1-\frac{1}{3}(\mathsf{X}_2+\mathsf{X}_3+\mathsf{X}_4),\qquad\Phi(t):=\frac{3}{4}e^{4/3}\mathbb{E}\Big[g\Big(\frac{4}{3}+t\mathsf{Z}\Big)\Big]\quad(t\ge 0).$$

Thus, if $\Phi(t)\le\frac{137}{72}$ for any $t>0$, then $F(a)\le\frac{137}{72}$ for any $a\in\mathbb{R}^4$ with $a_1+\cdots+a_4=0$.

Step 2. In this step we compute the density ψ of Z and a closed formula for $\Phi(t)$ when $0<t<3$.

The random variable $\frac{1}{3}(\mathsf{X}_2+\mathsf{X}_3+\mathsf{X}_4)$ is independent of X_1, and has density $\frac{27}{2}y^2e^{-3y}$ for $y>0$ and 0 for $y\le 0$. For $z\ge 0$,

$$\psi(z)=\int_0^{+\infty}\frac{27}{2}y^2e^{-3y}e^{-(z+y)}dy=\frac{27}{2}e^{-z}\int_0^{+\infty}y^2e^{-4y}dy=\frac{27}{64}e^{-z}.$$

For $z=-v$ with $v>0$,

$$\psi(-v)=\int_0^{+\infty}e^{-x}\frac{27}{2}(x+v)^2e^{-3(x+v)}dx=\frac{27}{2}e^{-3v}\int_0^{+\infty}(x+v)^2e^{-4x}dx=\frac{27}{64}(1+4v+8v^2)e^{-3v}.$$

Thus

$$\psi(z)=\begin{cases}\frac{27}{64}e^{-z}, & z\ge 0,\\ \frac{27}{64}(1-4z+8z^2)e^{3z}, & z<0.\end{cases}$$

Hence $\mathbb{P}(\mathsf{Z}<0)=1-\int_0^{+\infty}\psi(z)dz=\frac{37}{64}$. Moreover, since $\mathsf{X}_1,\dots,\mathsf{X}_4$ are independent with mean 1 and variance 1, we have $\mathbb{E}\mathsf{Z}=0$ and $\mathbb{E}\mathsf{Z}^2=1+\frac{3}{9}=\frac{4}{3}$.

Let $t>0$. Since $\frac{4}{3}+tz\le 0$ if and only if $z\le -\frac{4}{3t}$, we have

$$g\Big(\frac{4}{3}+tz\Big)=\begin{cases}1, & z\le -\frac{4}{3t},\\ \big(\frac{7}{3}+tz\big)e^{-4/3}e^{-tz}, & z>-\frac{4}{3t}.\end{cases}$$

We split $\mathbb{E}g(\frac{4}{3}+t\mathsf{Z})$ into the integrals over $z<-\frac{4}{3t}$, over $-\frac{4}{3t}\le z<0$, and over $z\ge 0$, and we substitute $z=-v$ in the first two. Since $\frac{3}{4}\cdot\frac{27}{64}=\frac{81}{256}$ and

$$\int_{4/(3t)}^{+\infty}(1+4v+8v^2)e^{-3v}dv=e^{-4/t}\Big(\frac{8}{3}\Big(\frac{4}{3t}\Big)^2+\frac{28}{9}\cdot\frac{4}{3t}+\frac{37}{27}\Big),$$

we obtain

$$\begin{aligned}\Phi(t)=\frac{81}{256}\Big[&e^{4/3-4/t}\Big(\frac{8}{3}\Big(\frac{4}{3t}\Big)^2+\frac{28}{9}\cdot\frac{4}{3t}+\frac{37}{27}\Big)\\&+\int_0^{4/(3t)}\Big(\frac{7}{3}-tv\Big)(1+4v+8v^2)e^{-(3-t)v}dv\\&+\frac{7}{3(1+t)}+\frac{t}{(1+t)^2}\Big],\end{aligned}$$

where the three summands in the bracket are the contributions of $z<-\frac{4}{3t}$, of $-\frac{4}{3t}\le z<0$, and of $z\ge 0$, respectively. Now let $0<t<3$. Integrating each term $v^je^{-(3-t)v}$, $0\le j\le 3$, a direct calculation shows that

$$\Phi(t)=R(t)+J(t)e^{4/3-4/t},$$

where

$$R(t)=-\frac{27(19t^2-14t-21)}{4(t-3)^4(t+1)^2},\qquad J(t)=\frac{3(37t^4-278t^3+341t^2+1392t+1152)}{256(t-3)^4}.$$

For the values at $t=0$ and $t=3$, we have $\Phi(0)=F(0)=\frac{7}{4}$, and the formula for $\Phi(t)$ with the integral at $t=3$ implies that $\Phi(3)=\frac{595655}{331776}$. Since $\frac{7}{4}=\frac{35}{20}$ and

$$5\cdot 595655=2978275<2985984=9\cdot 331776,$$

we have

$$\Phi(0)<\frac{37}{20},\qquad \Phi(3)<\frac{9}{5}<\frac{37}{20}.$$

Step 3. In this step we prove that $\Phi(\frac{k}{3})<\frac{37}{20}$ for $k=1,\dots,8$ by finitely many exact rational inequalities.

For $k=1,\dots,8$, let

$$\theta_k:=\frac{12}{k}-\frac{4}{3},$$

and write $\theta_k=\sigma_k/\tau_k$ with coprime positive integers σ_k and τ_k. Then $\theta_k>0$ and $\frac{4}{3}-\frac{4}{k/3}=-\theta_k$, so **Step 2** implies that

$$\Phi\Big(\frac{k}{3}\Big)=R\Big(\frac{k}{3}\Big)+J\Big(\frac{k}{3}\Big)e^{-\theta_k}.$$

Let $P_{16}(y):=\sum_{j=0}^{16}y^j/j!$. In Table 2, the second column lists 10^5U_k, so each U_k has five decimal places, and we show below that $e^{-\theta_k}<U_k$. The positive integers A_k and B_k enter the first two inequalities below through the terms $A_k/10^8$ and $B_k/10^6$. Only $A_k>0$ and $B_k>0$ are used.

For $k=1,\dots,8$, the numbers in Table 2 satisfy the exact rational inequalities

$$U_kP_{16}(\theta_k)\ge 1+\frac{A_k}{10^8},\qquad \frac{37}{20}-\Big(R\Big(\frac{k}{3}\Big)+J\Big(\frac{k}{3}\Big)U_k\Big)\ge\frac{B_k}{10^6},\qquad J\Big(\frac{k}{3}\Big)>0.$$

k	$10^5 U_k$	θ_k	A_k	B_k
1	3	32/3	22964058	81327
2	941	14/3	67619	47562
3	6949	8/3	9424	27930
4	18888	5/3	2328	22492
5	34416	16/15	1805	24873
6	51342	2/3	561	30962
7	68322	8/21	1401	36545
8	84649	1/6	977	16206

TABLE 2. The numbers U_k, θ_k, A_k, and B_k for the nodes $t = \frac{k}{3}$, $k = 1, \dots, 8$.

Since

$$P_{16}\Big(\frac{\sigma_k}{\tau_k}\Big) = \frac{\sum_{j=0}^{16} \frac{16!}{j!}\sigma_k^j \tau_k^{16-j}}{16!\tau_k^{16}},$$

after clearing the positive denominator, the first inequality is equivalent to the following inequality of integers

$$10^8(10^5 U_k)\sum_{j=0}^{16}\frac{16!}{j!}\sigma_k^j\tau_k^{16-j} \geq 10^5(10^8 + A_k)16!\tau_k^{16}.$$

The second and the third inequalities are inequalities of rational numbers, since $R(\frac{k}{3})$ and $J(\frac{k}{3})$ are given by the rational functions R and J of **Step 2** at the rational point $\frac{k}{3}$.

Since $\theta_k > 0$, all terms of the exponential series at θ_k are positive, so $e^{\theta_k} > P_{16}(\theta_k)$. Thus

$$U_k e^{\theta_k} > U_k P_{16}(\theta_k) \geq 1 + \frac{A_k}{10^8} > 1,$$

that is, $e^{-\theta_k} < U_k$. Since $J(\frac{k}{3}) > 0$, we obtain

$$\Phi\Big(\frac{k}{3}\Big) < R\Big(\frac{k}{3}\Big) + J\Big(\frac{k}{3}\Big)U_k \leq \frac{37}{20} - \frac{B_k}{10^6} < \frac{37}{20}.$$

Step 4. In this step we bound Φ between the nodes $t = \frac{k}{3}$, $k = 0, \dots, 9$, and prove that $\Phi(t) < \frac{137}{72}$ for $0 \leq t \leq 3$.

The function g is C^1, with $g'(y) = 0$ for $y \leq 0$ and $g'(y) = -ye^{-y}$ for $y > 0$. For $y > 0$, $g''(y) = (y-1)e^{-y}$, and

$$|(y-1)e^{-y}| \leq \max\{e^{-y}, ye^{-y}\} \leq 1.$$

Thus $|g'(y) - g'(y')| \leq |y - y'|$ for any $y, y' \in \mathbb{R}$. Since g' is bounded and $\mathbb{E}|\mathsf{Z}| \leq (\mathbb{E}\mathsf{Z}^2)^{1/2} < +\infty$, we may differentiate under the expectation, and $\Phi'(t) = \frac{3}{4}e^{4/3}\mathbb{E}[\mathsf{Z}g'(\frac{4}{3} + t\mathsf{Z})]$ for $t \geq 0$. Let $L := e^{4/3}$. For $s, s' \geq 0$,

$$|\Phi'(s') - \Phi'(s)| \leq \frac{3}{4}e^{4/3}\mathbb{E}\big[|\mathsf{Z}| \cdot |s' - s||\mathsf{Z}|\big] = \frac{3}{4}e^{4/3}\mathbb{E}\mathsf{Z}^2|s' - s| = L|s' - s|,$$

and $L < \frac{19}{5}$ by **Step 1**.

Let $k \in \{0, \dots, 8\}$, let $t_0 := \frac{k}{3}$ and $t_1 := \frac{k+1}{3}$, let ℓ be the affine function with $\ell(t_0) = \Phi(t_0)$ and $\ell(t_1) = \Phi(t_1)$, and let

$$\delta(t) := \Phi(t) - \ell(t) - \frac{L}{2}(t - t_0)(t_1 - t), \qquad t \in [t_0, t_1].$$

Then $\delta(t_0) = \delta(t_1) = 0$, and for $t_0 \leq s < s' \leq t_1$,

$$\delta'(s') - \delta'(s) = \Phi'(s') - \Phi'(s) + L(s' - s) \geq -L(s' - s) + L(s' - s) = 0.$$

Thus δ' is non-decreasing, so δ is convex on $[t_0, t_1]$, and $\delta \le 0$ on $[t_0, t_1]$. Since $t_1 - t_0 = \frac{1}{3}$, we have $(t-t_0)(t_1-t) \le \frac{1}{4}\cdot(\frac{1}{3})^2$, so for $t \in [t_0, t_1]$,

$$\Phi(t) \le \ell(t) + \frac{L}{2}(t-t_0)(t_1-t) \le \max\{\Phi(t_0), \Phi(t_1)\} + \frac{L}{8}\cdot\Big(\frac{1}{3}\Big)^2.$$

By **Steps 2** and 3, $\Phi(\frac{k}{3}) < \frac{37}{20}$ for $k = 0, \dots, 9$. Since $L < \frac{19}{5}$, we obtain

$$\Phi(t) < \frac{37}{20} + \frac{19/5}{72} = \frac{137}{72} \qquad \text{for } 0 \le t \le 3.$$

Step 5. In this step we prove that $\Phi(t) < \frac{137}{72}$ for $t \ge 3$.

Let $t \ge 3$. We bound g by 1 on $\mathsf{Z} < 0$, and we integrate exactly on $\mathsf{Z} \ge 0$ as in **Step 2**. Since $\mathbb{P}(\mathsf{Z} < 0) = \frac{37}{64}$ and $\frac{3}{4}\cdot\frac{37}{64} = \frac{111}{256}$, we obtain

$$\Phi(t) \le \frac{111}{256}e^{4/3} + \frac{81}{256}\Big(\frac{7}{3(1+t)} + \frac{t}{(1+t)^2}\Big) < \frac{111}{256}\cdot\frac{19}{5} + \frac{81}{256}\Big(\frac{7}{3(1+t)} + \frac{t}{(1+t)^2}\Big).$$

For $t \ge 1$,

$$\frac{d}{dt}\Big(\frac{7}{3(1+t)} + \frac{t}{(1+t)^2}\Big) = -\frac{7}{3(1+t)^2} + \frac{1-t}{(1+t)^3} < 0,$$

and $\frac{7}{3\cdot 4} + \frac{3}{16} = \frac{37}{48}$. Thus for $t \ge 3$,

$$\Phi(t) < \frac{111}{256}\cdot\frac{19}{5} + \frac{81}{256}\cdot\frac{37}{48} = \frac{38739}{20480} < \frac{19}{10} < \frac{137}{72}.$$

By **Steps 4** and 5, $\Phi(t) < \frac{137}{72}$ for any $t \ge 0$. Thus **Step 1** implies that $F(a) \le \frac{137}{72}$ for any $a \in \mathbb{R}^4$ with $a_1 + \cdots + a_4 = 0$, which is the inequality of Lemma 6.9. □

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